

Prepladder Version X Qbank

Subject: Psychiatry

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Basics of Psychiatry.

1. A 30-year-old male presented with complaints of decreased energy, feeling low and sad for most of the time since the last 3 months. The patient says that he occasionally feels anxious during certain situations. Which of the following terms best describes the patient's presentation?

- A. Mood
- B. Affect
- C. Abstract reasoning
- D. Perceptual disturbances

2. Choose the correct options regarding the statements given below Statement-1: Insight and judgment are intact in neurosis. Statement-2: Intact reality testing is seen in psychosis.

- A. Statement-1 is true and statement-2 is false
- B. Statement-1 is false and statement-2 is true
- C. Both statements are true
- D. Both statements are false

3. Which method is often used in clinical settings to evaluate recent memory?

- A. 24-Hour Recall Method
- B. Serial 7s Subtraction Method
- C. Mini-Mental State Examination (MMSE)
- D. Digit Span Test

4. Which test is commonly done to assess concentration in a person?

- A. Digit Span Test
 - B. Digit forward test
 - C. Serial 7s Subtraction Test
 - D. 24-Hour recall method
-

5. Which of the following is an example of a delusion, not a hallucination?

- A. Hearing voices telling you to harm yourself
 - B. Believing that you are being followed by secret agents without any evidence
 - C. Seeing people who are not present
 - D. Smelling smoke when there is none
-

6. Which of the following statements on obsession is false?

- A. Persistent thought
 - B. Causes great anxiety or guilt
 - C. Often repugnant to the individual
 - D. Individual is unaware that the thought is irrelevant
-

7. A 40-year-old female has come to a psychiatrist for low mood. This is her first visit. The psychiatrist decides to evaluate her mental function by mental status examination. Which among the following are components of this 1) Speech 2)

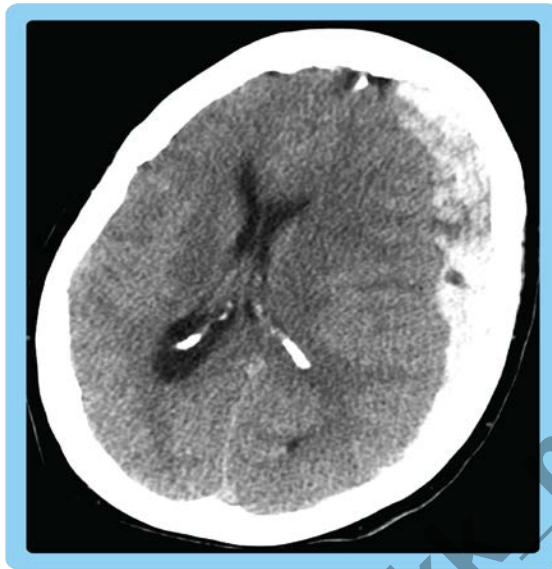
Mood and affect 3) Thought content 4) Judgment 5) Abstract reasoning 6) Motor activity

- A. 1,2,3
 - B. 2,4,5
 - C. 1,2,3,4,5
 - D. 1,2,3,4,5,6
-

8. A 45-year-old man is admitted to the hospital after experiencing disorientation and confusion. He describes when he looks in the mirror, he cannot see his own reflection. This phenomenon is best described as which of the following?

- A. Negative autoscopy
 - B. Internal autoscopy
 - C. Extracampine hallucination
 - D. Hypnagogic hallucination
-

9. A 10-year boy was brought by his mother to a psychiatric clinic. She says that the boy has been complaining of headaches and behaves abnormally saying that he stays silent most of the time and becomes violent occasionally for the last 1 month, his academic grades have dropped drastically, and his teachers have complained about his aggressive behavior towards other children in school. There is a history of falls a few months back for which the clinician ordered a CT brain and the report is as follows What group of disorders would you classify this patient's condition under?



- A. Personality disorder
- B. Organic brain disorder
- C. Functional neurotic disorder
- D. Functional psychotic disorder

10. Choose the appropriate match regarding the multiaxial system of DSM 1)
Assessment of patient functioning a) Axis I 2) Mental retardation b) Axis II 3)
Diabetes mellitus c) Axis III 4) Schizophrenia d) Axis V

1) Assessment of patient functioning	a) Axis I
2) Mental retardation	b) Axis II
3) Diabetes mellitus	c) Axis III
4) Schizophrenia	d) Axis V

- A. 1a 2b 3c 4d
- B. 1c 2d 3b 4a
- C. 1c 2d 3a 4b

D. 1d 2b 3c 4a

11. Match the following clinical scenarios with their appropriate grade of insight

1) Grade-I a) A 60 year male who has been diagnosed with narcolepsy, says that he has this condition because he lives in Shimla.

2) Grade-II b) 40 year old male who was a chronic smoker decided to give up smoking, attends rehabilitation sessions, takes medications and is focused on improving his lifestyle.

3) Grade-III c) Female patient says that she is a known case of bipolar disorder, but refuses to take medications.

4) Grade-IV d) 25 year old female who has been admitted to a psychiatric ward for schizophrenia says that she has come to the hospital for fever and cough and denies any psychiatric illness.

5) Grade-V

1) Grade-I	a) A 60 year male who has been diagnosed with narcolepsy, says that he has this condition because he lives in Shimla.
2) Grade-II	
b) 40 year old male who was a chronic smoker decided to give up smoking, attends rehabilitation sessions, takes medications and is focused on improving his lifestyle.	
3) Grade-III	
c) Female patient says that she is a known case of bipolar disorder, but refuses to take medications.	
4) Grade-IV	

d) 25 year old female who has been admitted to a psychiatric ward for schizophrenia says that she has come to the hospital for fever and cough and denies any psychiatric illness.	
5) Grade-V	

- A. 1d 3a 4c 5b
- B. 1b 2c 4a 5d
- C. 1d 2a 4c 5b
- D. 1b 3a 4c 5d

12. The clinician asked a patient 'You see a person bleeding profusely on the roadside, what would you do?', the patient replied 'I would make another cut with a knife on him'. What type of judgment is being evaluated here and what is the likely pathology?

- A. Impairment of social judgment - depression
- B. Impairment of personal judgment - mania
- C. Impairment of test judgment - schizophrenia
- D. Impairment of social judgment - schizophrenia

13. Which of the following tasks is most commonly used to assess abstract thinking abilities?

- A. Raven's Progressive Matrices
 - B. Digit Span Test
 - C. Serial 7s Subtraction Test
 - D. 24-Hour Recall Method
-

14. On psychiatric evaluation, a 30-year-old female when asked about what she did that morning. Said “ I went to a supermarket, the supermarket's logo was in yellow color, which reminded me of the sun. I like the sun a lot more than the moon. I don't understand how the astronauts are able to go to the moon with such pathetic costumes.....” went on to talk about it for a long period in a rapid pressurized manner. What term best describes the patient's presentation?

- A. Clang association
 - B. Circumstantiality
 - C. Loosening of association
 - D. Flight of ideas
-

15. Correctly match the following statements with True(T)/ False(F) 1) Continuity is disrupted in tangentiality 2) Goal is not reached in circumstantiality 3) Perseveration is associated with organic mental disorder 4) Continuity is preserved in perseveration 5) Thought blocking is associated with schizophrenia

- A. 1-F,2-T,3-F,4-T,5-F
 - B. 1-T,2-F,3-T,4-F,5-T
 - C. 1-F,2-F,3-T,4-F,5-T
 - D. 1-T,2-F,3-T,4-T,5-T
-

16. A 30-year-old female was referred to a psychiatry clinic with complaints of not taking food for the past one week. When asked about it, the patient said 'my intestines do not exist, How can I eat?'. She has been experiencing low mood, unable to sleep and decreased interest in surroundings. What is the content of delusion and its corresponding type seen in this patient?

- A. Cotard syndrome - mood congruent
- B. Cotard syndrome - mood incongruent
- C. Nihilistic delusion - mood congruent
- D. Nihilistic delusion - mood incongruent

17. A 35 y/o man insists that his wife has been replaced by an identical imposter, despite her behavior and appearance remaining unchanged. He is convinced that the person living with him is not his real wife. What syndrome is he most likely experiencing?

- A. Fregoli syndrome
- B. Capgras syndrome
- C. Reduplicative paramnesia
- D. Subjective doubles syndrome

18. Which of the following statements best describes formication?

- A. Feeling of small insects crawling on or under their skin
- B. Pricking sensation on the skin
- C. Feeling of limbs being moved
- D. A feeling of floating

19. A man hears a voice commanding him to lock all the doors in his house immediately, warning that he is in danger. The voice is persistent and direct, addressing him personally, saying, "You must lock the doors now, or something terrible will happen." Despite no visible threat, the man feels compelled to follow the voice's instructions. What is the diagnosis?

- A. First-person auditory hallucination
- B. Second-person auditory hallucination
- C. Third-person auditory hallucination
- D. Fourth-person auditory hallucination

20. Match the following: 1. Illusion a. Hearing voices that are perceived as coming from other people when no one is speaking 2. Hallucination b. Hearing voices inside one's head but recognizing them as not being from external sources. 3. Pseudo-hallucination c. Mistaking a shadow for an attacker on a dark road.

1. Illusion	a. Hearing voices that are perceived as coming from other people when no one is speaking
2. Hallucination	b. Hearing voices inside one's head but recognizing them as not being from external sources.
3. Pseudo-hallucination	c. Mistaking a shadow for an attacker on a dark road.

- A. 1c,2a,3b
- B. 1a,2b,3c
- C. 1c,2b,3a
- D. 1b,2c,3a

21. Which of the following best describes the concept of thought alienation?

- A. The individual recognizes that their obsessive thoughts are their own but finds them difficult to control
 - B. The individual experiences their thoughts as being controlled or influenced by an external force, with thoughts being inserted, withdrawn, or broadcasted to others
 - C. The individual has intrusive thoughts that are consistent with their self-image and desires
 - D. The individual experiences sudden lapses in memory, forgetting their thoughts and actions entirely
-

22. A 42 y/o woman is convinced she has cancer despite normal test results and reassurance from multiple doctors. She frequently searches online for symptoms and insists that her condition has been overlooked, even though there is no clinical evidence of illness. What is the probable diagnosis of this patient?

- A. Conversion Disorder
 - B. Delusion of reference
 - C. Hypochondriacal delusion
 - D. Somatic symptom disorder
-

23. A 45-year-old patient is asked, "What is the color of grass?" and responds, "White." When asked, "What is the color of snow?" the patient replies, "Green." This type of response, where the patient understands the question but provides incorrect yet related answers, is most likely an example of which phenomenon?

- A. Loosening of associations
 - B. Word Approximation
 - C. Word salad
 - D. Vorbeireden
-

24. A 30 y/o man with schizophrenia is noted to use the word "skycar" to describe an airplane and "watercar" to refer to a boat. He also invents new words when trying to explain his experiences. Which of the following best describes his speech pattern?

- A. Paraphasia
 - B. Neologism
 - C. Malapropism
 - D. Word Approximation
-

25. A psychiatrist asks a patient, "Have you been feeling anxious lately?" The patient responds, "I've been going on walks in the park and enjoying the weather." What speech pattern is this an example of?

- A. Circumstantiality
 - B. Tangentiality
 - C. Neologism
 - D. Loosening of association
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	1
Question 3	1
Question 4	3
Question 5	2
Question 6	4
Question 7	4
Question 8	1
Question 9	2
Question 10	4
Question 11	1
Question 12	3
Question 13	1
Question 14	4
Question 15	3
Question 16	3
Question 17	2
Question 18	1

Question 19	2
Question 20	1
Question 21	2
Question 22	3
Question 23	4
Question 24	2
Question 25	2

Solution for Question 1:

Correct answer: A) Mood

Explanation:

Feeling 'low', 'sad', 'anxious' (internal subjective experiences) by the patient is best described by the term "Mood".

Mood & Affect:

Mood (Option A)	Affect (Option B)
Subjective experience of patient's pervasive and persistent emotional state.	Expression of mood (emotions) in terms of facial expressions and body language.
Internal phenomenon	External phenomenon
Told by the patient	Experienced by the therapist
Long-term: persistent and sustained	Cross-sectional: observed at that moment
Example : A patient reports feeling sad and hopeless for weeks, indicating a depressed mood.	Example : The patient appears tearful, withdrawn, and downcast during a therapy session, even though they verbally describe feeling "okay."

Abstract reasoning (Option C) is the ability of a person to think conceptually and comprehend rather than relying on concrete examples. There is no such presentation in the given scenario.

Perceptual disturbances (Option D) are the disturbances in perception, such as illusions, hallucinations, etc. Feeling low/anxious (in the given scenario) does not fall into this term.

Solution for Question 2:

Correct answer: A) Statement-1 is true and statement-2 is false

Explanation:

Correct statements are:

- Insight and judgment are intact in neurosis. (Statement-1 true)
- Reality testing is absent in psychosis. (Statement-2 false)

Functional disorders are further classified into:

	Neurosis	Psychosis
Insight and Judgment	Intact	Impaired
Personality	Preserved	Affected
Reality testing	Present	Absent
Delusions/hallucinations	Absent	Present
Examples	Depression Obsessive compulsive disorder (OCD) Anxiety disorders.	Delusional disorders Schizophrenia Depression with psychosis Mania

Solution for Question 3:

Correct Answer: A) 24-Hour Recall Method

Explanations:

24-Hour Recall Method: This method involves asking patients to recall events, activities, or conversations that occurred within the last 24 hours, effectively assessing recent memory function.

Types of memory

Immediate memory/ Short-term memory	Recent Memory	Remote Memory
The ability to hold and manipulate information temporarily while performing cognitive tasks.	The ability to recall information or events that have occurred recently, typically within the last few days or weeks.	The ability to recall information or events from the distant past, often years or decades ago.
Seconds	Minutes, hours to days	Months, Years to a lifetime
Eg- Solving a math problem in your head, remembering a phone number long enough to dial it.	Eg- Recalling a conversation from earlier today or remembering what you ate for breakfast.	Eg- Remembering childhood experiences or significant historical events.
Test for concentration and attention to assess immediate memory. Digit Span Test, n-back task, and mental arithmetic tasks.	24-Hour Recall Method, Rey Auditory Verbal Learning Test (RAVLT).	Autobiographical memory questioning, asking personal information.

- **Serial 7s Subtraction Method (Option B):** This method assesses concentration and attention by having patients repeatedly subtract 7 from 100 but does not evaluate recent memory or recall of past events.
- **Mini-Mental State Examination (MMSE) (Option C):** General cognitive assessment tool that screens for various cognitive impairments but does not focus specifically on recent memory.
- **Digit Span Test (Option D):** This test assesses short-term and working memory by requiring individuals to repeat back sequences of numbers but does not evaluate recent memory related to past events.

Solution for Question 4:

Correct Answer: C) Serial 7s Subtraction Test

Explanation:

Serial 7s Subtraction Test: This test assesses concentration by requiring participants to continuously subtract 7 from 100, focusing on sustained attention and mental arithmetic.

Concentration	Orientation	Attention
Cognitive ability to focus one's mental efforts on a specific task, stimulus, or activity for an extended period while ignoring irrelevant information or distractions.	Awareness of oneself in relation to time, place, and person.	Cognitive process of selectively concentrating on specific aspects of the environment while ignoring others.
It involves: Sustained mental focus Efficient information processing Resisting cognitive fatigue	Components : Temporal Orientation: Understanding the time. Spatial Orientation: Awareness of one's position in relation to the environment. Personal Orientation: Recognition of one's identity.	It involves Focusing on relevant stimuli Sustaining concentration over time Dividing focus between multiple tasks Shifting focus when needed
Assessment method: Serial 7s Subtraction Method: Procedure : The subject starts with a number (typically 100). They subtract 7 repeatedly, stating each result aloud. This continues for a set number of subtractions or a specific time period. Example sequence: 100, 93, 86, 79, 72, 65.	Assessment : Often evaluated through direct questioning and observation. Questioning like: What is your name? Who am I? What place is this? Where is it located?	Assessment Methods: Continuous Performance Test (CPT) Trail Making Test Stroop Color and Word Test Digit Span Test- Digit forward (option B) and Digit backward test.

- Digit Span Test (Option A): This test evaluates short-term memory by requiring individuals to repeat back sequences of numbers, but it does not specifically focus on concentration.
- 24-Hour Recall Method (Option D): This method assesses memory by asking individuals to recall everything they ate and did over the past 24 hours.

Solution for Question 5:

Correct Answer: B) Believing that you are being followed by secret agents without any evidence

Explanation:

Delusions are fixed, false beliefs (like being followed by secret agents) that persist despite evidence to the contrary.

Delusion Vs Hallucination

Aspect	Delusions	Hallucinations
Definition	Fixed, false beliefs that are resistant to reasoning or contrary evidence.	False sensory perceptions without external stimuli.
Nature	Cognitive in nature; involves thought processes.	Sensory in nature; involves the senses (sight, sound, touch, etc.)
Examples	Believing one is being persecuted (delusion of persecution) or is famous (grandeur).	Hearing voices, seeing things that are not present, or feeling sensations on the skin.
Perception of Reality	Individuals may be convinced of their beliefs despite clear evidence against them.	Individuals experience sensations as real, even though they are not.
Associated Conditions	Commonly seen in schizophrenia, delusional disorder.	Common in schizophrenia, psychotic disorders, neurological conditions.

Hallucinations involve false sensory perceptions, such as hearing voices (Option A), seeing people (Option C), or smelling smoke (Option D) without any external stimuli.

Solution for Question 6:

Correct Answer: D) Individual is unaware that the thought is irrelevant

Explanation:

The individual experiencing obsessions is aware that the thought is irrelevant

Obsession/ rumination:

- Persistent thought that dominates one's thinking (Option A)
- The individual is aware that the thought is purposeless or irrelevant, but it continues to dominate their mind (Option D)
- Often cause great anxiety or guilt due to their content (Option B)

- The content is usually repugnant to the individual, such as sexual thoughts tormenting a prudish person or blasphemous thoughts disturbing a religious person (Option C)

Solution for Question 7:

Correct answer: D) 1,2,3,4,5,6

Explanation:

Mental status examination (MSE) is the psychiatric equivalent of general physical examination in the rest of medicine. All the above-mentioned parameters are components of MSE.

Mental Status Examination (MSE):

- The Mental Status Examination (MSE) is the psychiatric counterpart to a physical examination used in the rest of medicine.
- It examines all aspects of mental functioning and identifies signs and symptoms indicative of mental illness. It provides a “snapshot” of the patient’s mental state at the time of the assessment.
- Cognitive evaluation in MSE is usually done in the form of a Mini Mental Status Examination (MMSE).

Aspects of MSE:

General description:
Appearance and behavior Motor activity (Statement-6) Speech (Statement-1)
Emotional state:
Mood: subjective experience of patient’s internal and sustained emotional state Affect: expression of mood (that appears to the clinician) (Statement-2)
Thinking:
Thought content (Statement-3) Thought process Perceptual disturbances
Cognition (higher mental function): most commonly evaluated by Mini Mental Status Examination (MMSE)

General description:

Orientation
Registration
Attention & calculation
Memory (immediate, recent & long term)
Abstract reasoning (Statement-5)
Insight & Judgement (Statement-4)

Solution for Question 8:

Correct Answer: A) Negative autoscopy

Explanation:

Negative autoscopy involves the inability to see one's reflection despite being present before the mirror.

Hallucinations are false sensory perceptions that occur in the absence of any external stimulus.

Special type of hallucinations

Type of Hallucination	Description	Associated Conditions
Reflex Hallucination	A hallucination in one sensory modality triggered by a stimulus in another sensory modality (e.g- hearing a sound when seeing a light)	Sensory processing disorders
Functional Hallucination	A real external stimulus coexists with a hallucination in the same modality (e.g hearing voices while listening to real external sounds)	Schizophrenia, psychosis
Extracampine Hallucination (Option C)	Perception of an object or sensation outside the normal sensory field (e.g hearing someone talking behind you when no one is present)	Schizophrenia

Type of Hallucination	Description	Associated Conditions
Autoscopy	Seeing one's own body or image from an external perspective. Also known as Phantom mirror-image Negative autoscopy - Not seeing one's reflection in the mirror being present. Internal autoscopy - person can see his own internal organs.(Option B)	Can occur in schizophrenia, but more common in delirium. Other condition: Temporal lobe epilepsy
Hypnagogic Hallucination (Option D)	Hallucinations occurring during the transition from wakefulness to sleep.	Narcolepsy
Hypnopompic Hallucination	Hallucinations occurring during the transition from sleep to wakefulness.	Narcolepsy
Organic Hallucinations	Hallucinations can occur in any sensory modality and are caused by an identifiable physical condition.	Neurological disorders, substance abuse, brain lesions. Charles Bonnet syndrome: consists of visual hallucinations (no other psychiatric symptom) and reduced vision due to eye disease.

Solution for Question 9:

Correct answer: B) Organic brain disorder

Explanation:

In the given clinical scenario, the patient's symptoms are attributed to Subdural hematoma (concavo-convex shaped opacity in CT Brain) secondary to fall. Symptoms owing to a demonstrable/apparent cause are grouped under Organic brain disorders.

Previously psychiatric disorders were broadly divided into:

Organic disorders:	Psychiatric symptoms following a demonstrable/apparent cause. Most common cause: metabolic encephalopathies (presents with altered sensorium/behavior) Routine investigations to be done: CT/MRI Brain, serum investigations, CSF analysis (if required) History of head injury may be present in cases of hematoma, etc.
Functional disorders:	Psychiatric symptoms are present without any demonstrable apparent cause. More commonly seen. Further classified into two categories: Neurosis: intact insight and judgment. Psychosis: impaired insight and judgment.

Solution for Question 10:

Correct answer: D) 1d 2b 3c 4a

Explanation:

1) Assessment of patient functioning	d) Axis V
2) Mental retardation	b) Axis II
3) Diabetes mellitus	c) Axis III
4) Schizophrenia	a) Axis I

DSM-IV used dimensional measures to assess disorder thresholds and rate severity.

Axes of DSM (according to DSM-IV):

Axes	Description	Example
I	Psychiatric disorders	Schizophrenia
II	Personality disorders and mental retardation	All personality disorders
III	General medical conditions	Diabetes mellitus

Axi s	Description	Example
IV	Psychosocial/environmental problems	Patient going through a divorce
V	Global assessment of patient functioning	Score from 0 to 100 (0 is the minimum functioning and 100 is the highest functioning)

Recent update:

DSM-V removed the multiaxial system (described in DSM-IV) and is no longer used. Instead of Axis-V, WHO Disability Assessment is used.

Solution for Question 11:

Correct answer: A) 1d 3a 4c 5b

Explanation:

- A 60 year male who has been diagnosed with narcolepsy, says that he has this condition because he lives in Shimla (Statement-a): Here the patient is aware of the illness but attributes it to unrelated external factors - Grade-III.
- A 40 year male who was a chronic smoker decided to give up smoking, attends rehabilitation sessions, takes medications and is focused on improving his lifestyle (Statement-b): Patient is aware that the illness is due to his own irrational feelings, decided to change and take steps accordingly - Grade-V.
- Female patient says that she is a known case of bipolar disorder, but refuses to take medications (Statement-c): Patient is aware that the illness is due to his own irrational feelings, but refuses to make a change regarding the same - Grade-IV.
- A 25 year old female who has been admitted to a psychiatric ward for schizophrenia says that she has come to the hospital for fever and cough and denies any psychiatric illness (Statement-d): Patient denies any psychiatric illness completely - Grade-I.

Grades of insight:

Grade-I	Complete denial of illness
Grade-I I	Slight awareness of being ill and denying at the same time
Grade-I II	Awareness of illness is present, but the patient attributes the disease to other external/medical/unknown organic factors.
Grade-I V	Awareness of illness is due to his/her own irrational feelings or emotions but the patient refuses to apply this knowledge to change his/her behavior. Also known as intellectual insight.
Grade-V	Awareness of illness is due to his/her own irrational feelings or emotions and the patient agrees to apply this knowledge to change his/her behavior.

Solution for Question 12:

Correct answer: C) Impairment of test judgment - schizophrenia

Explanation:

In this question, a hypothetical scenario is being evaluated. This type of testing a patient in imaginary situations is known as Test judgment . Impairment of test judgment is seen in Schizophrenia.

Types of judgments:

Type	Description	Example
Social	Social behavior of the patient in day-to-day routine events . Contrary to acceptable behavior in culture.	Harmful behavior, like rash driving, fighting without a reason, or other unacceptable behavior.

Type	Description	Example
Personal	The ability of a patient to take care of one's own self. Categorized under social judgment.	If a patient says to start working again and continue with his normal life after discharge from the hospital suggests intact personal judgment. On the contrary, if he says he will buy a gun and start shooting people, indicates impaired personal judgment.
Test	Patient's prediction of what they would do in hypothetical situations. (Option C) An imaginary scenario is asked by the clinician and the reply is assessed.	The clinician asked a patient 'You see a person bleeding profusely on the roadside, what would you do?', the patient replied 'I would make another cut with a knife on him' indicates impaired test judgment, seen in Schizophrenia. (Option C)

Solution for Question 13:

Correct Answer: A) Raven's Progressive Matrices

Explanation:

Raven's Progressive Matrices : This test evaluates abstract reasoning and the ability to identify patterns and relationships among visual stimuli, making it a standard assessment for abstract thinking.

Abstract Thinking	Intelligence
The ability to understand concepts that are not tied to concrete objects or experiences; involves reasoning about ideas, principles, and relationships.	Broader construct that encompasses various cognitive abilities, including reasoning, problem-solving, learning, and adaptation to new situations.
Components: Pattern recognition Problem-solving skills Conceptualization Logical reasoning	Components: Logical reasoning Verbal comprehension Quantitative reasoning Spatial visualization

Abstract Thinking	Intelligence
Assessment Methods: Raven's Progressive Matrices Proverb testing Similarity testing	Assessment Methods: Wechsler Adult Intelligence Scale (WAIS) Stanford-Binet Intelligence Scales
Example : When asked the meaning of the proverb, "A stitch in time saves nine," a person explains that it means solving a small problem early can prevent it from becoming a bigger issue. This shows an ability to grasp the underlying concept rather than just the literal meaning.	Example : Navigating daily life challenges by adapting, learning, and making decisions also reflects intelligence.

- Digit Span Test (Option B): This test primarily focuses on assessing short-term and working memory rather than abstract thinking abilities.
- Serial 7s Subtraction Test (Option C): This method assesses concentration and attention by having participants subtract 7 from 100, not abstract reasoning.
- 24-Hour Recall Method (Option D): This method evaluates recent memory by asking patients to recall events or activities from the past 24 hours, not abstract thinking.

Solution for Question 14:

Correct answer: D) Flight of ideas

Explanation:

The given clinical scenario where the patient talks in a rapid hurried manner with thoughts seeming to move from idea to idea rapidly is suggestive of a "flight of ideas".

Disorders of the flow of thought - thought tempo/speed increased:

1) Flight of ideas: (Option D)

- Disorder of flow of thought with increased tempo/speed
- Succession of multiple associations of thoughts following each other rapidly.
- Goal is not usually reached . However, both the words and sentences make sense .
- Speech is rapid, pressurized but still partially understandable .
- Seen in mania, schizophrenia, and organic mental disorders .

- Clang association (Option A): form of Flight of Ideas but the words in the sentences are rather connected by rhyming . Eg. 'How high? I fly, look at the sky, feeling shy'.

2) Prolivity:

- Milder form of Flight of Ideas, where the speech is less rapid and better understandable.
- Also known as ordered flight of ideas.
- Goal is reached.
- Seen in hypomania.

Circumstantiality (Option B) : Disorder of flow of thought with tempo decreased . Here the speech and thoughts are slow (not rapid) but the continuity is preserved, characterized by the overinclusion of unnecessary and trivial details but ultimately reaches the answer .

Loosening of association (Option C) :

- Disorder of form of thought .
- Similar to the Flight of Ideas, words form meaning but the sentences do not form meaning and successive associations are not connected .
- Speech is dis coherent and completely non-understandable.
- Goal is never reached .
- Seen in Schizophrenia .
- Eg. 'Next day when I would be going out you know, I took control like I put bleach on my hair in Kolkata...'

Solution for Question 15:

Correct answer: C) 1-F,2-F,3-T,4-F,5-T

Explanation:

The correct statements are,

- 1) Continuity is maintained in tangentiality
- 2) Goal is reached in circumstantiality
- 3) Perseveration is associated with organic mental disorder
- 4) Continuity is disrupted in perseveration

5) Thought blocking is associated with schizophrenia

Hence, 1-F , 2-F , 3-T , 4-F , 5-T is the correct answer.

Disorders of flow/stream of thought:

Continuity maintained with increased speed:
Flight of ideas: rapid flow of ideas from one to another, connected and partially understandable Proximity: Milder form of flight of ideas Clang association: sentences with rhyming
Continuity maintained with decreased speed: (Statement-1)
Thought slowing/retardation: Decrease in speed of thought, seen in depression Circumstantiality: Overinclusion of trivial and unnecessary details, but ultimately reaches the answer. (Statement-2) Tangentiality: Flow of irrelevant thoughts that deviates from the main topic and never reaches the answer.
Continuity disrupted:
Perseveration: (Statement-4) Same thought persists beyond the point of relevance. Pathological repetition of the same response to different stimuli. Seen in organic mental disorder (Statement-3) and schizophrenia. Thought blocking: Sudden arrest of thought and then the person resumes with an entirely new thought. Seen in schizophrenia. (Statement-5)

Solution for Question 16:

Correct answer: C) Nihilistic delusion - mood congruent

Explanation:

- The type of delusion seen in this patient is Nihilistic delusion , often seen as a mood-congruent delusion in Depression with psychosis .
- Nihilistic delusions are seen in Cotard syndrome, which can also be associated with depression. But since specifically Content/type of delusion has been asked, Nihilistic delusion would be more appropriate.

Delusion:

Typically characterized as a fixed, false belief that persists despite evidence to the contrary and is inconsistent with the individual's social and cultural context.

Types(content) of delusion:

Delusion	Description
Delusion of Persecution	Belief of being harassed, persecuted, or mistreated by others. Most common type of delusion
Delusion of Reference	Belief that surrounding people are talking about them or spying on them. It is a form of delusion of persecution.
Delusion of Grandiosity	Exaggerated belief in one's own importance, power, or identity. Belief in being a significant figure who can help others or may claim to hear the voices of God, which reinforces their belief in their elevated status. Seen in mania (as Mood-congruent delusion)
Nihilistic Delusion	Belief that the body/part of body/world or everything related to it has ceased to exist. Also known as delusion of negation. seen in depression (most commonly) as (Mood-congruent delusion) Cotard syndrome: Patient with depression having delusions of nihilism (specifically related to parts of the body).
Delusion of guilt	Patient firmly believes that he is a sinner, though not having committed any sin. seen in depression (as Mood-congruent delusion)
Delusion of Poverty	Patient is convinced that he is poor and in a state of destitution although he is not.
Delusion of enormity	Patient believes that their action can produce catastrophe. Eg. the patient refuses to urinate, because he believes that it may flood the world.
Delusion of Infidelity	Belief that one's partner is unfaithful. Associated factor: prolonged alcohol use Also known as Othello syndrome/Morbid jealousy
De clerambault's syndrome/ Erotomania	The patient (often a single woman) believes that a person of higher status is in love with her. Maybe a feature of paranoid schizophrenia.
Hypochondriacal delusion/Delusion of ill health	Characteristically seen in depression, may also be seen in schizophrenia.
Various forms of Hypochondriacal delusion:	

Delusion	Description
Delusion of Infestation/ Parasitosis/ Ekbom syndrome: Patient falsely believes that he is infested with insects, he does not experience sensation. It affects both men and women equally and becomes more common with age. 'Empty matchbox sign' may be present (when the patient attempts to catch the insect to show it to the clinician). Delusional dysmorphophobia: Patient believes that his body/part of his body is deformed and wants it to be corrected. Delusional halitosis: Belief that the patient has a malodorous body odor.	

Cotard syndrome:

Nihilistic delusions typically related to parts of the body seen in patients with depression. Associated anxiety may also be present. Affects both men and women equally.

Mood-congruent and mood-incongruent delusions:

Mood-congruent delusion	Mood-incongruent delusion
Content of delusion that is consistent with the patient's mood.	Content of delusion that has no relation to the patient's mood.
E.g., depressed patients who believe that they are responsible for the destruction of the world.	E.g., depressed patients who believe that they are the new Messiah.

Solution for Question 17:

Correct Answer: B) Capgras syndrome

Explanation

Capgras syndrome is characterized by the belief that a close person has been replaced by an identical double.

Misidentification syndrome	
Capgras Syndrome (Option B)	Belief that a close person (often a family member) has been replaced by an identical double.

Misidentification syndrome	
Fregoli Syndrome (Option A ruled out)	Belief that various people are actually the same person in disguise. Eg. Patient believes her neighbor could change appearance, clothes and sex to spy on her.
Intermetamorphosis Syndrome	Delusion that people have swapped identities but kept their physical appearance.
Subjective Doubles Syndrome (Option D ruled out)	Belief that the patient has a double or doppelganger.
Reduplicative Paramnesia (Option C ruled out)	Delusional belief that identical places or events exist.

Solution for Question 18:

Correct Answer: A) Feeling of small insects crawling on or under their skin

Explanation:

- Formication: Sensation of small animals, like insects, crawling over the body. (Option A) Often seen in acute organic states and cocaine psychosis, where it is called the "cocaine bug."

Types of tactile hallucinations:

- Superficial Hallucinations: Feelings over the skin like hot, cold, pricking (Option B ruled out) etc. Formication is a type of superficial hallucination.
- Kinaesthetic Hallucinations: Affect muscles and joints, giving the sensation of limbs being twisted or moved. (Option C ruled out). Vestibular Hallucinations are regarded as a variant of kinesthetic hallucination and involve sensations of movement, such as floating in air (Option D ruled out) or sinking in bed. Often seen in delirium, alcohol intoxication, or benzodiazepine withdrawal, and can also be experienced after long periods of physical activity like sailing.
- Visceral hallucination: false perceptions of inner organs such as pain, heaviness, stretching, palpitation

Solution for Question 19:

Correct Answer: B) Second-person auditory hallucination

Explanation:

Since the voices are directly addressing the patient giving him commands, this is second-person auditory hallucination

Auditory Hallucination - False perception of sound without an external stimulus. Can range from elementary sounds like bells or whispers to fully-formed voices. Often associated with conditions like schizophrenia, severe depression, and organic states such as delirium or dementia.

- First-Person Auditory Hallucination: Hearing one's own thoughts spoken aloud (Option A ruled out)
- Second-Person Auditory Hallucination: Voices that directly address the patient, giving instructions or commands (Option B)
- Third-Person Auditory Hallucination: Voices that speak about the patient in the third person, often providing a running commentary on their actions or behaviors. (Option C ruled out)

Solution for Question 20:

Correct Answer: A) 1c,2a,3b

Explanation:

	Illusion	Hallucination	Pseudo-hallucination
Definition	A false perception combining a real stimulus with a mental image	A false perception without a real external stimulus, experienced as coming from outside the self	A vivid mental image perceived inside the mind, recognized as unreal but still involuntary
Stimulus	Requires an actual external stimulus	No external stimulus	No external stimulus

	Illusion	Hallucination	Pseudo-hallucination
Insight	The individual may or may not realize the misinterpretation	Often lacks insight	The individual recognizes the experience as unreal
Associated Conditions	Not necessarily pathological; can occur in normal individuals or in conditions like delirium	Usually indicates a serious mental illness, such as schizophrenia or psychosis	Can occur in mental illness
Example	Mistaking a shadow for an attacker on a dark road	Hearing voices that are perceived as coming from other people when no one is speaking	Hearing voices inside one's head but recognizing them as not being from external sources

Solution for Question 21:

Correct Answer: B) The individual experiences their thoughts as being controlled or influenced by an external force, with thoughts being inserted, withdrawn, or broadcasted to others

Explanation:

Thought alienation:

- Experience of one's thoughts being controlled or influenced by an external source
- Makes the thoughts feel foreign or externally controlled
- Types: Thought insertion - patient believes that foreign thoughts are being placed into their mind from an outside source Thought deprivation - patient believes that thoughts suddenly disappear and an external force is withdrawing them Thought broadcasting - patient believes their thoughts are being shared with or heard by others, sometimes imagining that everyone else is thinking in unison with them
- Thought insertion - patient believes that foreign thoughts are being placed into their mind from an outside source
- Thought deprivation - patient believes that thoughts suddenly disappear and an external force is withdrawing them
- Thought broadcasting - patient believes their thoughts are being shared with or heard by others, sometimes imagining that everyone else is thinking in unison with them

- Part of the diagnostic criteria for schizophrenia .
- Also referred to as Passivity of thought/ Delusions of control of thinking.
- Thought insertion - patient believes that foreign thoughts are being placed into their mind from an outside source
- Thought deprivation - patient believes that thoughts suddenly disappear and an external force is withdrawing them
- Thought broadcasting - patient believes their thoughts are being shared with or heard by others, sometimes imagining that everyone else is thinking in unison with them

Solution for Question 22:

Correct Answer: C) Hypochondriacal delusion

Explanation:

This patient displays an unfounded belief that she has a serious illness despite contrary medical evidence, characteristic of hypochondriacal delusions.

Types of Delusion	
Persecutory Delusions	Belief that one is being harmed or conspired against.
Delusions of Reference (Option B ruled out)	Belief that ordinary events are personally significant.
Delusions of Infidelity	Unshakable belief that a partner is unfaithful.
Erotomania	Belief that a high-status person is in love with the individual.
Grandiose Delusions	Inflated belief in one's own importance, power, or identity.
Hypochondriacal Delusions	Unfounded belief of suffering from a serious illness. (Option C)
Delusions of Guilt	Belief in having committed a grave sin, deserving punishment.
Nihilistic Delusions	Belief that oneself or the world is nonexistent or ending.

Types of Delusion	
Delusions of Poverty	False belief of impending poverty or destitution.

Conversion Disorder (Option A): Also called Functional Neurological Symptom Disorder, is a mental health condition where a person experiences neurological symptoms—such as paralysis, blindness, or seizures—without an identifiable physical cause. These symptoms are not intentionally produced and often occur after psychological stress or trauma .

Somatic symptom disorder (Option D): These are marked by physical symptoms resembling medical conditions, and are common in geriatric psychiatry due to frequent somatic complaints in older adults. Dismissing these symptoms as imaginary can cause resentment. Instead, clinicians should validate the patient's experience and address the issue with psychological or psychopharmacological treatment.

Actionable Delusions:

- Delusion of infidelity, guilt, illness (hypochondriacal)
- More likely to lead to actions, potentially causing harm to oneself or others

Solution for Question 23:

Correct Answer: D) Vorbeireden

Explanation:

Abnormal thought process	
Talking Past the Point (Vorbeireden) (Option D)	Patient responds with incorrect answers related to the topic, despite understanding the question. Example: Saying "white" for grass color and "green" for snow color. Seen in hysterical pseudodementia (dissociative disorder) and Ganser's syndrome (approximate answers). Also occurs in acute schizophrenia , especially in adolescents (hebephrenia).
Loosening of Associations (Option A ruled out)	Disorganized thought progression, shifting between unrelated topics, leading to incoherent communication.

Abnormal thought process	
Derailment	Sudden or gradual deviation in thought (without blocking), often used synonymously with loosening of associations.
Incoherence	Disconnected, disorganized, or incomprehensible speech. Word Salad : Incoherent, incomprehensible mixture of words and phrases, often seen in advanced schizophrenia. (Option C ruled out)
Word Approximation (Option B ruled out)	Using conventional words in an unconventional way (e.g., "hand shoes" for gloves) Distinguished from a neologism, which is a newly created word with an unclear or unknown origin.

Solution for Question 24:

Correct Answer: B) Neologism

Explanation

Neologism involves creating new words or using ordinary words in a new way, such as "skycar" for airplane, often seen in schizophrenia.

Neologism	
Neologisms	New words or ordinary words used in a new way, often linked to schizophrenia. Neologisms can result from: Completely new words with no clear derivation. Distorted versions of existing words. Words formed incorrectly using normal word-formation rules.
Causes of Neologisms	Formal thought disorder: Fusing words or concepts together. Derailment: Using incorrect words, e.g., "relativity" instead of "relationship." Technical neologisms: Words invented for private, unexplainable experiences. Hallucinatory Voices: Voices may influence neologisms, forcing the patient to create new words to communicate or defend.

Neologism	
Catatonia & Neologisms	Neologisms may be linked to stereotyped movements or mannerisms, distorting word pronunciation.
Stock Words	Using one word repeatedly in different contexts (e.g., "air car" for airplane, "sea car" for boat). (Option B)

Solution for Question 25:

Correct Answer: B) Tangentiality

Explanation:

- In tangentiality, the patient responds in a way that is related to the general topic but does not answer the question directly, which fits the scenario.
- Tangentiality is characterized by oblique, digressive, or irrelevant speech that fails to convey the central idea.
- Example: Doctor: "Have you had any trouble sleeping?" Patient: "I usually sleep in my bed, but now I'm sleeping on the sofa."
- Doctor: "Have you had any trouble sleeping?"
- Patient: "I usually sleep in my bed, but now I'm sleeping on the sofa."
- Doctor: "Have you had any trouble sleeping?"
- Patient: "I usually sleep in my bed, but now I'm sleeping on the sofa."

Circumstantiality (Option A):

- The patient eventually answers the question after giving excessive detail, which is not the case here.
- Example: When asked about their health, a patient begins by explaining their entire morning routine, describing what they had for breakfast and the weather, before finally answering the question about their health.

Neologism (Option C):

- New words or ordinary words used in a new way, are often linked to schizophrenia.
- Example: A patient says, "I need to go to the flibberflam store to get some boozleberries." These words are not real and have meaning only to the speaker.

Loosening of association (Option D):

- A thought or speech disturbance where ideas shift randomly, resulting in disorganized, unconnected thoughts that fail to convey a coherent message. Often associated with schizophrenia.
 - Example: "I like working as an accountant. The clouds today are so nice. Did you see that movie about space?"
-

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Schizophrenia And Other Psychotic Disorders

1. A 30-year-old female patient with treatment-resistant schizophrenia has started on clozapine. What is the appropriate duration and frequency of blood test monitoring for this patient taking clozapine?

- A. Weekly for 6 months, bi-weekly for 6 months, then monthly.
- B. Monthly for 6 months, then every 3 months.
- C. Weekly for 3 months, bi-weekly for 3 months, then monthly.
- D. Monthly for 1 year, then every 6 months.

2. A 28-year-old woman with a history of mood swings, manic episodes, and psychotic symptoms for 10 months presents for evaluation. She mentions that her psychotic symptoms occur both during mood episodes and independently. What is the most likely diagnosis?

- A. Bipolar I Disorder
- B. Schizophrenia
- C. Schizoaffective Disorder
- D. Mood disorder with Psychotic Features

3. A 38-year-old man, a software engineer in a stable five-year relationship, has recently become convinced that his partner is unfaithful. Despite having no evidence, he frequently accuses her, checks her phone, questions her whereabouts, and shows irrational jealousy when she is with friends or colleagues. What is the most likely diagnosis?

- A. Munchausen's syndrome

- B. Cotard syndrome
 - C. De Clerambault syndrome
 - D. Othello syndrome
-

4. A 45-year-old man who has lived a normal life and works as an accountant begins to believe that his wife and children have been replaced by impostors. He is convinced that those he interacts with daily are not his real family, leading to significant anxiety and fear. What is the most likely diagnosis?

- A. Intermetamorphosis
 - B. Capgras syndrome
 - C. Fregoli syndrome
 - D. Nihilistic delusion
-

5. A 32-year-old woman who has performed well in her personal and professional life believes her coworkers are whispering about her, hiding her belongings, and sabotaging her work projects. Despite having no evidence, she feels anxious and increasingly isolated. What is the most likely diagnosis?

- A. Delusion of persecution
 - B. Delusion of grandeur
 - C. Delusion of erotomania
 - D. Delusion of infidelity
-

6. A 28-year-old male with schizophrenia, who has been stable on antipsychotics for years, was recently switched to a higher dose of a first-generation antipsychotic. He then becomes increasingly agitated, develops a high fever, and

exhibits severe muscle stiffness, appearing less responsive and confused. The involved receptors and the drug of choice for managing this condition are:

- A. D2; Dantrolene sodium
 - B. D2; Bromocriptine
 - C. 5HT2A; Dantrolene sodium
 - D. 5HT2A; Bromocriptine
-

7. A 45-year-old woman with a diagnosis of schizophrenia has been on antipsychotic medication for over a decade. Her family has noticed that she has been experiencing involuntary movements for the past few months, including repetitive facial grimacing, lip-smacking, and jerky motions of her arms and legs. What is the drug of choice for treatment of this condition?

- A. Propranolol
 - B. Clonazepam
 - C. Tetrabenazine
 - D. Valbenazine
-

8. A 30-year-old male diagnosed with schizophrenia was prescribed a new antipsychotic medication two weeks ago to manage his symptoms. Since then, he has experienced intense restlessness, a strong urge to move, and difficulty sitting still. What is the most likely diagnosis for his condition?

- A. Dystonia
 - B. Akathisia
 - C. Tardive dyskinesia
 - D. Parkinsonism
-

9. A 25-year-old woman with a history of schizophrenia has been on antipsychotic treatment for three years. After being prescribed a new medication, she developed unusual muscle contractions in her neck and jaw within a week, resulting in difficulty controlling her head movements, which often twist to one side. Which pathway is primarily responsible for these symptoms?

- A. Mesolimbic pathway
 - B. Mesocortical pathway
 - C. Nigrostriatal pathway
 - D. Tuberoinfundibular pathway
-

10. An 8-year-old boy has been exhibiting significant behavioural changes over the past six months, including increased withdrawal, loss of interest in activities, declining academic performance, self-talk, and occasional reports of visual hallucinations. Which factor is most likely associated with a better prognosis in this case?

- A. Insidious in onset
 - B. Prominence of negative symptoms
 - C. Presence of precipitating factors
 - D. Absence of mood symptoms
-

11. A 32-year-old male with schizophrenia has been non-adherent to oral antipsychotic medication, resulting in multiple relapses. The psychiatrist decides to switch him to a long-acting injectable antipsychotic to improve adherence. Which of the following medications is preferred?

- A. Clozapine
- B. Chlorpromazine

- C. Zotepine
 - D. Risperidone
-

12. A 23-year-old woman has been having thoughts that her friends are prison escapees plotting against her for the past week. Her parents report that she has been talking to imaginary friends, becoming increasingly withdrawn, suspicious, and agitated over the last two weeks, with poor sleep and appetite. She has no prior psychiatric history. What is the most likely diagnosis?

- A. Schizoaffective disorder
 - B. Delusional disorder
 - C. Schizophrenia
 - D. Brief psychotic disorder
-

13. A 28-year-old man with treatment-resistant schizophrenia has been on multiple antipsychotic medications without significant improvement in his symptoms. He continues to experience severe hallucinations and delusions. What is the most appropriate next step in his management?

- A. Aripiprazole
 - B. Olanzapine
 - C. Clozapine
 - D. Risperidone
-

14. A 52-year-old obese male with type 2 diabetes presents with schizophrenia, primarily showing negative symptoms like social withdrawal and lack of motivation. Which antipsychotic medication is the most appropriate choice for his treatment?

- A. Aripiprazole
 - B. Olanzapine
 - C. Haloperidol
 - D. Chlorpromazine
-

15. A 34-year-old female with a diagnosis of schizophrenia has been treated with a low dose of olanzapine for the past three weeks. Despite this treatment, she continues to experience persistent positive symptoms with only minimal improvement, including hallucinations and delusions. What is the most appropriate next step in the management of this patient?

- A. Increase the dosage of olanzapine
 - B. Switch to another antipsychotic
 - C. Initiate Electroconvulsive therapy (ECT)
 - D. Initiate Cognitive behaviour therapy (CBT)
-

16. A 25-year-old male presents with a 6-month history of auditory hallucinations, paranoid delusions, and disorganised thinking, resulting in difficulty maintaining relationships and functioning at work. Physical and mental status examinations reveal hallucinations and delusions, with no history of mood disorders or substance abuse. What is the most appropriate first-line treatment for this patient?

- A. Fluphenazine
 - B. Risperidone
 - C. Lithium
 - D. Haloperidol
-

17. A 22-year-old male is evaluated for his first episode of psychosis. As part of the assessment, the clinician reviews various potential risk factors. Which of the following is not considered a risk factor for developing schizophrenia?

- A. Childhood trauma
 - B. Born in Winter
 - C. Urban upbringing
 - D. Male gender
-

18. A 25-year-old female with schizophrenia presents with symptoms such as lack of interest, alogia, and diminished facial expressions. Which of the following findings is commonly associated with this type of schizophrenia according to Timothy Crow's classification?

- A. Auditory hallucinations
 - B. Normal Ventricles
 - C. Increased dopamine levels
 - D. Enlarged ventricles
-

19. A 34-year-old woman is experiencing significant delusions of persecution and often fears that her food is being tampered with. She has auditory hallucinations urging her to remain vigilant and distrustful. While her cognitive abilities seem intact, her emotional reactions are inappropriate. What diagnosis should be considered?

- A. Catatonic Schizophrenia
 - B. Van Gogh syndrome
 - C. PfRopf Schizophrenia
 - D. Paranoid Schizophrenia
-

20. A 32-year-old woman diagnosed with schizophrenia has been increasingly neglecting her personal hygiene and no longer initiates activities she used to do before. She spends most of her time sitting in one place without doing anything productive. Which negative symptom does this describe?

- A. Anhedonia
 - B. Avolition
 - C. Alogia
 - D. Flat affect
-

21. A 30-year-old woman presents with symptoms indicating that her thoughts are not under her control. She states that she cannot think without external influence and frequently feels that someone is interfering in her thinking. Which of Schneider's first-rank symptoms does she display?

- A. Made volition
 - B. Auditory hallucinations
 - C. Delusional perception
 - D. Thought insertion
-

22. According to Eugen Bleuler, which of the following symptoms are diagnostic criteria for schizophrenia? Affect disturbances Association disturbances Autism Ambivalence

- A. 1,2,3
- B. 2,3,4
- C. 1,2,4
- D. 1, 2, 3, 4

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Correct Answers

Question	Correct Answer
Question 1	1
Question 2	3
Question 3	4
Question 4	2
Question 5	1
Question 6	1
Question 7	4
Question 8	2
Question 9	3
Question 10	3
Question 11	4
Question 12	4
Question 13	3
Question 14	1
Question 15	1
Question 16	2
Question 17	4
Question 18	4

Question 19	4
Question 20	2
Question 21	4
Question 22	4

Solution for Question 1:

Monitoring of absolute neutrophil count (ANC) is essential for patients on clozapine due to the risk of agranulocytosis . The recommended protocol is as follows:

- Weekly blood monitoring for the first 6 months.
- Biweekly monitoring for the next 6 months.
- Monthly monitoring thereafter.

Treatment is stopped if

- ANC is < 1500 /mm³
- WBC is < 3000 /mm³.

Clozapine is the most effective antipsychotic for treatment-resistant schizophrenia , but it carries risks such as agranulocytosis (approximately 0.3% incidence), seizures (up to 5% with doses exceeding 600 mg), and rare side effects like myocarditis.

Solution for Question 2:

Schizoaffective disorder involves both psychotic symptoms and mood episodes (mania or depression) occurring together and separately. At least a 2-week period of psychotic symptoms (delusions or hallucinations) should be there without mood symptoms to diagnose a case of Schizoaffective disorder.

Bipolar I Disorder (Option A):

In bipolar I disorder, psychotic symptoms may occur but only during mood episodes, and they are not present when the person is stable. This patient's psychotic symptoms occur independently, even when she is not experiencing a mood episode, suggesting schizoaffective disorder rather than bipolar disorder.

Schizophrenia (Option B):

The presence of prominent mood symptoms along with psychosis, as seen in this patient, makes schizoaffective disorder more likely than schizophrenia.

Mood disorder with Psychotic Features (Option D):

According to DSM-5 criteria, schizoaffective disorder can be differentiated from mood disorder with psychotic features by the presence of at least a 2-week period of psychotic symptoms (delusions or hallucinations) without mood symptoms, as seen in this case.

Solution for Question 3:

Othello syndrome, or Delusion of infidelity/jealousy, is a psychiatric condition where the person has a fixed belief that their partner is being unfaithful despite having no evidence.

- The individual often engages in obsessive behaviours such as checking their partner's phone or monitoring their activities, as seen in this case.
- It is more common in men.
- It can be associated with schizophrenia, alcohol abuse, or cocaine abuse.

Munchausen's syndrome (Option A) : A factitious disorder in which the person intentionally fabricates or induces illness in themselves for attention or sympathy; individuals may also engage in self-injury to secure admission to the hospital.

Cotard syndrome (Option B) is a delusion in which the individual believes they are dead, often accompanied by beliefs that they are rotting, emit a foul odour, or that certain body parts do not exist.

De Clerambault syndrome (Option C), or the delusion of erotomania, is a false belief that another person (often of higher status) is in love with them (more common in women).

Solution for Question 4:

The scenario describes Capgras Syndrome, a type of delusional misidentification where the individual irrationally believes that a familiar person, usually a close family member or spouse, has been replaced by an identical imposter.

Intermetamorphosis (Option A) is characterised by delusions that individuals have swapped identities (psychological and physical transformation) while keeping the same appearance.

Fregoli syndrome (Option C) is a delusional misidentification where the individual falsely believes that different people they encounter are actually the same person in disguise.



Nihilistic syndrome (Option D) is a depressive delusion in which the patient denies the existence of their body, their mind, their loved ones and the world around them.

Solution for Question 5:

- The scenario described, where an individual believes that others are attempting to harm them, is characteristic of a delusion of persecution, which falls under the category of Delusional disorders.

- Delusions are defined as fixed false beliefs that are resistant to reason or contrary evidence and are not aligned with the individual's cultural context.

Types of delusional disorders:

Type	Description
Persecutory (Delusion of persecution)	Believing others are trying to harm them (Option A)
Jealous (Delusion of infidelity or Othello syndrome)	Believing that a lover or partner is unfaithful (Option D)
Erotomaniac (Delusion of erotomania or De Clerambault's syndrome)	Believing that another person (often of higher status) is in love with them (Option C)
Somatic type	Believing he or she has some physical disorder that includes: Delusion of parasitosis (infested with parasites, such as bugs or worms) Delusion of dysmorphophobia (body or appearance is distorted or altered in a negative way) Delusion of halitosis (bad breath)
Grandiose (Delusion of grandeur)	Believing one has exceptional abilities, wealth, or fame (Option B)
Delusion of Guilt	Believing that they are an evil person and have ruined their family
Nihilistic delusion	Deny the existence of their body, mind, loved ones, or the world, claiming they have no intelligence, that parts of their body are absent, or even believing they are dead or that the world has ended
Delusion of Poverty	Believing they are impoverished and that destitution threatens them and their family.
Mixed and unspecified type (Delusion of misidentification)	Delusions that are a combination of types or not described by one of the categories: Capgras syndrome Fregoli syndrome Syndrome of metamorphosis Syndrome of subjective doubles

Solution for Question 6:

The scenario is indicative of Neuroleptic Malignant Syndrome (NMS), in which the blockage of D2 receptors leads to extrapyramidal side effects, and the drug of choice for treatment is Dantrolene sodium.

Neuroleptic Malignant Syndrome (NMS)		
Description	A rare, potentially fatal consequence of neuroleptic administration.	
Mechanism	D2 receptor blocking at: Striatum (muscle rigidity) Hypothalamus (hyperthermia) Spinal nerves (autonomic disturbances)	
Clinical manifestations	Motor and behavioural	Autonomic symptoms
Muscular rigidity (most common) Dystonia Akinesia, Mutism Obtundation Agitation	Fever (most common) Diaphoresis Raised pulse and blood pressure Sweating Altered consciousness	
Associated conditions	Renal failure and myoglobinuria	
Laboratory findings	Raised levels of creatinine phosphokinase (CPK) (most common) Leukocytosis (raised WBC) Raised liver enzymes Raised plasma myoglobin	
Management	Stop the drug (causing symptoms) Supportive care (maintain hydration and other vitals) Skeletal muscle relaxants: Dantrolene sodium (drug of choice) Dopamine agonists: Bromocriptine, Amantadine Second-generation antipsychotics (for restarting the treatment for Schizophrenia).	

Solution for Question 7:

The scenario is diagnosed as medication-induced Tardive dyskinesia, for which the drug of choice is Valbenazine.

Tardive dyskinesia	
Description	A delayed effect (abnormal involuntary movements) of antipsychotics typically occurs until after 6 months of treatment.
Cause	Supersensitization of dopamine (D) receptors due to prolonged exposure to antipsychotic medications.
Manifestations	Abnormal, involuntary movements such as: Lip smacking Chewing Darting, twisting, and protrusion of the tongue (catching fly) Choreiform hand movements (playing piano) Facial grimacing Finger movements and hand clenching In Severe cases: Torticollis (twisting of neck muscles to one side) Retrocollis (neck extension) Trunk twisting Pelvic thrusting
Management	Prevention: Using antipsychotic medications only when indicated and in the lowest effective doses. Regular monitoring: Patients taking antipsychotics should be regularly assessed for any signs of abnormal movements. Treatment: Reducing antipsychotic drugs (or stop if possible) Switching the medication to clozapine (least EPS-causing) or atypical antipsychotics VMAT (Vesicular Monoamine Transporter type 2) receptor inhibitor: Valbenazine (drug of choice) (Option D) Tetrabenazine- Lesser efficacy, more sedation than Valbenazine (Option C ruled out)

Propranolol (Option A) is a non-selective beta-blocker primarily used to treat hypertension, anxiety, and certain types of tremors. While it can help alleviate symptoms of akathisia and anxiety, it is ineffective for treating tardive dyskinesia .

Clonazepam (Option B) is a benzodiazepine used to treat anxiety disorders, panic disorders, and certain types of seizures but is not effective in managing tardive dyskinesia.

Solution for Question 8:

The scenario, characterised by inner restlessness and an uncontrollable need for constant motion, is indicative of medication-induced acute akathisia.

Akathisia			
Definition	A subjective feeling of restlessness, objective signs of restlessness, or both.		
Manifestations	Subjective: Patient reports feeling restless Objective: (noticed by others) Anxiety Inability to relax Jitteriness Rocking motion Pacing around Rapid alteration of sitting and standing	Anxiety Inability to relax Jitteriness Rocking motion	Pacing around Rapid alteration of sitting and standing
Anxiety Inability to relax Jitteriness Rocking motion	Pacing around Rapid alteration of sitting and standing		

Akathesia			
Management	Reduce the dosage of the current medication. Switch to atypical or low-potency typical antipsychotics. Treat with appropriate medications: Beta-blockers (Propranolol is the drug of choice) Anticholinergics Benzodiazepines Consider changing the neuroleptic if needed.		
Anxiety Inability to relax Jitteriness Rocking motion	Pacing around Rapid alteration of sitting and standing		

Dystonia (Option A) refers to brief or prolonged muscle contractions causing abnormal movements or postures, such as oculogyric crises (upward eye rolling), tongue protrusion, trismus, torticollis (neck muscle twisting), laryngeal-pharyngeal dystonias, and dystonic postures of the limbs and trunk. Anticholinergic drugs like, benztropine, diphenhydramine and promethazine can be used for treatment.

Tardive dyskinesia (Option C) is a delayed effect of antipsychotics that typically occurs after six months of treatment. It is characterised by abnormal, involuntary, irregular choreoathetoid movements in the muscles of the head, limbs, and trunk. Valbenazine is the drug of choice in its management.

Parkinsonism (Option D) refers to a group of movement disorders that mimic the symptoms of Parkinson's disease, including tremors, stiffness, bradykinesia (slowness of movement), and postural instability. It is treated with anticholinergics, levodopa-carbidopa, dopamine agonists and MAO-B inhibitors based on symptom severity.

Solution for Question 9:

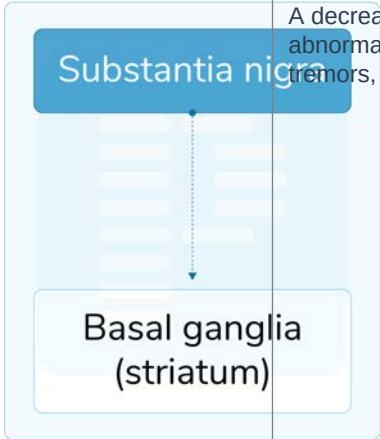
- The symptoms described in this case are indicative of Acute dystonia , a type of extrapyramidal side effect often seen with antipsychotic medications.

- This condition occurs when there is a disruption in the nigrostriatal pathway , which is primarily responsible for motor control.

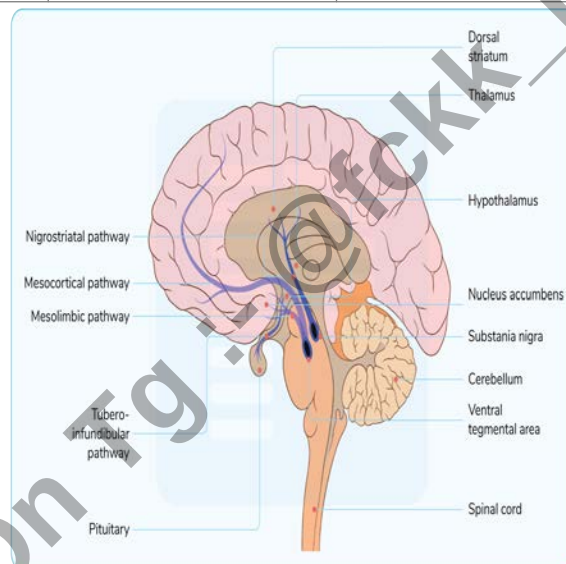
Acute Dystonia:

- It refers to brief, sudden muscle contractions that lead to abnormal movements or postures.
- Clinical manifestations: Oculogyric crises (upward rolling of eyes) Tongue protrusion Trismus (jaw muscles) Torticollis (neck muscles twisting to one side) Laryngospasm Dysarthria, Dysphasia (glossopharyngeal symptoms) Dystonic postures of the limbs and trunk.
- Oculogyric crises (upward rolling of eyes)
- Tongue protrusion
- Trismus (jaw muscles)
- Torticollis (neck muscles twisting to one side)
- Laryngospasm
- Dysarthria, Dysphasia (glossopharyngeal symptoms)
- Dystonic postures of the limbs and trunk.
- Oculogyric crises (upward rolling of eyes)
- Tongue protrusion
- Trismus (jaw muscles)
- Torticollis (neck muscles twisting to one side)
- Laryngospasm
- Dysarthria, Dysphasia (glossopharyngeal symptoms)
- Dystonic postures of the limbs and trunk.
- Management: It can be effectively managed using anticholinergics. Parenteral (IV or IM) is for immediate treatment, while oral anticholinergics are used for prophylaxis to prevent future episodes. Drugs include Benztropine, Diphenhydramine, and Promethazine.
- It can be effectively managed using anticholinergics.
- Parenteral (IV or IM) is for immediate treatment, while oral anticholinergics are used for prophylaxis to prevent future episodes.
- Drugs include Benztropine, Diphenhydramine, and Promethazine.
- It can be effectively managed using anticholinergics.
- Parenteral (IV or IM) is for immediate treatment, while oral anticholinergics are used for prophylaxis to prevent future episodes.
- Drugs include Benztropine, Diphenhydramine, and Promethazine.

Neural pathways:

Pathway	Extent	Manifestations
Nigrostriatal pathway (Option C)	 Substantia nigra Basal ganglia (striatum)	A decrease in dopamine leads to abnormal movement symptoms like tremors, rigidity, dystonias, etc.
Mesolimbic pathway (Option A)		An increase in dopamine levels leads to positive symptoms such as: Delusion Hallucination
Mesocortical pathway (Option B)		A decrease in dopamine levels leads to negative symptoms such as: Anhedonia (loss of interest) Avolition (loss of will) Affective blunting (inability to express emotions). Asociality (decrease in drive to socialise). Alogia (decrease in verbal communication). Attention deficits Apathy (lack of concern).

Pathway	Extent	Manifestations
Tuberoinfundibular pathway (Option D)		A decrease in dopamine levels leads to hyperprolactinemia, which manifests as: Decreased libido (sex drive) Decreased sexual dysfunction Galactorrhea and amenorrhea (females)



Solution for Question 10:

The scenario described is a case of Schizophrenia, where the presence of precipitating factors suggests a better prognosis. In contrast, an insidious onset, prominence of negative symptoms, and absence of mood symptoms indicate a poorer prognosis.

Prognostic factors of schizophrenia

Factors	Good Prognosis	Poor Prognosis
Onset	Acute (within two weeks) or Abrupt (within two days)	Insidious (Option A)
Age of onset	Late	Early
Subtypes	Catatonic schizophrenia Paranoid schizophrenia	Simple (Worst) schizophrenia Hebephrenic schizophrenia
Gender	Female	Male
Prominent symptoms	Positive symptoms	Negative symptoms (Option B)
Mood symptoms	Present	Absent (Option D)
Family history of Schizophrenia	Absent	Present
Marital status	Married	Unmarried or divorced
Social support	Good	Poor
Premorbid personality disorder	Absent	Present
Career	Employed	Unemployed
Precipitating factors or Stressors	Present (Option C)	Absent
Substance abuse	Absent	Present
Compliance to medication	Compliant	Non-compliant

Solution for Question 11:

- Risperidone is a long-acting injectable commonly used for managing schizophrenia in patients with poor adherence to oral medications. It is administered as IM injections (every two weeks, once a month, or even longer).
- Clozapine and Zotepine are only available in oral formulations.
- Chlorpromazine is available in short-acting injectables and oral formulations.

Long-Acting Injectable (LAI) Antipsychotics or DEPOT Antipsychotics:

Characteristic	1st Generation (Typical) LAIs	2nd Generation (Atypical) LAIs
Examples	Fluphenazine Haloperidol Flupenthixol Zuclopenthixol Pipotiazine	Risperidone (Option D) Paliperidone Aripiprazole Olanzapine
Formulation	Oil-based	Aqueous-based
Absorption rate	Slower	Quicker
Test dose requirement	Often required	Not required
Special note	All drugs are esterified forms of the parent compound except Risperidone, which is encapsulated in polymer microspheres.	

Clozapine (Option A) , an oral, atypical antipsychotic, is not available as a long-acting injectable. It is primarily used for treatment-resistant schizophrenia due to its risk of serious side effects, including agranulocytosis, which requires regular blood monitoring.

Chlorpromazine (Option B) is a typical antipsychotic that blocks dopamine D2 receptors to alleviate psychotic symptoms. It is available in oral and short-acting injectable forms but not as a long-acting injectable. It may also cause anticholinergic side effects like dry mouth, blurred vision, and constipation.

Zotepine (Option C) is an atypical antipsychotic that blocks dopamine D2 and serotonin 5-HT2 receptors, available only in oral forms and not as a long-acting injectable.

Solution for Question 12:

- Acute psychotic disorder is characterised by the sudden onset of psychotic symptoms such as delusions, hallucinations, disorganised speech, or grossly disorganised or catatonic behaviour that lasts for < 1 month (ICD 11) or < 6 months (DSM 5) .
- DSM-5 classifies further:

Brief Psychotic disorder: Symptoms < 1 month

Schizophreniform disorder: Symptoms > 1 month- < 6 month

- It is an acute condition that is common in the 20s to 30s, especially in women comparatively and in lower socio-economic populations.

Disorder	ICD-11 Criteria	DSM-5 Criteria
Schizoaffective Disorder (Option A)	Symptoms of both schizophrenia and mood disorders for ≥ 1 month	Psychotic symptoms without mood disturbances for ≥ 2 weeks.
Delusional Disorder (Option B)	Delusions present for more than three months.	Delusions are present for at least one month.
Schizophrenia (Option C)	Two or more symptoms (e.g., delusions, hallucinations) present for $\geq$ one month	Two or more symptoms present for $\geq$ six months
Schizophrenia requires two or more symptoms, with at least one from delusions, hallucinations, thought disorder, or distortion of self-experience . Additional symptoms include negative symptoms, disorganised behaviour, or psychomotor disturbances.		

Solution for Question 13:

Clozapine is the drug of choice for treatment-resistant schizophrenia, especially after the failure of at least two antipsychotics , including both typical and atypical agents like risperidone and aripiprazole, given in adequate dosage and duration with good compliance. It is highly effective in managing persistent symptoms like hallucinations and delusions.

Mechanism of Action:

Clozapine primarily works as a dopamine D2 receptor antagonist , but it also has significant effects on serotonin receptors (5-HT_{2A} and 5-HT_{2C}), which helps reduce symptoms in treatment-resistant cases.

Side Effects:

- Severe risks: Agranulocytosis: Occurs in approximately 0.3% of users, necessitating regular monitoring of white blood cell counts. Seizures: Affects up to 5% of patients on doses

exceeding 600 mg.

- Agranulocytosis: Occurs in approximately 0.3% of users, necessitating regular monitoring of white blood cell counts.
- Seizures: Affects up to 5% of patients on doses exceeding 600 mg.
- Agranulocytosis: Occurs in approximately 0.3% of users, necessitating regular monitoring of white blood cell counts.
- Seizures: Affects up to 5% of patients on doses exceeding 600 mg.
- Common side effects: Hypersalivation Sedation Tachycardia Weight gain Postural hypotension
- Hypersalivation
- Sedation
- Tachycardia
- Weight gain
- Postural hypotension
- Rare side effects include myocarditis.
- Hypersalivation
- Sedation
- Tachycardia
- Weight gain
- Postural hypotension

Solution for Question 14:

- Atypical antipsychotics are generally preferred for treating negative symptoms associated with schizophrenia.
- Aripiprazole is the most suitable choice, and it effectively addresses negative symptoms while minimising the risk of weight gain and metabolic issues, as the patient is already obese and diabetic.

Medications for patients with schizophrenia primarily exhibiting negative symptoms:

- Clozapine (Most Effective)
- Olanzapine

- Risperidone
- Aripiprazole
- Ziprasidone

Metabolic Effects of Antipsychotics: Chronic therapy with certain antipsychotics can elevate blood sugar and triglyceride levels.

ANTIPSYCHOTIC-INDUCED WEIGHT GAIN

High risk of weight gain	Moderate risk of weight gain	Low risk of weight gain
Clozapine Olanzapine	Chlorpromazine Iloperidone Sertindole Quetiapine Risperidone Paliperidone	Amisulpride Asenapine Brexiprazole Aripiprazole Cariprazine Haloperidol Lumateperone Lurasidone Sulpiride Trifluoperazine Ziprasidone

Olanzapine (Atypical antipsychotic) (Option B) is effective for schizophrenia with predominantly negative symptoms but is associated with significant weight gain and a high risk of metabolic syndrome, which can worsen obesity and diabetes.

Haloperidol (1st generation anti-psychotic) (Option C) is primarily effective for positive symptoms and is not effective for negative symptoms, and it also has a higher risk of extrapyramidal side effects.

Chlorpromazine (1st generation anti-psychotic) (Option D) is not effective in managing negative symptoms, and it can cause sedation and weight gain, posing additional risks for the patient.

Solution for Question 15:

- In cases of inadequate response to a low dose of an antipsychotic like olanzapine after an appropriate trial period (typically 4-6 weeks), the most appropriate next step is to increase the dosage of the current medication.
- Antipsychotics often require dose adjustments to achieve therapeutic levels.

- Switching to another antipsychotic may be considered if the patient fails to respond to a higher dose or experiences intolerable side effects . (Option B ruled out)

Selecting Second Treatment Options for Acute Schizophrenia:

1. Treatment Change Criteria

- Patients who do not respond even slightly by the second week are less likely to benefit from the current medication, indicating that a treatment change may be necessary.
- Assessing the degree of symptom improvement is more informative than labelling patients as responders or nonresponders.

2. Adequate Trial:

- Consists of a 4-6 week period on a proper dose of antipsychotic medication.
- Patients showing even mild improvement during this time may continue to benefit for 3 to 6 months.

3. Monitoring and Compliance:

- Plasma concentration monitoring can help confirm adequate dosing.
- Low plasma levels may indicate non-compliance or inadequate absorption; increasing the dose may be beneficial in these cases.
- High plasma levels might suggest that side effects are hindering the therapeutic response.

4. Adjusting Treatment:

- While increasing the dose above the standard therapeutic range may be considered, switching to another medication is generally preferred if no improvement is observed.
- Clozapine is particularly effective for patients unresponsive to other antipsychotics, showing significant advantages in those with severe symptoms.

Initiate Electroconvulsive therapy (ECT) (Option C) : ECT is generally considered for severe cases of schizophrenia, particularly when there is treatment resistance or severe mood symptoms (such as depression) accompanying the psychosis.

Initiate Cognitive Behaviour Therapy (CBT) (Option D) : CBT can be beneficial as an adjunctive treatment for schizophrenia, especially in managing residual symptoms and improving coping mechanisms. However, it is not a substitute for pharmacological treatment.

Solution for Question 16:

The given scenario is consistent with Schizophrenia in which Risperidone, a second-generation antipsychotic, is considered first-line due to its efficacy in managing both positive symptoms (e.g., hallucinations and delusions) and negative symptoms (e.g., social withdrawal), with a lower risk of extrapyramidal side effects compared to first-generation antipsychotics like Fluphenazine, Haloperidol.

Management of Schizophrenia:

Pharmacotherapy:

- Acute Treatment Phase: Address severe symptoms, lasting typically 4-8 weeks.
- Antipsychotic medications are the primary treatment, with guidelines favouring second-generation (Atypical) antipsychotics for their broader efficacy and improved side effect profiles.
- Antipsychotic Medications:

First Generation Antipsychotics (Typical)	Second Generation Antipsychotics (Atypical)	Second Generation Antipsychotics (Atypical)
D2 receptor antagonist	Partial D2 Agonists	D2/5-HT _{2A} Antagonists
Haloperidol (Option D) Fluphenazine (Option A) Perphenazine Chlorpromazine	Aripiprazole Brexpiprazole Cariprazine (Option D)	Risperidone Olanzapine Quetiapine Ziprasidone Paliperidone Lurasidone Clozapine

- Treatment Duration: For first-episode patients, a minimum of 1-2 years of treatment is recommended. For those with multiple episodes, 5 years - indefinite treatment is often advised.
- For first-episode patients, a minimum of 1-2 years of treatment is recommended.
- For those with multiple episodes, 5 years - indefinite treatment is often advised.
- Side Effects:
- For first-episode patients, a minimum of 1-2 years of treatment is recommended.
- For those with multiple episodes, 5 years - indefinite treatment is often advised.

Extrapyramidal Side Effects (EPS)	Common with first-generation antipsychotics, including symptoms like Acute dystonia - Earliest, can occur within minutes of the injection Pseudo Parkinsonism - Occurs within days to weeks Akathisia - Occurs within days to weeks Alternatives for management include dose reduction, antiparkinson medications, or switching to medications with lower EPS risk.
Tardive Dyskinesia (TD)	Affects 20-30% of patients on long-term first-generation antipsychotics, with higher risk in the elderly and those with cognitive or mood disorders. Prevention strategies include using the lowest effective doses and switching to other medications like Clozapine.
Sedation & Postural hypotension	Especially with low-potency antipsychotics.
Long-term antipsychotic use can elevate prolactin levels, affecting sexual function and potentially leading to osteoporosis.	

Lithium (mood stabiliser) (Option C) is primarily used for bipolar disorder and is not effective for treating schizophrenia symptoms.

Solution for Question 17:

Male gender is not a risk factor for developing schizophrenia as it has equal prevalence in both sexes.

Schizophrenia demographics:

- Equal prevalence in men and women.
- Men tend to have an earlier onset (ages 10-25) than women (ages 25-35).
- Men often experience more negative symptoms; women generally have better social functioning and outcomes.
- Late-onset schizophrenia occurs after age 45 (3-10% of women).
- 90% of treated patients are aged 15-55.

Risk factors of Schizophrenia:

- Seasonality of birth : Increased prevalence in those born in winter/early spring (Option B) (Northern Hemisphere: January-April; Southern Hemisphere: July-September).
- Maternal factors : Complications during delivery, malnutrition & maternal illness.
- Early life experiences : Childhood trauma (Option A) , social isolation, and deprivation may increase risk.
- Urban upbringing (Option C) : Risk is higher for those raised in cities.
- Cannabis use : Associated with increased risk of schizophrenia, particularly in heavy users (up to 40% increase).
- Cognitive deficits : Poor verbal learning, memory issues, and slower processing speed may predict impending psychosis.

Prognostic factors:

Positive factors	Negative factors
Acute onset Female sex Living in a developed country Presence of mood symptoms Family history of mood disorder Married	Insidious onset Childhood or adolescent onset Poor premorbid functioning Cognitive impairment Absence of mood symptoms Negative symptoms Family History of schizophrenia Single/ separated/ divorced

Solution for Question 18:

Enlarged ventricles are a structural brain finding frequently seen in schizophrenia, mainly when negative symptoms are prominent, as in this case. This aligns with type II in Timothy Crow's classification.

Timothy Crow's classification	
Type 1	Type 2
Associated with positive symptoms like hallucinations, delusions, and bizarre behaviour (Option A)	Presents with negative symptoms like Alogia, avolition, anhedonia
High dopamine levels (Option C)	Low dopamine levels
Normal ventricles (Option B)	Enlarged ventricles

Timothy Crow's classification	
Good response to medication	Poor response to medication

Solution for Question 19:

- Paranoid Schizophrenia is characterised by the presence of delusions, particularly delusions of persecution (where the individual believes that they are being targeted or harmed by others) and hallucinations .
- The patient's fear that her food is being tampered with, along with auditory hallucinations urging her to remain vigilant and distrustful, aligns with this diagnosis, but cognitive functions remain intact.

Types of Schizophrenia:

Type of Schizophrenia	Characteristics
Paranoid Schizophrenia	Characterised by delusions and hallucinations. Individuals may have a fixed belief that others are out to harm them, often leading to mistrust and vigilance. Cognitive functions usually remain intact, while emotional responses may be inappropriate.
Catatonic Schizophrenia (Option A)	Characterised by three or more of the following symptoms: Motor immobility (stupor) Excessive, odd movement (stereotypies, or mannerisms) Waxy flexibility (bending like a wax candle) Extreme negativism (resisting instructions or maintaining rigidity) Unusual postures (bizarre posturing, catalepsy) Echolalia or echopraxia (Mimicking speech or movements)
Disorganised (Hebephrenic) Schizophrenia	Involves disorganised speech and behavior. Inappropriate emotional responses and poor functioning. Typically associated with a poorer prognosis.

Type of Schizophrenia	Characteristics
Residual Schizophrenia	History of schizophrenia with delusions and hallucinations (in the past 1 year) with lingering negative symptoms but minimal or no positive symptoms at present. No prominent delusions or hallucinations Individuals may exhibit social withdrawal and reduced emotional expression.
Simple Schizophrenia	Involves prominent negative symptoms
Undifferentiated Schizophrenia	Diagnosis for patients who do not fit into other categories presenting a mix of symptoms.

Van Gogh Syndrome (Option B): It is associated with self-mutilation behaviours in individuals with schizophrenia.



PfRopf Schizophrenia (Option C): Occurs in individuals with learning disabilities and mental retardation.

Solution for Question 20:

Avolition is characterised by a marked reduction in the motivation to initiate and perform purposeful, goal-directed activities . It involves a significant lack of drive, which results in the individual's inability to complete daily tasks or maintain personal care.

Schizophrenia Symptoms:

1. Positive Symptoms: These reflect exaggerations or distortions of normal functions; respond to medications and have a good prognosis.

- Delusions : False beliefs not based in reality, e.g., persecutory or grandiose delusions.
- Hallucinations: Perceptions without external stimuli, typically auditory (e.g., hearing voices).
- Disorganised speech: Incoherent or illogical speech, such as loose associations or "word salad."
- Disorganised behaviour: Erratic or inappropriate actions, including agitation or inappropriate affect.
- Avolition (Option B) : Lack of motivation to perform goal-directed activities, such as self-care.
- Anhedonia (Option A): Inability to experience pleasure from activities which were previously pleasurable.
- Alogia (Option C) : Poverty of speech, brief or uninformative responses.
- Flat affect or Blunting actions (Option D) : Reduced emotional expression, monotone speech, or lack of facial expressions.
- Social withdrawal (Asociality) : Lack of interest in social relationships.
- Impaired attention: Difficulty focusing or sustaining attention.
- Impaired working memory: Trouble holding and manipulating information.
- Executive dysfunction: Difficulty with planning, organising, and problem-solving.
- Slowed processing speed: Slower response to information and tasks.

Solution for Question 21:

Thought insertion refers to the belief that one's thoughts are not their own but are being placed into their mind by an external source.

Schneider's First Rank Symptoms:

Category	Symptom	Description
Thought Alienation Phenomenon	Thought Insertion	A belief that others are placing thoughts into the patient's mind.
Thought Withdrawal	A belief that thoughts are being removed from the patient's mind.	
Thought Broadcasting	A belief that others can know or access the patient's thoughts.	
Made Phenomenon	Made Volition (Option A)	A belief that actions are controlled by an external force.
Made Affect	A belief that external forces are altering the patient's emotions.	
Made Impulses	Sudden urges or impulses that are perceived as implanted by others.	
Auditory Hallucinations (Option B)	Voices Arguing/Discussing	The patient hears voices discussing or arguing about them, often in the third person. Example: The patient hears two voices arguing, one saying, "He's a failure," and the other replying, "No, he's not."
Voices Commenting	The patient hears a voice providing commentary on their actions, usually in the third person. Example: While cooking, a patient hears a voice say, "She's burning the food," reflecting their actions.	
Audible Thoughts	The patient hears their own thoughts spoken aloud by a voice. Example: A patient thinks, "I need to go to the store," and immediately hears a voice say the same phrase.	
Somatic Passivity	Physical sensations are perceived as imposed by external agents. Example: A patient feels a burning sensation in their arm and believes it is caused by invisible rays from a satellite.	

Category	Symptom	Description
Delusional Perception (Option D)	A normal perception is interpreted with a delusional belief Example : A patient sees a red car parked outside and believes it indicates the police are monitoring them for criminal activity.	

Solution for Question 22:

According to Eugen Bleuler, the diagnostic criteria for schizophrenia include (4 'A's):

- Affect (emotional expression) disturbances : Patients often exhibit diminished emotional expression, which can be observed through a lack of facial expressions and emotional responsiveness.
- Association (thought processes) disturbances: refers to disorganised thinking, where patients may form unusual or illogical connections in their thoughts, such as catastrophic beliefs about the world.
- Autism (social withdrawal) refers to detachment from reality, evident in patients who may appear silent around others and frequently engage in self-directed conversation.
- Ambivalence (mixed feelings) : Patients experience conflicting emotions or attitudes, which can lead to indecisiveness and difficulty making choices.

Previous Year Questions

1. What is the recommended medical approach for a psychotic patient who experiences voluntary aimless motions and has been observed standing motionless for extended periods with waxy flexibility and negativism?

- A. Haloperidol
- B. Lorazepam
- C. Clonidine
- D. Propanolol

2. A patient shows flat affect. What does it indicate?

- A. Positive symptom
- B. Negative symptom
- C. Affect symptom
- D. Cognitive symptom

3. A 28-year-old woman diagnosed with schizophrenia had been compliant with her prescribed olanzapine for several months. However, she discontinued the treatment. Which of the following is the most likely reason for her discontinuation?

- A. Weight gain
- B. Acute dystonia
- C. Akathisia
- D. Tardive dyskinesia

4. A doctor asked the patient what her age is and she told him that she is married to an older man, and she has 2 kids of 15 and 18 years, and she is 39 years old. What is the disorder present?

- A. Derailment
 - B. Circumstantial speech
 - C. Flight of ideas
 - D. Tangentiality
-

5. All of the following are features of pseudohallucinations, except?

- A. They occur without external stimulus
 - B. They are perceived in the outer objective space
 - C. They are not vivid
 - D. They are done voluntarily
-

6. What is the most frequently encountered acute organic mental disorder?

- A. Delirium
 - B. Dementia
 - C. Amnesia
 - D. Anxiety disorder
-

7. Which of the following is not considered a negative symptom of schizophrenia?

- A. Over activity

- B. Anhedonia
 - C. Alogia
 - D. Apathy
-

8. What is the probable diagnosis for a young male who displays social withdrawal, shyness, emotional aloofness, and lives in solitude, without any reported instances of hallucinations or delusions?

- A. Paranoid personality disorder
 - B. Schizoid personality disorder
 - C. Antisocial personality disorder
 - D. Emotionally unstable personality disorder
-

9. Which of the following is not considered a core symptom of schizophrenia?

- A. Autism
 - B. Automatism
 - C. Association defect
 - D. Ambivalence
-

10. What is the probable diagnosis for a middle-aged man who visits the psychiatry OPD claiming that he has been hearing voices from aliens instructing him to harm his family members for the last seven months?

- A. Mania
- B. Delusional disorder
- C. Schizophrenia

D. Brief psychotic disorder

11. Which of the following factors does not indicate a favorable prognosis in schizophrenia?

- A. Positive symptoms
 - B. Late age of onset
 - C. Insidious onset of symptoms
 - D. Associated with depression
-

12. Which lobes may be impacted in a 40-year-old woman who has a history of urinating in public without feeling any remorse, along with difficulties in motivation and mild memory deficits?

- A. Parietal
 - B. Occipital
 - C. Frontal
 - D. Temporal
-

13. What is the probable diagnosis for a 37-year-old man who experienced symptoms after a sudden financial loss? He reported feeling fearful of harm, having sleep and appetite disturbances, and hearing abusive voices. These symptoms lasted for about a week. The patient has no relevant medical history, drug abuse history, or past medical history.

- A. Schizophrenia
- B. Schizophreniform disorder
- C. Brief psychotic episode

D. Schizoaffective disorder

14. A 21-year-old girl suspects that the college principal is making a conspiracy against her and is planning to fail her in the upcoming exam. She feels that other teachers have also joined the conspiracy and students are helping them too. What is the diagnosis?

- A. Thought insertion
 - B. Delusion of control
 - C. Delusion of persecution
 - D. Delusion of reference
-

15. What is the recommended treatment for a 35-year-old female patient presenting to the psychiatric outpatient department with symptoms of reduced sleep, excessive elation, and increased talkativeness lasting for one month? The patient is exhibiting disruptive behavior by shouting at the nurses and is difficult to control. She also reports irritability and difficulty in staying focused. There is no known medical history for this patient.

- A. Lithium
 - B. Lorazepam
 - C. Carbamazepine
 - D. Haloperidol
-

16. Which one of the options does not contribute as a risk factor for delusional disorder?

- A. Recent immigration

- B. Social isolation
 - C. Family history
 - D. Young age
-

17. What is the probable diagnosis for a 43-year-old individual who arrives at the emergency department accompanied by the police, alleging that his daughter intends to harm him, but subsequent investigation by the police reveals it to be a false accusation? The patient's physical examination does not reveal any notable findings.

- A. Delusional disorder
 - B. Illusions
 - C. Cotard syndrome
 - D. Hallucinations
-

18. What is the term used to describe a person experiencing a reduced sense of enjoyment in activities that they previously found pleasurable?

- A. Tangentiality
 - B. Anhedonia
 - C. Alexithymia
 - D. Asociality
-

19. What is the probable diagnosis for a 16-year-old girl who exhibits symptoms such as excessive familiarity, rapid flow of thoughts, heightened mood, heightened sexual desire, and experiencing pseudohallucinations?

- A. Mania

- B. Hypomania
 - C. Cyclothymia
 - D. Schizomania
-

20. A patient is diagnosed with schizophrenia and started on risperidone. After taking the medication last night, the patient rushed to OPD today, complaining of upward rolling of his eyes. What is the management of the condition?

- A. Lorazepam
 - B. benztropine
 - C. Olanzapine
 - D. Stop antipsychotic.
-

21. A schizophrenic patient was prescribed drug A after he did not respond to haloperidol and thioridazine. He now presents with excessive salivation, an increase in blood glucose, and hyperlipidemia. What is drug?

- A. Ziprasidone
 - B. Risperidone
 - C. Clozapine
 - D. Aripiprazole
-

22. Which of the following does not suggest a medical cause of mental illness?

- A. Elderly age
- B. Loss of consciousness
- C. Auditory hallucination

D. Acute onset

23. Atypical antipsychotic with least metabolic side effects?

- A. Haloperidol
 - B. Risperidone
 - C. Ziprasidone
 - D. Quetiapine
-

24. Patient who is on antipsychotic medication for his schizophrenia started to have dystonia. Management of this patient is by?

- A. Benzhexol
 - B. Replace with fluphenazine
 - C. Increase the dose of haloperidol
 - D. Clozapine
-

25. Due to the requirement of regular blood monitoring, what rare side effect must be considered for a patient undergoing treatment with clozapine, who is diagnosed with schizophrenia?

- A. Agranulocytosis
 - B. Myocarditis
 - C. Seizures
 - D. Cerebral bleed
-

26. A young male who is shy and prefers social isolation appears to be emotionally cold and resides alone. There is no history of hallucinations or delusions. The most likely diagnosis is:

- A. Paranoid personality disorder
 - B. Schizoid personality disorder
 - C. Antisocial personality disorder
 - D. Emotionally unstable personality disorder
-

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Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	1
Question 4	2
Question 5	2
Question 6	1
Question 7	1
Question 8	2
Question 9	2
Question 10	3
Question 11	3
Question 12	3
Question 13	3
Question 14	3
Question 15	4
Question 16	4
Question 17	1
Question 18	2

Question 19	1
Question 20	2
Question 21	3
Question 22	3
Question 23	3
Question 24	1
Question 25	1
Question 26	2

Solution for Question 1:

The correct answer is B,

- Lorazepam. In the given scenario, the patient exhibits voluntary purposeless movements, waxy flexibility, and negativism, which are suggestive of catatonia. Catatonia is a syndrome associated with various psychiatric and medical conditions. Lorazepam, a benzodiazepine, is a commonly used medication to treat catatonia and can help alleviate symptoms such as motor abnormalities and agitation.

Incorrect options:

Option A . Haloperidol: Haloperidol is an antipsychotic medication commonly used to treat psychotic symptoms, but it may not be the ideal choice for managing catatonia. In fact, the use of antipsychotics alone may worsen catatonic symptoms in some cases.

Option C . Clonidine: Clonidine is primarily used to treat conditions such as hypertension and ADHD. It is not a first-line medication for managing catatonia.

Option D . Propranolol: Propranolol is a beta-blocker medication used for conditions such as hypertension, angina, and tremors. It is not typically used in the management of catatonia

Solution for Question 2:

Flat affect refers to a lack of emotional expression, where the patient shows little or no facial expressions, gestures, or vocal tone variations despite the context. This is classified as a negative symptom in psychiatric disorders, especially schizophrenia. Negative symptoms represent a loss or reduction of normal functions, such as diminished emotional expression, social withdrawal, and lack of motivation.

Types:

- Delusions: False beliefs not based in reality, e.g., persecutory or grandiose delusions.
- Hallucinations: Perceptions without external stimuli, typically auditory (e.g., hearing voices).
- Disorganised speech: Incoherent or illogical speech, such as loose associations or "word salad."
- Disorganised behaviour: Erratic or inappropriate actions, including agitation or inappropriate affect.

2. Negative Symptoms: These involve loss or reduction of normal functions.

- Avolition: Lack of motivation to perform goal-directed activities, such as self-care.
- Anhedonia: Inability to experience pleasure from activities which were previously pleasurable.
- Alogia: Poverty of speech, brief or uninformative responses.
- Flat affect or Blunting actions (Option B): Reduced emotional expression, monotone speech, or lack of facial expressions.
- Social withdrawal: Lack of interest in social relationships.

3. Cognitive Symptoms (Option D): These affect thinking and memory and have a moderate prognosis.

- Impaired attention: Difficulty focusing or sustaining attention.
- Impaired working memory: Trouble holding and manipulating information.
- Executive dysfunction: Difficulty with planning, organising, and problem-solving.
- Slowed processing speed: Slower response to information and tasks.

Other Options:

Affect symptom (Option C): While flat affect is related to emotional expression, "affect symptom" is not a standard classification in psychiatric terminology. Instead, it falls under negative

symptoms .

Solution for Question 3:

Olanzapine , an atypical antipsychotic , is commonly associated with significant weight gain and metabolic side effects . Many patients discontinue the medication due to excessive weight gain, increased appetite, and risk of metabolic syndrome (including diabetes and dyslipidemia) . This is one of the most frequent reasons for non-compliance with olanzapine treatment.

- Antipsychotic Medications:

First Generation Antipsychotics (Typical)	Second Generation Antipsychotics (Atypical)	Second Generation Antipsychotics (Atypical)
D2 receptor antagonist	Partial D2 Agonists	D2/5-HT2A Antagonists
Haloperidol Fluphenazine Perphenazine Chlorpromazine	Aripiprazole Brexipiprazole Cariprazine	Risperidone Olanzapine Quetiapine Ziprasidone Paliperidone Lurasidone Clozapine

- Treatment Duration: For first-episode patients, a minimum of 1-2 years of treatment is recommended. For those with multiple episodes , indefinite treatment is often advised.
- For first-episode patients, a minimum of 1-2 years of treatment is recommended.
- For those with multiple episodes , indefinite treatment is often advised.
- Side Effects:
- For first-episode patients, a minimum of 1-2 years of treatment is recommended.
- For those with multiple episodes , indefinite treatment is often advised.

Extrapyramidal Side Effects (EPS)	Common with first-generation antipsychotics , including symptoms like Acute dystonia - Earliest, can occur within minutes of the injection (Option B) Parkinson's - Occurs within days to weeks Akathisia - Occurs within days to weeks (Option C) Alternatives for management include dose reduction, antiparkinson medications, or switching to medications with lower EPS risk.
Tardive Dyskinesia (TD) (Option D)	Affects 20-30% of patients on long-term first-generation antipsychotics , with higher risk in the elderly and those with cognitive or mood disorders. Prevention strategies include using the lowest effective doses and switching to other medications like Clozapine.
Sedation & Postural hypotension	Especially with low-potency antipsychotics.
Long-term antipsychotic use can elevate prolactin levels, affecting sexual function and potentially leading to osteoporosis.	

Solution for Question 4:

The patient was asked a direct question about her age but instead provided excessive and unnecessary details about her marriage and children before eventually reaching the point. This is characteristic of circumstantial speech , where the person includes irrelevant details but eventually answers the question .

Other Options:

Derailment (Option A): This refers to the loosening of associations , where the person's speech lacks logical connection between ideas and shifts randomly to unrelated topics. In this case, the response is still somewhat related to age, making derailment unlikely.

Flight of Ideas (Option C): Seen in mania , this involves rapidly shifting from one topic to another , often with pressured speech and little coherence. The patient's response does not exhibit this disorganized, racing thought pattern.

Tangentiality (Option D): In this disorder, the patient never returns to the original question and goes off on unrelated tangents. However, in circumstantial speech , the person eventually answers the question, which is seen in this case.

Solution for Question 5:

Correct Option B- They are perceived in the outer objective space:

- This statement is incorrect. Pseudohallucinations are not perceived in the objective space. They are typically perceived in the subjective or internal space of the individual's mind. These perceptions might seem real to the person experiencing them, but they are not projected into the external environment.

Incorrect Options:

Option A- They occur without external stimulus: Pseudohallucinations are defined as perceptions that occur without an external stimulus. In other words, the individual experiences sensory perceptions that seem real but are not triggered by any actual external input. This can be contrasted with true hallucinations, which are perceptions without external stimuli but are experienced as if they are real.

Option C- They are not vivid: hallucinations are often described as vivid and intense sensory experiences. While they are not based on external stimuli, they can still be very detailed and realistic to the person experiencing them. This vividness contributes to the confusion between pseudohallucinations and true hallucinations.

Option D- They are done voluntarily: Pseudohallucinations, like true hallucinations, are under the control of the individual experiencing them. They occur voluntarily and are often distressing to the person. The lack of control distinguishes hallucinations from imagination or deliberate thought processes.

Solution for Question 6:

Correct option. A

- Delirium is the most common acute organic mental disorder. It is an acute confusional state characterized by a disturbance in attention and awareness. Delirium often has a rapid onset

and fluctuating course, and it is commonly caused by underlying medical conditions, such as infections, metabolic imbalances, drug intoxication, or withdrawal. Delirium is a medical emergency that requires prompt evaluation and treatment to address the underlying cause and manage the symptoms.

Incorrect option:

Option B. Dementia: Dementia is a chronic and progressive neurocognitive disorder characterized by a decline in memory, thinking, behavior, and the ability to perform daily activities. Unlike delirium, dementia develops slowly and gradually over time, and it is not considered an acute disorder. While dementia can cause cognitive impairment, it is not the most common acute organic mental disorder.

Option C. Amnesia: Amnesia refers to the loss of memory, either partial or complete, due to various causes such as brain injury, trauma, or psychological factors. While amnesia can occur acutely, it is not as common as delirium as an acute organic mental disorder. Delirium involves not only memory impairment but also disturbances in attention, awareness, and other cognitive functions.

Option D. Anxiety disorder: Anxiety disorder is a broad term that encompasses various mental health conditions characterized by excessive worry, fear, and anxiety. While anxiety disorders can occur acutely, they are not classified as acute organic mental disorders. Acute organic mental disorders typically refer to conditions with an acute onset and organic (physiological) etiology, such as delirium caused by medical factors.

Solution for Question 7:

Correct option. A

- Negative symptoms of schizophrenia are characterized by deficits or reductions in normal emotional, cognitive, or behavioral functioning. They reflect a loss or diminishment of typical functioning rather than an excess or overactivity. Therefore, "Over activity" is the correct answer as it is not considered a negative symptom of schizophrenia.

Incorrect option:

Option B. Anhedonia: Anhedonia is a negative symptom of schizophrenia. It refers to the loss of interest or pleasure in previously enjoyable activities. Individuals with schizophrenia may have difficulty experiencing joy or finding satisfaction in things they used to find pleasurable.

Option C. Alogia: Alogia is a negative symptom of schizophrenia. It is characterized by reduced speech output or impoverished speech content. Individuals with alogia may have difficulty expressing themselves or providing detailed responses in conversations.

Option D. Apathy: Apathy is a negative symptom of schizophrenia. It refers to a lack of interest, motivation, or enthusiasm in activities or goals. Individuals with apathy may have reduced drive or find it difficult to initiate and sustain activities.

Solution for Question 8:

Correct Option B:

Schizoid personality disorder. Schizoid personality disorder is characterized by a pervasive pattern of detachment from social relationships and a restricted range of emotional expression. Individuals with this disorder often prefer solitude and have difficulty forming close relationships. They may appear emotionally cold and detached, and they typically do not experience hallucinations or delusions.

Incorrect Options:

Option A: Paranoid personality disorder: Paranoid personality disorder is characterized by a pervasive distrust and suspicion of others. Individuals with this disorder may be overly suspicious and have a tendency to interpret others' actions as malicious or threatening.

Option C: Antisocial personality disorder: Antisocial personality disorder is characterized by a disregard for the rights of others and a pattern of irresponsible and impulsive behavior. Individuals with this disorder may engage in illegal activities and show a lack of remorse or empathy.

Option D: Emotionally unstable personality disorder: Emotionally unstable personality disorder, also known as borderline personality disorder, is characterized by instability in mood, self-image, and relationships. Individuals with this disorder often experience intense and unstable emotions and have difficulty maintaining stable relationships.

Solution for Question 9:

Correct Option. B

- Automatism: Automatism is not a fundamental symptom of schizophrenia. Automatism refers to automatic, involuntary movements or behaviors without conscious control or awareness. In the context of schizophrenia, the primary symptoms include positive symptoms (e.g., hallucinations, delusions), negative symptoms (e.g., reduced emotional expression, social withdrawal), and cognitive impairments (e.g., disorganized thinking, difficulty concentrating). Automatism is not considered a core feature or defining symptom of schizophrenia.

Incorrect options

Option A. Autism: Autism is a developmental disorder characterized by difficulties in social interaction, communication challenges, and repetitive behaviors. Autism and schizophrenia are separate diagnostic entities with distinct symptomatology. While some symptoms may overlap between autism and schizophrenia, they are considered distinct disorders with different underlying causes and diagnostic criteria.

Option C. Association defect: Association defect, also known as formal thought disorder or disorganized thinking, is a fundamental symptom of schizophrenia. It refers to disruptions in the logical and coherent flow of thoughts, resulting in difficulties in organizing and expressing ideas coherently. Association defects can manifest as tangential thinking, loose associations, or incoherent speech patterns. Option D. Ambivalence: Ambivalence, or the coexistence of conflicting feelings or attitudes, can be present in schizophrenia but is not considered a fundamental symptom. Ambivalence can manifest as mixed emotions, contradictory thoughts, or indecisiveness. It is more commonly associated with other psychiatric conditions, such as borderline personality disorder, rather than being specific to schizophrenia.

Solution for Question 10:

correct answer: option c

- The case described in the question presents a middle-aged man reporting delusions that aliens are speaking to him and telling him to kill his relatives over a period of 7 months. This is consistent with a delusional, fixed, false belief not based on reality. Additionally, hallucinations (i.e., hearing voices) are consistent with the diagnosis of schizophrenia.

- Schizophrenia is a chronic and severe mental illness characterized by a range of symptoms that may include delusions, hallucinations, disorganized thinking and speech, and diminished emotional expression.

- Symptoms : The symptoms of schizophrenia can be grouped into three main categories: positive, negative, and cognitive symptoms. Positive symptoms include delusions (false beliefs), hallucinations (seeing, hearing, or feeling things that are not actually there), disordered thinking and speech, and grossly disorganized behavior. A loss or decrease in normal functions characterizes negative symptoms. They can include Avolition, Apathy, Anhedonia, Asociality, Affective flattening, and Alogia Cognitive symptoms: Cognitive symptoms are characterized by difficulties with attention, memory, and executive functioning (the ability to plan, prioritize, and initiate tasks). Diagnosis: Acc to DSM-5 Delusions Hallucinations Disorganized speech (or FTD) Disorganized or catatonic behavior Negative symptoms 2 out of these 5 and at least 1 out of the first 3, present for 1 month Duration of illness At least 6 months At least 1 out of the first 3 symptoms are present for atleast a month. For rest 5 months, abnormal behavior / other symptoms are seen Treatment: Schizophrenia is a chronic illness that requires ongoing treatment. Treatment typically involves a combination of medication and psychotherapy. Medication: Antipsychotic medications are the cornerstone of treatment for schizophrenia. Psychotherapy: Cognitive-behavioral therapy (CBT) and family therapy are two types of psychotherapy that have been shown to be effective in treating schizophrenia.

- The symptoms of schizophrenia can be grouped into three main categories: positive, negative, and cognitive symptoms.

- Positive symptoms include delusions (false beliefs), hallucinations (seeing, hearing, or feeling things that are not actually there), disordered thinking and speech, and grossly disorganized behavior.

- A loss or decrease in normal functions characterizes negative symptoms. They can include Avolition, Apathy, Anhedonia, Asociality, Affective flattening, and Alogia

- Cognitive symptoms: Cognitive symptoms are characterized by difficulties with attention, memory, and executive functioning (the ability to plan, prioritize, and initiate tasks).

- Diagnosis:

- Acc to DSM-5 Delusions Hallucinations Disorganized speech (or FTD) Disorganized or catatonic behavior Negative symptoms 2 out of these 5 and at least 1 out of the first 3, present for 1 month Duration of illness At least 6 months At least 1 out of the first 3 symptoms are present for atleast a month. For rest 5 months, abnormal behavior / other symptoms are seen Treatment: Schizophrenia is a chronic illness that requires ongoing treatment. Treatment typically involves a combination of medication and psychotherapy. Medication: Antipsychotic medications are the cornerstone of treatment for schizophrenia. Psychotherapy: Cognitive-behavioral therapy (CBT) and family therapy are two types of psychotherapy that have been shown to be effective in treating schizophrenia.

- Delusions
 - Hallucinations
 - Disorganized speech (or FTD)
 - Disorganized or catatonic behavior
 - Negative symptoms
 - 2 out of these 5 and at least 1 out of the first 3, present for 1 month
 - Duration of illness At least 6 months At least 1 out of the first 3 symptoms are present for atleast a month. For rest 5 months, abnormal behavior / other symptoms are seen
- Treatment: Schizophrenia is a chronic illness that requires ongoing treatment. Treatment typically involves a combination of medication and psychotherapy. Medication: Antipsychotic medications are the cornerstone of treatment for schizophrenia. Psychotherapy: Cognitive-behavioral therapy (CBT) and family therapy are two types of psychotherapy that have been shown to be effective in treating schizophrenia.
- At least 6 months
 - At least 1 out of the first 3 symptoms are present for atleast a month.
 - For rest 5 months, abnormal behavior / other symptoms are seen Treatment:
 - Schizophrenia is a chronic illness that requires ongoing treatment. Treatment typically involves a combination of medication and psychotherapy.
 - Medication: Antipsychotic medications are the cornerstone of treatment for schizophrenia.
 - Psychotherapy: Cognitive-behavioral therapy (CBT) and family therapy are two types of psychotherapy that have been shown to be effective in treating schizophrenia.
 - The symptoms of schizophrenia can be grouped into three main categories: positive, negative, and cognitive symptoms.
 - Positive symptoms include delusions (false beliefs), hallucinations (seeing, hearing, or feeling things that are not actually there), disordered thinking and speech, and grossly disorganized behavior.
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 - Cognitive symptoms: Cognitive symptoms are characterized by difficulties with attention, memory, and executive functioning (the ability to plan, prioritize, and initiate tasks).
 - Diagnosis:
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- Delusions
 - Hallucinations
 - Disorganized speech (or FTD)
 - Disorganized or catatonic behavior
 - Negative symptoms
 - 2 out of these 5 and at least 1 out of the first 3, present for 1 month
 - Duration of illness At least 6 months At least 1 out of the first 3 symptoms are present for atleast a month. For rest 5 months, abnormal behavior / other symptoms are seen
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- At least 6 months
 - At least 1 out of the first 3 symptoms are present for atleast a month.
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- Medication: Antipsychotic medications are the cornerstone of treatment for schizophrenia.
 - Psychotherapy: Cognitive-behavioral therapy (CBT) and family therapy are two types of psychotherapy that have been shown to be effective in treating schizophrenia.

Diagnosis:

- Delusions
 - Hallucinations
 - Disorganized speech (or FTD)
 - Disorganized or catatonic behavior
 - Negative symptoms
 - 2 out of these 5 and at least 1 out of the first 3, present for 1 month
 - Duration of illness At least 6 months At least 1 out of the first 3 symptoms are present for atleast a month. For rest 5 months, abnormal behavior / other symptoms are seen
- Treatment: Schizophrenia is a chronic illness that requires ongoing treatment. Treatment

typically involves a combination of medication and psychotherapy. Medication: Antipsychotic medications are the cornerstone of treatment for schizophrenia. Psychotherapy: Cognitive-behavioral therapy (CBT) and family therapy are two types of psychotherapy that have been shown to be effective in treating schizophrenia.

- At least 6 months
- At least 1 out of the first 3 symptoms are present for at least a month.
- For rest 5 months, abnormal behavior / other symptoms are seen Treatment:
- Schizophrenia is a chronic illness that requires ongoing treatment. Treatment typically involves a combination of medication and psychotherapy.
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- Psychotherapy: Cognitive-behavioral therapy (CBT) and family therapy are two types of psychotherapy that have been shown to be effective in treating schizophrenia.
- At least 6 months
- At least 1 out of the first 3 symptoms are present for at least a month.
- For rest 5 months, abnormal behavior / other symptoms are seen Treatment:
- Schizophrenia is a chronic illness that requires ongoing treatment. Treatment typically involves a combination of medication and psychotherapy.
- Medication: Antipsychotic medications are the cornerstone of treatment for schizophrenia.
- Psychotherapy: Cognitive-behavioral therapy (CBT) and family therapy are two types of psychotherapy that have been shown to be effective in treating schizophrenia.

Incorrect Choices:

- Option a . Mania is a state of elevated or irritable mood, energy, and activity levels that can occur in bipolar disorder or as a standalone diagnosis in the form of mania or hypomania. While a person experiencing mania may have delusions and hallucinations, the content of the delusion in this case (i.e., aliens telling him to kill his relatives) is not typical of mania.
- Option b . Delusional disorder is a mental illness characterized by persistent, non-bizarre delusions that are not due to another mental illness, such as schizophrenia or bipolar disorder. While the case described in the question does involve delusions, the presence of hallucinations suggests a diagnosis of schizophrenia rather than delusional disorder.
- Option d . Brief psychotic disorder is a mental illness characterized by the sudden onset of psychotic symptoms, such as delusions, hallucinations, disorganized speech, or grossly disorganized or catatonic behavior, that last for at least one day but less than one month. While the duration of symptoms in the case described in the question is consistent with brief psychotic disorder, the fact that the symptoms have persisted for 7 months suggests a diagnosis of schizophrenia, which is a longer-lasting condition.

Solution for Question 11:

The correct answer is C,

- Insidious onset of symptoms. In schizophrenia, an insidious onset of symptoms refers to a gradual and subtle development of psychotic symptoms over an extended period. This is generally considered a poor prognostic factor because it indicates a longer duration of untreated psychosis, which may lead to more severe and chronic illness.

Incorrect options:

Option A . Positive symptoms: Positive symptoms in schizophrenia refer to the presence of abnormal experiences or behaviors that are not typically seen in healthy individuals, such as hallucinations, delusions, disorganized speech, and grossly disorganized or catatonic behavior. The presence of positive symptoms is not considered a good prognostic factor.

Option B . Late age of onset: Late age of onset refers to the development of schizophrenia symptoms later in life, typically after the age of 40. While late-onset schizophrenia is associated with better premorbid functioning and less severe cognitive impairment compared to early-onset schizophrenia, it is generally considered a better prognostic factor.

Option D . Associated with depression: Comorbid depression in schizophrenia is common and can significantly impact a person's overall functioning and quality of life. However, the presence of depression alone is not considered a good prognostic factor in schizophrenia.

Solution for Question 12:

Correct Option C.

- Frontal lobe: The frontal lobe plays a crucial role in executive functions, which include decision-making, motivation, impulse control, emotional regulation, and social behavior. Dysfunction or lesions in the frontal lobe can lead to changes in behavior, impulsivity, disinhibition, impaired judgment, and difficulties in motivation. The symptoms described in the scenario, such as urinating in public without remorse and difficulty in getting motivated,

are more consistent with frontal lobe dysfunction.

Incorrect Options:

Option A. Parietal lobe: The parietal lobe is primarily involved in processing sensory information and spatial awareness. It plays a role in integrating sensory input, perception of touch, spatial orientation, and body awareness. However, the symptoms described in the scenario, such as urinating in public without remorse, difficulty in motivation, and mild memory deficits, are not typically associated with lesions or dysfunction in the parietal lobe. Option B. Occipital lobe: The occipital lobe is primarily responsible for processing visual information. It is involved in visual perception, object recognition, and visual memory. However, the symptoms mentioned in the scenario do not directly align with occipital lobe dysfunction, as they are not primarily related to visual processing or visual disturbances.

Option D. Temporal lobe: The temporal lobe is involved in various functions, including memory, language processing, auditory perception, and emotional regulation. While mild memory deficits can be associated with temporal lobe dysfunction, the other symptoms described in the scenario, such as urinating in public without remorse and difficulty in motivation, are not commonly associated with isolated temporal lobe dysfunction.

Solution for Question 13:

Correct Option: C

- In the above case, sudden financial loss is the stressor present. The fear that someone is about to harm him is a delusion of persecution. He was unable to sleep or eat properly. The voices of a female are auditory hallucinations. The duration of symptoms is only one week. Here psychotic symptoms are present for only one week. It is a case of a brief psychotic episode.

Incorrect Choices:

Option A: According to ICD 11 the duration should be 1 month for Schizophrenia and . 6 months according to DSM.

Option B: Here, the duration of symptoms must be more than one month but less than 6 months, according to DSM.

Option D: If a patient has schizophrenia symptoms for 8 months and most of the time also has mood symptoms - schizoaffective disorder.

Solution for Question 14:

Correct Option: C

- A person who has this misconception thinks that they are being harassed or persecuted by someone or some organization. The individual may feel as though they are in danger and that they are being watched, listened to, or followed. It is significant to emphasize that these views are founded on a distortion of reality rather than any objective evidence or logical reasoning.
- The kid appears to have a strong conviction in her hallucination, as seen by her distrust of the principal and other instructors, as well as her assumption that classmates are complicit in the scheme. Even when there isn't any genuine support for her suspicions, she can misinterpret these people's harmless behaviors or statements as more proof of their participation in the plot. This terrible misconception may significantly impact a person's life, and it may result in stress, despair, loneliness, and even violence.

Incorrect Choices:

Option A. Thought insertion: A person who has this kind of hallucination thinks that someone else is forcing thoughts into their head. The person may feel that they are being controlled by someone else and that these ideas are not their own. Thought insertion is a frequent psychotic symptom that can also be seen in a wide range of other medical diseases and mental illnesses. Thought insertion is most frequently linked to schizophrenia, though.

Option B. Delusion of control: Another sign of schizophrenia is the delusion of control. It entails the conviction that an outside force or agency is in charge of or manipulating one's ideas, behaviors, or body. The individual could feel that they are not in control of their own lives and that they are being pushed to do things they don't want to do.

Option D. Delusion of reference: A delusion of reference is a conviction that commonplace occurrences, things, or actions in the environment have a unique importance aimed toward oneself. The individual thinks that unrelated things or occurrences have particular significance for them. The individual may think that these occurrences or items are attempting to communicate with them in some manner or sending them messages. For instance, a person can think that haphazard events or other people's acts are purposefully meant to reveal hidden messages or meanings to them.

Solution for Question 15:

Correct Option: D

- The 35-year-old woman appears to be experiencing symptoms typical of a manic episode, which is a hallmark of bipolar disorder, based on the descriptions. An elevated or irritable mood, increased energy, decreased need for sleep, excessive talking, impulsivity, irritability, and distractibility are all symptoms of manic episodes. Considering this show, the treatment of decisions for dealing with the intense side effects of insanity is normally a temperament stabilizer or antipsychotic medicine.
- Haloperidol is an antipsychotic prescription that is generally used to treat intense hyper episodes. It aids in the reduction of mania-related agitation, aggression, and psychotic symptoms. Haloperidol works by blocking dopamine receptors in the brain. This helps to control the overproduction of dopamine, which is thought to be a factor in manic symptoms. It very well may be powerful in rapidly quieting down the patient and decreasing their hyper side effects.

Incorrect Choices:

Option A. Lithium: Another often used mood stabilizer for bipolar illness is lithium, which is typically more efficient for managing and preventing manic and depressive episodes over the long term than for treating immediate symptomatology. Lithium needs to be regularly checked in the bloodstream to keep therapeutic levels and avoid toxicity.

Option B. Lorazepam: Another mood stabilizer frequently used in treating bipolar illness is lithium, which is typically more helpful for chronic management and avoidance of manic and depressed episodes than for acute symptom control. To keep therapeutic levels of lithium and avoid toxicity, blood levels must be regularly monitored.

Option C. Carbamazepine: Trigeminal neuralgia, acute manic and mixed episodes of bipolar I disorder, as well as epilepsy, are all managed and treated with carbamazepine. However, it is not normally the therapy of choice in the case of an acute manic episode, such as the one described in this example. The entire therapeutic benefit of carbamazepine often takes several weeks to become apparent. It is frequently vital to provide prompt symptom alleviation in the acute condition described in order to protect the patient's safety and well-being.

Solution for Question 16:

Correct Option D. Young age.

- Delusional disorder is a psychiatric condition characterized by the presence of persistent delusions without significant impairment in other cognitive functions. Delusions are fixed, false beliefs that are maintained despite evidence to the contrary. While various factors can contribute to the development of delusional disorder, young age is not considered a significant risk factor.

Incorrect options:

Option A. Recent immigration: Recent immigration can be a risk factor for delusional disorder. The stressors and challenges associated with adapting to a new environment, cultural differences, and social isolation can potentially contribute to the development of delusions.

Option B. Social isolation: Social isolation or lack of social support is considered a risk factor for delusional disorder. It can lead to feelings of loneliness, exacerbate existing delusions, and contribute to the persistence of abnormal beliefs.

Option C. Family history: Family history of psychiatric disorders, including delusional disorder, is considered a risk factor. There may be a genetic predisposition or shared environmental factors within families that contribute to the development of delusions.

Solution for Question 17:

Correct Option A:

A delusion is a fixed false belief based on an inaccurate interpretation of an external reality despite evidence to the contrary. The diagnosis of a delusional disorder is made when a person has one or more non-bizarre (situations that are not real but also not impossible) delusional thoughts for one month or more that cannot be explained by any other condition.

- The following are the types of delusional disorders: Persecutory type: Delusion of persecution. Jealous type: Delusion of infidelity. Erotomanic type: Delusion of love. Somatic type: The patient may have a delusion that he is infested by parasites (delusional

parasitosis), that he has misshaped body parts (delusion of dysmorphophobia), or that his body has a foul odor (delusion of halitosis). Grandiose type : Delusion of grandiosity. Unspecified type: In patients where the above-mentioned categories are not applicable. Delusion of misidentification is an example of an unspecified type.

- Persecutory type: Delusion of persecution.
- Jealous type: Delusion of infidelity.
- Erotomaniac type: Delusion of love.
- Somatic type: The patient may have a delusion that he is infested by parasites (delusional parasitosis), that he has misshaped body parts (delusion of dysmorphophobia), or that his body has a foul odor (delusion of halitosis).
- Grandiose type : Delusion of grandiosity.
- Unspecified type: In patients where the above-mentioned categories are not applicable.
- Delusion of misidentification is an example of an unspecified type.
- Risk factors Advanced age Sensory impairment or isolation Family history Social isolation Personality features like unusual interpersonal sensitivity.
- Advanced age
- Sensory impairment or isolation
- Family history
- Social isolation
- Personality features like unusual interpersonal sensitivity.
- Persecutory type: Delusion of persecution.
- Jealous type: Delusion of infidelity.
- Erotomaniac type: Delusion of love.
- Somatic type: The patient may have a delusion that he is infested by parasites (delusional parasitosis), that he has misshaped body parts (delusion of dysmorphophobia), or that his body has a foul odor (delusion of halitosis).
- Grandiose type : Delusion of grandiosity.
- Unspecified type: In patients where the above-mentioned categories are not applicable.
- Delusion of misidentification is an example of an unspecified type.
- Advanced age
- Sensory impairment or isolation
- Family history
- Social isolation
- Personality features like unusual interpersonal sensitivity.

Treatment

- May be resistant to treatment.
- Antipsychotics is the drug of choice.

Incorrect Options :

Option B: illusion is a disturbance of perception. it is the false perception of a real object.

Option C: Cotard's syndrome: also known as Nihilistic delusion/delusion of negation. Here, the patient may deny the existence of their body, their mind, or the world in general. They may claim that everybody is dead, the world has stopped, and the "end of existence".

Option D: hallucination is the false perception without any stimulus.

Solution for Question 18:

Correct option:

Option B: Anhedonia

- Anhedonia refers to the decreased ability to experience pleasure or a lack of interest in activities that were previously enjoyable. It is a symptom commonly associated with various mental health conditions, particularly depressive disorders such as major depressive disorder (MDD). Other symptoms of depression may also be present alongside anhedonia, such as changes in appetite, sleep disturbances, low mood, fatigue, and difficulty concentrating.
- There are two types of anhedonia: Social anhedonia: This refers to a reduced ability to experience pleasure from social interactions or relationships. People with social anhedonia may find it difficult to engage in and enjoy social activities, leading to feelings of isolation and disconnection. Physical or sensory anhedonia: This refers to a diminished pleasure response to sensory stimuli, such as taste, touch, or other physical sensations. Individuals may have difficulty deriving pleasure from activities such as eating delicious food, engaging in sexual experiences, or experiencing the enjoyment of touch or warmth.
- Social anhedonia: This refers to a reduced ability to experience pleasure from social interactions or relationships. People with social anhedonia may find it difficult to engage in and enjoy social activities, leading to feelings of isolation and disconnection.
- Physical or sensory anhedonia: This refers to a diminished pleasure response to sensory stimuli, such as taste, touch, or other physical sensations. Individuals may have difficulty deriving pleasure from activities such as eating delicious food, engaging in sexual

experiences, or experiencing the enjoyment of touch or warmth.

- Treatment for anhedonia often involves a combination of psychotherapy and medication. Psychotherapeutic approaches such as cognitive-behavioral therapy (CBT) and behavioral activation can help individuals identify and modify negative thought patterns and engage in pleasurable activities. Antidepressant medications, such as selective serotonin reuptake inhibitors (SSRIs) or other classes of antidepressants, may also be prescribed to address anhedonia as part of overall symptom management.

Incorrect Options:

Option A. Tangentiality is a formal thought disorder where the person responds to questions with irrelevant details that never reach a goal.

Option C. Alexithymia is difficulty in identifying and expressing emotions.

Option D. Asociality includes a lack of interest or motivation to engage in social interactions.

Solution for Question 19:

Correct Option A:

- Mania is a mood disorder characterized by a distinct period of abnormally elevated, expansive, or irritable mood and increased energy or activity levels. The symptoms mentioned, such as overfamiliarity, flight of ideas, elevated mood, increased sexual desire, and pseudohallucination, are consistent with the manic episode seen in bipolar disorder.
- Mania is a period of at least 1 week where a person's behavior drastically changes and affects their functioning.
- Individuals experiencing mania engage in heightened goal-directed activity and may show psychomotor agitation.
- Mania is a characteristic of Bipolar I disorder .
- Symptoms
 - [Mnemonic to remember: My Asia FAST GDP] M - Mood elevation A - Activity levels Increased F - Flight of ideas A - Abnormally increased levels of activity/energy S - Sleep decreased (Need for sleep is also decreased) T - Talkativeness (Overtalkativeness) G - Grandiose ideas and increased self-esteem D - Distractibility P - Painful consequences (Involvement in such activities
- M - Mood elevation

- A - Activity levels Increased
- F - Flight of ideas
- A - Abnormally increased levels of activity/energy
- S - Sleep decreased (Need for sleep is also decreased)
- T - Talkativeness (Overtalkativeness)
- G - Grandiose ideas and increased self-esteem
- D - Distractibility
- P - Painful consequences (Involvement in such activities
- Treatment
- Antidepressant drugs and mood stabilizers like Lithium, Valproate, Carbamazepine, Oxcarbazepine, and Lamotrigine.

Symptoms

- M - Mood elevation
- A - Activity levels Increased
- F - Flight of ideas
- A - Abnormally increased levels of activity/energy
- S - Sleep decreased (Need for sleep is also decreased)
- T - Talkativeness (Overtalkativeness)
- G - Grandiose ideas and increased self-esteem
- D - Distractibility
- P - Painful consequences (Involvement in such activities

Treatment

Incorrect Options:

Option B: Hypomania is a less severe form of mania.

Option C: Cyclothymia is a chronic mood disorder characterized by cycling between periods of hypomanic and depressive symptoms.

Option D: Schizomania is a mood disorder with manic and mixed episodes in the bipolar type of schizoaffective disorder.

Solution for Question 20:

Correct Option B - benztropine :

- Risperidone is used in the acute treatment of schizophrenia and as a maintenance treatment of schizophrenia. The above-mentioned symptoms suggest an oculogyric crisis , an uncommon side effect of certain antipsychotics. Characterized by neck and body stiffness and the upward deviation of eyes. Mainly due to dopamine hyperactivity within the basal ganglia when there is a fall in the levels of antipsychotic drugs in the CNS.
- entropine is preferred in oculogyric crisis induced by risperidone due to its potent antimuscarinic properties, effectively counteracting dopamine hyperactivity in the basal ganglia. Its specificity for muscarinic receptors makes it a targeted treatment with fewer sedative side effects compared to promethazine. Additionally, its rapid onset of action and availability in both oral and injectable forms contribute to its suitability as a first-line intervention.

Incorrect Options:

Option A - Lorazepam: Intravenous diazepam can also treat oculogyric crises but is not used as first-line management.

Option C - Olanzapine: Olanzapine can be used in treating oculogyric crisis; however, it is not the ideal first-line management. Intramuscular anticholinergics or intramuscular diphenhydramine relieves these symptoms.

Option D - Stop antipsychotic: If medical management fails, the antipsychotic can be stopped if the patient is concerned about the recurrence of its symptoms.

Solution for Question 21:

Correct Option C:

- Clozapine is an atypical antipsychotic medication that is often used as a last resort in the treatment of schizophrenia when other drugs have proven ineffective. It is known for causing several side effects, including hypersalivation (excessive salivation), hyperglycemia (an increase in blood glucose), and hyperlipidemia (an increase in blood lipids), as described in the patient's presentation.
- Clozapine
- 1st atypical antipsychotic discovered.

- DOC for TRS (Treatment-resistant schizophrenia)
- Antagonism D > D (Least EPS) 4 2 5HT , D , 2A 3 a
- D > D (Least EPS) 4 2
- 5HT , D , 2A 3 a
- Only antipsychotics with anti-suicidal properties (shown ↓ suicidal ideas in Schizophrenia patients)
- Used for treatment-resistant Schizophrenia Clozapine is not the 1st drug to be used in a Schizophrenia patient. Used only when a patient doesn't respond to at least 2 antipsychotics.
- Clozapine is not the 1st drug to be used in a Schizophrenia patient.
- Used only when a patient doesn't respond to at least 2 antipsychotics.
- Side effects
- Sedation (most common)
- Weight gain (maximum weight gain of all antipsychotics)
- Hypotension
- Tachycardia
- Sialorrhea - Hypersalivation
- Nausea
- Vomiting
- Constipation
- Serious side effects (SAM)
- S - Seizures Dose dependant > 600 mg has more seizures
- Dose dependant
- > 600 mg has more seizures
- A - Agranulocytosis Idiosyncratic ↓ WBC (Neutrophils) The main reason for deaths and withdrawal from the market Weekly monitoring of WBC and Neutrophil count for 1st 6 months is important
- Idiosyncratic
- ↓ WBC (Neutrophils)
- The main reason for deaths and withdrawal from the market
- Weekly monitoring of WBC and Neutrophil count for 1st 6 months is important
- M - Myocarditis Idiosyncratic Chest pain Dyspnea Abnormal ECG
- Idiosyncratic
- Chest pain
- Dyspnea

- Abnormal ECG
- Contraindications -
- WBC count - $< 3500/\text{mm}$
- Previous bone marrow disorder
- H/o of Agranulocytosis during Clozapine treatment
- Use of another bone marrow suppressant (carbamazepine)
- Clozapine monitoring TLC and ANC Once a week for the first 6 months Once in two weeks for next 6 months Once a month till the patient continues
- TLC and ANC Once a week for the first 6 months Once in two weeks for next 6 months Once a month till the patient continues
- Once a week for the first 6 months
- Once in two weeks for next 6 months
- Once a month till the patient continues

Clozapine

- D > D (Least EPS) 4 2
- 5HT_{2A} , D₂ , 2A 3 a
- Clozapine is not the 1st drug to be used in a Schizophrenia patient.
- Used only when a patient doesn't respond to at least 2 antipsychotics.

Side effects

Serious side effects (SAM)

- Dose dependant
- > 600 mg has more seizures
- Idiosyncratic
- ↓ WBC (Neutrophils)
- The main reason for deaths and withdrawal from the market
- Weekly monitoring of WBC and Neutrophil count for 1st 6 months is important
- Idiosyncratic
- Chest pain
- Dyspnea
- Abnormal ECG

Contraindications -

- TLC and ANC Once a week for the first 6 months Once in two weeks for next 6 months Once a month till the patient continues

- Once a week for the first 6 months
- Once in two weeks for next 6 months
- Once a month till the patient continues
- Once a week for the first 6 months
- Once in two weeks for next 6 months
- Once a month till the patient continues

Incorrect Options:

Option A . Ziprasidone: Ziprasidone is an atypical antipsychotic medication that is less likely to cause the mentioned side effects.

Option B . Risperidone: Risperidone is another atypical antipsychotic, and while it can cause some metabolic side effects, it is not typically associated with hypersalivation.

Option D . Aripiprazole: Aripiprazole is also an atypical antipsychotic, but hypersalivation is not a typical side effect of this medication.

Solution for Question 22:

correct answer:

- Option c : Auditory hallucination
- The correct answer is Auditory hallucination. This is because auditory hallucinations, or hearing voices or sounds that are not actually there, can be a symptom of various mental illnesses, and typically cannot be explained by underlying medical disorder.

Incorrect Choices:

Option a . Older adults are at increased risk for certain medical conditions that can cause or exacerbate mental illness, such as dementia, stroke, and Parkinson's.

Option b . Loss of consciousness or fainting can be caused by various conditions, such as seizures, low blood pressure, or heart problems that may also have mental health implications.

Option d . An abrupt or sudden onset of mental illness symptoms may indicate a medical emergency, such as a stroke, brain injury, or infection.

Solution for Question 23:

Correct Option C- Ziprasidone:

Atypical antipsychotic / 2nd generation antipsychotics

- MOA: D2 and 5HT2A receptor antagonists
- Less extrapyramidal side effects, hyperprolactinemia and with more metabolic side effects
- Among the atypical antipsychotics given, Ziprasidone has the least metabolic side effects comparatively
- Ziprasidone can cause increased QT interval

Incorrect Options:

Option A- Haloperidol:

- This drug falls under the category of typical or 1st generation antipsychotics

Option B- Risperidone:

- It is a atypical antipsychotic drug which is known to have more of extrapyramidal symptoms and hyperprolactinemia
- It has more metabolic side effects than Ziprasidone

Option D- Quetiapine:

- Adverse effect: somnolence, headache, dizziness, constipation, weight gain, elevated plasma triglycerides and cholesterol

Solution for Question 24:

Correct Option A- Benzhexol:

Explanation

- Acute dystonia symptoms include: Eyes: oculogyric crisis Neck: torticollis Jaw muscles: trismus Larynx: laryngospasm Glossopharyngeal: dysarthria, dysphasia, dystonia

- Eyes: oculogyric crisis
- Neck: torticollis
- Jaw muscles: trismus
- Larynx: laryngospasm
- Glossopharyngeal: dysarthria, dysphasia, dystonia
- Treatment for acute dystonia: Parenteral (IV/IM) anticholinergics
- The only anticholinergic drug given in the option is Benzhexol.
- This drug appears to act on parasympathetic nervous system by inhibiting efferent pulses directly.
- Eyes: oculogyric crisis
- Neck: torticollis
- Jaw muscles: trismus
- Larynx: laryngospasm
- Glossopharyngeal: dysarthria, dysphasia, dystonia

Incorrect Options B, C & D:

- Other given drugs are different classes of antipsychotic medication which cannot resolve the dystonic symptoms.

Solution for Question 25:

Correct Option A - Agranulocytosis:

- Agranulocytosis is a severe condition characterized by a significant decrease in the number of white blood cells, specifically neutrophils. It can weaken the immune system, making individuals more susceptible to infections. Clozapine has been associated with a small but significant risk of agranulocytosis, so regular blood monitoring is required to detect any decrease in white blood cell count.

Incorrect Options:

Option B - Myocarditis: Myocarditis refers to inflammation of the heart muscle. While myocarditis is a potential side effect of some medications, it is not a well-established side effect of clozapine.

Option C - Seizures: Seizures are a known side effect of clozapine, but regular blood monitoring is not primarily conducted to monitor seizures. Clozapine carries a risk of lowering the seizure

threshold, especially at higher doses. However, blood monitoring is primarily done to monitor the risk of agranulocytosis rather than seizures.

Option D - Cerebral bleed: Cerebral bleed, or intracerebral hemorrhage, is not a recognized or common side effect of clozapine. While antipsychotic medications can have cardiovascular and cerebrovascular effects, including an increased risk of stroke, cerebral bleed specifically is not associated with clozapine.

Solution for Question 26:

Correct Option B - Schizoid personality disorder:

- Schizoid personality disorder: Schizoid personality disorder is characterized by a pattern of social detachment and emotional coldness. Individuals with this disorder typically have limited interest in social relationships, prefer solitary activities, and often appear emotionally distant. They tend to avoid close relationships and may show a lack of desire for social interaction. The description of a young male who is shy, prefers social isolation, and appears emotionally cold aligns more closely with the features of schizoid personality disorder.

Incorrect Options:

Option A - Paranoid personality disorder: Paranoid personality disorder is characterized by a pervasive distrust and suspicion of others, interpreting their motives as malevolent. Individuals with paranoid personality disorder often exhibit a pattern of suspicion, hyper-vigilance, and a reluctance to confide in others.

Option C - Antisocial personality disorder: Antisocial personality disorder is characterized by a disregard for the rights of others and a lack of empathy. Individuals with this disorder often engage in impulsive and irresponsible behavior, have difficulties conforming to social norms, and may exhibit a pattern of deceit and manipulation.

Option D - Emotionally unstable personality disorder: Emotionally unstable personality disorder, also known as borderline personality disorder, is characterized by a pervasive pattern of instability in mood, self-image, and interpersonal relationships. Individuals with this disorder often experience intense and rapidly shifting emotions, engage in impulsive behavior, and have difficulties maintaining stable relationships.

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Mood disorders

1. A 45-year-old male with a history of depression and anxiety presents with worsening shortness of breath. He admits to heavy smoking over the past 20 years, which has worsened his respiratory symptoms. Despite attempts to quit, he finds smoking alleviates his anxiety. His doctor recommends cessation due to respiratory difficulties. Which medication is best for smoking cessation while addressing his depression?

- A. Varenicline
- B. Bupropion
- C. Nicotine Replacement Therapy (NRT)
- D. Sertraline

2. A 45-year-old male suffering from depression, being treated with lithium carbonate 300 mg thrice daily, develops vomiting, abdominal pain, and dizziness. On examination, he has nystagmus, muscle weakness, and slurred speech. What would be the serum concentration of lithium to cause these symptoms?

- A. > 2.5 mEq/L
- B. 1.5 to 2 mEq/L
- C. 2 to 2.5 mEq/L
- D. 0.5 to 1.2 mEq/L

3. Which of the following statements is true regarding lithium-induced tremors?

- A. D2 blockers may worsen lithium tremor.
- B. Coarse tremor is typical of early lithium use.

- C. Fine tremor from lithium is most noticeable during rest.
 - D. High-dose steroids are commonly used to treat lithium tremors.
-

4. A patient with mood disorder is on treatment with lithium. She is suffering from pain in her hand after a sprain. Which of the following NSAIDs can be advised to this patient in this situation?

- A. Naproxen
 - B. Indomethacin
 - C. Ibuprofen
 - D. Aspirin
-

5. A patient with major depressive disorder is advised with lithium therapy. Which of the following statements is false about lithium?

- A. The commonly used form is lithium carbonate (Li_2CO_3)
 - B. Lithium modulates levels of neurotransmitters and neuropeptides.
 - C. Lithium is a drug only used for the treatment of psychiatric conditions
 - D. Lithium is rapidly and completely absorbed on oral administration
-

6. A 15-year-old child is brought to the psychiatric department for attempting suicide. On further questioning, he states that one of his peers in school died of suicide, followed by which a few of the others, including him, attempted the same method. What is this condition called?

- A. Copycat suicide
- B. Paradoxical suicide
- C. Parasuicide

D. Major depressive disorder

7. A 25-year-old woman visits her doctor 3 weeks postpartum. She reports feeling persistently sad, irritable, and exhausted, has trouble sleeping, and has lost interest in activities she once enjoyed. She also confides that she has thoughts of not wanting to be alive but hasn't acted on them. What condition is she most likely experiencing?

- A. Postpartum Blues
 - B. Postpartum Psychosis
 - C. Postpartum Depression
 - D. Postpartum Anxiety
-

8. A 28-year-old woman presents with significant mood swings, irritability, and a lack of interest in her usual activities in the week before her period. She also mentions difficulty in falling asleep, breast tenderness and easy fatigability. These symptoms improve shortly after her period begins and minimal in the week post menses. What condition is she most likely experiencing?

- A. Dysthymia
 - B. Cyclothymia
 - C. Premenstrual Syndrome
 - D. Premenstrual Dysphoric Disorder
-

9. A 32-year-old woman with a history of bipolar disorder (BD) is planning to conceive. She has been stable on her current medication but is concerned about the risks to her future child. Which of the following is the safest drug for managing BD during pregnancy?

- A. Lithium
 - B. Valproate
 - C. Lamotrigine
 - D. Carbamazepine
-

10. A 31-year-old male with bipolar disorder often engages in impulsive behaviors during manic episodes and struggles with depressive episodes afterwards. He attends therapy sessions where he is taught skills to manage his symptoms, improve interpersonal skills, build routine and recognize patterns in his mood changes. Which therapy would best address his needs?

- A. Cognitive-behavioral therapy (CBT)
 - B. Family-Focused Therapy (FFT)
 - C. Interpersonal & Social Rhythm Therapy (IPSRT)
 - D. Acceptance and Commitment Therapy (ACT)
-

11. A 40-year-old woman with a history of bipolar disorder is brought to the hospital with severe agitation, grandiosity, and hyperactivity. The patient's manic episode has not responded to lithium monotherapy, and her condition continues to deteriorate. What would be an appropriate next step in management?

- A. Add haloperidol to lithium
 - B. Switch to carbamazepine monotherapy
 - C. Continue lithium and add divalproex
 - D. Administer chlorpromazine
-

12. A 32-year-old female presents with a history of frequent mood swings over the past year. She reports episodes of depression lasting several weeks, followed by periods of elevated mood where she feels overly energetic and needs very little sleep. She has had at least five of these episodes in the past 12 months. She is currently on an antidepressant. What is the next best step in her management?

- A. Increase the dose of fluoxetine
 - B. Add lithium + fluoxetine
 - C. Withdraw the antidepressant and add valproate
 - D. Prescribe a higher dose of the antidepressant and add a mood stabilizer
-

13. A 28-year-old male presents with excessive talkativeness, distractibility, and overconfidence for the past 5 days. He has been making impulsive decisions and seems elated. However, he is functioning well at his job and does not require hospitalization. What is the most likely diagnosis?

- A. Hypomania
 - B. Mania
 - C. Mixed Episode
 - D. Cyclothymia
-

14. A 28-year-old woman has had multiple episodes of feeling euphoric, highly energetic, and needing little sleep lasting 3-4 days. However, her symptoms never impair her work or social functioning. She has also had episodes of prolonged low mood and anhedonia lasting several weeks. What is the most likely diagnosis?

- A. Bipolar I Disorder
- B. Bipolar II Disorder
- C. Cyclothymic Disorder
- D. Major Depressive Disorder

15. A 16-year-old female patient recently separated from her boyfriend presents with severe depression with suicidal ideation and has attempted self-harm by slitting her wrist. What is the best possible management care offered to the patient in this situation?

- A. Cognitive Behavioral Therapy (CBT)
- B. Electroconvulsive Therapy (ECT)
- C. Selective Serotonin Reuptake Inhibitors (SSRIs)
- D. Family Therapy

16. A 45-year-old male with a history of severe depression was involved in a construction accident where a nail gun discharged and injured his jaw. As a result, he now experiences significant feeding difficulties. What alternative treatment options available for intranasal administration can be provided to manage his condition?

- A. Esketamine
- B. Fluoxetine
- C. Lithium
- D. Trazodone

17. A 35-year-old man lost his spouse a year ago and has been experiencing depressive disorder. He is consistent with his medications and shows improvement. Which of the following statements is incorrect about the biological factors that contribute to depression?

- A. Neurogenesis hypothesis states that brain abnormalities cause depression

- B. Deficiency or dysregulation of brain monoamines leads to depressive symptoms
 - C. The atrophy of already-developed neurons can cause depression
 - D. Brain dysfunction affecting the amygdala or hippocampus plays no role in depressive disorders
-

18. A 36-year-old divorced man with a history of depression is being treated with moclobemide; on the day of his wedding anniversary, he starts reminiscing about his ex-wife and consumes a large quantity of red wine. Shortly afterwards, he presents with severe headaches, palpitations, sweating, and hypertension. What is the most appropriate management for this patient?

- A. Wait for 30 minutes for symptoms to resolve on their own
 - B. Intravenous Phentolamine
 - C. Discontinuation of moclobemide and administration of intravenous fluids
 - D. Intravenous Labetalol
-

19. A 16-year-old male presents to the clinic with concerns about persistent bedwetting (nocturnal enuresis). His history reveals that this issue has led to depression and social withdrawal. After ruling out any organic causes, the physician decides to start pharmacological therapy. What is the most appropriate class of drug to manage both his depression and bedwetting?

- A. Selective Serotonin Reuptake Inhibitors (SSRIs)
 - B. Tricyclic Antidepressants (TCAs)
 - C. Monoamine Oxidase Inhibitors (MAOIs)
 - D. Benzodiazepines
-

20. A 32-year-old male patient visits a psychiatrist complaining of recurrent panic attacks and significant anxiety in social situations. The patient is also previously diagnosed with depressive disorder. After a thorough assessment, the psychiatrist decides to start him on an SNRI. Which of the following medications is most appropriate for this patient's condition?

- A. Venlafaxine
- B. Desvenlafaxine
- C. Milnacipran
- D. Levomilnacipran

21. A 27-year-old nulliparous woman presents to the clinic with a history of recurrent mood disorder. As she and her partner plan to conceive in the near future, which of the following medications should be avoided in her treatment?

- A. Sertraline
- B. Paroxetine
- C. Fluoxetine
- D. Citalopram

22. A 32-year-old male presents with altered mental status, hyperreflexia, diarrhea and diaphoresis. He recently increased his dose of fluoxetine for depression and has started taking a new medication for his diffuse headaches. On examination, his vital signs reveal tachycardia and mild hypertension. Considering his recent medication changes, which of the following medications is most likely contributing to his current symptoms?

- A. Ibuprofen
- B. Sumatriptan
- C. Metoprolol

D. Loratadine

23. A 28-year-old, recently married male presents with low self-esteem and concerns about premature ejaculation, which has caused distress and contributed to his depression and anxiety. His physician decides to prescribe a drug to manage his condition. Which of the following is the most appropriate drug in this case?

- A. Dapoxetine
 - B. Sildenafil
 - C. Clomipramine
 - D. Alprazolam
-

24. A 35-year-old woman visits the clinic after experiencing persistent sadness, increased anxiety, and sleep disturbances for the past month. She recently went through a traumatic event involving the loss of a loved one. She reports feeling overwhelmed and unable to cope with her emotions. What type of depression is she most likely experiencing?

- A. Endogenous Depression
 - B. Exogenous Depression
 - C. Persistent Depressive Disorder
 - D. Major Depressive Disorder with Psychotic Features
-

25. A 32-year-old female patient has been complaining of persistent sadness, fatigue, and difficulty concentrating over the past few months. Upon examination, you notice several physical signs on her face, which are shown below (asterisk). Identify the sign shown in the image.



- A. Veraguth's Fold
- B. Omega Sign
- C. Chandler's Sign
- D. Dennie-Morgan Lines

26. A 32-year-old female presents to the clinic with persistent low mood, increased appetite, and excessive sleeping. She reports that her mood tends to improve temporarily in response to positive events and that she has heightened sensitivity to rejection. Which of the following treatments is least appropriate for her condition?

- A. Monoamine oxidase inhibitors (MAOIs)
- B. Selective serotonin reuptake inhibitors (SSRIs)
- C. Tricyclic antidepressants (TCAs)
- D. Cognitive-behavioral therapy (CBT)

27. A 50-year-old woman presents with severe depression, marked by sadness, loss of interest, and recent episodes of immobility and unresponsiveness. She maintains bizarre postures and shows minimal verbal response, with low mood and agitation during episodes. Which diagnosis is most appropriate?

- A. Depression with melancholic features
- B. Atypical depression
- C. Depression with catatonic features
- D. Major depressive disorder with psychotic features

28. A 50-year-old woman presents with a 2-year history of sadness and loss of interest in activities, along with recent episodes of hallucinations lasting 1-2 weeks. Her mood remains low between these episodes. Which of the following statements in this scenario is/are true? Statement 1: Monotherapy with tricyclic antidepressants (TCA) is effective. Statement 2: A combination of antidepressants and antipsychotics is the treatment of choice. Statement 3: Patient likely to have schizoaffective disorder since she experiences both mood and psychotic symptoms. Statement 4: Electroconvulsive therapy is the treatment of choice with severe psychotic symptoms and high suicide risk.

- A. 1, 2, and 4
- B. 2 and 4
- C. 2, 3, and 4
- D. 1 and 4

29. A 22-year-old patient is experiencing sadness, hopelessness, anhedonia, fatigue, and low energy. During the examination, she reports a sad mood and she appears agitated. What is the required duration of these symptoms for a diagnosis of depression?

- A. 3 weeks

- B. 2 months
- C. 1 week
- D. 2 weeks

30. Which of the following is not a component of the “Beck’s cognitive triad”?

- A. Negative self-perception
- B. Perception of the environment as hostile
- C. Expectations of suffering and failure in the future
- D. Introjection of lost significant object

31. Match the following genes involved in the development of depression with their normal functions. Genes Encoded for A) SLC6A4 i) Serotonin receptor B) DRD4 ii) Dopamine transporter C) HTR1A iii) Serotonin transporter D) SLC6A3 iv) Dopamine receptor

Genes	Encoded for
A) SLC6A4	i) Serotonin receptor
B) DRD4	ii) Dopamine transporter
C) HTR1A	iii) Serotonin transporter
D) SLC6A3	iv) Dopamine receptor

- A. a-ii, b-iv, c-i, d-iii
- B. a-iii, b-iv , c-i, d-ii
- C. a-ii, b-i, c-iv , d-iii
- D. a-i, b-ii, c-iii, d-iv

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Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	1
Question 4	4
Question 5	3
Question 6	1
Question 7	3
Question 8	4
Question 9	3
Question 10	3
Question 11	3
Question 12	3
Question 13	1
Question 14	2
Question 15	2
Question 16	1
Question 17	4
Question 18	2

Question 19	2
Question 20	1
Question 21	2
Question 22	2
Question 23	1
Question 24	2
Question 25	2
Question 26	3
Question 27	3
Question 28	2
Question 29	4
Question 30	4
Question 31	2

Solution for Question 1:

Correct Answer: B) Bupropion

Explanation:

- Bupropion is a dual-action medication that serves both as an antidepressant and as a smoking cessation aid.
- It works by inhibiting the reuptake of dopamine and norepinephrine, reducing cravings and withdrawal symptoms from nicotine.
- This makes it ideal for patients who suffer from depression and anxiety, like this patient, as it can manage both mood disorders and smoking cessation simultaneously.

Atypical antidepressants:

Category	Mechanism of Action	Drugs	Indications	Common Side Effects
SARI (Serotonin Antagonist and Reuptake Inhibitors)	Antagonizes serotonin receptors (5-HT _{2A} , 5-HT _{2C}) and inhibits serotonin reuptake	Trazodone	Major depressive disorder; off-label for insomnia	Sedation, dizziness, dry mouth, priapism
		Nefazodone	Major depressive disorder	Hepatotoxicity, sedation, dizziness
NaSSA (Noradrenergic and Specific Serotonin Antidepressants)	Antagonizes presynaptic alpha-2 adrenergic receptors; blocks 5-HT ₂ and 5-HT ₃ receptors	Mirtazapine	Major depressive disorder, especially with insomnia	Sedation, weight gain, minimal sexual side effects
NDRI (Norepinephrine-Dopamine Reuptake Inhibitors)	Inhibits the reuptake of norepinephrine and dopamine	Bupropion	Major depressive disorder, anxiety disorders, smoking cessation, Preventive treatment of seasonal affective disorder	Insomnia, tremors, nausea, seizures (at high doses) minimal sexual side effects

Varenicline (Option A) is a partial agonist at the nicotine receptor and is highly effective for smoking cessation by reducing cravings and withdrawal symptoms. However, it does not directly address the patient's underlying depression or anxiety, making it a less ideal option in this scenario where mood management is also crucial.

Nicotine Replacement Therapy (NRT) (Option C) (e.g., nicotine patches, gums, or lozenges) is commonly used to ease nicotine withdrawal but does not address the underlying mood disorders. It may be helpful as part of a smoking cessation strategy but would not treat the patient's depression and anxiety.

Sertraline (Option D) is an SSRI, commonly used for anxiety and depression. While it would help with mood disorders, it is not effective in treating nicotine addiction or facilitating smoking cessation, making it less suitable than bupropion in this case.

Solution for Question 2:

Correct Answer: B) 1.5 to 2 mEq/L

Explanation:

The clinical scenario here points to the diagnosis of mild to moderate lithium toxicity which is caused by a lithium level of 1.5 to 2 mEq/L.

Lithium toxicity is a medical emergency and can cause permanent neuronal damage and death.

Signs and symptoms of lithium toxicity:

Mild to moderate intoxication (lithium level, 1.5–2.0 mEq/L)
Gastrointestinal: Vomiting Abdominal pain Dryness of mouth Neurologic: Ataxia Dizziness Slurred speech Nystagmus Lethargy or excitement Muscle weakness
Moderate to severe intoxication (lithium level: 2.0–2.5 mEq/L) (Option C)
Gastrointestinal: Anorexia Persistent nausea and vomiting Neurologic: Muscle fasciculations Clonic limb movements Hyperactive deep tendon reflexes Choreoathetoid movements Convulsions Delirium Syncope Electroencephalographic changes Stupor Coma Circulatory failure: low BP, arrhythmias, and conduction abnormalities)
Severe lithium intoxication (lithium level >2.5 mEq/L) (Option A)
Generalized convulsions Oliguria and renal failure Death

Mild to moderate intoxication (lithium level, 1.5–2.0 mEq/L)

Management

Discontinue lithium and treat dehydration

Removal of unabsorbed lithium: Emesis, gastric lavage, and absorption with Kayexalate (sodium polystyrene sulfonate) or GoLYTELY (polyethylene glycol solution)

For any patient with a serum lithium level greater than 4.0 mEq/L, hemodialysis

Therapeutic level of lithium:

- 0.5 to 1.2 mEq/L (Option D) is the therapeutic level, but even a slightly higher amount may lead to toxicity, especially in elderly patients.
- For acute mania, 1 to 1.2 mEq/L is the required serum concentration and 0.4 to 0.8 mEq/L for maintenance.

Serum level monitoring of lithium:

- Monitoring should be done regularly every 2 to 6 months.
- Baseline ECG study yearly.
- Weekly monitoring for dose adjustments, signs of toxicity, and noncompliance.
- Samples are taken 12 hours post-dose at steady-state (usually after 5 days of consistent dosing).

Solution for Question 3:

Correct answer: A) D2 blockers may worsen lithium tremor.

Explanation: D2 blockers may exacerbate tremors when used alongside lithium.

Lithium Induced Tremors:

Fine Tremor:

- Prevalence: Tremor of the hands is a common side effect of lithium, affecting around 25% of patients.
- Characteristics: Lithium tremor is generally fine symmetric and resembles essential or physiologic tremor. It is most noticeable during intentional posture (e.g., writing, holding objects). (Option C)
- Lithium tremor is generally fine symmetric and resembles essential or physiologic tremor.

- It is most noticeable during intentional posture (e.g., writing, holding objects). (Option C)
- Additive Effects: Tremor severity can increase due to anxiety, alcohol withdrawal, caffeine, or idiopathic/familial tremors. D2 blockers and some antidepressants may exacerbate tremors when used alongside lithium. (Option A)
- Tremor severity can increase due to anxiety, alcohol withdrawal, caffeine, or idiopathic/familial tremors.
- D2 blockers and some antidepressants may exacerbate tremors when used alongside lithium. (Option A)
- Timing: Typically presents early in treatment but may appear later. In some cases, the tremor improves spontaneously after years of lithium use.
- Typically presents early in treatment but may appear later.
- In some cases, the tremor improves spontaneously after years of lithium use.
- Risk Factors: Higher lithium levels are associated with an increased risk of tremor. Older patients are more susceptible due to the additive effect of age-related essential tremors.
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- Older patients are more susceptible due to the additive effect of age-related essential tremors.
- Lithium tremor is generally fine symmetric and resembles essential or physiologic tremor.
- It is most noticeable during intentional posture (e.g., writing, holding objects). (Option C)
- Tremor severity can increase due to anxiety, alcohol withdrawal, caffeine, or idiopathic/familial tremors.
- D2 blockers and some antidepressants may exacerbate tremors when used alongside lithium. (Option A)
- Typically presents early in treatment but may appear later.
- In some cases, the tremor improves spontaneously after years of lithium use.
- Higher lithium levels are associated with an increased risk of tremor.
- Older patients are more susceptible due to the additive effect of age-related essential tremors.

Coarse Tremor:

- Occurs during lithium toxicity, more irregular, widespread, and severe. (Option B)

Management of Tremors associated Lithium:

- Conservative management includes reducing caffeine intake and maintaining lithium levels in the low–medium range.
- Mild tremors are often tolerated by patients, but moderate or socially embarrassing tremors may require treatment.

Treatment:

- Beta-blockers (e.g., propranolol) are the most common treatment for lithium-induced tremor (Option D).

Other options include:

- Primidone
- Benzodiazepines
- Vitamin B6

Solution for Question 4:

Correct Answer: D) Aspirin

Explanation:

The NSAID that can be advised here is aspirin as that is the only one without any interaction with lithium whereas the others cause an increase in lithium concentrations.

Precautions and Adverse Effects of Lithium:

- Water and Salt Balance: Changes in body water and sodium levels can affect lithium excretion. Excess Sodium: High sodium intake can lower lithium concentrations. Low Sodium: Insufficient sodium (e.g., from fad diets) can lead to toxic lithium levels.
- Changes in body water and sodium levels can affect lithium excretion.
- Excess Sodium: High sodium intake can lower lithium concentrations.
- Low Sodium: Insufficient sodium (e.g., from fad diets) can lead to toxic lithium levels.
- Decreased body fluid (e.g., excessive sweating) can cause dehydration and lithium intoxication.
- Changes in body water and sodium levels can affect lithium excretion.
- Excess Sodium: High sodium intake can lower lithium concentrations.
- Low Sodium: Insufficient sodium (e.g., from fad diets) can lead to toxic lithium levels.

Side effects of Lithium	
Neurologic	Benign, nontoxic: Dysphoria, Lack of spontaneity, Slowed reaction time , Memory difficulties Toxic: Coarse tremor, Dysarthria, Ataxia, Neuromuscular irritability, Seizures, Coma, Death Miscellaneous: Peripheral neuropathy, Benign intracranial hypertension, Myasthenia gravis-like syndrome, Lower seizure threshold
Endocrine	Thyroid: Goiter, hypothyroidism , exophthalmos, hyperthyroidism Parathyroid: Hyperparathyroidism , adenoma
Cardiovascular	Benign T-wave changes Sinus node function
Renal	Concentrating defect, Morphologic changes, Polyuria (nephrogenic diabetes insipidus) , Reduced GFR, Nephrotic syndrome, Renal tubular acidosis
Dermatologic	Acne, Hair loss, Psoriasis, Rash
Gastrointestinal	Loss of appetite, Nausea, Vomiting, Diarrhea
Miscellaneous	Altered carbohydrate metabolism Weight gain Fluid retention
Drug interactions	

Side effects of Lithium	
Generally safe and effective. Higher doses of dopamine-releasing agents (DRAs) may increase: Lithium-induced neurologic side effects. Neuroleptic extrapyramidal symptoms. Rare cases of encephalopathy were reported. Interactions with Other Medications: Carbamazepine, Lamotrigine, Valproate, Clonazepam: ■ Li concentrations. Diuretics: Thiazide and Potassium-Sparing Diuretics: ■ Li concentrations. Osmotic and Loop Diuretics, Carbonic Anhydrase Inhibitors, Xanthines: ■ Li concentrations. ACE Inhibitors: ■ Li concentrations. Angiotensin II Receptor Inhibitors (e.g., Losartan, Irbesartan): Do not affect lithium concentrations. NSAIDs: Many (e.g., Indomethacin, Ibuprofen, Naproxen) can ■ Li concentrations (decrease lithium clearance). (Option A, B & C) Aspirin and Sulindac: Do not affect lithium concentrations. Quetiapine (Seroquel): May cause somnolence but is generally well tolerated. Ziprasidone (Geodon): May modestly increase the incidence of tremors. Calcium Channel Inhibitors: Avoid coadministration due to the risk of fatal neurotoxicity. Electroconvulsive Therapy (ECT): Discontinue lithium 2 days before ECT to reduce delirium risk.	

Solution for Question 5:

Correct Answer: C) Lithium is a drug only used for the treatment of psychiatric conditions

Explanation:

Lithium has a proven use in the treatment of non-psychiatric disorders, including neurological disorders like epilepsy, hematological like aplastic anemia, endocrine like thyroid cancer, etc.

Lithium:

- Dosage form: Commonly used as lithium carbonate (Li_2CO_3). (Option A)

Pharmacokinetics

- Absorption: rapid and complete after oral administration; peak serum levels in 1–1.5 hours (standard) and 4–4.5 hours (slow-release). (Option D)
- Pregnancy: Excretion increases during pregnancy, decreases post-delivery, and is excreted in breast milk.

Mechanisms of Action

- Mood Stabilization: the exact mechanism is unclear; theories include: Ion transport: alterations in ion transport across cell membranes. Neurotransmitter effects: modulate levels of neurotransmitters and neuropeptides. (Option B) Signal transduction: influences signal transduction pathways and second messenger systems.
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- Neurotransmitter effects: modulate levels of neurotransmitters and neuropeptides. (Option B)
- Signal transduction: influences signal transduction pathways and second messenger systems.
- Ion transport: alterations in ion transport across cell membranes.
- Neurotransmitter effects: modulate levels of neurotransmitters and neuropeptides. (Option B)
- Signal transduction: influences signal transduction pathways and second messenger systems.

Uses of Lithium

Psychiatric uses	Nonpsychiatric uses *
Well-established (FDA approved): Bipolar disorder Evidence benefit in particular groups: Schizophrenia Aggression (episodic), explosive behavior, and self-mutilation Conduct disorder in children & adolescents Cognitive disorders Intellectual disability Aggressive prisoners Anecdotal, controversial, unresolved, or doubtful: Alcohol and substance-related disorders PTSD OCD ADHD Phobias Eating disorders Cocaine abuse Sexual disorders Transvestism Exhibitionism Pathologic hypersexuality Personality disorders	Neurologic Epilepsy Headache Huntington disease Hematologic Aplastic anemia Cancer- chemotherapy-induced/ radiotherapy-induced Felty syndrome Endocrine Thyroid cancer Cardiovascular Antiarrhythmic agent Dermatologic Genital herpes Eczematoid & seborrheic dermatitis Gastrointestinal Cyclic vomiting Gastric ulcers Pancreatic cholera Ulcerative colitis Respiratory Asthma Cystic fibrosis Other Bovine spastic paresis
	*(Do not have FDA approved labeling)

Solution for Question 6:

Correct Answer: A) Copycat suicide

Explanation:

The condition here is copycat suicide where one individual's suicidal behavior influences others within the same peer group to attempt suicide. It can be due to an influence from a famous personality as well.

Modified SAD PERSONS Score:

- Mnemonic for 10 criteria for assessment of potential suicidal behavior.

Letter	Criteria	Number of Points Assigned
S	Sex: Male	1
A	Age: <19 or >45 years	1
D	Depression or hopelessness	2
P	Previous attempts or psychiatric care	1
E	Excessive alcohol or drug use	1
R	Rational thinking loss	2
S	Separated/divorced/widowed	1
O	Organized or serious attempt	2
N	No social supports	1
S	Stated future intent	2

- Patients with a score of ≤ 5 : Outpatient psychiatric evaluation
- Patients with a score of ≥ 6 : Emergent psychiatric consultation and a possible hospital admission

Protective Factors:

- Secure connections to school and peers can mitigate risk, even with other risk factors.

Paradoxical suicide (Option B)	Copycat suicide	Parasuicide (Option C)
Patients recovering from suicidal depression are at increased risk as their mood improves. Gaining energy can lead to the ability to act on suicidal plans which is known as paradoxical suicide. Some individuals may suddenly appear at peace, indicating a hidden decision to commit suicide. Clinicians should be vigilant regarding significant changes in behavior, as they may signal an impending suicide attempt.	Copycat suicide refers to the phenomenon where one individual's suicidal behavior influences others within the same peer group to attempt suicide. Clusters of suicides can occur among adolescents who know each other or attend the same school. Suicidal behavior in one adolescent can lead to subsequent attempts by peers through identification. Media Influence : Certain media portrayals, especially of adolescent suicide, can lead to transient increases in suicide attempts. The phenomenon known as "Werther syndrome" illustrates how publicized suicides can inspire imitative behavior among vulnerable individuals. Historical Context : The term "Werther syndrome" originates from Goethe's novel The Sorrows of Young Werther , which led to a spike in suicides among young men after its publication. Impact of exposure: Direct exposure to a peer's suicide is more likely to result in increased risks of depression and PTSD rather than directly causing more suicides.	A parasuicide is nothing but an attempted suicide. History of parasuicide significantly increases suicide risk by up to 40 times, remaining elevated for up to 20 years. Highest risk of suicide occurs within the first months post-psychiatric care, with elevated risk persisting for 1-2 years after parasuicide. More severe and frequent parasuicide attempts indicate higher risk; detailed planning of attempts is a concerning sign. Co-occurring mental disorders (especially depression) and substance abuse during parasuicide increase the risk for poor outcomes. The rates are 60% for young men and 80% for young women.

Major depressive disorder (Option C) is where there are feelings of sadness, hopelessness, and worthlessness; described as "blue" or "in the dumps" with anhedonia and other symptoms like fatigue, insomnia, loss of libido, etc which may lead to suicide attempts but not always.

Solution for Question 7:

Correct Answer: C) Postpartum Depression

Explanation: Symptoms such as depressed mood, irritability, insomnia, lack of interest in usual activities (anhedonia), and suicidal thoughts at three weeks postpartum indicate a diagnosis of Postpartum Depression.

	Postpartum Blues (Option A)	Postpartum Depression (Option C)	Postpartum Psychosis (Option B)
Occurrence	Affects up to 80% of women after childbirth	Affects 10- 15% of women 70% risk of recurrence in next pregnancy	Incidence is about 1-2 in 1000 deliveries 50% risk of recurrence in next pregnancy
Timing	Onset 3-5 days postpartum, resolves by days to weeks	Within the first 6 weeks postpartum or after abortion	Symptoms appear within 2-3 weeks postpartum, last 2-3 months
Symptoms	Transient symptoms such as Emotionally labile Insomnia Weepiness	Loss of energy Depressed mood Loss of interest or pleasure (anhedonia) Loss of appetite Insomnia Social withdrawal Irritability Suicidal thoughts	Initial symptoms: insomnia, tearfulness, mood lability, fatigue, followed by Hallucinations Delusions
Risk Factors	Not specific; linked to biochemical changes	Young maternal age Antenatal depression Family history of mood disorder Unmarried status Smoking Stress Newborns in intensive care Physical or verbal abuse Serious adverse obstetrical events	Nulliparas Obstetrical complications Family history of mood disorder History of psychotic illness
Treatment	Supportive treatment and reassurance	Antidepressants and psychotherapy	Hospitalization, pharmacological treatment (lithium, antipsychotics, antidepressants), long-term psychiatric care Temporary separation from the infant and nursing supervision may be necessary

- Brexanolone: New FDA (in 2019) approved drug for the treatment of postpartum depression. Intravenous formulation of allopregnanolone administered over 60 hours. It is a

neuroactive steroid and functions as an allosteric modulator of GABA■.

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- It is a neuroactive steroid and functions as an allosteric modulator of GABA■.
- Zuranolone/ Zurzuva: First FDA approved oral medication indicated to treat postpartum depression (PPD) in adults.

Postnatal Anxiety (Option D) negatively affects breastfeeding, early interactions, infant temperament, sleep, and cognitive development, while the symptoms of persistent sadness, irritability, and suicidal thoughts are more indicative of postpartum depression.

Solution for Question 8:

Correct Answer: D) Premenstrual Dysphoric Disorder

Explanation: Presence of at least five specific symptoms , including mood swings, irritability, lack of interest, insomnia and easy fatigability, that occur in the luteal phase and resolve after menstruation , align with diagnosis of Premenstrual Dysphoric Disorder (PMDD).

Premenstrual Dysphoric Disorder (PMDD):

Definition	Characterized by debilitating mood and behavioral change in the week preceding menstruation (luteal phase) that interferes with normal functioning. Disappears within a few days of the onset of menses.
Diagnosis	At least 5 of the following symptoms must be present: Affective lability (e.g., mood swings) Irritability or anger or increased interpersonal conflicts Depressed mood, feeling of hopelessness, or self-deprecating thoughts Anxiety, tension and/or feeling of being keyed up or on edge Decreased interest in usual activities (e.g., work, school) Difficulty in concentration Lethargy, easy fatigability, or marked lack of energy Change in appetite; overeating; specific food cravings Hypersomnia or insomnia Sense of being overwhelmed or out of control Physical Symptoms: Breast tenderness or swelling Joint or muscle pain Sensation of bloating Weight gain Diagnosis: Confirmed if these symptoms present for at least 2 symptomatic cycles.
Differential Diagnosis	Premenstrual Syndrome (PMS) (Option C) Dysmenorrhea Bipolar disorder, major depressive disorder, and dysthymia
Treatment	SSRI: Citalopram, escitalopram, fluoxetine, paroxetine and sertraline Benzodiazepines: Alprazolam

PMS vs PMDD:

Features	PMS	PMDD
Severity of symptoms	Mild to moderate	Severe
Mood symptoms	Persistent, but not dominant	Severe emotional & mood symptoms dominate
Functional impairment	Minimal to mid; daily life remains manageable	Significant; affects work, social, and personal life
DSM-5 diagnostic criteria	No specific criteria	Requires at least 5 symptoms

Dysthymia (Dysthymic Disorder/Persistent Depressive Disorder): (Option A)

- Chronic mild depressive symptoms lasting most of the day and almost continuously.

- According to DSM-5 , persistent depressive disorder is diagnosed when depressive symptoms persist for 2 or more years (or 1 year for children), with no symptom-free periods lasting longer than 2 months during the illness.
- Double depression: Diagnosed if, after the initial 2 years of dysthymia, a person then suffers from major depressive episodes.

Cyclothymia: (Option B)

- Milder form of bipolar disorder .
- According to DSM-5, cyclothymic disorder is characterized by depressive and hypomanic symptoms lasting at least 2 years (or 1 year for children), with symptoms present for at least 50% of the time during this period.

Solution for Question 9:

Correct Answer: C) Lamotrigine

Explanation: Lamotrigine is considered one of the safest medications for pregnant individuals with bipolar disorder, with a low risk of fetal malformations and no adverse effects on postpartum cognitive development.

Treatment of Bipolar Disorder (BD) in Pregnancy:

The safety of pharmacotherapeutic agents for treating bipolar disorder must be considered in terms of the following:

Medication	Potential Fetal/Newborn risk	Clinical Recommendation
Mood Stabilizers		
Lithium (Option A)	Low risk of Ebstein's anomaly: 1 in 1000 with first-trimester exposure No evidence of behavioral or cognitive problems in children exposed to lithium till 5 years of age	Fetal echocardiography Frequent serum lithium monitor in view of fluid shifts over pregnancy Should be well hydrated to reduce chances of lithium toxicity

Medication	Potential Fetal/Newborn risk	Clinical Recommendation
Valproic acid (Option B)	Increased rate of congenital malformations (approximate 10%), including neural tube defects Linked to neurodevelopmental disorders in children with in-utero exposure	Must be avoided in favor of safer agents Prior to conception, women with BD preferably to be shifted to another safer prophylactic agent High-dose folic acid supplementation (4 mg/day) advised prior to conception
Carbamazepine (Option D)	Congenital malformations: Around 3%–6% Neural tube defect, cleft palate, fingernail hypoplasia Risk of bleeding diathesis in neonates due to Vitamin K deficiency	If possible, use in pregnancy should be avoided High-dose folic acid supplementation (4 mg/day) advised prior to conception Oral Vitamin K (20mg/day) supplementation in the last month of pregnancy in women taking carbamazepine Parenteral Vitamin K (1mg IM) supplementation for newborn, shortly after birth.
Lamotrigine (Option C)	Risk of major congenital malformations (2%–3%) No significant neurobehavioral sequelae	Preferred safer agent to use in pregnancy Dose may be increased due to enhanced clearance rates across pregnancy stages
Other Drugs for Bipolar Disorder Safe in Pregnancy		
First Generation Antipsychotics (FGA): Haloperidol, Trifluoperazine Second Generation Antipsychotics (SGA): Olanzapine, quetiapine, aripiprazole Selective Serotonin Reuptake Inhibitors (SSRI): Fluoxetine, escitalopram, sertraline, fluvoxamine		

Solution for Question 10:

Correct answer: C) Interpersonal & Social Rhythm Therapy (IPSRT)

Explanation:

Interpersonal & Social Rhythm Therapy (IPSRT) is the most appropriate choice for this patient as it directly addresses the dual challenges of managing impulsivity during manic episodes, improve interpersonal skills, build routine and recognizing mood patterns to mitigate depressive episodes, aligning closely with his treatment needs.

- Lithium
- Anticonvulsants: Lamotrigine, valproate
- Antipsychotics: Quetiapine (Best evidence for efficacy) , olanzapine, lurasidone, cariprazine
- Antidepressants: May induce mania , but some antidepressants (e.g., fluoxetine) may be useful in combination with mood stabilizers but should not be used as monotherapy.
- Electroconvulsive Therapy (ECT): Effective in treatment-resistant cases or where there is a strong suicidal tendency.

Lithium

Anticonvulsants: Lamotrigine, valproate

Antipsychotics: Quetiapine (Best evidence for efficacy) , olanzapine, lurasidone, cariprazine

- Maintenance Treatment : Goal : Prevent recurrences without causing unwanted side effects like sedation, weight gain, or cognitive impairment. Key Medications: Lithium, carbamazepine, valproic acid, and lamotrigine (better for depressive phases).
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- Key Medications: Lithium, carbamazepine, valproic acid, and lamotrigine (better for depressive phases).
- Psychotherapy: Essential for treatment adherence and emotional stability.
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- Goal : Prevent recurrences without causing unwanted side effects like sedation, weight gain, or cognitive impairment.
- Key Medications: Lithium, carbamazepine, valproic acid, and lamotrigine (better for depressive phases).
- Essential for treatment adherence and emotional stability.

Therapy Type	Conceptualization of Disorder Etiology	Interventions
Cognitive- Behavioral Therapy (CBT) (Option A)	Biological vulnerability interacting with stress, along with a lack of skills , limits the ability to manage symptoms effectively.	Identify and challenge automatic thoughts, Engage in rewarding activities for routine and stability Practice communication skills with providers
Interpersonal & Social Rhythm Therapy (Option C)	Interpersonal vulnerabilities from early attachment and learned relationship patterns, disrupting social rhythms	Develop awareness of relationship patterns Track and stabilize social rhythms . Interpersonal skills training, communication analysis
Family-Focused Therapy (Option B)	Biological vulnerability worsened by negatively expressed emotions in the family environment	Education on disorder and risk factors, establishing a relapse prevention plan with family, Communication and problem-solving skills training

CBT (Option A) addresses cognitive distortions and negative thought patterns. While effective for symptom management, it does not specifically target social rhythms

FFT (Option B) educates families about bipolar disorder and improves communication within the family. It does not directly address individual mood management, or impulsive behaviors.

ACT (Option D) promotes psychological flexibility and acceptance of thoughts and feelings. However, it lacks a direct focus on mood stabilization or recognizing patterns specific to bipolar disorder.

Solution for Question 11:

Correct answer: C) Continue lithium and add Divalproex Explanation:

When lithium monotherapy fails to control acute mania, the next appropriate step is to use combination with antipsychotics like olanzapine, risperidone. Since this is not mentioned in the options, add divalproex (valproate) as a second-line combination therapy. This combination is effective in managing manic symptoms when monotherapy is insufficient.

Pharmacologic Treatment of Acute Mania	
First line	Monotherapy : Lithium , divalproex, and atypical antipsychotics such as olanzapine, risperidone, quetiapine, aripiprazole, cariprazine, asenapine, paliperidone, iloperidone, ziprasidone.
Adjunctive therapy with lithium or divalproex: Risperidone, quetiapine, olanzapine, aripiprazole, asenapine	
Second line	Monotherapy: Carbamazepine , electroconvulsive therapy (ECT), haloperidol.
Combination therapy: Lithium + divalproex (Option C)	
Third line	Monotherapy : Chlorpromazine , Clozapine, Tamoxifen
Combination therapy: lithium or divalproex + haloperidol, lithium + carbamazepine, adjunctive tamoxifen	

Novel agent:

- Endoxifen: for acute mania in Bipolar I disorder
- Metabolite of tamoxifen.
- MOA: Protein kinase C (PKC) inhibitor
- Lithium: It remains a cornerstone in mood stabilization, with therapeutic levels ranging from 0.6 to 1.2 mEq/L . However, due to its side effects and the need for frequent monitoring, newer drugs are often preferred in acute settings.

Anticonvulsants:

- Valproate (divalproex) is now more commonly used than lithium for acute mania .
- Carbamazepine.

Add haloperidol to lithium (Option A) : Haloperidol is a third-line option due to its side effect profile (e.g., extrapyramidal symptoms), and is typically reserved for patients who have not responded to second-line treatment.

Switch to carbamazepine monotherapy (Option B) : While carbamazepine can be used as a second-line monotherapy, the scenario calls for augmentation, not switching, making the addition of Divalproex a more appropriate choice.

Administer chlorpromazine (Option D) : Chlorpromazine is a third-line treatment for acute mania and is not preferred due to its sedative effects and risk of tardive dyskinesia.

Solution for Question 12:

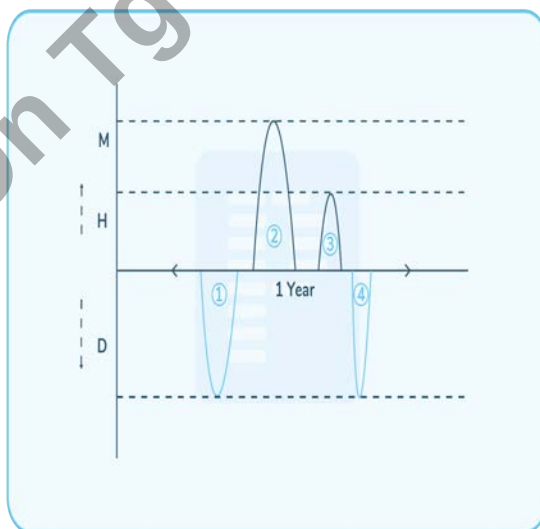
Correct answer: C) Withdraw the antidepressant and add valproate

Explanation:

The patient's five distinct mood episodes in the past year indicate rapid cycling bipolar disorder, likely worsened by her antidepressant use. Withdrawing the antidepressant and initiating mood stabilizers like lithium and valproate or lamotrigine is essential to stabilize her mood and prevent mania.

Rapid cycling bipolar disorder:

- Rapid cycling involves experiencing at least four mood episodes (manic, depressive, or hypomanic) within 1 year .
- A minimum 2-month period of partial or full remission must occur between episodes .
- It is more common with bipolar II disorder than bipolar I disorder.
- More common in females , typically accompanied by depressive and hypomanic episodes.



Factors favoring occurrence of Rapid cycling:

- Long term, aggressive antidepressant usage
- Hypothyroidism

- Menopause
- Temporal lobe dysrhythmias
- Circadian rhythm disturbances
- Kindling
- Substance abuse

Management:

- Withdrawal of antidepressant
- Evaluate possible precipitant(hypothyroidism, external stressors)
- Mood stabilizers like Valproate (DOC) / lamotrigine
- Antipsychotics

Increase the dose of fluoxetine (Option A) : Increasing the dose of the antidepressant (fluoxetine) could exacerbate her mood instability and potentially trigger manic episodes, especially in rapid cycling bipolar disorder.

Add lithium + fluoxetine (Option B) : Combining lithium with an antidepressant like fluoxetine may not be safe in a patient with rapid cycling bipolar disorder, as antidepressants can provoke mania. A more effective approach is to withdraw the antidepressant altogether.

Prescribe a higher dose of the antidepressant and add a mood stabilizer (Option D) : Although adding a mood stabilizer is appropriate, increasing the antidepressant dose could further exacerbate the rapid cycling.

Solution for Question 13:

Explanation:

The patient has symptoms of hypomania (elation, impulsivity, and distractibility) lasting for 5 days , with no marked impairment in functioning, consistent with the diagnostic criteria for hypomania.

Diagnostic criteria for Mania

Mood Changes: Elation/ irritability
Persistently increased Energy
+ ≥3 of the following symptoms (4 if irritable mood) for at least 1 week :
Distractibility.
Impulsivity/ high-risk activities
Grandiosity/ Inflated Self-Esteem
Flight of Ideas (Racing and expansive thoughts)
Activity increased (Hyperactivity)
Sleep deficit (decreased need for sleep)
Talkativeness/ Pressured speech

Mania with Psychotic Symptoms:

- May include delusions or hallucinations .
- Can be mood-congruent or mood-incongruent.

Mood-congruent: When psychotic symptoms are mood-congruent, the content of a person's delusions or hallucinations aligns with the person's mood state.

Mood-incongruent: The content of a person's delusions or hallucinations does not match their mood.

Hypomania (Option A)

- Symptoms are similar to mania but less severe .
- NO Marked impairment in functioning and NO hospitalization necessitated
- No psychotic symptoms
- Flight of ideas usually not seen
- Duration of hypomania is ≥4 days .

Mixed Episode :

- Manic and depressive symptoms alternate throughout the day.
- Duration is ≥7 days .

Mania (Option B) is more severe than hypomania, involving significant functional impairment or the need for hospitalization. Here, the patient is functioning well.

Mixed Episode (Option C) involves simultaneous symptoms of mania and depression. The patient has no depressive symptoms.

Cyclothymia (Option D) is characterized by chronic mood fluctuations over at least 2 years, including periods of hypomanic and depressive symptoms. The patient's symptoms have lasted only 5 days and do not include depressive episodes.

Solution for Question 14:

Correct answer: B) Bipolar II Disorder

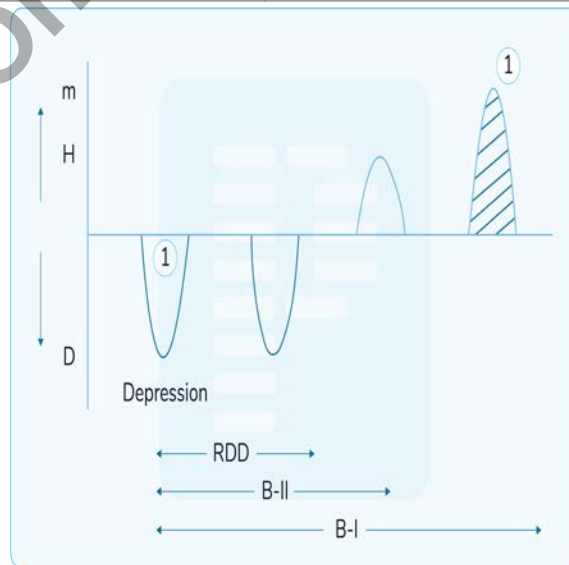
Explanation :

The presence of hypomanic episodes (euphoria without significant impairment) alongside depressive episodes indicates Bipolar II Disorder .

Bipolar disorder :

- Bipolar disorder is a type of mood disorder characterized by episodes of depression, mania, hypomania, mixed episodes.
- Even a single episode of mania is considered bipolar disorder.
- ICD-11 classifies two types of bipolar disorder.

Bipolar I Disorder	Bipolar II Disorder
Characterized by at least 1 episode of mania and 1 episode of depression	Characterized by at least 1 episode of hypomania and 1 episode of depression (Option B)
M=F	F>M
Mean age of onset is 30yrs	Mean age of onset is slightly higher



Epidemiology	Annual incidence is less than 1% . Equal prevalence of Bipolar I disorder in men and women . Manic episodes are more common in men . Depressive episodes are more common in women . Women are more prone to mixed episodes and rapid cycling . Age of Onset: Onset ranges from 5 to 50 years, with a mean age of 30 . Prevalence in adolescents rises from 2% at ages 13-14 to 4% by age 18 . Socioeconomic Factors: Higher incidence in divorced or single individuals. More common in upper socioeconomic groups and non-college graduates.
Genetics	Family history increases the risk for mood disorders. 1st-degree relatives have a 10-fold risk of developing bipolar disorder. Genetic overlap with schizophrenia , especially on chromosomes 18q and 22q.
Brain Imaging	Increased hyperintensities in subcortical areas (basal ganglia, thalamus) in bipolar I disorder . Imaging studies show increased dopamine transporter levels. Functional imaging shows abnormal brain activation, with hyperactivity in frontal areas and hypoactivity in the amygdala and basal ganglia .

Bipolar I Disorder (Option A) : Requires at least one manic episode causing significant impairment or psychosis. Since the patient has only experienced hypomania, this diagnosis is incorrect.

Cyclothymic Disorder (Option C) : Involves periods of hypomanic and depressive symptoms that don't meet full criteria for either, lasting at least 2 years.

Major Depressive Disorder (Option D) : Involves only depressive episodes without any history of hypomanic or manic episodes.

Solution for Question 15:

Correct Answer: B) Electroconvulsive Therapy (ECT)

Explanation:

Electroconvulsive Therapy (ECT) is indicated for providing rapid relief of suicidal ideation in patients experiencing severe depression . It is also effective for treatment-resistant depression that does not respond to multiple antidepressant trials. Additionally, ECT is beneficial for severe psychosis, food refusal secondary to depression, catatonia, bipolar depressive and manic patients.

Treatment Approaches for Depression:

Type of Therapy	Description	Indication
Combination Therapy	Pharmacotherapy + Psychotherapy Combines medications (e.g., antidepressants) with psychotherapy for a higher response rate.	Moderate to severe depression
Single Therapy	Psychotherapy alone may be sufficient for mild cases.	Mild depression

Psychotherapy :

Psychotherapy Type	Description	Focus
Cognitive Behavioral Therapy (CBT) (Option A)	Evidence-based therapy focusing on correcting cognitive distortions and maladaptive behaviors.	Developing healthier thought patterns and coping strategies.
Interpersonal Therapy (IPT)	Addresses current interpersonal problems, including family dynamics and relationship issues.	Enhancing social support and resolving conflicts.
Family Therapy (Option D)	Involves family members to improve communication and resolve conflicts.	Family dynamics and relationships.
Psychoanalytically Oriented Therapy	Uncovers unconscious motivations and past experiences affecting behavior.	Insight into past influences on current behavior.
Behavior Therapy	Utilizes reinforcement techniques to modify problematic behaviors.	Behavior modification and improvement.

Somatic Treatments :

Treatment	Mechanism/Procedure	Indication	Considerations
Electroconvulsive Therapy (ECT)	Induces controlled seizures under anesthesia and muscle relaxant. Used to rapidly alleviate severe depressive symptoms.	Severe depression, especially with suicidal risk or psychotic features.	Safe for pregnant patients.
Transcranial Magnetic Stimulation (TMS)	Uses short pulses of magnetic energy to stimulate brain nerve cells involved in mood regulation.	Patients unresponsive to at least one antidepressant.	Non-invasive, no anesthesia required.
Vagal Nerve Stimulation (VNS)	The electrodes are implanted to stimulate the left vagus nerve and regulate mood.	Chronic or treatment-resistant depression.	Considered after other treatments have failed.
Deep Brain Stimulation (DBS)	Implantation of leads in specific brain areas associated with mood regulation.	Chronic or intractable depression.	Considered after multiple treatment failures.
Phototherapy	Uses bright light (1500-10,000 Lux) from a specialized lightbox for 1 to 2 hours daily to regulate circadian rhythms.	Seasonal Affective Disorder (SAD).	Effective for seasonal mood changes.
Sleep Deprivation	Provides significant but short-lived benefits in alleviating depressive symptoms through controlled sleep loss.	Temporary relief from depressive symptoms.	Effects reverse after normal sleep resumes.

Cognitive Behavioral Therapy (CBT) (Option A): While CBT is effective for mild to moderate depression, it may not be sufficient alone for severe cases, particularly when there is a risk of self-harm.

Selective Serotonin Reuptake Inhibitors (SSRIs) (Option C): SSRIs are commonly prescribed for depression, but they may take several weeks to show effectiveness, which is not suitable for someone in acute danger of self-harm.

Family Therapy (Option D): Family therapy can be beneficial in the long-term management of depression but does not address the immediate safety concerns or the severity of the patient's current condition.

Solution for Question 16:

Correct Answer: A) Esketamine

Explanation:

Esketamine is an NMDA receptor antagonist approved for treatment-resistant depression. It is administered intranasally, making it suitable for patients who cannot take oral medications due to feeding difficulties, such as this patient with a jaw injury.

Novel Agents:

Medication	Mechanism of Action	Indications	Route of Administration	Common Side Effects
Esketamine	NMDA receptor antagonist; modulates glutamate	Treatment-resistant depression	Nasal spray	Sedation, dissociation; requires monitoring
Gepirone	Agonist/ partial agonist at 5HT _{1A}	Major depressive disorder	Oral	Dizziness, nausea, insomnia, abdominal pain. C/I in Prolonged QTc interval (>450 msec) or congenital long QT syndrome
GABA Modulators: Brexanolone Zuranolone	GABA-A receptor positive allosteric modulator (synthetic analogue of neuroactive steroid allopregnanolone)	Postpartum depression	Brexanolone- iv Zuranolone- oral	Drowsiness, dizziness, diarrhea. Advice not to drive or engage in hazardous activities at least 12 hours after administration.

Monitoring: Patients receiving Esketamine should be observed for at least 2 hours post-administration for sedation and dissociation effects.

Combination: Often used in conjunction with oral antidepressants.

Fluoxetine (Option B) is an SSRI commonly used to treat depression. However, it is typically administered orally, which may be problematic for this patient due to his feeding difficulties. Additionally, it has a slower onset of action compared to Esketamine.

Lithium (Option C) is often used as a mood stabilizer, particularly in bipolar disorder, but it also requires oral administration. Lithium also necessitates regular blood monitoring due to its narrow therapeutic index, which may complicate treatment in this patient, given his current physical condition.

Electroconvulsive therapy (ECT) (Option D) is an option for severe depression, but it is usually reserved for cases where other treatments fail or in acute emergencies. Esketamine provides a less invasive option that can be considered earlier in treatment.

Solution for Question 17:

Correct Answer: D) Brain dysfunction affecting the amygdala or hippocampus plays no role in depressive disorders

Explanation:

Brain dysfunctions affecting the amygdala, hippocampus, prefrontal cortex, and anterior cingulate cortex are factors that lead to depressive disorder in many cases.

Epidemiology of Depressive Disorders:

- General trend: Prevalence increases with age. Females > males (Children: Boys = Girls). Mean age of onset (40 yrs). May be more common in Rural > Urban areas. Low socio-economic status > High socio-economic status. 20% experience major depression by age 18.
- Females > males (Children: Boys = Girls).
- Mean age of onset (40 yrs)
- May be more common in Rural > Urban areas
- Low socio-economic status > High socio-economic status
- 20% experience major depression by age 18.
- Family History: ~3 times higher risks if a first-degree relative has major depression.
- Persistent Depressive Disorder/Dysthymia/Dysthymic Disorder: According to DSM-5, persistent depressive disorder is diagnosed when depressive symptoms persist for 2 or more years (or 1 year for children), with no symptom-free periods lasting longer than 2 months during the illness.

- Double Depression: ~9.9% develop major depression within 6 months of persistent depressive disorder.
- Females > males (Children: Boys = Girls).
- Mean age of onset (40 yrs)
- May be more common in Rural > Urban areas
- Low socio-economic status > High socio-economic status
- 20% experience major depression by age 18.

Etiology of Depressive Disorders:

Environmental factors	Biological factors
Acute negative life events Many cases are preceded by such events Loss of a significant person or relation like a spouse Immigrant or refugees due to loss, isolation, and trauma of migration Chronic stress: Poverty, unemployment, single motherhood, poor housing, poor residential conditions, parents of disabled children Childhood adversity: Early exposure to trauma (loss of a parent) Social and institutional barriers: Racism, discrimination, and socioeconomic status	Monoamine hypothesis: deficiency or dysregulation of brain monoamines leads to depressive symptoms (Option B) Neuroendocrine: HPA axis hyperactivity Hypocortisolemia Undetected thyroid dysfunction Low brain - derived neurotrophic factor Neurotransmitter deficits: Norepinephrine : decreased sensitivity of beta-adrenergic receptors Low serotonin precipitates depression Dopamine may be reduced Decreased GABA levels due to chronic stress Decreased 5-hydroxy indole acetic acid (5-HIAA) Increased level of ACh and glutamate. Brain dysfunction affecting the amygdala, hippocampus, prefrontal cortex, and anterior cingulate cortex Neurogenesis hypothesis stating brain abnormalities causing depression is due to developmental anomalies leading to a deficiency of neurons (Option A) Neuroplasticity hypothesis states that the atrophy of already-developed neurons can cause depression (Option C)
Immune system processes Chronic stress may activate immune responses (CRF, cytokines, and interleukins), leading to inflammation. Elevated proinflammatory cytokines (IL-6, CRP), indicate a bidirectional relationship between depression and inflammatory responses. Genetic vulnerability Family history significantly increases risk.	

Solution for Question 18:

Correct Answer: B) Intravenous Phentolamine

Explanation: The patient is experiencing a hypertensive crisis due to the "cheese reaction," where tyramine from red wine interacts with moclobemide, a monoamine oxidase inhibitor (MAOI), leading to excessive norepinephrine release resulting in hypertensive crisis. Intravenous phentolamine is the most appropriate treatment, as it rapidly lowers blood pressure by blocking norepinephrine effects.

Monoamine Oxidase Inhibitors (MAOIs):

Types	Irreversible, Non-Selective MAOIs: Phenelzine Tranylcypromine Isocarboxazid Reversible MAO-A Inhibitors: Moclobemide MAO-B Inhibitors: Selegiline Rasagiline
Primary Use	Depression (especially atypical depression)
Other Uses	Anxiety disorders, panic disorder, social phobia, treatment-resistant depression, Parkinson's disease (e.g., selegiline)
Mechanism of Action	Inhibits monoamine oxidase enzymes, preventing the breakdown of neurotransmitters such as serotonin, norepinephrine, and dopamine, leading to increased concentrations of these neurotransmitters in the brain.
Irreversible MAOIs	Bind permanently to MAO enzymes and inhibit both MAO-A and MAO-B, leading to elevated levels of serotonin, norepinephrine, and dopamine in the brain.
Reversible MAO-A Inhibitors	Reversibly inhibit only MAO-A, primarily increasing serotonin and norepinephrine without affecting dopamine as much.

Types	Irreversible, Non-Selective MAOIs: Phenelzine Tranylcypromine Isocarboxazid Reversible MAO-A Inhibitors: Moclobemide MAO-B Inhibitors: Selegiline Rasagiline
Common MAOI Drugs	Phenelzine: Non-selective, irreversible; used for depression Tranylcypromine: Non-selective, irreversible; rapid onset of action Isocarboxazid: Non-selective, irreversible; for treatment-resistant depression Moclobemide: Reversible, selective for MAO-A; used for major depression Selegiline: Used for Parkinson's disease and depression Rasagiline: Used primarily for Parkinson's disease
Side Effects	CNS Effects: Anxiety, insomnia, agitation, restlessness, dizziness, tremors CVS: Orthostatic hypotension, hypertensive crisis (especially when interacting with tyramine-containing foods) GI Effects: Nausea, constipation Sexual Dysfunction: Decreased libido, anorgasmia Weight Gain: Common with irreversible MAOIs Serotonin Syndrome: When combined with other serotonergic drugs (SSRIs, TCAs, etc.)
Contraindications	Use with serotonergic agents (SSRIs, SNRIs, TCAs, etc.) due to risk of serotonin syndrome Use with sympathomimetic drugs due to hypertensive crisis risk Certain foods (high in tyramine) due to the risk of hypertensive crisis (cheese reaction)
Cheese Reaction/ Tyramine induced Hypertensive Crisis	MAOIs inhibit the breakdown of tyramine, an amino acid found in aged cheeses, cured meats, fermented products, and certain alcoholic beverages (e.g., red wine, beer) High levels of tyramine can trigger a release of norepinephrine, leading to a hypertensive crisis characterized by severe headache, palpitations, stiff neck, nausea, vomiting, and elevated blood pressure. In emergencies, treatment options include phentolamine (an alpha-blocker) or nitroprusside to manage the crisis.

Wait for 30 minutes for symptoms to resolve on their own (Option A): This is inappropriate because hypertensive crises need immediate intervention to prevent complications like stroke or myocardial infarction.

Discontinuation of moclobemide and administration of intravenous fluids (Option C): While discontinuing the MAOI (moclobemide) is essential, IV fluids alone are insufficient to manage the acute hypertensive crisis.

Intravenous Labetalol (Option D): Although labetalol can lower blood pressure by blocking both alpha and beta receptors, in cases of tyramine-MAOI interactions, pure alpha-adrenergic blockade (like with phentolamine) is more appropriate to target the cause of the hypertension directly. Beta-blockers alone may lead to unopposed alpha-receptor stimulation, worsening vasoconstriction.

Solution for Question 19:

Correct Answer: B) Tricyclic Antidepressants (TCAs)

Explanation:

Tricyclic Antidepressants (TCAs) such as imipramine are commonly used in the treatment of nocturnal enuresis in children and adolescents. They have anticholinergic properties that reduce bladder activity, thus helping with bedwetting. TCAs are also effective in treating depression, addressing both the emotional and enuretic aspects in this case.

Tricyclic and Tetracyclic Antidepressants (TCAs and TeCAs):

Primary Use	Depression
Other Uses	Anxiety disorders, chronic pain, insomnia, OCD, nocturnal enuresis
Mechanism of Action	Inhibits reuptake of serotonin (5-HT) and norepinephrine (NE) by blocking transporters (SERT and NET)
Receptor Antagonism	Muscarinic receptors: Anticholinergic effects (dry mouth, blurred vision, constipation) H1 histamine receptors: Sedation, weight gain Alpha-1 adrenergic receptors: Postural hypotension Alpha-2 adrenergic receptors: NE feedback modification Voltage-sensitive sodium channels: Cardiotoxicity

Primary Use	Depression
Tricyclic Antidepressants	Clomipramine: Most serotonin-selective, used in OCD Amitriptyline: Depression, neuropathic pain Doxepin: Anxiety, insomnia Imipramine: Formerly used for nocturnal enuresis Desipramine: Most norepinephrine-selective Nortriptyline: Depression, chronic pain, diabetic neuropathy Viloxazine: Bicyclic antidepressant
Tetracyclic Antidepressants	Maprotiline Amoxapine: Also blocks D2 receptors, and may cause EPS Mianserin
Side Effects	Anticholinergic: Dry mouth, constipation, blurred vision, urinary retention, delirium Cardiac Side Effects: Tachycardia, flattened T waves, prolonged QT interval, ST depression, arrhythmias, cardiotoxicity Autonomic Side Effects: Postural hypotension (alpha-1 receptor blockade), risk of falls in elderly Neurological Side Effects: Tremors, seizures, extrapyramidal symptoms (EPS) from amoxapine. Endocrine Side Effects: Galactorrhea, gynecomastia Other Side Effects: Sedation, weight gain (H1 histamine receptor antagonism)

Selective Serotonin Reuptake Inhibitors (SSRIs) (Option A) are primarily used for depression but have no direct benefit in managing nocturnal enuresis.

Monoamine Oxidase Inhibitors (MAOIs) (Option C) are less commonly used in treating depression and are not indicated for nocturnal enuresis.

Benzodiazepines (Option D) are primarily anxiolytics and sedatives, not suitable for treating nocturnal enuresis or depression in this context.

Solution for Question 20:

Correct Answer: A) Venlafaxine

Explanation: Venlafaxine is indicated for both panic disorder and social phobia, making it the most appropriate choice for this patient's symptoms.

Selective Serotonin-Norepinephrine Reuptake Inhibitors (SNRI):

- Also referred to as dual reuptake inhibitors.

Mechanism of Action	Inhibit both serotonin transporters (SERT) and norepinephrine transporters (NET). Causes enhanced serotonergic and noradrenergic neurotransmission.
Drugs and Uses	Venlafaxine: (Option A) Major depression Generalized anxiety Panic disorder Social phobia Desvenlafaxine: (Option B) Major depression Duloxetine: Major depression Generalized anxiety Diabetic peripheral neuropathy Fibromyalgia Musculoskeletal pain Levomilnacipran: (Option D) Major depression Milnacipran: (Option C) Fibromyalgia

Mechanism of Action	Inhibit both serotonin transporters (SERT) and norepinephrine transporters (NET). Causes enhanced serotonergic and noradrenergic neurotransmission.
Side Effects	Neurological Side Effects: Insomnia Irritability Sexual Dysfunction: Decreased libido Erectile dysfunction Anorgasmia Ejaculatory delay Gastrointestinal Effects: Nausea Vomiting Diarrhea Withdrawal Symptoms: Dizziness Headache Nervousness Nausea Insomnia Dry mouth

Solution for Question 21:

Correct answer: B) Paroxetine

Explanation:

Paroxetine has been linked to an increased risk of congenital heart defects when taken during the first trimester. It is classified as a Category D drug, making it the least safe option during pregnancy.

SSRI in Pregnancy

- Discontinuing or modifying antidepressant regimens during pregnancy has relapse rates of major depression between 68% and 100%. Therefore it is necessary to continue antidepressants during pregnancy and postpartum to prevent depression relapse.
- SSRIs are used if postpartum blues persist or to prevent postpartum depression during pregnancy in at-risk women.

Fetal outcomes

- SSRIs generally do not increase the risk of major congenital malformations, except for paroxetine . (Option B)
- Paroxetine , a Category D drug, is associated with increased heart defect risks when taken during the first trimester .
- Late pregnancy SSRI use can cause Poor neonatal adaptation syndrome (PNAS)- short term effect, may represent discontinuation syndrome with features such as poor muscle tone, sleep problems, crying, irritability, hypoglycemia, seizures.
- Late pregnancy SSRI use also carries a small risk (1-2 per 1,000 births) of Persistent pulmonary hypertension in newborns.
- Low Apgar score
- ADHD, Autism spectrum disorder

Maternal outcomes

- SSRIs may impair platelet function and myometrial contractions, disrupting normal hemostasis and vasoconstriction, which can increase the risk of Postpartum hemorrhage .
- SSRIs can influence serotonin levels and inhibit nitric oxide synthesis , leading to vascular changes that elevate the risk of pre-eclampsia .
- SSRIs may worsen insulin resistance, increasing the risk of GDM during pregnancy.
- Minimal amounts of SSRIs are present in breast milk, with no reported harmful effects in breastfed babies.
- It is essential to inform patients of the potential risks and benefits of SSRI use during pregnancy and breastfeeding.

Sertraline (Option B) : Considered one of the safer SSRIs during pregnancy, with no significant risks in the first trimester.

Fluoxetine (Option B) : Regarded as relatively safe, with no pronounced risks highlighted for first-trimester use.

Citalopram (Option B) : Shares a similar safety profile to sertraline and fluoxetine, with no specific risks compared to paroxetine.

Solution for Question 22:

Correct Answer: B) Sumatriptan

Explanation:

The patient's symptoms, including altered mental status, hyperreflexia, diaphoresis, tachycardia, and mild hypertension, suggest a potential serotonin syndrome, which can occur with the combination of fluoxetine (SSRI) and sumatriptan (a triptan used for migraine treatment). Sumatriptan can increase serotonin levels and, when used with other serotonergic agents like fluoxetine, may lead to serotonin syndrome.

Ibuprofen, metoprolol, and loratadine do not have a significant impact on serotonin levels and are unlikely to contribute to the patient's symptoms.

Serotonin syndrome

Definition	Serotonin syndrome is a potentially life-threatening condition caused by excessive serotonergic activity in the central nervous system, typically due to the concurrent use of medications that increase serotonin levels or enhance serotonergic neurotransmission.
Causes	The syndrome is often triggered by concurrent use of SSRIs (Selective Serotonin Reuptake Inhibitors) with other medications: SSRIs: Selective Serotonin Reuptake Inhibitors that increase serotonin levels. MAOIs: Monoamine Oxidase Inhibitors that inhibit the breakdown of serotonin, leading to increased serotonin levels. L-Tryptophan: An amino acid precursor to serotonin that enhances serotonin synthesis. Lithium: Enhances the effects of serotonergic drugs, potentially leading to increased serotonin activity. Triptans: Medications for migraines that act on serotonin receptors (5-HT_{1B} and 5-HT_{1D}), increasing serotonin levels. Herbal Supplements: St. John's Wort can induce serotonin syndrome when taken with SSRIs or other serotonergic agents.
Pathophysiology	The condition arises when the balance of serotonin production, release, and degradation is disrupted, leading to elevated serotonin levels in the synaptic cleft. This overstimulation of serotonin receptors, particularly 5-HT _{1A} and 5-HT _{2A} , results in various symptoms.
Symptoms	Autonomic Symptoms: Diarrhea Increased heart rate (tachycardia) High blood pressure (hypertension) Hyperthermia (elevated body temperature)

Definition	Serotonin syndrome is a potentially life-threatening condition caused by excessive serotonergic activity in the central nervous system, typically due to the concurrent use of medications that increase serotonin levels or enhance serotonergic neurotransmission.
Neuromuscular Symptoms: Restlessness and extreme agitation Myoclonus (rapid muscle contractions) Hyperreflexia (exaggerated reflex responses) Rigidity	
Cognitive Symptoms: Delirium or altered mental status Coma in severe cases	
Severe Outcomes: Cardiovascular collapse Seizures Potentially death if not treated promptly	
Treatment	Discontinuation of Serotonergic Agents: All contributing medications should be stopped immediately to prevent further serotonin accumulation. Immediate Supportive Care: Ensure patient safety and monitor vital signs. Implement cooling measures to manage hyperthermia. Pharmacological Interventions: Cyproheptadine : Antihistamine with 5-HT _{2A} antagonist properties that can help counteract serotonin effects. Benzodiazepines : May be used to manage agitation and myoclonus.

Discontinuation syndrome

Definition	Discontinuation syndrome occurs following the abrupt cessation of certain medications, particularly those affecting serotonergic activity.
Causes	Short-acting Drugs: Venlafaxine: SNRI associated with withdrawal symptoms if discontinued abruptly. Fluvoxamine: SSRI linked to discontinuation symptoms upon abrupt withdrawal. Paroxetine: SSRI that often leads to withdrawal symptoms when stopped suddenly. Long-acting Drugs: Fluoxetine: Does not cause discontinuation syndrome due to its extended half-life.
Symptoms	Flu Like symptoms (Lethargy, weakness, headache) Insomnia Nausea Imbalance Sensory disturbances (paraesthesia) Hyperarousal (rebound anxiety, irritability)

Solution for Question 23:

Correct Answer: A) Dapoxetine

Explanation: Selective Serotonin Reuptake Inhibitors (SSRIs), such as dapoxetine, are the most common treatment for premature ejaculation. They delay ejaculation by increasing serotonin levels, which affects the ejaculation reflex.

Pharmacotherapy of depression:

The pharmacotherapy of depression primarily involves antidepressant medications. These drugs work by influencing neurotransmitters in the brain, particularly serotonin, norepinephrine, and dopamine, which are thought to be imbalanced in depressive disorders.

Major classes and medications used in treating depression :

Class	Examples	Use
SSRIs	Fluoxetine, Sertraline, Escitalopram, Citalopram, Paroxetine	First-line due to safety and tolerability

Class	Examples	Use
SNRIs	Venlafaxine, Duloxetine	For those not responding to SSRIs
Atypical Antidepressants	Bupropion, Mirtazapine	Less sexual dysfunction (Bupropion); helps insomnia (Mirtazapine)
Tricyclic Antidepressants (TCAs)	Amitriptyline, Nortriptyline	For treatment-resistant cases
Monoamine oxidase inhibitors (MAOIs)	Phenelzine, Tranylcypromine	Reserved for severe, resistant depression
Adjunct Therapy	Atypical antipsychotics, Lithium	For treatment-resistant depression
Treatment Duration: Acute Phase: Minimum of 6 months or length of previous episode (whichever is longer) Maintenance/Prophylaxis: Indicated for 3+ episodes or chronic depression (>2 years)		

SSRIs in the treatment of depression:

Definition	Selective Serotonin Reuptake Inhibitors (SSRIs) are a class of drugs primarily used to treat depression and other mental health conditions. They increase serotonin levels in the brain, which helps regulate mood, emotion, and sleep.	
Common SSRIs	Sertraline Fluoxetine Paroxetine Citalopram Escitalopram Vilazodone (SPARI- Serotonin Partial Agonist and Reuptake Inhibitor) Fluvoxamine (mainly for OCD)	

Definition	Selective Serotonin Reuptake Inhibitors (SSRIs) are a class of drugs primarily used to treat depression and other mental health conditions. They increase serotonin levels in the brain, which helps regulate mood, emotion, and sleep.	
Side Effects	Gastrointestinal	Nausea (most common) Diarrhea Anorexia Dry mouth Weight gain Constipation (especially with paroxetine)
CNS	Anxiety Insomnia Sedation Vivid dreams Emotional blunting Seizures (rare) Extrapyramidal symptoms	
Sexual Dysfunction (most common side effect in long-term use)	Decreased libido Delayed ejaculation Inhibited orgasm	
Hematological	Impaired platelet aggregation Increased risk of GI bleeding (especially with NSAIDs)	
Miscellaneous	Hyponatremia Allergic rashes Hyperprolactinemia (leading to galactorrhea or menstrual irregularities)	

Sildenafil (Option B) is a phosphodiesterase type 5 Inhibitor used for erectile dysfunction and does not address premature ejaculation.

Clomipramine (Option C) is a tricyclic antidepressant that helps with premature ejaculation, but not the first-line treatment due to potential side effects; SSRIs are generally preferred.

Alprazolam (Option D) is a benzodiazepine, primarily used for anxiety disorders, they do not target premature ejaculation and may only help with anxiety without addressing ejaculation timing.

Solution for Question 24:

Correct answer: B) Exogenous Depression

Explanation: Symptoms are triggered by a specific life event (the loss of a loved one), aligning with the characteristics of Exogenous (reactive) depression.

Endogenous Depression and Exogenous (Reactive) Depression

Feature	Endogenous Depression (Option A)	Exogenous (Reactive) Depression (Option B)
Definition	Primarily caused by biological factors.	Arises due to psychological or environmental factors.
Precipitating Events	Manifests without identifiable precipitating events.	Triggered by specific life events (e.g., loss, trauma).
Symptoms	Characterized by melancholic or psychotic features.	Exhibits multiple somatic complaints and emotional distress.
Mood Regulation	Linked to genetic or neurochemical factors affecting mood.	Relates to an individual's emotional response to their circumstances.
Treatment Response	Often requires pharmacological treatment (e.g., antidepressants).	May respond well to psychotherapy and counseling focused on coping strategies.
Duration	Symptoms can be persistent and chronic.	Symptoms may be more acute and linked to the duration of the triggering event.

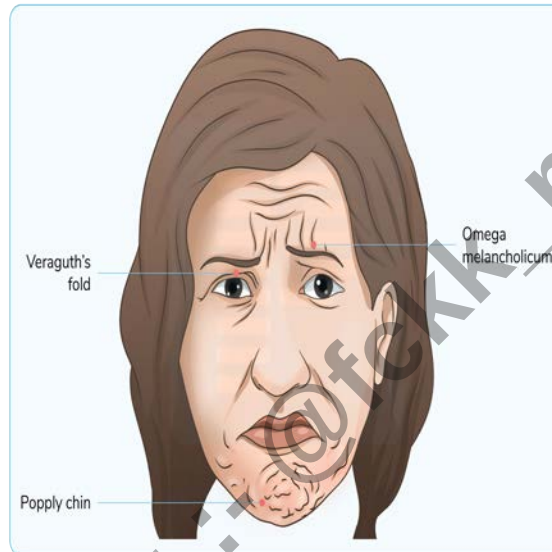
Persistent Depressive Disorder (Option C) requires symptoms to persist for 2 or more years without symptom-free periods lasting longer than 2 months, whereas the patient's symptoms are acute and linked to a specific recent triggering event.

Major Depressive Disorder with Psychotic Features (Option D) requires the presence of psychotic symptoms (such as delusions or hallucinations) alongside major depressive symptoms. The patient's symptoms do not indicate psychosis, making this option inappropriate.

Solution for Question 25:

Correct Answer: B) Omega Sign

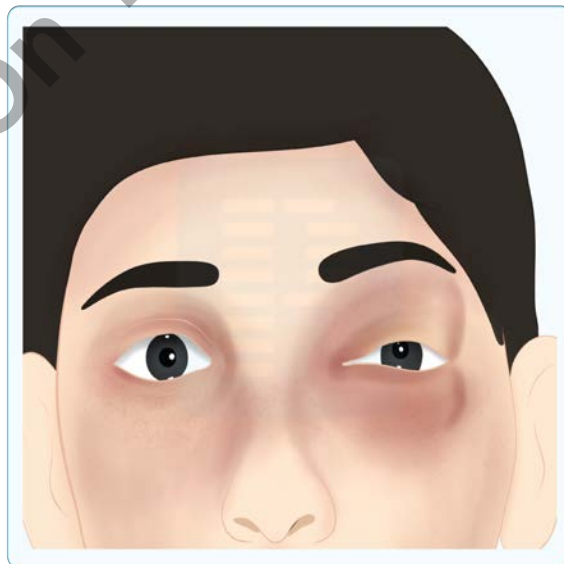
Explanation: The omega sign, an omega-shaped fold on the forehead from excessive use of the corrugator muscle, reflects emotional stress and is indicative of depressive disorders.



Physical Signs of Depression:

Veraguth's Fold (Option A)	Omega Sign (Option B)
 A triangular fold in the inner corner of the upper eyelid due to changes in the corrugator and zygomatic muscles from prolonged muscle tension associated with depression. {{caption_text}}	 An omega-shaped fold on the forehead reflecting excessive use of the corrugator muscle, indicative of emotional stress related to depressive disorders.

Chandler's Sign (Option C): Is associated with ocular conditions, specifically describing a finding related to an eye movement or appearance, particularly in certain cases of ophthalmic disorders like orbital cellulitis. It refers to the presence of a swollen, edematous eyelid along with proptosis and restricted ocular movement, often seen in serious orbital infections. This sign is not relevant to psychiatric or depressive disorders.



Dennie-Morgan Lines (Option D): These are associated with allergic conditions, particularly atopic dermatitis, and are characterized by extra folds of skin beneath the eyes. They are not

indicative of depressive disorders.



Solution for Question 26:

Correct Answer: C) Tricyclic antidepressants (TCAs)

Explanation:

Symptoms of mood reactivity, increased appetite, hypersomnia, and rejection sensitivity align with diagnosis of Atypical Depression.

TCAs are less effective in treating atypical depression compared to other antidepressants. They are associated with more significant side effects, such as sedation and weight gain, which can worsen the symptoms of this subtype of depression.

Atypical Depression :

It is a subtype of major depressive disorder (MDD) characterized by mood reactivity (the ability to feel a temporary mood lift in response to positive events) and a distinct set of other symptoms. Despite the name, it is one of the more common forms of depression. It tends to begin earlier in life and is more common in women. According to DSM-5, atypical features can occur in both MDD and persistent depressive disorder (dysthymia).

Features of Atypical Depression (DSM-5):

- **Mood Reactivity:** A defining feature of atypical depression, where the individual's mood brightens significantly in response to positive events (e.g., good news).
- **Leadens Paralysis:** A heavy, lead-like feeling in the limbs that results in extreme fatigue and physical sluggishness.
- **Hypersomnia:** Excessive sleeping, usually more than 10 hours per day, or difficulty staying awake during the day, which contrasts with the early morning awakening seen in melancholic depression.
- **Increased Appetite and Weight Gain:** Unlike other forms of depression that may cause weight loss, individuals with atypical depression often experience an increase in appetite and significant weight gain.
- **Interpersonal Rejection Sensitivity:** The patient experiences extreme sensitivity to interpersonal rejection, often feeling hurt or rejected by others, which may interfere with their social or occupational functioning.

Distinctive Characteristics:

- **Mood Reactivity:** Unlike melancholic depression, mood improves temporarily after positive events.
- **Comorbid Anxiety:** Higher rates of co-occurring anxiety disorders.
- **Earlier Onset:** Typically begins earlier in life than other depressive subtypes.

Treatment:

- **Monoamine Oxidase Inhibitors (MAOIs)** (Option A), like phenelzine, were historically effective for atypical depression but are less used now due to dietary restrictions (cheese reaction).
- **Selective Serotonin Reuptake Inhibitors (SSRIs)** (Option B), such as fluoxetine, are more commonly prescribed and effective for many.
- **Cognitive Behavioral Therapy (CBT)** (Option D) is also beneficial, especially for addressing rejection sensitivity and mood instability in atypical depression.

Solution for Question 27:

Correct Answer: C) Depression with catatonic features

Explanation:

The most appropriate diagnosis is Depression with catatonic features, as the patient exhibits marked immobility, bizarre postures, and minimal verbal response, characteristic of this subtype.

Depression with catatonic features (Option C) is a rare but severe subtype of major depressive disorder characterized by marked motor disturbances. According to DSM-5, catatonic features can occur in the context of several mental disorders, including major depressive disorder (MDD), and involve extreme physical immobility or excessive motor activity. These features are linked with increased morbidity and require prompt treatment.

Features of Depression with Catatonic Features:

- Stupor: A marked decrease in reactivity to the environment, where the patient may appear unresponsive and immobile.

Catalepsy: Passive induction of a posture held against gravity where the patient holds a fixed posture for an extended period.



- Waxy flexibility: slight, even resistance to positioning by examiner.
- Mutism: Little or no verbal response despite prompts.
- Negativism: An opposition or lack of response to instructions or external stimuli.
- Posturing: Spontaneous maintenance of a rigid or bizarre posture.
- Stereotypies: Repetitive, non-goal-directed movements (e.g., hand flapping or grimacing).
- Mannerisms: repetitive, odd, purposeful movements.
- Agitation: Excessive and purposeless motor activity not influenced by external stimuli.
- Echolalia or Echopraxia: Repetition of another's words (echolalia) or movements (echopraxia).

Distinctive Characteristics:

Motor Symptoms: The defining aspect of catatonic depression is its effect on movement, with patients either becoming immobile or, in contrast, exhibiting hyperactivity.

Treatment:

- First-line treatment involves benzodiazepines (e.g., lorazepam), which are effective in alleviating catatonic symptoms.
- Electroconvulsive therapy (ECT) is also highly effective and often used in patients who do not respond to pharmacotherapy or when rapid resolution is required.

Depression with melancholic features (Option A) is a severe subtype of major depressive disorder, often seen in older adults. It is marked by a profound lack of pleasure in all or nearly all activities and a failure to respond to pleasurable stimuli (anhedonia).

Melancholic features include:

- A distinct quality of depressed mood- profound despondency, despair, empty mood.
- Psychomotor Disturbance: Either agitation or retardation.
- Excessive Guilt: Feelings of worthlessness, often out of proportion to reality.
- Early Morning Awakening: Waking up at least two hours earlier than usual.
- Mood Worse in the Morning: A clear diurnal variation, where the mood is worse in the early part of the day.
- Anorexia/Weight Loss: Often severe, leading to significant weight loss.

Patients with melancholic features do not experience mood reactivity, which distinguishes this subtype from atypical depression. Other notable characteristics include psychomotor disturbance and increased risk of suicide.

Treatment typically involves SNRIs (serotonin-norepinephrine reuptake inhibitors) , as they are often more effective than SSRIs for this group.

Atypical depression (Option B): Mood reactivity is present (patients can feel better in response to positive events), and symptoms such as increased sleep, increased appetite, and weight gain are more common. These are absent in this patient.

Major depressive disorder with psychotic features (Option D) involves the presence of delusions or hallucinations, which are absent in this patient's presentation.

Solution for Question 28:

Correct Answer: B) 2 and 4

Explanation:

The clinical scenario here points to major depressive disorder with psychotic symptoms. The two preferred choices of treatment are the combination of antidepressants and antipsychotics, and electroconvulsive therapy (with high remission rates of up to 95%).

Patients with Major Depressive Disorder with Psychotic Features experience psychotic symptoms only during mood episodes. In contrast, schizoaffective disorder requires at least 2 weeks of only psychotic symptoms (delusions and hallucinations) without mood symptoms (Statements 3 ruled out).

Definition	Major depression disorder (MDD) with psychotic features includes mood disturbance with delusions, hallucinations, or both.
Prevalence	Occurs in approximately 18.5% of MDD patients and increases with age.
Characteristics	Higher treatment resistance More suicide attempts Longer illness duration Greater functional impairment Higher relapse rates
Geriatric Considerations	Patients often show significant cognitive dysfunction; depressive pseudodementia may develop.
Cognitive Impairment	Combination therapy : antidepressants + antipsychotics or ECT is the standard of care.
Treatment Approaches	Monotherapy with SSRIs or TCAs is less effective . (Statements 1 ruled out)
Efficacy of ECT	ECT shows high remission rates (up to 95% in psychotic depression) compared to pharmacotherapy. ECT is first line treatment in case of severe psychotic symptoms and high suicide risk.

Solution for Question 29:

Correct Answer: D) 2 weeks

Explanation:

The clinical scenario has more than five symptoms of major depressive disorder with both depressed mood and anhedonia, and for its diagnosis, the symptoms must persist for the next 2 weeks .

Symptoms

- Usually, insomnia (especially early morning awakening, prolonged sleep latency, increased intermittent awakenings)
- Reduced latency of REM sleep

Atypical Depression: Mood reactivity (ability to feel better with positive events) and at least two of the following: hyperphagia, hypersomnia, leaden paralysis, or long-standing rejection sensitivity.

Diagnosis:

- Five or more of the symptoms present during the same 2 - week period and represent a change from previous functioning; At least one of the symptoms is either depressed mood or loss of interest or pleasure .

Recurrent depressive disorder:

- When there are 2 or more episodes of symptoms of depression, with a minimum of 2 months between episodes, it is called recurrent depressive disorder.

Solution for Question 30:

Correct Answer: D) Introjection of lost significant object

Explanation:

The introjection of lost objects is a part of the psychodynamic theory of depression, while the cognitive triad is a component of the cognitive theory, involving self - perception , environment, and the future .

Psychosocial Theories of Depression

Life Events and Environmental Stress

- Stressful life events often precede the first episode of mood disorders.
- Initial stress may lead to long-lasting biological changes, increasing the risk for future episodes even without further stressors.

Psychodynamic Theories

- Early disturbances in the infant-mother relationship predispose to depression.
- Depression is linked to real or imagined loss of significant objects.
- Introjection of lost objects serves as a defense mechanism.
- Feelings of love and hate toward the lost object result in self-directed anger.

Cognitive Theory

- Depression arises from cognitive distortions known as depressogenic schemata.
- Cognitive Triad (by Aaron Beck):
 - Worthlessness: Negative self-perception. (Option A)
 - Helplessness: Perception of the environment as hostile and demanding. (Option B)
 - Hopelessness: Expectation of suffering and failure in the future. (Option C)
- Worthlessness: Negative self-perception. (Option A)
- Helplessness: Perception of the environment as hostile and demanding. (Option B)
- Hopelessness: Expectation of suffering and failure in the future. (Option C)
- Therapy focuses on modifying these cognitive distortions.
- Worthlessness: Negative self-perception. (Option A)
- Helplessness: Perception of the environment as hostile and demanding. (Option B)
- Hopelessness: Expectation of suffering and failure in the future. (Option C)

Learned Helplessness

- Developed from studies showing that exposure to uncontrollable stressors (e.g., electric shocks in dogs) leads to passive behavior.
- Humans may internalize helplessness and lose self-esteem after adverse events.
- Improvement in depression involves learning a sense of control and mastery over one's environment.

Evolutionary Theory

- Depression is viewed as an adaptive response to perceived environmental threats.
- Withdrawal from social situations and decreased activity may serve as protective mechanisms.

Solution for Question 31:

Correct Answer: B) a-iii, b-iv, c-i, d-ii

Explanation:

Genetic Influence	About one-third of major depression risk is attributed to genetic factors. The gene encoding of serotonin receptor : HTR1A The gene for serotonin transporter : 5HTTP / SLC6A4 The gene for dopamine receptor : DRD4 The gene for dopamine transporter : SLC6A3
Family History	Risk increases 2.5–3 times if a first-degree relative has depression. Life events can raise risk from 5 to 16 times shortly after occurrence.
Neurobiological Mechanisms	Studies show that individuals with the short allele of 5-HTTLPR (serotonin transporter gene) exhibit amygdala hyperreactivity to emotional threats

Previous Year Questions

1. A young man with a psychiatric disorder has become very aggressive. He speaks very rapidly and sleeps very little. He exhibits hyperactive behavior after discontinuing his prescribed medication. What is the most likely diagnosis?

- A. Depression
- B. Bipolar disorder - manic episode
- C. Schizophrenia
- D. Substance- induced mood disorder

2. A patient was started on amitriptyline and was found unconscious. Which of the following does not suggest TCA toxicity?

- A. Arrhythmia
- B. Confusion
- C. Hypothermia
- D. Pupillary dilatation

3. A pregnant female with bipolar disorder with a previous history of 4 episodes and no episode for the past 1 year well controlled with lithium 750mg, now in the third trimester, came to a psychiatrist. What should be done next in the management of this patient?

- A. Stop lithium
- B. Decrease Dose
- C. Increase Dose
- D. Change to Valproate

4. A patient on a mood stabilizer has developed hyperammonemia and hepatitis /raised hepatic enzymes. Which of the following is likely to be responsible?

- A. Lithium
 - B. Valproate
 - C. Carbamazepine
 - D. Topiramate
-

5. A 35-years-old patient presents with complaints of sudden and irresistible urges to take action. This is the characteristic of which symptom?

- A. Impulse
 - B. Hallucination
 - C. Delusion
 - D. Pseudohallucination
-

6. A 30-year-old patient presents with persistent feelings of sadness, fatigue, loss of appetite and lack of motivation for the past several weeks. On further assessment, the patient reports a decreased interest in activities that were once enjoyable and a general sense of hopelessness. Which of the following is the key symptom of depression?

- A. Suicidal ideations
- B. Low mood
- C. Loss of interest in pleasurable activity
- D. Loss of Appetite

7. A 36-year-old male patient on psychiatric medication developed a sudden hypertensive episode after taking wine. He is on antidepressants. Which of the following drug the person be taking?

- A. MAO inhibitors
 - B. Amitriptyline
 - C. SSRI
 - D. Trazodone
-

8. A 38-year-old female presented to a psychiatrist with symptoms of low mood, loss of interest, loss of appetite and difficulty sleeping for 2 months. The diagnosis of a major depressive disorder is made by the psychiatrist. Which drug is best for the above-mentioned disorder?

- A. Buspirone
 - B. Lithium
 - C. Divalproex
 - D. Fluoxetine
-

9. Which of the following is not true about rapid cyclers in bipolar disorder?

- A. Occurs commonly in men
 - B. Commonly associated with concomitant hypothyroidism
 - C. At least 4 distinct episodes per year
 - D. Antidepressant's use can increase the likelihood
-

10. Which of the following is not a component of Beck's cognitive theory of depression?

- A. Cognitive distortions
 - B. Automated thoughts
 - C. Introjection
 - D. Dysfunctional belief
-

11. A patient with low mood, no mood to eat, always sleeping and low energy presented to you with such symptoms for 2 months. She had similar symptoms 5 months ago which subsided with medications. Identify the disorder?

- A. Depression
 - B. Recurrent depressive disorder
 - C. Dysthymia
 - D. Cyclothymia
-

12. Which of the following adverse effects is the least probable in a patient prescribed with escitalopram?

- A. Nausea
 - B. Vivid dreams
 - C. Anorgasmia
 - D. Sialorrhea
-

13. Which of the following statements accurately describes lithium?

- A. It can cause fine postural tremors at therapeutic dosage
 - B. It is not absorbed from the gut
 - C. It is not teratogenic
 - D. It is also used for treatment of absence seizures
-

14. True for lithium:-

- A. Delayed absorption
 - B. Narrow therapeutic index
 - C. Protein-bound
 - D. Can be given safely in renal dysfunction
-

15. In OCD, which of the options below is associated with an unfavorable outcome when treated with exposure and response prevention?

- A. Pathological doubt
 - B. Magical thinking
 - C. Hoarding
 - D. Contamination obsession
-

16. What is the likely diagnosis for a 35-year-old woman who experiences heightened anxiety and recurrent thoughts about her hands being contaminated, leading her to frequently wash her hands? Despite finding these thoughts distressing, she is unable to resist them, and attempting to refrain from washing only worsens her anxiety.

- A. Phobia
 - B. Forced thinking
 - C. Obsessive-compulsive disorder
 - D. Generalized anxiety disorder
-

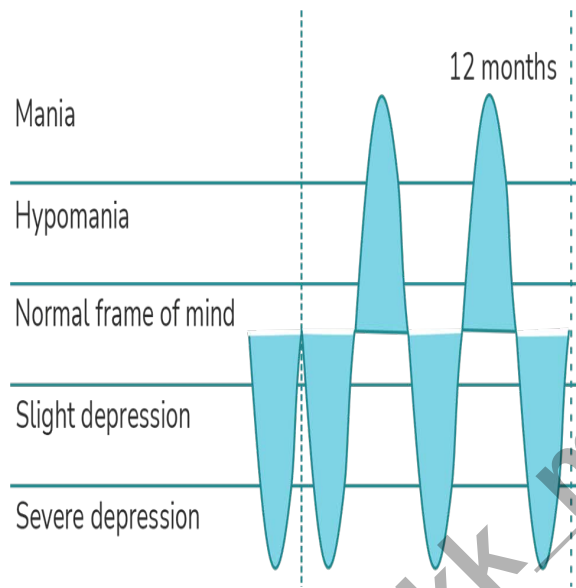
17. Which of the following adverse effects is the least probable in a patient prescribed with escitalopram?

- A. Nausea
 - B. Vivid dreams
 - C. Anorgasmia
 - D. Sialorrhea
-

18. Which of the options below does not characterize melancholia?

- A. Severe anhedonia
 - B. Weight loss
 - C. Sleep-onset insomnia
 - D. Profound feelings of guilt over trivial events
-

19. A 40-year-old patient comes to the OPD with a history significant for a mood disorder. Her moods can be charted as done in the image given below. What is the diagnosis?



- A. Cyclothymia
- B. Dysthymia
- C. Rapid cycling disorder
- D. Bipolar disorder

20. Double depression is:-

- A. Major depressive disorder from 2 years
- B. Depression with OCD
- C. Depression with dysthymia
- D. Depression with anxiety attack

21. What is the preferred treatment option for postpartum blues?

- A. Fluoxetine
- B. Cognitive behavioral therapy

- C. Lithium carbonate
 - D. Support and Education
-

22. Lithium usage in pregnancy can lead to which of the following teratogenic effects?

- A. Cardiac defects
 - B. Facial defects
 - C. CNS defects
 - D. Urogenital defects
-

23. In your OPD, a young male patient with a paternal family history of bipolar disorder has presented with three episodes of mania. Considering the most effective treatment, which of the following mood stabilizers can be prescribed to him?

- A. Valproate
 - B. Lamotrigine
 - C. Olanzapine
 - D. Lithium
-

24. Which antidepressant would you recommend for a 38-year-old professor experiencing depression, who is concerned about potential sexual dysfunction as a side effect?

- A. Bupropion
- B. Escitalopram
- C. Venlafaxine

D. Fluoxetine

25. Which statement is incorrect regarding bipolar disorder?

- A. It most commonly affects ages 35-45 years
 - B. Type 1 is common in both sexes
 - C. Prevalence in the general population is 1-2%
 - D. Attempt to commit suicide seen in around 15% of people
-

26. A pregnant mother was on SSRIs. The risk of which of the following is increased in the newborn? Low APGAR score ADHD Persistent pulmonary hypertension Delayed developmental milestones

- A. A, C
 - B. A, B, C, D
 - C. B, C, D
 - D. A, B, C
-

27. What could be the probable diagnosis for a woman who has been brought to the psychiatrist 5 weeks post-delivery, displaying symptoms of fearfulness, irritability, and suicidal ideation?

- A. Post-partum blues
 - B. Post-partum psychosis
 - C. Post-partum depression
 - D. Panic disorder
-

28. What is the diagnosis for a 21-year-old female who avoids using elevators due to fear and always opts for stairs instead?

- A. Social phobia
 - B. Agoraphobia
 - C. Specific phobia
 - D. Panic disorder
-

29. A Clinician asks a patient “how is your mood”. Patient replies by saying “ It is going up & down”. What is the phenomenon seen here?

- A. Derailment
 - B. Tangentiality
 - C. Clang association
 - D. Neologism
-

30. What is the most probable diagnosis for a 60-year-old man who is grieving the loss of his wife and expressing guilt, believing his intestines have deteriorated and that he deserves to be imprisoned? He also reports persistent sadness and a lack of interest in daily activities since his wife's passing.

- A. Normal grief reaction
 - B. Psychotic depression
 - C. Delusional disorder
 - D. Schizophrenia
-

31. What is the probable diagnosis for a woman who is 4 days postpartum and experiencing tearfulness, mood swings, and occasional insomnia?

- A. Postpartum depression
 - B. Postpartum blues
 - C. Postpartum psychosis
 - D. Postpartum anxiety
-

32. What diagnosis according to DSM-V would be appropriate for a middle-aged female patient who has been experiencing symptoms such as loss of interest in daily activities, weight loss, insomnia, and persistent sadness for a period of six months? Additionally, she reports a single incident of hearing a voice instructing her to commit suicide.

- A. Major depressive disorder with psychotic features
 - B. Schizophrenia
 - C. Schizoaffective disorder
 - D. Delusional disorder
-

33. What is the recommended treatment for a recently diagnosed female patient with mild depression?

- A. Cognitive behavioral therapy
 - B. Cognitive behavioral therapy and antidepressants
 - C. Mild antidepressants
 - D. Vagus nerve stimulation
-

34. Which of the following disorders may present with symptoms similar to Addison's disease?

- A. Depression
 - B. Generalised anxiety disorder
 - C. Bipolar disorder
 - D. Panic disorder
-

35. What should be the next course of action for a 31-year-old woman who recently went through a divorce and is currently experiencing symptoms such as palpitations, sweating, chest heaviness, and pain? She has had several similar episodes over the past few weeks, but her blood test results, ECG, and chest X-ray show no abnormalities.

- A. Admit for detailed investigations
 - B. Start counseling sessions
 - C. Use beta-blockers
 - D. Use benzodiazepines for symptom relief
-

36. Abrupt stoppage of which of the following drugs causes irritability, rebound anxiety, and insomnia?

- A. Valproate
 - B. Fluoxetine
 - C. Olanzapine
 - D. Venlafaxine
-

37. Which one of the following characteristics is not typically associated with atypical depression?

- A. Increased sleep
 - B. Reactive mood response to positive stimulus
 - C. The heaviness of the body
 - D. Responds better to TCAs than MAO-I and SSRI
-

38. What was the intended function of thiopentone and succinylcholine given to a patient receiving ECT treatment for severe depression?

- A. To bring calming effect
 - B. To produce mood elevation
 - C. To induce anesthesia and relax muscles
 - D. To prevent memory loss after the ECT
-

39. Which neurotransmitters are implicated in depression?

- A. GABA and dopamine
 - B. Serotonin and norepinephrine
 - C. Dopamine and serotonin
 - D. GABA and norepinephrine
-

40. Which of the following side effects would most likely be caused by a drug that specifically decreases the reabsorption of serotonin?

- A. Constipation
- B. Sexual dysfunction

C. Sialorrhea

D. Visual disturbance

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Correct Answers

Question	Correct Answer
Question 1	2
Question 2	3
Question 3	2
Question 4	2
Question 5	1
Question 6	2
Question 7	1
Question 8	4
Question 9	1
Question 10	3
Question 11	2
Question 12	4
Question 13	1
Question 14	2
Question 15	3
Question 16	3
Question 17	4
Question 18	3

Question 19	3
Question 20	3
Question 21	4
Question 22	1
Question 23	4
Question 24	1
Question 25	1
Question 26	2
Question 27	3
Question 28	3
Question 29	2
Question 30	2
Question 31	2
Question 32	1
Question 33	1
Question 34	1
Question 35	4
Question 36	4
Question 37	4
Question 38	3
Question 39	2

Solution for Question 1:

The patient's symptoms, including aggression, rapid speech, decreased need for sleep, and hyperactivity, strongly suggest a manic episode. Manic episodes are characterized by elevated or irritable mood, increased energy, impulsivity, distractibility, and decreased need for sleep, typically lasting at least one week. The fact that these symptoms worsened after discontinuing prescribed medication further supports the diagnosis, as stopping mood stabilizers or antipsychotics can trigger a manic relapse.

Bipolar disorder

- Bipolar disorder is a type of mood disorder characterized by episodes of depression, mania, hypomania, mixed episodes.
- Even a single episode of mania is considered bipolar disorder.
- ICD-11 classifies only two types of bipolar disorder.

Bipolar I Disorder	Bipolar II Disorder
Characterized by at least 1 episode of mania and 1 episode of depression	Characterized by at least 1 episode of hypomania and 1 episode of depression (Option B)
M=F	F>M
Mean age of onset is 30yrs	Mean age of onset is slightly higher

Diagnostic criteria for Mania

Mood Changes: Elation/ irritability
 + ≥3 of the following symptoms (4 if irritable mood) for at least 1 week :
 Distractibility.
 Impulsivity/ high-risk activities
 Grandiosity/ Inflated Self-Esteem
 Flight of Ideas (Racing and expansive thoughts)
 Activity increased (Hyperactivity)
 Sleep deficit (decreased need for sleep)
 Talkativeness/ Pressured speech

Mania with Psychotic Symptoms:

- May include delusions or hallucinations.
- Can be mood-congruent or mood-incongruent.

Mood-congruent: When psychotic symptoms are mood-congruent, the content of a person's delusions or hallucinations aligns with the person's mood state.

Mood-incongruent: The content of a person's delusions or hallucinations does not match their mood.

Other Options:

Depression (Option A): This condition typically presents with low mood, fatigue, decreased energy, and sleep disturbances, but not hyperactivity, aggression, or rapid speech. The patient's symptoms are more suggestive of mania rather than a depressive episode.

Schizophrenia (Option C): While schizophrenia can involve disorganized speech and behavior, it is primarily characterized by delusions, hallucinations, and cognitive impairment, which are not mentioned in this case. The episodic nature of the patient's symptoms also favors bipolar disorder over schizophrenia.

Substance-Induced Mood Disorder (Option D): While certain drugs (e.g., stimulants) can cause hyperactivity and aggression, there is no mention of substance use, withdrawal, or intoxication in the case history, making this diagnosis less likely.

Solution for Question 2:

Correct Option C - Hypothermia:

- TCA toxicity is associated with hyperthermia, not hypothermia. Hyperthermia occurs because of the blockage of thermoregulatory sweating due to the anticholinergic effects of TCAs, disrupting normal thermoregulation.

Incorrect option:

Option A - Arrhythmia: TCA toxicity often leads to cardiac toxicity, including arrhythmias. The inhibitory effect on sodium channels can result in conduction disturbances and arrhythmias.

Option B - Confusion: Central nervous system effects, such as confusion, are common in TCA toxicity due to the drug's anticholinergic properties.

Option D - Pupillary dilatation: Mydriasis (pupillary dilation) in TCA toxicity is attributed to the anticholinergic effects of tricyclic antidepressants. TCAs block M3 receptors, leading to the disruption of the normal pupillary constriction mediated by the parasympathetic nervous system.

Solution for Question 3:

Correct Option B - Decrease the dose:

- Adjustments to medication during pregnancy should be done considering both the well-being of the mother and the potential risks to the fetus .
- Lithium use should be closely monitored.
- Dose reduction may be considered especially as the patient is approaching her expected delivery date. Lithium dose should be adjusted during this time to prevent potential toxicity in the newborn.

Incorrect Options:

Option A - Stop lithium: Rapid discontinuation can lead to the recurrence of the patient's manic symptoms. This may lead to lead to worsening of overall outcomes.

Option C - Increase Dose: If the patient is well-controlled on the current dose and there are no signs of relapse, increasing the dose is generally not necessary and could expose the fetus to higher levels of lithium.

Option D - Change to Valproate: The patient is currently in her third trimester, and her symptoms are effectively managed with lithium. The teratogenic effects of lithium are usually observed in the first trimester of pregnancy and it is relatively safe for use in the third trimester (with routine monitoring). There is no need to change her mood stabilizer to valproate at present.

Solution for Question 4:

Correct Option B - Valproate (Sodium Valproate):

- Hyperammonemia: Valproate can cause hyperammonemia by inhibiting the urea cycle, leading to impaired ammonia detoxification. This can result in elevated levels of ammonia in the blood.

- Elevated transaminases: Valproate is known to be associated with hepatotoxicity. It can cause liver damage, leading to elevated liver enzymes and, in severe cases, hepatitis.

Hyperammonemia: Valproate can cause hyperammonemia by inhibiting the urea cycle, leading to impaired ammonia detoxification. This can result in elevated levels of ammonia in the blood.

Elevated transaminases: Valproate is known to be associated with hepatotoxicity. It can cause liver damage, leading to elevated liver enzymes and, in severe cases, hepatitis.

Incorrect Options:

Option A - Lithium:

- Lithium is primarily associated with renal side effects, such as nephrogenic diabetes insipidus, Hypothyroidism, leukocytosis, alopecia, and chronic kidney disease, it is not typically linked to hyperammonemia or hepatitis.

Lithium is primarily associated with renal side effects, such as nephrogenic diabetes insipidus, Hypothyroidism, leukocytosis, alopecia, and chronic kidney disease, it is not typically linked to hyperammonemia or hepatitis.

Option C - Carbamazepine:

- Used in bipolar disorder and mania.
- Adverse Effects of the carbamazepine:
- It can cause HAT: H -Hepatotoxic, hyponatremia A -Allergy, SJS T -Teratogenicity (neural tube defects, cleft lip and palate) and thrombocytopenia.
- H -Hepatotoxic, hyponatremia
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H -Hepatotoxic, hyponatremia

A -Allergy, SJS

T -Teratogenicity (neural tube defects, cleft lip and palate) and thrombocytopenia.

Option D - Topiramate:

- Topiramate is known to be associated with renal stones.

Topiramate is known to be associated with renal stones.

Solution for Question 5:

Correct option A - Impulse:

- Impulse is a sudden and irresistible urge to take a specific action.
- It is a strong and often uncontrollable desire to perform an action without much forethought or consideration of the consequences.

Incorrect options:

Option B - Hallucination: Hallucination refers to perceiving something that is not actually present in the external environment.

Option C - Delusion: Delusion is a fixed, false belief that is resistant to reason or contrary evidence.

Option D - Pseudohallucination: Pseudohallucination is similar to hallucinations but are recognized by the person as being generated by their own mind.

Solution for Question 6:

Correct Option B- Low mood:

- This is the key symptom of depression. Persistent feelings of sadness, low mood, or emptiness are often the hallmark of depressive episodes. It's not just feeling down occasionally; it's a pervasive and lasting emotional state that significantly impacts a person's overall well-being.

Incorrect Options:

Option A- Suicidal ideations: While suicidal ideations can be a serious symptom of depression, they are not the key symptom that defines depression. Many individuals with depression may not experience suicidal thoughts, but they can still exhibit other characteristic symptoms.

Option C - Loss of interest in pleasurable activity: This is indeed a symptom of depression, but it is not the key defining symptom. Anhedonia, or the loss of interest in activities that were once pleasurable, is commonly seen in depression, but it is usually accompanied by other symptoms like low mood and fatigue.

Option D- Loss of Appetite: Changes in appetite, including both overeating and loss of appetite, are common in depression. However, like option C, this is a symptom often seen in conjunction with other symptoms rather than being the key defining symptom.

Solution for Question 7:

Correct option A - MAO inhibitors (monoamine oxidase inhibitors):

- MAO inhibitors are a class of antidepressant medications that work by inhibiting the activity of the enzyme monoamine oxidase, which breaks down neurotransmitters like serotonin, dopamine and norepinephrine.
- These medications can interact with certain foods and beverages, including wine, that contain tyramine.
- Consuming tyramine-rich foods and beverages while on MAO inhibitors can lead to a sudden and severe increase in blood pressure, known as a hypertensive crisis.

Incorrect options : Option B, C and D do not cause hypertensive crisis when combined with wine.

Solution for Question 8:

Correct option D - Fluoxetine:

- Fluoxetine is a selective serotonin reuptake inhibitor (SSRI).

- SSRIs are recommended as a first-line treatment for major depressive disorder due to their efficacy in alleviating depressive symptoms.
- It works by increasing the availability of serotonin in the brain, which can help improve mood, appetite, and sleep patterns.

Incorrect options: Option A, B and D are not used in treatment for major depressive disorder.

Solution for Question 9:

Correct Option A - Occurs commonly in men:

- Rapid cycling in bipolar disorder is actually more common in women, not men making it the correct answer for the question

Incorrect Options:

Option B - Commonly associated with concomitant hypothyroidism: Hypothyroidism is a known risk factor for rapid cycling in individuals with bipolar disorder..

Option C - At least 4 distinct episodes per year: Rapid cycling is defined by the occurrence of four or more mood episodes in a year.

Option D - Antidepressant use can increase the likelihood: Antidepressant use can indeed trigger or increase the likelihood of rapid cycling in individuals with bipolar disorder.

Solution for Question 10:

Correct Option C - Introjection:

- Introjection refers to an unconscious process where individuals internalize the beliefs and attitudes of others.
- It is more associated with psychoanalytic theory rather than Beck's cognitive theory of depression.
- Therefore, it is not considered a component of Beck's cognitive model.

- It is considered as a defence mechanism.

Incorrect Options:

Option A - Cognitive distortions:

- Systematic errors in thinking affecting perception of self, environment, and future
- Arbitrary inference, selective abstraction, overgeneralization, magnification, and minimization.
- Central to Beck's theory; they contribute to negative thinking patterns in depression.

Option B - Automated thoughts:

- These are spontaneous, involuntary thoughts that occur in response to specific stimuli.
- In depression, these thoughts are typically negative and automatic, playing a crucial role in sustaining depressive symptoms. They are a key component of Beck's cognitive model.

Option D - Dysfunctional belief:

- Rigid, unhealthy beliefs about oneself, others, and the world.
- Often negative and unrealistic, contributing to depression.
- Beck's theory emphasizes identifying and modifying these beliefs to treat depression

Solution for Question 11:

Correct Option B - Recurrent depressive disorder:

- 2 or more episodes of depression.
- Minimum duration of normal period between 2 episodes is 2 months

Incorrect Options:

Option A – Depression: Because there is duration of normal period between 2 episodes is 2 months.

Option C & D - Dysthymia & Cyclothymia: There should be persistence of symptoms for 2 years

Solution for Question 12:

Correct Option D - Sialorrhea:

- Sialorrhea refers to excessive salivation or drooling, and it is not commonly associated with the use of escitalopram. Sialorrhea, or excessive salivation, is not typically associated with the use of escitalopram. Other medications or medical conditions may be more commonly linked to this side effect.

Incorrect Options:

Option A - Nausea: Nausea is a common side effect of escitalopram and can occur during the initial stages of treatment. It usually subsides over time.

Option B - Vivid dreams: Escitalopram can cause changes in dream patterns, including vivid dreams. This is a relatively common side effect.

Option C - Anorgasmia: Anorgasmia refers to the difficulty or inability to achieve orgasm. It is a known side effect of selective serotonin reuptake inhibitors (SSRIs) such as escitalopram.

Solution for Question 13:

Correct Option A - It can cause fine postural tremors at therapeutic dosage:

- This statement is true. One of the common side effects of lithium, even at therapeutic dosages, is the development of fine postural tremors. These tremors typically occur during tasks that require fine motor skills and can be dose-dependent.

Incorrect Options:

Option B - It is not absorbed from the gut: This statement is false. Lithium is well absorbed from the gastrointestinal tract after oral administration. Once absorbed, it is distributed throughout the body and excreted primarily by the kidneys.

Option C - It is not teratogenic: This statement is false. Lithium has been associated with an increased risk of teratogenic effects when taken during pregnancy. It is generally not recommended for use during pregnancy, particularly during the first trimester, due to the potential risks to the developing fetus.

Option D - It is also used for the treatment of absence seizures: This statement is false. Lithium is not a recognized or established treatment for absence seizures.

Solution for Question 14:

Correct Option B - Narrow therapeutic index:

- Narrow therapeutic index: This statement is true for lithium. Lithium has a narrow therapeutic index, meaning that the difference between an effective dose and a toxic dose is relatively small. Close monitoring of blood lithium levels is necessary to maintain therapeutic efficacy while avoiding toxicity.

Incorrect Options:

Option A - Delayed absorption: This statement is not true for lithium. Lithium is generally rapidly and almost completely absorbed from the gastrointestinal tract after oral administration. Peak blood levels are typically reached within 1 to 4 hours.

Option C - Protein-bound: This statement is not true for lithium. Lithium is primarily a monovalent cation and is not extensively bound to plasma proteins. It is mainly distributed in the extracellular fluid, including intracellular compartments. Option D - Can be given safely in renal dysfunction: This statement is not entirely accurate. Lithium is primarily eliminated by the kidneys, and renal dysfunction can affect its clearance. In patients with renal impairment, the clearance of lithium may be decreased, leading to higher blood levels and an increased risk of toxicity. Therefore, lithium use in renal dysfunction requires careful monitoring and dosage adjustments to avoid excessive accumulation and potential toxicity.

Solution for Question 15:

Correct Option C - Hoarding:

- Hoarding refers to the excessive accumulation of items and difficulty discarding them, often resulting in cluttered living spaces and significant distress .

- Hoarding behaviors are typically resistant to change and can be more challenging to treat with ERP compared to other OCD symptoms.

Incorrect Options:

Option A - Pathological doubt: Pathological doubt refers to excessive and irrational doubts that individuals with OCD experience, often leading them to seek reassurance repeatedly. While pathological doubt can be challenging to treat, it generally responds well to ERP and has a more favorable prognosis compared to hoarding.

Option B - Magical thinking : Magical thinking involves beliefs that one's thoughts or actions can influence events in unrealistic or supernatural ways. While it can be a symptom of OCD, it doesn't necessarily indicate a poor prognosis with ERP. The effectiveness of ERP treatment depends on the individual and the specific nature of their OCD symptoms.

Option D - Contamination obsession : Contamination obsessions involve an intense fear of contamination or germs and often result in excessive cleaning or avoidance behaviors. Contamination obsessions tend to respond well to ERP treatment, and individuals with this symptom subtype generally have a better prognosis compared to hoarding.

Solution for Question 16:

Correct Option C - Obsessive-compulsive disorder:

- The patient presents with repetitive behavior of thoughts of washing her hands and doubts of them being contaminated, which are features of Obsessive Compulsive Disorder.

Incorrect Options:

Option A - Phobia: Phobia involves a morbid and irrational fear with acute anxiety and panic-like symptoms.

Option B - Forced thinking: Forced thinking is seen in the pre-ictal stage and is relieved after a seizure.

Option D - Generalized anxiety disorder: Generalized anxiety disorder involves continuous, free-floating anxiety and is not situation dependent.

Solution for Question 17:

Correct Option D - Sialorrhea:

- Sialorrhea, also known as excessive salivation or drooling, is not typically associated with the use of escitalopram.

Incorrect Options:

Option A - Nausea: Nausea is a common side effect associated with the use of escitalopram and other SSRIs. It refers to the sensation of unease or discomfort in the stomach, often accompanied by a desire to vomit. Approximately 20% of patients may experience nausea when starting treatment with an SSRI like escitalopram. However, it's important to note that this side effect is usually temporary and tends to improve over time as the body adjusts to the medication.

Option B - Vivid dreams: Vivid dreams, also known as intense or abnormal dreams, can occur as a side effect of escitalopram. Some individuals may experience more vivid or unusual dreams while taking this medication. These dreams can be highly detailed, emotional, or even bizarre. While vivid dreams are not as common as nausea, they are still a recognized side effect of escitalopram and other SSRIs.

Option C - Anorgasmia: Anorgasmia refers to the difficulty or inability to achieve orgasm. It is a potential side effect of escitalopram and other SSRIs. Anorgasmia can affect both men and women and is thought to be related to the medication's impact on serotonin levels in the brain. It's important to note that not all individuals will experience this side effect, and for those who do, it may resolve over time or with dosage adjustments.

Solution for Question 18:

Correct Option C - Sleep-onset insomnia:

- Sleep-onset insomnia: This symptom is not typically associated with melancholia. Instead, individuals with melancholia may experience early morning awakening.
- Melancholia, also known as melancholic depression, is a subtype of major depressive disorder.

Incorrect Options:

Option A - Severe anhedonia: Anhedonia refers to the inability to experience pleasure or enjoy activities that were previously enjoyable. In melancholia, this symptom is typically severe.

Option B - Weight loss: Melancholia often leads to significant weight loss due to reduced appetite and decreased interest in eating.

Option D - Profound feelings of guilt over trivial events: Melancholia is often accompanied by excessive guilt and self-blame, even for minor or trivial events.

Solution for Question 19:

Correct Option C - Rapid cycling disorder:

- The finding of the quick cycling problem depends on the accompanying standards: The patient has had at least four episodes of craziness, hypomania, or sorrow in a year's time span. A substance or another ailment does not bring about episodes of craziness, hypomania, or sadness. The episodes of madness, hypomania, or discouragement cause huge impedance in friendships, work-related, or other significant areas of life.
- The patient has had at least four episodes of craziness, hypomania, or sorrow in a year's time span.
- A substance or another ailment does not bring about episodes of craziness, hypomania, or sadness.
- The episodes of madness, hypomania, or discouragement cause huge impedance in friendships, work-related, or other significant areas of life.
- The patient in the picture meets these measures. She has had at least four episodes of lunacy, hypomania, or wretchedness over a year. A substance or another ailment does not bring about these episodes. Moreover, these episodes have caused critical disability in her social, work-related, and other significant areas of life.
- The patient has had at least four episodes of craziness, hypomania, or sorrow in a year's time span.
- A substance or another ailment does not bring about episodes of craziness, hypomania, or sadness.
- The episodes of madness, hypomania, or discouragement cause huge impedance in friendships, work-related, or other significant areas of life.

Incorrect Options:

Option A - Cyclothymia: The person will experience periods of low mood followed by periods of extreme joy and excitement (hypomania), where one doesn't need much sleep and feel like they have a lot of energy. A diagnosis of clinical depression is not warranted because neither the mood swings nor their duration are severe enough. Cyclothymia, otherwise called cyclothymic turmoil, causes mood swings that reach from low to high. Because their symptoms are so mild or because the emotional highs feel good, the majority of people do not seek treatment for mental health issues. As a result, they are unaware that there is a problem and do not wish to seek assistance. Cyclothymia is frequently not diagnosed or treated as a result.

Option B - Dysthymia: Dysthymia is a milder form of depression that lasts longer. It's similarly called a persistent difficult issue. Major depressive episodes may also occur from time to time in people with this condition.

Option D - Bipolar disorder: Bipolar disorder, formerly known as manic depression, is a mental health condition that can cause extreme mood swings like mania or hypomania as well as depression. When one develops depression, they may experience feelings of sadness or hopelessness and may lose interest in or enjoyment in most activities. In the case of milder forms of manic depression, the mood may change to one that is euphoric, full of energy, or unusually irritable.

Solution for Question 20:

Correct Option C - Depression with dysthymia:

- Double depression refers to the coexistence of major depressive disorder and dysthymia
- Dysthymia is a chronic depressive disorder, and when an individual with dysthymia experiences a major depressive episode, it is considered double depression. In this case, the individual would have ongoing depressive symptoms for at least two years (dysthymia) and then experience a distinct period of more severe depressive symptoms meeting the criteria for major depressive disorder.

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Incorrect Options:

Option A - Major depressive disorder from 2 years: While it is possible to have a chronic course of major depressive disorder, the specific term "double depression" is not used to describe this condition. Major depressive disorder itself can be a recurrent and chronic condition, but double depression specifically refers to the co-occurrence of MDD and dysthymia.

Option B - Major depression with OCD: Obsessive-Compulsive Disorder (OCD) is a separate psychiatric disorder characterized by intrusive thoughts (obsessions) and repetitive behaviors or mental acts (compulsions). While individuals with OCD may also experience depressive symptoms, the term "double depression" is not used to describe this combination. Double depression specifically involves the coexistence of major depressive disorder and dysthymia.

Option D - Depression with anxiety attack: While anxiety symptoms can often co-occur with depressive symptoms, the term "double depression" is not used to describe this combination. Double depression specifically refers to the combination of major depressive disorder and dysthymia.

Solution for Question 21:

Correct Option D - Support and Education:

- Postpartum blues, also known as "baby blues," is a common condition that affects many women after childbirth. It is characterized by mild mood swings, tearfulness, irritability, and emotional lability. These symptoms typically appear within the first few days after delivery and resolve spontaneously within a few weeks.
- Support and education are the mainstays of treatment for postpartum blues. This involves providing reassurance, empathy, and understanding to the woman experiencing the condition. . Education about postpartum blues and its self-limiting nature can help alleviate anxiety and promote an understanding of the normal emotional changes that occur during the postpartum period.

Incorrect Options:

Option A - Fluoxetine: Fluoxetine is a selective serotonin reuptake inhibitor (SSRI) antidepressant. While SSRIs may be used in the treatment of postpartum depression, they are not the first-line treatment for postpartum blues, which is a milder and self-limiting condition.

Option B - Cognitive behavioral therapy (CBT): CBT is a form of psychotherapy that can be effective for various mental health conditions, including postpartum depression. However, it is

not typically the primary treatment for postpartum blues, as this condition tends to resolve without formal therapeutic intervention.

Option C - Lithium carbonate: Lithium carbonate is a mood-stabilizing medication primarily used in the treatment of bipolar disorder. It is not indicated for the treatment of postpartum blues, as this condition does not involve significant mood disturbances or require pharmacological intervention in most cases.

Solution for Question 22:

Correct Option A - Cardiac defects:

- Lithium usage during pregnancy has been associated with an increased risk of cardiac defects in the developing fetus.
- Specifically, the use of lithium in the first trimester has been linked to an increased incidence of Ebstein's anomaly, a congenital heart defect where the tricuspid valve is abnormally formed.

Incorrect Options:

Option B - Facial defects: Lithium usage during pregnancy is not commonly associated with facial defects in the developing fetus. Various factors can cause facial defects, but lithium is not typically implicated in this teratogenic effect.

Option C - CNS defects: While lithium has been associated with certain risks to the developing fetus, such as cardiac defects, it is not commonly linked to central nervous system (CNS) defects. CNS defects can result from genetic, environmental, and other medication-related factors, but the evidence does not suggest a strong association between lithium usage during pregnancy and CNS defects.

Option D - Urogenital defects: Urogenital defects involving the urinary and genital systems are not typically associated with lithium usage during pregnancy. Lithium's teratogenic effects primarily revolve around cardiac abnormalities rather than urogenital defects.

Solution for Question 23:

Correct Option D - Lithium:

- Mood stabilizers are the primary class of medications used to treat bipolar disorder. They work to help stabilize mood and prevent episodes of mania or depression.
- In this case, the patient has a family history of bipolar disorder and has already experienced three episodes of mania. Lithium is considered the most effective mood stabilizer for treating bipolar disorder, particularly for preventing future episodes of mania.

Incorrect Options:

Option A - Valproate: Valproate is a mood stabilizer that can be used to treat bipolar disorder. However, it is typically more effective at preventing future episodes of depression than mania.

Option B - Lamotrigine: Lamotrigine is a mood stabilizer that can be used to treat bipolar disorder. However, it is less effective at preventing manic episodes than lithium.

Option C - Olanzapine: Olanzapine is an atypical antipsychotic medication that can be used to treat bipolar disorder. It is effective in treating manic episodes but is not typically used as a first-line treatment for bipolar disorder.

Solution for Question 24:

Correct Option A - Bupropion:

- Among the given options, Bupropion is least likely to cause sexual dysfunction as compared to the other antidepressants.
- This is because Bupropion is a dopamine-norepinephrine reuptake inhibitor (DNRI), whereas the other options are selective serotonin reuptake inhibitors (SSRIs) or serotonin-norepinephrine reuptake inhibitors (SNRIs).
- SSRIs and SNRIs increase serotonin levels in the brain, which can lead to sexual dysfunction as a side effect.
- On the other hand, Bupropion does not affect serotonin levels but increases dopamine and norepinephrine levels. It has been shown to improve sexual function in some cases.

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SSRIs and SNRIs increase serotonin levels in the brain, which can lead to sexual dysfunction as a side effect.

On the other hand, Bupropion does not affect serotonin levels but increases dopamine and norepinephrine levels. It has been shown to improve sexual function in some cases.

Incorrect Options:

Option B - Escitalopram: Escitalopram is a selective serotonin reuptake inhibitor (SSRI) antidepressant. SSRIs work by increasing serotonin levels in the brain, which can improve mood but can also cause sexual dysfunction as a side effect. In some studies, up to 60% of patients taking SSRIs reported sexual dysfunction as a side effect.

Option C - Venlafaxine: Venlafaxine is a serotonin-norepinephrine reuptake inhibitor (SNRI) antidepressant. SNRIs work by increasing the levels of both serotonin and norepinephrine in the brain, which can improve mood but can also cause sexual dysfunction as a side effect. Studies have shown that up to 30% of patients taking SNRIs reported sexual dysfunction as a side effect. Venlafaxine is also known to have a higher risk of causing sexual dysfunction compared to some other antidepressants, such as bupropion.

Option D - Fluoxetine: Fluoxetine is an SSRI antidepressant commonly prescribed for depression and anxiety disorders. Like other SSRIs, it can cause sexual dysfunction as a side effect. However, it is less likely to cause sexual dysfunction than other SSRIs such as paroxetine or sertraline. Fluoxetine has a long half-life, which means it stays in the body longer and can potentially have a prolonged effect on sexual function. Therefore, it may not be the best option for a patient concerned about sexual dysfunction.

Solution for Question 25:

Correct Option A - It most commonly affects ages 35-45 years:

- Bipolar disorder usually affects the age groups of 15-25 years. Bipolar disorder is classified into 2 types: Bipolar I (Mania + Depression) and Bipolar II (Hypomania + Depression). The

mean age of onset for Bipolar type I is 18 years . The mean age of onset for Bipolar type II is mid-20s. Male to female ratio is 1.1: 1.

- Bipolar I (Mania + Depression) and Bipolar II (Hypomania + Depression). The mean age of onset for Bipolar type I is 18 years . The mean age of onset for Bipolar type II is mid-20s. Male to female ratio is 1.1: 1.
- Bipolar I (Mania + Depression) and Bipolar II (Hypomania + Depression). The mean age of onset for Bipolar type I is 18 years . The mean age of onset for Bipolar type II is mid-20s. Male to female ratio is 1.1: 1.

Incorrect Options:

Option B - Type 1 is common in both sexes: bipolar disorders are equally prevalent among both sexes. Manic episodes are most commonly seen among males, and depressive episodes are more common among women.

Option C - Prevalence in the general population is 1-2%: Annual incidence of bipolar disorders is usually less than 1%, but the milder forms are usually missed. The prevalence rate of bipolar I disorder is 0-2.4%, and the prevalence of bipolar II disorder is 0.3-4.8%.

Option D - Attempt to commit suicide is seen in around 15% of people: About 2/3rd of all patients with major depression contemplates committing suicide, and around 10-15% of individuals commit suicide.

Solution for Question 26:

Correct Option B - A, B, C, D:

- Babies born to mothers who took selective serotonin reuptake inhibitors during pregnancy are at a high risk of developing pulmonary hypertension, low APGAR score, ADHD, and delayed developmental milestones.
- Similarly, paroxetine should be avoided in pregnancy as it is known to cause serious side effects (heart defects) in the baby (ASD, VSD).

Incorrect Options:

- Option A, C & D are incorrect
-

Solution for Question 27:

Correct Option C - Postpartum depression:

- Based on the symptoms described, the likely diagnosis for the female in question is Postpartum depression (PPD). Postpartum depression is a mood disorder that occurs after childbirth, typically within the first few weeks to months.

Incorrect Options:

Option A - Postpartum blues/Baby blues result from psychological factors such as the emotional stress of childbirth and the feelings of added responsibility. It gets resolved within 7-10 days of delivery.

Option B - Post-partum psychosis: Tearfulness, sleeplessness, and even delusions and hallucinations (the baby is dead, she didn't give birth) are all symptoms of postpartum psychosis .

Option D - Panic disorder: Panic disorder is a different condition characterised by continuous episodes of panic attacks.

Solution for Question 28:

Correct Option C - Specific phobia:

- Specific phobias involve strong, persistent, and irrational fear of an object or situation. The phobia of elevators or lifts given in the question is a situational phobia .

Incorrect Options:

Option A - Social Phobia: Involves intense fear and anxiety in social situations. It is also known as a social anxiety disorder. Individuals with this phobia usually fear embarrassment or humiliation in front of others.

Option B - Agoraphobia: Individuals with Agoraphobia have the fear of places which are difficult to escape. For example, fear of open spaces, traveling alone, crowded places, or public transport.

Option D - Panic disorder: Involves recurring panic attacks, but the fear here is specifically related to elevators rather than panic attacks.

Solution for Question 29:

Correct Option B - Tangentiality:

- Tangentiality is a formal thought disorder. Individuals with tangentiality wander from the topic and never reach a particular goal. As mentioned in the question, the patient is unable to reach a conclusion about his mood.

Incorrect Options:

Option A - Derailment: Is a disorder of thought. Patients with this disorder shift from the main topic to unrealistic ideas.

Option C - Clang association: Involves the association of words based on sound rather than meaning. It is not applicable here as the patient's response does not involve using words based on sound patterns.

Option D - Neologism: Is associated with creating words that lead to the formation of a sentence with no actual meaning.

Solution for Question 30:

Correct Option B - Psychotic depression:

- Psychotic depression, also known as a major depressive disorder with psychotic features, is a subtype of depression characterized by the presence of psychotic symptoms alongside depressive symptoms. It is a severe form of depression that can involve delusions, hallucinations, or both.
- In the given scenario, the patient is experiencing feelings of guilt and responsibility for his wife's death, which are indicative of depressive symptoms. Additionally, he has developed delusions of his intestines being rotten and feeling deserving of imprisonment, suggesting

the presence of psychotic symptoms. These delusions are not based on reality and are inconsistent with the features of psychotic depression.

Incorrect Options:

Option A - Normal grief reaction: While it is normal to experience grief and sadness after the loss of a loved one, the presence of delusions and severe depressive symptoms suggests a more severe condition than a typical grief reaction.

Option C - Delusional disorder: Delusional disorder is characterized by persistent delusions without prominent mood symptoms such as depression. The presence of depressive symptoms in addition to delusions points more towards psychotic depression.

Option D - Schizophrenia: Schizophrenia is a chronic psychiatric disorder characterized by a combination of symptoms, including hallucinations, delusions, disorganized thinking, and negative symptoms. While the patient in the scenario has delusions, there is no mention of other typical symptoms of schizophrenia.

Solution for Question 31:

Correct Option B - Postpartum blues:

- The symptoms of tearfulness, occasional insomnia, and mood swings in a woman who is 4 days postpartum indicate the diagnosis of postpartum blues. Postpartum blues, also known as "baby blues," is a common condition that affects up to 80% of women after giving birth. It is a mild and self-limiting condition characterized by symptoms such as mood swings, tearfulness, anxiety, irritability, and sleep disturbances. The symptoms usually appear within the first few days after delivery and resolve within a week or two without requiring any treatment.

Incorrect Options :

Option A - Postpartum depression: After giving birth, postpartum depression can also happen, although it often manifests with more severe and enduring symptoms, including depressed mood, loss of interest, feelings of guilt or worthlessness, and suicidal thoughts. Postpartum depression typically begins within the first few weeks following delivery and, if ignored, can linger for months to years.

Option C - Postpartum psychosis: Postpartum psychosis is a rare but severe condition that can occur within the first few weeks after childbirth. It is characterized by symptoms such as

delusions, hallucinations, disorientation, and severe mood swings. Women with postpartum psychosis require urgent medical attention and may need to be hospitalized.

Option D - Postpartum anxiety: While anxiety can be a symptom of postpartum blues, it is not typically the primary presenting symptom. Postpartum anxiety can also occur after childbirth and is characterized by excessive worry, restlessness, and physical symptoms such as palpitations and shortness of breath.

Solution for Question 32:

Correct Option A - Major depressive disorder with psychotic features:

- Based on the information provided, the patient is experiencing symptoms of both depression and psychosis. The presence of symptoms such as loss of interest, weight loss, insomnia, and sadness for six months indicates a depressive episode. Additionally, the patient's report of hearing a voice that ordered her to kill herself suggests the presence of psychotic symptoms.

Incorrect Options:

Option B - Schizophrenia: Schizophrenia is a separate psychiatric disorder characterized by persistent symptoms such as hallucinations, delusions, disorganized thinking, and negative symptoms. The presence of a depressive episode and the lack of sufficient symptoms to meet the criteria for schizophrenia make this diagnosis less likely.

Option C - Schizoaffective disorder: Schizoaffective disorder is characterized by a combination of symptoms of both schizophrenia and a mood disorder (such as depression or mania). However, the information provided suggests a primary diagnosis of major depressive disorder rather than a combination of schizophrenia and mood disorder symptoms.

Option D - Delusional disorder: Delusional disorder is characterized by the presence of one or more delusions for at least one month. However, the patient's symptoms of loss of interest, weight loss, insomnia, and sadness are more consistent with a depressive episode, making major depressive disorder a more appropriate diagnosis.

Solution for Question 33:

Correct Option A - Cognitive behavioral therapy:

- Treatment of choice of depression: Combination of pharmacotherapy + psychotherapy has a higher response. Single therapy alone is sufficient for most of the part.
- Combination of pharmacotherapy + psychotherapy has a higher response.
- Single therapy alone is sufficient for most of the part.
- Combination of pharmacotherapy + psychotherapy has a higher response.
- Single therapy alone is sufficient for most of the part.

Mild	Moderate /severe
CBT	CBT + drugs

Incorrect Options:

Option B - Cognitive behavioral therapy(CBT) and antidepressants: For mild depression, CBT alone will be sufficient to treat. For moderate/severe depression there is a combination of CBT and drugs.

Option C - Mild antidepressants : It is usually indicated in chronic moderate to severe depression and not mild depression

Option D - Vagus nerve stimulation: Not indicated for mild depression. Vagus nerve stimulation is wherein the Left vagus nerve is stimulated by electrodes via a pulse generator implanted in the chest. It is indicated in resistant depression.

Solution for Question 34:

Correct Option A - Depression:

- Addison's disease, also known as adrenal insufficiency, can present with symptoms similar to depression.
- Addison's disease occurs when the adrenal glands do not produce enough cortisol and sometimes aldosterone.

- The symptoms of Addison's disease can include fatigue, loss of appetite, weight loss, low mood, and a lack of interest or pleasure in activities, which can overlap with depressive symptoms.

Incorrect Options:

Option B - Generalized anxiety disorder: Generalized anxiety disorder (GAD) is a mental health condition characterized by excessive and persistent worry or anxiety about various aspects of life. While there may be some overlap in symptoms such as fatigue, difficulty concentrating, and irritability between Addison's disease and GAD, the core features and causes of the two conditions are distinct. Addison's disease primarily results from adrenal gland dysfunction, while GAD is a psychological disorder characterized by excessive and chronic worry.

Option C - Bipolar disorder: Bipolar disorder is a mood disorder characterized by alternating periods of depression and mania or hypomania. While there may be some overlap in symptoms such as fatigue, changes in appetite, and altered mood between Addison's disease and bipolar disorder, the underlying causes and nature of the conditions are different. Addison's disease results from adrenal gland dysfunction, whereas bipolar disorder involves disturbances in mood regulation.

Option D - Panic disorder: Panic disorder is a mental health condition characterized by recurrent and unexpected panic attacks, which are intense periods of fear and discomfort. While there may be some overlap in symptoms such as fatigue, dizziness, and feelings of unreality between Addison's disease and panic disorder, the core features and causes of the two conditions are distinct. Addison's disease primarily results from adrenal gland dysfunction, whereas panic disorder is characterized by recurrent panic attacks and associated anxiety symptoms.

Solution for Question 35:

Correct Option D - Use benzodiazepines for symptom relief:

- It appears that the patient is having a panic attack or an acute anxiety attack for which the drug of choice is benzodiazepines for immediate symptomatic relief. It is only acutely used but for long term treatment SSRI and CBT is used.

Incorrect Options:

Option A - Admit for detailed investigations: Not required as the investigations in the past showed no abnormality.

Option B - Start counseling sessions: Counseling sessions are started only once the panic attack is under control.

Option C - Use beta-blockers: They are more so used in social anxiety and do not play a large role in panic attacks.

Solution for Question 36:

Correct Option D - Venlafaxine:

- Discontinuation syndrome occurs with abrupt cessation of certain short-acting drugs like venlafaxine, fluvoxamine, and paroxetine.
- It is not seen with long-acting drugs such as fluoxetine.
- Symptoms include flu-like symptoms (lethargy, weakness, headache), insomnia, nausea, dizziness, sensory disturbances like paraesthesia, and hyperarousal (rebound anxiety, irritability).

Incorrect Options:

- Option A,B and C are examples of long acting drugs

Solution for Question 37:

Correct Option D - Responds better to TCAs than MAO-I and SSRI:

- Responds better to TCAs than MAO-Is and SSRIs: This option is not a feature of atypical depression. Selective serotonin reuptake inhibitors (SSRIs) and monoamine oxidase inhibitors (MAO-Is) are generally considered first-line treatments for depression, including atypical depression. Tricyclic antidepressants (TCAs) are not typically preferred as first-line treatment for depression, including atypical depression, due to their side effect profile and the availability of newer, better-tolerated medications.

Incorrect Options:

Option A - Increased sleep: Increased sleep is indeed a feature of atypical depression. Individuals with atypical depression may experience hypersomnia (excessive sleepiness) and have an increased need for sleep. They may feel excessively tired and find it difficult to get out of bed in the morning. Option B - Reactive mood response to a positive stimulus: Mood reactivity is a feature of atypical depression. Mood brightens in response to a positive event. Option C - The heaviness of the body: Feeling a physical heaviness or leaden paralysis is a common feature of atypical depression. Individuals may describe a sensation of their limbs feeling heavy or feeling physically weighed down. This symptom is typically more pronounced in the morning and may improve as the day progresses.

Solution for Question 38:

Correct Option C - To induce anesthesia and relax muscles:

- Electroconvulsive therapy (ECT) is a procedure used to treat severe depression, and it involves the administration of a general anesthetic and a muscle relaxant. Thiopental (thiopentone) is a short-acting barbiturate used to induce anesthesia, while succinylcholine is a depolarizing muscle relaxant that helps to prevent convulsions during the procedure.

Incorrect Options:

Option A - To bring a calming effect: ECT is not administered to bring a calming effect but rather to induce a controlled seizure for therapeutic purposes.

Option B - To produce mood elevation: ECT is not directly administered to produce mood elevation but rather to alleviate severe depressive symptoms.

Option D - To prevent memory loss after ECT: The medications given in ECT are not intended to prevent memory loss associated with the procedure, but rather to induce anesthesia and muscle relaxation.

Solution for Question 39:

Correct Option B - Serotonin and norepinephrine:

- Serotonin and norepinephrine are two key neurotransmitters implicated in depression. Serotonin is involved in mood regulation, sleep, appetite, and other functions. Norepinephrine, also known as noradrenaline, is involved in the body's stress response and plays a role in mood regulation. Imbalances in serotonin and norepinephrine levels have been associated with depression, and many antidepressant medications target these neurotransmitters to alleviate depressive symptoms.

Incorrect Options:

Option A - GABA and dopamine: GABA (gamma-aminobutyric acid) is an inhibitory neurotransmitter that helps regulate neuronal excitability in the brain. Dopamine is a neurotransmitter involved in reward, motivation, and pleasure. While imbalances in GABA and dopamine levels have been implicated in certain mental health conditions, such as anxiety disorders and schizophrenia, they are not considered the primary neurotransmitters involved in depression. However, dopamine dysregulation may play a role in specific subtypes of depression, such as atypical depression.

Option C - Dopamine and serotonin: While both dopamine and serotonin are involved in mood regulation, dopamine is not considered one of the primary neurotransmitters involved in depression. Dopamine dysregulation is more closely associated with conditions like schizophrenia, substance abuse disorders, and certain mood disorders. Option D - GABA and norepinephrine: Similar to the first option, GABA and norepinephrine are not considered the primary neurotransmitters involved in depression. While norepinephrine plays a role in mood regulation and stress response, GABA dysregulation is more commonly associated with anxiety disorders and epilepsy rather than depression.

Solution for Question 40:

Correct Option B - Sexual dysfunction:

- Selective serotonin reuptake inhibitors (SSRIs) are a class of antidepressant medications that work by inhibiting the reuptake of serotonin in the brain. This leads to increased levels of serotonin, which can have various effects on different systems in the body. One common side effect of SSRIs is sexual dysfunction, which can include decreased libido, difficulty achieving orgasm, and erectile dysfunction.

Incorrect Options:

Option A - Constipation: While constipation can be a side effect of certain medications, it is not a commonly reported side effect of SSRIs.

Option C - Sialorrhea: Sialorrhea refers to excessive salivation or drooling, and it is not typically associated with the use of SSRIs.

Option D - Visual disturbance: Visual disturbances are not commonly associated with the use of SSRIs as a side effect.

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Neurotic, stress related, Obsessive compulsive and related disorders

1. A 23-year-old female presents with complaints of sudden episodes of intense fear and discomfort. She reports palpitations, shortness of breath, chest tightness, and tremors during these episodes. They occur unpredictably and last for 20–30 minutes. She has been free of symptoms between attacks. Her mother had a similar condition in her teens. Vitals are within normal limits, and ECG is unremarkable. Which of the following medications can help her during the acute attacks?

- A. Sertraline
- B. Venlafaxine
- C. Clonazepam
- D. Mirtazapine

2. A 42-year-old woman reports that for the past year, she has been experiencing severe anxiety whenever she leaves her home alone and while using public transport. She fears that if she experiences dizziness or shortness of breath, she will not be able to find help. Due to this fear, she has been unable to visit friends, shop, or attend social events without a companion. She has never had a panic attack. What is the most likely diagnosis?

- A. Panic Disorder
- B. Specific Phobia
- C. Social Anxiety Disorder
- D. Agoraphobia

3. A 19-year-old college student reports a persistent, irrational fear of blood. She states that seeing blood, even in movies, makes her feel faint and nauseous. This has caused her to avoid medical procedures and prevented her from volunteering for first aid training, which is required for her degree. Her symptoms have been present since childhood. Which of the following would be the most effective initial treatment for this condition?

- A. Behavioral therapy with exposure therapy
 - B. Serotonin-Norepinephrine Reuptake Inhibitor (SNRI)
 - C. Short term benzodiazepines
 - D. Antipsychotic medication
-

4. In Psychiatry rotations you are asked by the consultant to take a history of a 19-year-old college student who is referred to a mental health professional due to severe anxiety related to social interactions. What is a characteristic feature that can help you distinguish social anxiety disorder from other anxiety disorders?

- A. Persistent worry about future events leading to avoidance behavior.
 - B. Intense fear of being embarrassed or judged in social situations.
 - C. Experiencing panic attacks in multiple, unpredictable situations.
 - D. Marked psychosocial distress and/or impairment in functioning.
-

5. A 45-year-old man has been experiencing chronic anxiety about various aspects of his life, including his job, health, and finances. He expresses a pervasive sense of dread and finds himself worrying about things that might happen in the future. What is the most likely duration of his symptoms that would support a diagnosis of Generalized Anxiety Disorder (GAD)?

- A. More than a month
- B. 1–3 months
- C. 6 months or more

D. More than a year

6. During Psychiatry rotations, the consultant presents a case study of a 20-year-old female patient diagnosed with Obsessive-Compulsive Disorder (OCD). The patient has a history of excessive handwashing, intrusive thoughts about contamination, and significant anxiety regarding the cleanliness of her environment. He emphasizes the importance of understanding common comorbidities associated with OCD, as they can impact treatment planning and patient outcomes. Which is the most common comorbid condition observed in patients with Obsessive-Compulsive Disorder (OCD)?

- A. Generalized Anxiety Disorder
 - B. Major Depressive Disorder
 - C. Panic Disorder
 - D. Social Anxiety Disorder
-

7. A 24-year-old woman presents with concerns about her facial appearance. She believes her nose is too large, although others reassure her it looks normal. She spends hours every day checking her appearance in the mirror and often compares herself to others. She has recently started avoiding social gatherings due to feeling self-conscious and is considering plastic surgery. What is the most likely diagnosis?

- A. Social Anxiety Disorder
 - B. Body Dysmorphic Disorder
 - C. Delusional Dysmorphophobia
 - D. Generalized Anxiety Disorder
-

8. A 55-year-old man presents with complaints from his family that his home is excessively cluttered. He has difficulty discarding any of his belongings, even items with no apparent value, and refuses to let others help him organize. The clutter has begun to block walkways and interfere with his daily activities like cooking and sleeping. What is the most effective treatment for this condition?

- A. Cognitive Behavioral Therapy (CBT)
- B. Antipsychotic medications
- C. Exposure therapy
- D. Psychoanalysis

9. A 30-year-old woman expresses distress about her habit of pulling her hair. She reports that she frequently pulls hair from her scalp using her fingers and sometimes tweezers, leading to noticeable bald spots. She admits to feeling the urge to pull her hair during stressful situations and has even swallowed some of the hair, which has caused her digestive discomfort. What is the most appropriate treatment for her condition?

- A. N-acetylcysteine
- B. Cognitive Behavioral Therapy (CBT) with habit reversal training
- C. Exposure Therapy
- D. SSRI

10. A 29-year-old woman has persistent worry that she has an unpleasant body odor that others can smell. Despite reassurances from her friends and family that she does not smell bad, she feels convinced that her odor is noticeable and offensive. This concern has led her to avoid social situations and spend excessive time checking her appearance and using strong fragrances to mask the smell. What is the most likely diagnosis for her condition?

- A. Social Anxiety Disorder

- B. Body Dysmorphic Disorder
 - C. Olfactory Reference Syndrome
 - D. Olfactory hallucinations
-

11. Which of the following is not a symptom of Post-Traumatic Stress Disorder (PTSD)?

- A. Flashbacks and nightmares
 - B. Avoidance of reminders related to the trauma
 - C. Intense feelings of guilt and fear
 - D. Increased sleepiness
-

12. A 32-year-old man visits you following a traumatic experience during military service. He reports experiencing persistent flashbacks, nightmares, and heightened anxiety when reminded of the event. He has also been avoiding situations that trigger his memories and finds it difficult to concentrate at work. The clinician decides to discuss treatment options with him. What is the treatment of choice for Post-Traumatic Stress Disorder (PTSD)?

- A. SSRIs
 - B. Cognitive Behavioral Therapy (CBT)
 - C. Eye Movement Desensitization and Reprocessing (EMDR)
 - D. Group Therapy
-

13. A 45-year-old man presents with symptoms following a series of traumatic experiences during his childhood, including neglect and abuse. He reports ongoing difficulties with emotional regulation, feelings of hopelessness, and

problems with relationships. Additionally, he experiences flashbacks and nightmares related to his past but finds it hard to identify specific triggers. His symptoms have persisted for many years, and he struggles with a strong sense of shame. Which diagnosis is most likely applicable to this patient?

- A. Post-Traumatic Stress Disorder (PTSD)
 - B. Complex Post-Traumatic Stress Disorder (C-PTSD)
 - C. Acute Stress Disorder
 - D. Generalized Anxiety Disorder
-

14. A 27-year-old woman recently experienced a significant life change after moving to a new city for a job opportunity. Since the move, she has found it difficult to adapt and has been feeling overwhelmed, anxious, and sad. She struggles with concentration at work, and often feels tearful. Her symptoms began within three months of the move and are affecting her daily functioning. What is the most likely diagnosis for her condition?

- A. Major Depressive Disorder
 - B. Generalized Anxiety Disorder
 - C. Adjustment Disorder
 - D. Acute Stress Disorder
-

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	4
Question 3	1
Question 4	2
Question 5	3
Question 6	2
Question 7	2
Question 8	1
Question 9	2
Question 10	3
Question 11	4
Question 12	2
Question 13	2
Question 14	3

Solution for Question 1:

Answer: C) Clonazepam

Explanation:

- In this scenario, the patient is most likely having panic disorder based on the clinical presentation.
- Clonazepam is a benzodiazepine that is highly effective for acute management of panic attacks due to its rapid onset of action. They are not recommended for long-term use due to the risk of dependency.

Panic Disorder

- It is the recurrent acute intense attack of anxiety accompanied by a feeling of impending doom.
- More common in females than males.
- Median age of onset: late adolescence or early adulthood.
- Familial risk: Highest for early-onset panic disorder.
- Patients with panic disorder presents commonly with agoraphobia
- Panic attacks for more than a month are termed as a Panic disorder.
- Neurotransmitters that are implicated in the development of panic disorders are norepinephrine, Serotonin, GABA, and Cholecystokinin.

Symptoms:

Panic attacks	Unexpected i.e., not restricted to any particular situation. Usually last for 20 to 30 minutes, rarely they may last more than one hour.
Cardiopulmonary	Feeling short of breath Palpitations Chest discomfort
Gastrointestinal	Nausea GI discomfort
Skin and systemic	Sweating Chills Feeling flushed
Neurologic	Tremors Dizziness Numbness and tingling sensation
Psychiatric	Derealization Depersonalization Fear of losing control Fear of dying

- Symptoms are random and unpredictable.

- The patient is free from anxiety symptoms in between attacks.
- Anticipatory anxiety: Fearing further attacks Recurrent attacks of severe anxiety or panic.
- Unpredictable Physiologic symptoms.
- Avoidance behaviors: Trying to avoid real or imagined triggers to prevent another attack.

Treatment:

Pharmacological, psychological, and combined treatments.

Pharmacotherapy:

- Drug of choice – Selective serotonin reuptake inhibitors, However they are not used for an acute attack (Option A ruled out)
- Other first-line: Serotonin-norepinephrine reuptake inhibitor (SNRI) venlafaxine, however they are not used for an acute attack (Option B ruled out)
- Second-line: Tricyclic antidepressants, MAOIs, Other Psychotropics - Agomelatine, Bupropion, Mirtazapine. However they are not used for an acute attack (Option D ruled out)
- Benzodiazepines: Short-term use as an adjunct to SSRIs.
- Acute phase – Benzodiazepines such as Lorazepam & Clonazepam . (Option C)

Psychosocial therapies:

- Cognitive-behavioral therapy (CBT)
- Interpersonal therapy.

Note: Treatment of choice - SSRIs + Psychotherapy (CBT)

Solution for Question 2:

Answer: D) Agoraphobia

Explanation:

Agoraphobia is characterized by the fear and avoidance of situations where escape might be difficult or help unavailable if panic-like symptoms occur for more than 6 months. This often includes avoiding crowded places, public transportation, or even leaving the house alone.

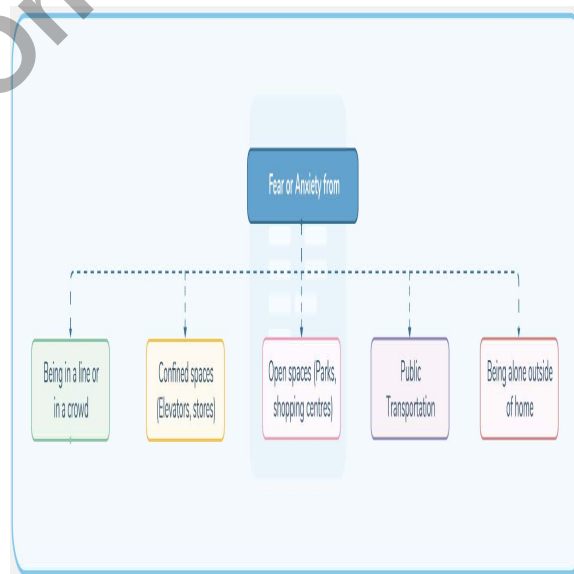
Agoraphobia

- It is the fear of or anxiety regarding places from which escape might be difficult.

- Persistence of this anxiety for six months - Agoraphobia.

Symptoms:

- Marked fear or anxiety about two (or more) of the following five situations: Using public transportation (e.g., automobiles, buses, trains, planes). Being in open spaces (e.g., parking lots, marketplaces, bridges). Being in enclosed places (e.g., shops, theatres, cinemas). Standing in line or being in a crowd. Being outside of the home alone.
- Using public transportation (e.g., automobiles, buses, trains, planes).
- Being in open spaces (e.g., parking lots, marketplaces, bridges).
- Being in enclosed places (e.g., shops, theatres, cinemas).
- Standing in line or being in a crowd.
- Being outside of the home alone.
- Avoidance of the situations, due to fear of having a panic attack while there or no access to a companion.
- Fear and avoidance are out of proportion to the potential threat.
- It causes marked distress or impairment of daily function and living.
- The most common comorbid psychiatric disorder with agoraphobia is panic disorder
- Using public transportation (e.g., automobiles, buses, trains, planes).
- Being in open spaces (e.g., parking lots, marketplaces, bridges).
- Being in enclosed places (e.g., shops, theatres, cinemas).
- Standing in line or being in a crowd.
- Being outside of the home alone.



Treatment:

Pharmacological, psychological, and combined treatments.

Psychological therapies:

- Behavioral therapy: For rapid and complete reduction of agoraphobia, such as systematic desensitization, graded exposure.
- Cognitive Behavioral therapy
- Virtual therapy
- Relaxation techniques

Pharmacotherapy:

- Limited role.
- Used to treat comorbid conditions like panic disorder.
- SSRIs, along with Benzodiazepines (short-term).

Panic Disorder (Option A): Characterized by recurrent, unexpected panic attacks and persistent concern about having future attacks lasting more than a month. Patients often develop anticipatory anxiety and behavioral changes, such as avoiding places where attacks have previously occurred. In this case, the primary concern is about avoiding situations due to fear of panic attacks, not recurrent panic attacks themselves. This option is incorrect.

Specific Phobia (Option B): Specific phobia refers to an intense fear of a particular object or situation (e.g., heights, animals, flying). The avoidance behavior in this case is not related to a specific object but to broader situations where escape might be difficult, which is more characteristic of agoraphobia. This option is incorrect.

Social anxiety disorder (Option C): It involves an intense fear of being judged or embarrassed in social or performance situations. The avoidance behavior in this case is not driven by fear of social scrutiny but by fear of being in places where escape might be difficult in case of a panic-like situation. This option is incorrect.

Solution for Question 3:

Answer: A) Behavioral therapy with exposure therapy

Explanation:

Behavioral therapy with exposure therapy is the most effective treatment for specific phobias, including blood phobia (hemophobia). In exposure therapy, the individual is gradually exposed to the feared object or situation in a safe, controlled manner, helping to reduce the intensity of the fear response over time.

Specific phobias

- Strong, persistent, irrational fear of a specific object/situation.
- Specific phobia is the persistence of this fear for 6 months or more.

Symptoms:

- Exposure to the object/situation causes immediate fear, panic, and/or anxiety.
- Avoidance behaviors: of the object or situation.
- The fear/anxiety is out of proportion to the likely threat.
- Marked distress or impairment psychosocially.
- DSM-5 Animal (e.g., Dogs, Cats, Spiders) Natural environment (e.g., Water, heights) Blood-Injection-Needle (e.g., Needles, Invasive Medical Procedures) Situational (e.g., Closed places, Planes) Others
- Animal (e.g., Dogs, Cats, Spiders)
- Natural environment (e.g., Water, heights)
- Blood-Injection-Needle (e.g., Needles, Invasive Medical Procedures)
- Situational (e.g., Closed places, Planes)
- Others
- Animal (e.g., Dogs, Cats, Spiders)
- Natural environment (e.g., Water, heights)
- Blood-Injection-Needle (e.g., Needles, Invasive Medical Procedures)
- Situational (e.g., Closed places, Planes)
- Others

Common phobias	
Acrophobia	Fear of height
Ailurophobia	Cats
Cynophobia	Dogs
Claustrophobia	Closed spaces
Mysophobia	Dirt and Germs
Hydrophobia	Water

Common phobias	
Thanatophobia	Death
Nyctophobia	Dark
Xenophobia	Strangers
Pyrophobia	Fire
Hemophobia	Blood
Arachnophobia	Spiders

Treatment:

Psychosocial therapy:

- The strategy of choice is behavioral therapy (Option A) Exposure therapy - systematic desensitization Reinforcing the realization that the phobic stimulus is safe and not dangerous. Techniques to deal with anxiety - relaxation and breathing control. The patients keep moving up the scale until the most anxiety-provoking stimuli no longer elicit any fear or anxiety (Desensitized). Flooding (Implosion): Increasing exposure to the stimulus to induce habituation and decrease anxiety- no longer preferred.
- Exposure therapy - systematic desensitization
- Reinforcing the realization that the phobic stimulus is safe and not dangerous.
- Techniques to deal with anxiety - relaxation and breathing control.
- The patients keep moving up the scale until the most anxiety-provoking stimuli no longer elicit any fear or anxiety (Desensitized).
- Flooding (Implosion): Increasing exposure to the stimulus to induce habituation and decrease anxiety- no longer preferred.
- Virtual therapy
- Hypnosis
- Supportive therapy
- Family therapy
- Exposure therapy - systematic desensitization
- Reinforcing the realization that the phobic stimulus is safe and not dangerous.
- Techniques to deal with anxiety - relaxation and breathing control.
- The patients keep moving up the scale until the most anxiety-provoking stimuli no longer elicit any fear or anxiety (Desensitized).

- Flooding (Implosion): Increasing exposure to the stimulus to induce habituation and decrease anxiety- no longer preferred.

Pharmacological therapy:

- Treat comorbid conditions.
- For example, beta-blockers and benzodiazepines for panic attacks.

Serotonin-Norepinephrine reuptake Inhibitor (SNRI) (Option B): They are used in treating generalized anxiety disorders, depression, and panic disorders, but they are not first-line treatments for specific phobias. Medications are generally not as effective for treating specific phobias as psychotherapy, particularly exposure-based techniques. This option is incorrect.

Short-term benzodiazepines (Option C): They can be used for temporary relief of acute anxiety in phobia-related situations, but they are not considered a long-term treatment or the most effective approach for specific phobias. This option is incorrect.

Antipsychotic medication (Option D): They are used as second-line drugs to treat anxiety-related disorders. They are not indicated for the treatment of specific phobias such as hemophobia. This option is incorrect.

Solution for Question 4:

Answer: B) Intense fear of being embarrassed or judged in social situations.

Explanation:

Intense fear of being embarrassed or judged in social situations is a hallmark feature of social anxiety disorder.

Social Anxiety Disorder (Social Phobia)

- Fear of social situations, including situations that involve contact with strangers.
- This fear persisting 6 months or more- Social anxiety disorder.
- It is equally prevalent in males and females (while other anxiety disorders: panic disorder, agoraphobia, specific phobia, GAD- more common in females than males)

Symptoms:

- Fear of being judged or scrutinized in a social situation. (Option B)

- Fear that others will notice the anxiety, causing additional judgment, embarrassment, or rejection.
- Fear of embarrassing oneself in front of others.
- Avoidance behaviors: of the feared social situations.
- The fear/anxiety is out of proportion to the social risk.
- Marked distress and/or impairment psychosocially.

Treatment:

Pharmacological, psychosocial, and combined therapies.

- Pharmacotherapy First-line: SSRIs along with Benzodiazepines (short-term) SNRIs- Venlafaxine. Beta-blockers (Propranolol) for performance anxiety. Second line: MAOIs Other Psychotropics —Agomelatine, Buspirone, Mirtazapine Benzodiazepines Anticonvulsant— Pregabalin
- First-line: SSRIs along with Benzodiazepines (short-term) SNRIs- Venlafaxine. Beta-blockers (Propranolol) for performance anxiety.
- SSRIs along with Benzodiazepines (short-term)
- SNRIs- Venlafaxine.
- Beta-blockers (Propranolol) for performance anxiety.
- Second line: MAOIs Other Psychotropics —Agomelatine, Buspirone, Mirtazapine Benzodiazepines Anticonvulsant— Pregabalin
- MAOIs
- Other Psychotropics —Agomelatine, Buspirone, Mirtazapine
- Benzodiazepines
- Anticonvulsant— Pregabalin
- Psychosocial therapies: Cognitive-behavioral therapy (CBT) Behavioral therapies Interpersonal therapy- interpersonal skills training for social anxiety Virtual Therapy Supportive Psychotherapy
- Cognitive-behavioral therapy (CBT)
- Behavioral therapies
- Interpersonal therapy- interpersonal skills training for social anxiety
- Virtual Therapy
- Supportive Psychotherapy
- First-line: SSRIs along with Benzodiazepines (short-term) SNRIs- Venlafaxine. Beta-blockers (Propranolol) for performance anxiety.
- SSRIs along with Benzodiazepines (short-term)

- SNRIs- Venlafaxine.
- Beta-blockers (Propranolol) for performance anxiety.
- Second line: MAOIs Other Psychotropics —Agomelatine, Buspirone, Mirtazapine Benzodiazepines Anticonvulsant— Pregabalin
- MAOIs
- Other Psychotropics —Agomelatine, Buspirone, Mirtazapine
- Benzodiazepines
- Anticonvulsant— Pregabalin
- SSRIs along with Benzodiazepines (short-term)
- SNRIs- Venlafaxine.
- Beta-blockers (Propranolol) for performance anxiety.
- MAOIs
- Other Psychotropics —Agomelatine, Buspirone, Mirtazapine
- Benzodiazepines
- Anticonvulsant— Pregabalin
- Cognitive-behavioral therapy (CBT)
- Behavioral therapies
- Interpersonal therapy- interpersonal skills training for social anxiety
- Virtual Therapy
- Supportive Psychotherapy

Persistent worry about future events leading to avoidance behavior (Option A): This is common in anxiety disorders and not specific to social anxiety disorder. Other anxiety disorders, like agoraphobia, also involve persistent worry and avoidance.

Experiencing panic attacks in multiple, unpredictable situations (Option C): This is mostly associated with panic disorder, where individuals experience unexpected panic attacks and may develop a fear of situations where they fear having another attack.

Marked psychosocial distress and/or impairment in functioning (Option D): It can occur in various anxiety disorders, including social anxiety disorder, this option is not specific enough to distinguish it from other anxiety disorders.

Solution for Question 5:

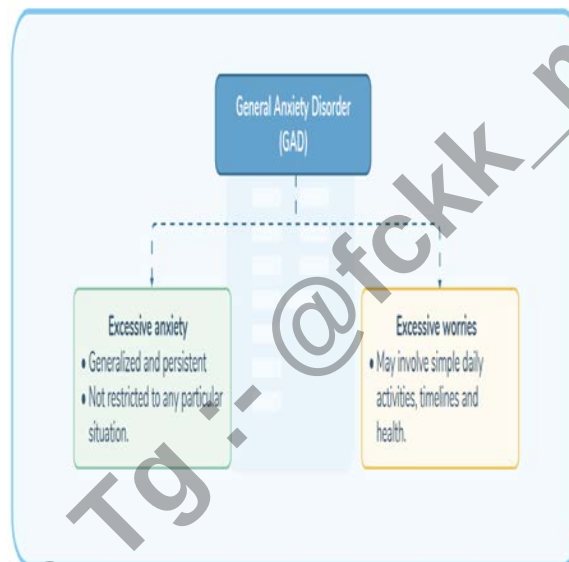
Answer: C) 6 months or more

Explanation:

According to the DSM-5, for a diagnosis of Generalized Anxiety Disorder (GAD), symptoms must persist for 6 months or more. This chronic nature of anxiety is a hallmark of GAD, distinguishing it from other anxiety disorders.

Generalized Anxiety Disorder

- Excessive anxiety and worries about several events or activities



Symptoms:

- Excessive anxiety or worry
- Difficulty controlling or managing worry
- Associated with somatic symptoms: Restlessness Fatigue Poor concentration Irritability Muscle tension Insomnia
- Restlessness
- Fatigue
- Poor concentration
- Irritability
- Muscle tension
- Insomnia
- Marked distress and/or impairment psychosocially.
- These symptoms persisting for 6 months- Generalised anxiety disorder. (Option C)

- GAD commonly has multiple relapses.
- Restlessness
- Fatigue
- Poor concentration
- Irritability
- Muscle tension
- Insomnia

Treatment:

- Pharmacotherapy: First-line: SSRIs along with Benzodiazepines (short-term) SNRIs- Venlafaxine. Second line: Tricyclic antidepressants (TCAs) Other Psychotropics —Agomelatine, Buspirone, Mirtazapine Atypical Antipsychotics— Quetiapine Benzodiazepines Anticonvulsant— Pregabalin
- First-line: SSRIs along with Benzodiazepines (short-term) SNRIs- Venlafaxine.
- SSRIs along with Benzodiazepines (short-term)
- SNRIs- Venlafaxine.
- Second line: Tricyclic antidepressants (TCAs) Other Psychotropics —Agomelatine, Buspirone, Mirtazapine Atypical Antipsychotics— Quetiapine Benzodiazepines Anticonvulsant— Pregabalin
- Tricyclic antidepressants (TCAs)
- Other Psychotropics —Agomelatine, Buspirone, Mirtazapine
- Atypical Antipsychotics— Quetiapine
- Benzodiazepines
- Anticonvulsant— Pregabalin
- Psychosocial therapies: Cognitive-behavioral therapy (CBT) Supported psychotherapy Insight-oriented therapy
- Cognitive-behavioral therapy (CBT)
- Supported psychotherapy
- Insight-oriented therapy
- First-line: SSRIs along with Benzodiazepines (short-term) SNRIs- Venlafaxine.
- SSRIs along with Benzodiazepines (short-term)
- SNRIs- Venlafaxine.
- Second line: Tricyclic antidepressants (TCAs) Other Psychotropics —Agomelatine, Buspirone, Mirtazapine Atypical Antipsychotics— Quetiapine Benzodiazepines Anticonvulsant— Pregabalin

- Tricyclic antidepressants (TCAs)
- Other Psychotropics —Agomelatine, Bupropion, Mirtazapine
- Atypical Antipsychotics— Quetiapine
- Benzodiazepines
- Anticonvulsant— Pregabalin
- SSRIs along with Benzodiazepines (short-term)
- SNRIs- Venlafaxine.
- Tricyclic antidepressants (TCAs)
- Other Psychotropics —Agomelatine, Bupropion, Mirtazapine
- Atypical Antipsychotics— Quetiapine
- Benzodiazepines
- Anticonvulsant— Pregabalin
- Cognitive-behavioral therapy (CBT)
- Supported psychotherapy
- Insight-oriented therapy

Solution for Question 6:

Answer: B) Major Depressive Disorder

Explanation:

Major Depressive Disorder is the most common comorbidity seen in individuals with OCD. It is crucial to evaluate for depression in these patients, as the first line of treatment for patients of OCD with depression is pharmacotherapy.

Obsessive-Compulsive Disorder

- Patient has recurrent obsessions and compulsions.
- Recurrent obsessions or compulsions may cause severe distress or anxiety to the person.

Obsessions	Compulsions
Intrusive and unwanted repetitive thoughts, urges, or impulses. Most common: Contamination	Repetitive actions or thoughts carried out in a strict, regulation-bound manner or reaction to obsessions Performed to relieve anxiety or fear. Most common: Checking

- These are time-consuming (at least 1 hr/day) and significantly interfere with the person's routine, occupational functioning, usual social activities, or relationships.
- Patients can have only obsessions, only compulsions, or both obsessions and compulsions (most common).
- Most common comorbidity associated with OCD is major depressive disorder (67%). (Option B)
- Major symptom patterns:

Contamination / Cleansing	Most common presentation. An obsession with contamination followed by washing and avoidance of the presumably contaminated object.
Pathological doubt /checking	Second most common presentation. Obsession of doubt, usually followed by compulsion of checking.
Intrusive/forbidden thoughts	Third most common presentation. Obsessed with intrusive thoughts of aggressive or sexual acts without a compulsion.
Symmetry/ordering	Fourth most common presentation. Patient needs symmetry/precision which results in the compulsion of slowness.
Other symptoms	Religious obsessions and compulsive hoarding.

Epidemiology:

- Lifetime prevalence- 2 to 3%.
- More common in females than males (Adolescents M>F).
- Mean age of onset in OCD is 19 years, early onset is more in males.

Etiology:

- OCD has been linked to the striatum, orbitofrontal cortex, anterior cingulate, and corticostriatal–thalamic–cortical (CSTC) circuit.
- Neurotransmitters implicated are neurotransmitter systems that contribute to OCD, including serotonergic, dopaminergic, glutamatergic, and GABAergic systems.
- Genetic susceptibility plays a role in OCD.

Treatment:

Pharmacotherapy:

- First line: SSRIs (preferred) and Clomipramine Treatment should be continued for 1-2 years
- SSRIs (preferred) and Clomipramine
- Treatment should be continued for 1-2 years

- Augmenting agents: Antipsychotics: Risperidone, Aripiprazole. Others: Memantine, riluzole, ketamine, lamotrigine, and N-acetylcysteine (NAC).
- Antipsychotics: Risperidone, Aripiprazole.
- Others: Memantine, riluzole, ketamine, lamotrigine, and N-acetylcysteine (NAC).
- SSRIs (preferred) and Clomipramine
- Treatment should be continued for 1-2 years
- Antipsychotics: Risperidone, Aripiprazole.
- Others: Memantine, riluzole, ketamine, lamotrigine, and N-acetylcysteine (NAC).

Psychotherapy:

- Cognitive Behaviour Therapy (CBT): Exposure and response prevention (ERP). Exposure to feared situations is given. The patient is asked not to engage in a compulsive response.
- Exposure and response prevention (ERP).
- Exposure to feared situations is given. The patient is asked not to engage in a compulsive response.
- Other therapies are desensitization, thought-stopping, flooding, aversive conditioning, psychodynamic psychotherapy, and family therapy.
- In younger patients, cognitive-behavioral therapy (CBT) is preferred first.
- In more severe symptoms or comorbid depression, pharmacotherapy is preferred first.
- Combination of CBT and pharmacotherapy is particularly useful.
- Exposure and response prevention (ERP).
- Exposure to feared situations is given. The patient is asked not to engage in a compulsive response.

Somatic Therapies:

- Anterior cingulotomy and capsulotomy.
- Deep brain stimulation with implantation of electrodes.

Solution for Question 7:

Answer: B) Body Dysmorphic Disorder

Explanation:

Body Dysmorphic Disorder (BDD) involves an obsessive preoccupation with perceived physical flaws that are unnoticeable or minor to others. In this case, the patient's fixation on her nose, frequent mirror-checking, and social avoidance are key signs of BDD. Many patients seek plastic surgery.

Body dysmorphic disorder:

Definition	Preoccupation with 1 or more perceived defects or flaws in physical appearance
Features	Compare themselves with others Checking the mirror frequently/ avoiding it all together Trying to hide perceived flaws Ideas or delusions of reference Social avoidance May seek plastic surgery Differential diagnosis- Delusional dysmorphophobia: fixed belief without insight that they are "disfigured" (Option C ruled out)
Treatment	SSRI CBT

Social Anxiety Disorder (Option A): This involves intense fear of social situations due to fear of negative evaluation but without the specific focus on perceived physical flaws.

Generalized Anxiety Disorder (GAD) (Option D): GAD involves excessive worry about various aspects of life, but it is not focused on physical appearance or perceived flaws.

Solution for Question 8:

Answer: A) Cognitive Behavioral Therapy (CBT)

Explanation:

Cognitive Behavioral Therapy (CBT) is considered the most effective treatment for Hoarding Disorder.

Hoarding disorders:

Definition	Persistent and profound difficulty in discarding their possessions, which leads to congestion, clutter, and impairment of normal function
Features	Inability to organize possessions and keep them organized Distorted belief about or an emotional attachment to possessions Leads to poor sanitation It is a safety risk Disturbs sleeping, eating, and other activities of daily living
Treatment	Challenging to treat CBT (Most effective) (Option A) SSRIs Poor response to exposure and response prevention

Antipsychotic medications (Option B) are not typically used as a primary treatment for hoarding disorder.

Exposure therapy (Option C) is a component of CBT that involves gradual exposure to feared situations (like discarding items), but it is not a standalone treatment for hoarding.

Psychoanalysis (Option D) focuses on uncovering unconscious conflicts and is generally not effective for treating hoarding disorder.

Solution for Question 9:

Answer: B) Cognitive Behavioral Therapy (CBT) with habit reversal training

Explanation:

Cognitive Behavioral Therapy (CBT) combined with habit reversal training is the most effective treatment for Trichotillomania (Hair-pulling Disorder) . Habit reversal training helps individuals become aware of their hair-pulling behaviors and learn alternative, healthier responses to the urges.

	Trichotillomania/Hair-pulling disorder	Excoriation/Skin picking disorder
Definition	Recurrent pulling of own hair, resulting in hair loss	Repeatedly pick their skin, enough to cause lesions

	Trichotillomania/Hair-pulling disorder	Excoriation/Skin picking disorder
Features	Hair pulling by hands, tweezers or other tools Hair from the scalp is the most common to be pulled but can be from anywhere Sense of tension builds and is relieved after pulling hair Some may eat the hair after plucking results in trichobezoar, intestinal obstruction, and malnutrition Punch biopsy can rule out alopecia areata, tinea capitis, traction alopecia	Sense of tension builds and is relieved after picking at their skin Unsuccessful attempts to stop Most common areas involved- Face, hands, fingers, arms, and legs Use makeup, clothing, or bandages to hide picking Associated with mood and anxiety disorders
Treatment	CBT and habit reversal training (1st line) (Option B) SSRIs (especially with comorbid conditions like depression) (Option D ruled out) N-acetylcysteine-may be used as a first line agent in adults. (Option A ruled out)	Habit Reversal Therapy SSRIs

Trichobezoar

- Definition : A trichobezoar is a mass of ingested hair that accumulates in the stomach, forming an indigestible lump.



Rapunzel Syndrome

- Definition : A rare variant of trichobezoar where the hair mass extends from the stomach into the small intestine or even the colon, resembling the long hair of the fairy-tale character Rapunzel.

Exposure Therapy (Option C): while useful for anxiety-related conditions, is not the primary treatment for Trichotillomania, which involves compulsive hair-pulling rather than a specific fear.

Solution for Question 10:

Answer: C) Olfactory Reference Syndrome

Explanation:

Olfactory Reference Syndrome (ORS) is characterized by a preoccupation with the belief that one has a foul or unpleasant odor that is noticeable to others, despite evidence to the contrary. In this scenario, the woman's persistent concern about having a body odor that her friends and family reassure her she does not have aligns with ORS.

Olfactory Reference Syndrome:

Definition	Preoccupation with the belief that one is emitting foul body odour or breath
Features	Associated with ideas or delusions of reference that others are noticing their bad odour May avoid social interaction as a result More common in males Compulsive or repetitive behavior such as washing clothes, or rechecking own breath, etc. Differential diagnosis: Those with temporal lobe epilepsy may have olfactory hallucinations
Treatment	SSRIs Antipsychotics Anxiolytics- to help reduce social anxiety Psychotherapy

Social Anxiety Disorder (Option A): While this patient exhibits social avoidance, social anxiety primarily involves fear of negative evaluation in social situations rather than a specific concern about body odor.

Body Dysmorphic Disorder (Option B): This disorder involves a preoccupation with perceived defects in physical appearance, but it does not specifically pertain to an odor and usually

involves more visual aspects.

Olfactory Hallucinations (Option D): This condition refers to experiencing smells that are not present in the environment. Individuals with olfactory hallucinations perceive an odor that others do not detect, but this is different from the fixed belief in ORS that one has an unpleasant odor.

Solution for Question 11:

Answer: D) Increased sleepiness

Explanation:

Insomnia is more commonly seen in PTSD rather than an increase in sleepiness.

Post-Traumatic Stress Disorder: stress disorder occurring after a traumatic event that may be exposure to actual or threatened death, serious injury, or sexual violence and symptoms occurs for a month or more.

Duration Criteria for PTSD Diagnosis (DSM-5)

To diagnose Post-Traumatic Stress Disorder (PTSD) according to the DSM-5 criteria, the duration of symptoms plays a critical role.

- Symptoms must persist for more than 1 month after exposure to a traumatic event.
- Symptoms typically begin within 3 months of the trauma but can have delayed onset , appearing months or even years later .
- The symptoms must cause significant distress or impairment in social, occupational, or other important areas of functioning.

Acute Stress Disorder (ASD)

- Acute Stress Disorder (ASD) is a mental health condition that can develop shortly after a traumatic event.
- It is characterized by intense fear, helplessness, or horror, similar to Post-Traumatic Stress Disorder (PTSD). Symptoms last from 3 days to 1 month .

Symptoms of Post-Traumatic Stress Disorder (PTSD)

- Mood (Option C ruled out) Negative feelings like fear, guilt, anger
- Negative feelings like fear, guilt, anger

- Avoidance (Option B ruled out) Avoiding people, places, or objects that relate to the traumatic event Social withdrawal
- Avoiding people, places, or objects that relate to the traumatic event
- Social withdrawal
- Hyperarousal (Option D) Hypervigilant Poor sleep Exaggerated startle response Poor concentration
- Hypervigilant
- Poor sleep
- Exaggerated startle response
- Poor concentration
- Intrusive symptoms Flashbacks (Option A ruled out) Dreams and memories Distressing re-experiencing of the traumatic event
- Flashbacks (Option A ruled out)
- Dreams and memories
- Distressing re-experiencing of the traumatic event
- Negative feelings like fear, guilt, anger
- Avoiding people, places, or objects that relate to the traumatic event
- Social withdrawal
- Hypervigilant
- Poor sleep
- Exaggerated startle response
- Poor concentration
- Flashbacks (Option A ruled out)
- Dreams and memories
- Distressing re-experiencing of the traumatic event

Solution for Question 12:

Answer: B) Cognitive Behavioral Therapy (CBT)

Explanation:

Cognitive Behavioral Therapy (CBT) is considered one of the most effective treatments for Post-Traumatic Stress Disorder (PTSD).

PTSD Treatment:

Pharmacotherapy	Psychotherapy
SSRI (most efficacious) drug SNRI (some initial evidence)	Trauma-focused cognitive-behavioral therapy inclusive of prolonged exposure (PE) therapy- requires reexperiencing of traumatic event Eye movement desensitization and reprocessing therapy (EMDR) Cognitive processing therapy (CPT) corrects overgeneralizations of seeing the world as dangerous Group therapy Family therapy

SSRIs (Selective Serotonin Reuptake Inhibitors) (Option A): are often used as a first-line pharmacological treatment for PTSD. They can help alleviate symptoms of depression and anxiety associated with PTSD, but they are not first line. They may be combined with CBT.

Eye Movement Desensitization and Reprocessing (EMDR) (Option C): is also an effective treatment for PTSD. It involves processing traumatic memories through guided eye movements, but it may not be considered the first choice.

Group Therapy (Option D): can provide support and help individuals share their experiences with others who have similar trauma backgrounds, but it is not the primary treatment of choice for PTSD compared to CBT or EMDR.

Solution for Question 13:

Answer: B) Complex Post-Traumatic Stress Disorder (C-PTSD)

Explanation:

- Complex Post-Traumatic Stress Disorder (C-PTSD) is a diagnosis used to describe the symptoms that arise from prolonged or repeated exposure to traumatic events, particularly in the context of relationships, such as childhood abuse or neglect or other repetitive events from which escape is difficult or impossible (e.g. torture, slavery, genocide campaigns, prolonged domestic violence)

- In this scenario, the man's history of ongoing trauma during childhood, along with his emotional regulation difficulties, feelings of hopelessness, and problems in relationships, aligns with the characteristics of C-PTSD.

Complex PTSD has been newly recognized in ICD-11 as a separate entity

- Occurs after repeated severe stressors such as torture, intimate partner abuse, sexual violence, domestic violence
- Diagnosed similar to PTSD
- Additional symptoms of: Poor emotional regulation Diminished self-belief difficulties in sustaining relationships and in feeling close to others
- Poor emotional regulation
- Diminished self-belief
- difficulties in sustaining relationships and in feeling close to others
- Significant social, family, educational, and occupational impairment
- Poor emotional regulation
- Diminished self-belief
- difficulties in sustaining relationships and in feeling close to others

Treatment:

- Trauma-focused cognitive behavioral therapy
- eye movement desensitization and reprocessing

Post-Traumatic Stress Disorder (PTSD) (Option A) : Would be more applicable if the individual had a single traumatic event without the additional complexities of emotional regulation and interpersonal issues commonly seen in C-PTSD.

Acute Stress Disorder (Option C) : This diagnosis is made when symptoms arise shortly after a traumatic event and last for at least three days but less than a month.

Generalized Anxiety Disorder (Option D) : This involves excessive worry about various aspects of life that are not directly related to a traumatic event.

Solution for Question 14:

Answer: C) Adjustment Disorder

Explanation:

Adjustment Disorder is characterized by the development of emotional or behavioral symptoms in response to a specific stressor, such as a significant life change, within three months of its onset. In this scenario, the woman's difficulties adapting to her new job and city, along with her feelings of anxiety, sadness, and social withdrawal, align with the symptoms of adjustment disorder.

Adjustment disorders:

Definition	Excessively distressing emotional or behavioral symptoms that occur in response to a stressor Stressors are mild to moderate such as: financial, medical, or relationship issues
Symptoms	Low mood and tearfulness Anxiety Aggression Symptoms lead to impairment and poor functioning Occur within 3 months of the stressor and usually resolve within 6 months of the resolution of the stressor Common in hospitalised patients
Differential s	Depression Generalized anxiety disorder Post-traumatic stress disorder
Treatment	Psychotherapy- group therapy, CBT Crisis intervention Pharmacotherapy with SSRIs and anxiolytics are not typically employed

Major Depressive Disorder (Option A): This diagnosis requires the presence of more pervasive and persistent symptoms, such as anhedonia (loss of interest in pleasurable activities) and significant impairment over a longer duration with/ without a clear external stressor.

Generalized Anxiety Disorder (Option B): This involves excessive worry about various aspects of life that is not necessarily linked to a specific stressor and lasts for at least six months.

Acute Stress Disorder (Option D): This is diagnosed when symptoms arise within three days to one month after a traumatic event, which does not apply to this scenario.

Previous Year Questions

1. A lady presents to the hospital after having a fight with her husband. She reports that she is unable to speak. However, all neurological and physical examinations are normal, and she does not seem concerned about her health status. What is the most likely diagnosis?

- A. Malingering
- B. La belle indifférence
- C. Adaptation
- D. Somatic symptom

2. A young girl presents with multiple episodes of loss of consciousness, each lasting about 20 minutes. These episodes occur exclusively in front of her family and only during the daytime. There is no history of tongue biting or incontinence associated with the episodes. Both EEG and MRI brain scans are normal. What is the appropriate management for this patient?

- A. Treat with aversive therapy
- B. Insight-oriented psychotherapy
- C. Valproate
- D. Ketogenic diet

3. A 40-year-old woman has consulted several plastic surgeons, seeking correction for a facial deformity that she perceives. Despite being examined and reassured by the surgeons and her family that no deformity is present, she remains convinced that her face requires surgery. She frequently checks her appearance and persists in requesting surgical intervention. What is the most appropriate management for

her condition?

- A. SSRI
- B. Atypical antipsychotic
- C. Insight therapy
- D. SNRI

4. A 45-year-old woman presents with a long-standing history of “feeling tensed” along with an upset stomach, diarrhea, and heartburn. She mentions that her family members also experience similar symptoms. Which of the following symptoms is most likely to be observed in this patient?

- A. Neologisms
- B. Tingling at extremities
- C. Hallucinations
- D. Idea of reference

5. Match the following: A. Depression 1. The feeling goes away as soon as you leave the hospital B. Suicide 2. State of emotional and physical exhaustion caused by prolonged stress, relieved after relaxation C. Burnout 3. Difficult to diagnose due to the shell of physician's pride D. Overworking 4. Women physicians, substance abuse, relationship troubles

A. Depression	1. The feeling goes away as soon as you leave the hospital
B. Suicide	2. State of emotional and physical exhaustion caused by prolonged stress, relieved after relaxation
C. Burnout	3. Difficult to diagnose due to the shell of physician's pride
D. Overworking	4. Women physicians, substance abuse, relationship troubles

- A. A-3, B-4, C-1, D-2
 - B. A-4, B-3, C-1, D-2
 - C. A-3, B-4, C-2, D-1
 - D. A-2, B-1, C-3, D-4
-

6. What is the medical term for a person who has a fear of seeing tall buildings and looking down from heights?

- A. Agoraphobia
 - B. Acrophobia
 - C. Claustrophobia
 - D. Nomophobia
-

7. All are stages of grief or loss, except:

- A. Denial
 - B. Anger
 - C. Bargaining
 - D. Agitation
-

8. What is the likely diagnosis for a 39-year-old man who frequently washes his hands (15-20 times a day), leading to a negative impact on his work performance? He attributes this behavior to persistent thoughts that his hands are contaminated with dirt.

- A. Bipolar disorder

- B. Obsessive compulsive disorder
 - C. Schizophrenia
 - D. Mania
-

9. In which condition is La belle indifference observed?

- A. Conversion disorder
 - B. Panic disorder
 - C. Schizophrenia
 - D. Depression
-

10. A patient believes his feelings and thoughts are controlled/influenced by an external agency?

- A. Delusion of nihilism
 - B. Delusion of reference
 - C. Othello syndrome
 - D. Passivity
-

11. A patient arrived at the hospital, claiming to be experiencing a cardiac arrest. After a comprehensive medical examination, it was determined that the patient was in normal health. However, he is refusing to leave the hospital and return home. He mentioned that he had been a victim of a robbery where he was held at gunpoint. What is the likely diagnosis in this case?

- A. Acute stress disorder
- B. Schizophrenia

- C. Avoidant personality disorder
 - D. Delirium
-

12. What is the diagnosis of a 40-year-old man who has been experiencing excessive and uncontrollable worry about daily life activities, finances, and his health for the past two years? He also presents with restlessness, fatigue, disturbed sleep, and waking up once due to a fear that his family was involved in an accident.

- A. Depression
 - B. Illness anxiety disorder
 - C. Somatoform disorder
 - D. Generalized anxiety disorder
-

13. A concerned grandmother brings an 18-year-old boy to you, reporting that he had multiple episodes of body jerking the day before his final exam. There were no signs of confusion or tongue biting after the episodes, and his medical tests came back normal. Which of the following conditions should be taken into consideration?

- A. Malingering
 - B. Factitious disorder
 - C. Somatisation syndrome
 - D. Hypochondriasis
-

14. Among the following, what is the most common symptom of adult obsessive-compulsive disorder?

- A. Aggressive symptoms

- B. Need for symmetry
 - C. Pathological doubt
 - D. Sexual symptoms
-

15. What could be the likely diagnosis for a male patient who recently lost his job and is experiencing persistent irritability, sadness, and worry about the future? He also displays irritability towards his family members. Although he is able to temporarily enjoy watching movies with friends, upon returning home, he experiences the same symptoms.

- A. Generalised anxiety disorder
 - B. Adjustment disorder
 - C. Mixed anxiety with depression
 - D. Moderate depression
-

16. In the psychiatry OPD, a male patient aged 40 presents with a concern regarding persistent thoughts. Specifically, he experiences a constant belief that his hands are dirty, despite them being clean. Consequently, he feels uneasy and is compelled to repeatedly wash his hands. The condition can be classified as a disorder involving thought processes known as _____.

- A. Flow
 - B. Form
 - C. Possession
 - D. Content
-

17. What is the probable diagnosis for a 25-year-old woman who presents with symptoms such as anxiety, palpitations, sweating, breathlessness, chest pain, and a sense of impending doom? The patient mentions having encountered approximately 5-6 comparable episodes in the last half-year, with each episode persisting for around 20-30 minutes.

- A. Depression
- B. Panic disorder
- C. Generalized anxiety disorder
- D. Phobia

18. What is the probable diagnosis for a 24-year-old woman who experienced sudden chest pain and palpitations lasting approximately 20 minutes, with a history of three similar episodes in the past, despite normal investigation results?

- A. Acute psychosis
- B. Panic attack
- C. Post-traumatic stress disorder
- D. Mania

19. Which of the subsequent options is not classified as an obsessive-compulsive disorder (OCD) related ailment?

- A. Hair pulling disorder
- B. Temper tantrums
- C. Hoarding disorder
- D. Skin picking

20. How can we describe the emotional release and discharge that occurs when a person consciously relieves a repressed painful experience?

- A. Catharsis
 - B. Abreaction
 - C. Venting out
 - D. Guided relaxation
-

21. Which of the following conditions does the 40-year-old woman, who was involved in a car accident 2 months ago, experience when she wakes up screaming at night due to recurring nightmares of the same incident?

- A. Acute stress reaction
 - B. Adjustment disorder
 - C. Mania
 - D. Post- traumatic stress disorder
-

22. Which of the following symptoms is least likely to be associated with a panic attack?

- A. Palpitations
 - B. Fear of dying
 - C. Chest discomfort
 - D. Suicide attempt
-

23. What is the fear of heights known as?

- A. Acarophobia

- B. Algophobia
 - C. Agoraphobia
 - D. Acrophobia
-

24. According to ICD 11, which of the options below is not categorized as Obsessive-Compulsive Disorder (OCD)?

- A. Trichotillomania
 - B. Body dysmorphic disorder
 - C. Hypochondriac disorder
 - D. PTSD
-

25. What is the most appropriate treatment for a patient who frequently experiences a sense of contamination in their hands and compulsively washes them, leading to distress and impaired performance?

- A. DBT
 - B. Exposure and response prevention
 - C. Systematic desensitization
 - D. CBT
-

26. Which of the following statements accurately describes obsessive-compulsive disorder (OCD)?

- A. Atypical antipsychotics are the first line of treatment
- B. Prevalence is 7-10 percent in the general population
- C. Depression is a common comorbidity

D. Contamination is an uncommon obsession

27. What is the probable diagnosis for a 20-year-old female who exhibits abrupt palpitations, sweating, breathlessness, and a sensation of being choked? She had contracted COVID-19 a year ago and is worried about her life due to her prior positive status. All physical examinations, ECG, and laboratory tests have yielded no significant findings. This episode, lasting approximately 20 minutes, is the first occurrence of its kind.

- A. Post-traumatic stress disorder
 - B. Bipolar disorder
 - C. Depression
 - D. Panic attack
-

28. What is the most probable diagnosis for a first-year MBBS student who presents at the OPD with intermittent episodes of intense sweating and a sensation of imminent danger, commonly experienced before her examinations?

- A. Panic disorder
 - B. Conversion disorder
 - C. Social anxiety disorder
 - D. Generalized anxiety disorder
-

29. What is the probable medical condition for a 25-year-old male who experienced a road traffic accident and is now experiencing anxiety about driving and going to the site of the incident for a period of 6 weeks? He is also having recurring memories of the accident and struggles with attending work.

- A. Acute stress disorder

- B. Post-traumatic stress disorder
 - C. Panic disorder
 - D. Major depressive disorder
-

30. What is the correct statement regarding the condition in a 14-year-old girl who has experienced sudden-onset blindness for the past 4 hours, but does not express concern about it, yet expresses regret over not spending enough time with her recently deceased mother?

- A. In children, equally among males and females
 - B. In adults, equally among males and females
 - C. In adults, it occurs more in males than in females
 - D. In children, it occurs more in females than in males
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	1
Question 4	2
Question 5	3
Question 6	2
Question 7	4
Question 8	2
Question 9	1
Question 10	4
Question 11	1
Question 12	4
Question 13	1
Question 14	3
Question 15	2
Question 16	3
Question 17	2
Question 18	2

Question 19	2
Question 20	2
Question 21	4
Question 22	4
Question 23	4
Question 24	4
Question 25	2
Question 26	3
Question 27	4
Question 28	1
Question 29	2
Question 30	4

Solution for Question 1:

Explanation:

The patient presents with an inability to speak (mutism) after an emotional conflict with her husband. Despite this, neurological and physical examinations are normal, and she does not seem concerned about her symptoms. This lack of distress about a serious symptom is known as "La belle indifférence", which is often seen in conversion disorder (functional neurological symptom disorder). It reflects an apparent indifference to significant neurological symptoms like paralysis, blindness, or mutism that have no medical cause.

Other Options:

Malingering (Option A): In malingering, the patient intentionally fakes or exaggerates symptoms for external gain, such as financial compensation or avoiding responsibilities. In this case, the

patient does not appear to have an external motive.

Adaptation (Option C): Adaptation refers to a normal psychological adjustment to stress or changes in life. The patient's inability to speak suggests a conversion disorder, not a normal adaptive response.

Somatic symptom disorder (Option D): This condition involves excessive distress and anxiety about physical symptoms, often leading to frequent medical visits. However, the patient in this case shows no concern about her condition, which is characteristic of La belle indifférence.

Solution for Question 2:

Correct Option B - Insight-oriented psychotherapy:

- The clinical scenario described suggests that the patient is experiencing pseudoseizures, which are often a manifestation of a conversion disorder. Insight-oriented psychotherapy is the treatment of choice for conversion disorder.

Incorrect Options:

Options A, C, and D are not preferred in the management of conversion disorder.

Solution for Question 3:

Correct Option A – SSRI:

- The woman in the scenario is likely suffering from Body Dysmorphic Disorder (BDD). Selective Serotonin Reuptake Inhibitors are the first-line pharmacological treatment for BDD.

Incorrect Options:

Options B, C, and D are not the first-line management of body dysmorphic disorder.

Solution for Question 4:

Correct Option B - Tingling at extremities:

- The patient's symptoms and her family members experiencing similar symptoms suggest a diagnosis of Generalized Anxiety Disorder (GAD). Tingling in the extremities is a physical symptom associated with GAD.

Incorrect Options:

Options A, C, and D are not associated with generalized anxiety disorder.

Solution for Question 5:

Correct Option C: A-3, B-4, C-2, D-1:

A. Depression	3. Difficult to diagnose due to the shell of physician's pride
B. Suicide	4. Women physicians, substance abuse, relationship troubles
C. Burnout	2. State of emotional and physical exhaustion caused by prolonged stress, relieved after relaxation
D. Overworking	1. The feeling goes away as soon as you leave the hospital

Incorrect Options: Options A, B, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 6:

Correct Option B - Acrophobia:

Some Common Phobias	
Acrophobia	Fear of height
Ailurophobia	Cats
Cynophobia	Dogs
Claustrophobia	Closed spaces
Mysophobia	Dirt and Germs
Hydrophobia	Water
Thanatophobia	Death
Nyctophobia	Dark
Xenophobia	Strangers
Pyrophobia	Fire

Incorrect Options:

Option A - Agoraphobia: Fear of entering open or crowded places or being in places from which it is difficult to escape

Option C - Claustrophobia: Its fear of closed spaces

Option D - Nomophobia: Fear or anxiety about not having mobile phone

Solution for Question 7:

Correct Option D - Agitation:

- Agitation - Agitation is not considered one of the recognized stages of death or grief.

Incorrect Options:

Option A - Denial: The initial stage where individuals may refuse to believe or accept the reality of the situation.

Option B - Anger: The stage where individuals may experience feelings of anger and frustration.

Option C - Bargaining: The stage where individuals may attempt to negotiate or make deals to change the outcome.

Solution for Question 8:

Correct Option B - Obsessive compulsive disorder:

- The most probable diagnosis in this case is obsessive-compulsive disorder (OCD). OCD is a mental health disorder characterized by repetitive, intrusive thoughts (obsessions) and/or repetitive behaviors (compulsions) that are difficult to control and interfere with daily life.
- In this case, the patient's repetitive hand washing behavior is a compulsive behavior that is likely driven by obsessive thoughts about contamination. This behavior is interfering with his workplace performance and is causing distress.
- Therefore, the correct choice is a, obsessive-compulsive disorder.

Incorrect Options:

Option A - Bipolar disorder: Bipolar disorder is a mental health disorder characterized by episodes of mania and depression. It is not typically associated with the specific symptoms described by the patient.

Option C - Schizophrenia: Schizophrenia is a mental health disorder characterized by a range of symptoms, including delusions, hallucinations, disorganized speech, and disorganized behavior. It is not typically associated with the specific symptoms described by the patient.

Option D - Mania: Mania is a symptom of bipolar disorder and is characterized by elevated or irritable mood, increased energy, decreased need for sleep, and other symptoms. It is not typically associated with the specific symptoms described by the patient.

Solution for Question 9:

Correct Option A - Conversion disorder:

- La belle indifference: Absence of psychological distress despite serious medical illness or symptoms.
- Most commonly associated with conversion disorder (CD) or functional neurologic symptom disorder (FNSD) in DSM 5.
- CD characterized by neurological deficits without medical or neurological etiology, incompatible with known disorders.
- Psychological distress can manifest as physical symptoms in CD patients.
- La belle indifference not formally diagnosed in DSM 5, used in conjunction with CD.

Incorrect Options:

Option B - Panic disorder:

- Panic disorder is characterized by recurrent, unexpected panic attacks, which are sudden episodes of intense fear or discomfort accompanied by physical symptoms such as palpitations, sweating, trembling, and feelings of impending doom or losing control.
- Individuals with panic disorder typically experience significant distress and anxiety during panic attacks, which is contrary to the indifference seen in la belle indifference.

Option C - Schizophrenia:

- Schizophrenia is a chronic mental disorder that involves a range of symptoms, including delusions, hallucinations, disorganized thinking, and emotional dysregulation. Individuals with schizophrenia usually experience distress and impairment due to their symptoms and are unlikely to exhibit the characteristic indifference associated with la belle indifference.

Option D - Depression:

- Depression is a mood disorder characterized by persistent feelings of sadness, loss of interest or pleasure, and other symptoms such as changes in appetite, sleep disturbances, and thoughts of death or suicide. Individuals with depression typically experience emotional distress and are unlikely to display the indifference associated with la belle indifference.

Solution for Question 10:

Correct Option D- Passivity:

- It is also known as Thought alienation or passivity phenomenon
- Patient claims that thoughts are under outside influence.

- It is one of the hallmark symptom of Schizophrenia.
- Passivity phenomenon includes Thought insertion Thought withdrawal Thought broadcast
- Thought insertion
- Thought withdrawal
- Thought broadcast
- Thought insertion
- Thought withdrawal
- Thought broadcast

Incorrect Options:

Option A- Delusion of nihilism:

- Delusion of negation
- Patient denies the existence of his own body, loved ones, the world, and everything
- Seen in disorders such as schizophrenia, depression

Option B- Delusion of reference:

- Patient believes that someone is talking about him or spying on him.

Option C- Othello syndrome:

- Patient believes that the partner or spouse is having an affair even though that is not true.
- Prolonged alcohol use is linked to this condition.

Solution for Question 11:

Correct Option A- Acute stress disorder:

- The question describes a stressful event a few days ago, which is the main basis for the diagnosis of this patient
- Exposure to actual or threatened death, serious injury, or sexual violence
- Symptoms are similar to PTSD (like mood and cognitive symptoms, avoidance, hyperarousal, and intrusion symptoms), and occur at an interval of 3 days to 1 month of the attack/threat.
- Treatment: Cognitive behavioral therapy and SSRIs.

Incorrect Options:

Option B- Schizophrenia:

- One of the most common severe mental illness with auditory hallucinations, thought alienation phenomenon, somatic passivity, delusional perception.
- These symptoms are not present in the given case.

Option C- Avoidant personality disorder:

- Extremely sensitive to rejection
- Fear of criticism, disapproval , or rejection
- Unwilling to get involved with people unless certain of being liked by them
- View oneself as socially inferior.
- These above mentioned symptoms or descriptions are not seen in the patient mentioned in the question.

Option D- Delirium:

- Most common organic mental disorder
- Impairment of consciousness is depicted by clouding of consciousness, confused state, altered sensorium and reduced orientation.
- Impairment of attention is depicted by reduced ability to focus, sustaining/ shifting attention.

Solution for Question 12:

Correct Option D - Generalized anxiety disorder:

- A sign of generalized anxiety disorder may be excessive, persistent anxiety and worry that are difficult to control and disrupt daily activities. It is possible for a child or an adult to develop generalized anxiety disorder.
- Generalized anxiety has side effects like frenzy issues, fanatical impulsive problems, and different sorts of nervousness. It frequently coexists with other mood or anxiety disorders. Psychotherapy or medication typically improves generalized anxiety disorder.

Incorrect Options:

Option A - Depression: Depression is a mental illness that makes people feel worthless, hopeless, and sad all the time. Additionally, it may alter sleep patterns, appetite, energy levels,

and concentration. The man in the situation makes reference to no side effects of gloom, likepity, sadness, or uselessness.

Option B - Illness anxiety disorder: Illness Anxiety Disorder is a sort of uneasiness problem where individuals have a steady feeling of dread toward fostering a difficult sickness. Even when there is no sign of illness, they may still be concerned about their health. The man in the scenario doesn't talk about any of the symptoms of anxiety disorder, like the constant fear of getting a serious illness.

Option C - Somatoform disorder: Somatoform Disorder is a mental condition in which people suffer from physical symptoms brought on by psychological factors like stress or anxiety. These side effects can be genuine and incapacitating, yet they are not brought about by an actual disease. The man in the scenario doesn't mention any signs of somatoform disorder, like physical signs that are caused by mental factors.

Solution for Question 13:

Correct Option A - Malingering:

- Malingering is the falsification or profound exaggeration of illness (physical or mental) to gain external benefits such as avoiding work or responsibility, seeking drugs, avoiding trial (law), seeking attention, avoiding military services, leave from school, paid leave from a job, among others.
- It is not a psychiatric illness. Malingering is associated with an antisocial personality disorder and histrionic personality trait. The boy's symptoms, which only occurred the day before his final exam, suggest malingering. He is likely faking his symptoms to avoid taking the exam.

Incorrect Options:

Option B - Factitious disorder: A condition known as a factitious disorder is one in which a person deliberately causes or feigns symptoms in order to assume the sick role. Dissimilar to malingering, individuals with factitious turmoil have no outer inspiration for their way of behaving. They might do it for the consideration, the compassion, or the sensation of being really focused.

Option C - Somatization syndrome: People with somatization syndrome experience a wide range of physical symptoms that are unrelated to any medical condition. A person with somatic symptom disorder (SSD) experiences extreme, exaggerated anxiety regarding physical

symptoms. The person experiences symptoms-related thoughts, feelings, and behaviors that are so intense that they believe they are unable to perform some of their day-to-day activities.

Option D - Hypochondriasis: An illness anxiety disorder, also known as hypochondriasis or health anxiety, is excessive worry about getting or staying ill. There may be no physical symptoms. Or on the other hand, one might accept that typical body sensations or minor side effects are indications of extreme disease, although an intensive clinical test doesn't uncover a serious ailment. One might worry a lot that body sensations like muscle twitching or fatigue are related to a particularly serious illness. Rather than the physical symptom itself, excessive anxiety causes severe distress that can disrupt their life.

Solution for Question 14:

Correct Option C - Pathological doubt:

- Pathological doubt: Pathological doubt, or excessive doubt and uncertainty, is indeed a common symptom seen in adult OCD. Individuals with OCD may experience persistent and distressing doubts, even when there is no rational basis for them. They may repeatedly seek reassurance or engage in checking behaviors to alleviate their doubts. Pathological doubt is a core feature of OCD and often contributes to the individual's ritualistic behaviors.

Incorrect Options:

Option A - Aggressive symptoms: Aggressive symptoms can be present in some cases of obsessive-compulsive disorder (OCD), but they are not the most common symptom. Aggressive obsessions, such as intrusive thoughts or images of harm or violence, coupled with compulsions aimed at reducing these thoughts, can occur in individuals with OCD. However, not all individuals with OCD experience aggressive symptoms, and they are generally not the most prevalent symptom.

Option B - Need for symmetry: The need for symmetry, or a strong desire for things to be arranged or organized in a particular way, is a common symptom seen in OCD. Individuals may feel compelled to arrange objects symmetrically, follow strict routines, or repeatedly check and recheck things to ensure symmetry or order. While it is a common symptom, it is not the most prevalent symptom in adult OCD.

Option D - Sexual symptoms: While sexual obsessions or intrusive thoughts related to sexual content can occur in OCD, they are generally not the most common symptom. Sexual obsessions can lead to distress and discomfort, causing individuals to engage in repetitive

behaviors or mental rituals to neutralize or suppress these thoughts. However, it's important to note that the most common symptoms of adult OCD typically involve doubt, intrusive thoughts, and repetitive behaviors rather than primarily focusing on sexual content.

Solution for Question 15:

Correct Option B - Adjustment disorder:

- Adjustment disorder. Adjustment disorder is characterized by the development of emotional or behavioral symptoms in response to a stressor within three months of the onset of the stressor. The symptoms are disproportionate to the stressor and significantly impair the individual's functioning. In this case, the patient's recent job loss has triggered feelings of irritability, sadness, worry about the future, and interpersonal conflicts, which are consistent with adjustment disorder.

Incorrect Options:

Option A - Generalized anxiety disorder: Generalized anxiety disorder is characterized by excessive and uncontrollable worry and anxiety about various aspects of life. While the patient in this case does experience worry, the symptoms are more directly related to the specific stressor (job loss) and do not meet the criteria for generalized anxiety disorder.

Option C - Mixed anxiety with depression: Mixed anxiety with depression refers to the co-occurrence of symptoms of both anxiety and depression. While the patient may experience some overlapping symptoms, the primary diagnosis in this case is adjustment disorder due to the clear temporal relationship between the stressor (job loss) and the onset of symptoms.

Option D - Moderate depression: While the patient may exhibit some depressive symptoms, the severity of the symptoms is not specified in the question stem. Furthermore, the symptoms are more consistent with adjustment disorder, which is a distinct diagnostic category related to a specific stressor.

Solution for Question 16:

Correct Option C - Possession:

- Possession. The described disorder of repetitive thoughts about having dirty hands and the compulsive need to wash them is characteristic of obsessive-compulsive disorder (OCD). OCD is characterized by the presence of obsessions (intrusive and unwanted thoughts) and compulsions (repetitive behaviors or mental acts performed in response to obsessions). In this case, the patient's obsessive thought is related to possession or contamination, leading to the compulsive behavior of handwashing.

Incorrect Options:

Option A - Flow: "Flow" is not a term or concept related to disorders of thought.

Option B - Form: "Form" is not a term or concept related to disorders of thought.

Option D - Content: Content refers to the specific themes or topics of a person's thoughts, but it does not capture the essence of the described disorder in this question.

Solution for Question 17:

Correct Option B - Panic disorder:

- Panic disorder is refined as recurrent and unexpected panic episodes, which are sudden surges of great fear or discomfort that reach a peak within minutes.
- This patient exhibits several symptoms that can occur during an attack, including palpitations, sweating, shaking, shortness of breath, chest pain, nausea, and fear of passing out or losing control.

Incorrect Options:

Option A - Depression: Depression is characterized by persistent feelings of sadness, hopelessness, and a lack of interest or pleasure in activities. The patient does not show the symptoms associated with depression.

Option C - Generalized anxiety disorder: Generalized anxiety disorder involves excessive and uncontrollable and unrealistic worry about various domains of life (e.g., work, health, family) occurring more days than not for at least 6 months. This patient's symptoms are not consistent with generalized anxiety disorder as they occur in distinct episodes.

Option D - Phobia: Phobia refers to an excessive and irrational fear reaction of a person, object, animal, activity, or situation. If the patient had a phobia, we would expect her symptoms to be

triggered by a specific object or situation, but that's not the case here.

Solution for Question 18:

Correct Option B - Panic attack:

- The patient's sudden onset of chest pain and palpitations, along with previous similar episodes and normal investigations, are suggestive of a panic attack. Panic attacks are characterized by intense feelings of fear or discomfort and can present with symptoms such as chest pain, palpitations, shortness of breath, and a sense of impending doom. Other medical conditions should be ruled out, but in this case, since all investigations were normal, panic attack is the likely diagnosis.

Incorrect Options:

Option A - Acute psychosis: Acute psychosis refers to a severe mental condition characterized by a loss of contact with reality. The symptoms described in the scenario are not consistent with acute psychosis.

Option C - Post-traumatic stress disorder: Post-traumatic stress disorder (PTSD) is a mental health condition that occurs after experiencing or witnessing a traumatic event. While anxiety and panic symptoms can be present in PTSD, the sudden onset of symptoms and the absence of a clear traumatic event make it less likely in this case.

Option D - Mania: Mania is a symptom of bipolar disorder and is characterized by an elevated mood, increased energy, and impulsive behavior. The symptoms described in the scenario are not consistent with mania.

Solution for Question 19:

Correct Option B - Temper tantrums:

- Temper tantrums are intense and often disruptive outbursts of anger or frustration, typically seen in children.

- Temper tantrums are not considered an OCD-related disorder.
- They are more associated with emotional regulation difficulties and are commonly seen in typically developing children as a part of their normal development.

Incorrect Options:

Option A - Hair pulling disorder (trichotillomania): Trichotillomania, also known as hair pulling disorder, is a disorder characterized by recurrent pulling out of one's hair, resulting in noticeable hair loss. It is considered an OCD-related disorder because it shares similarities with obsessive-compulsive disorder (OCD) in terms of repetitive behaviors and urges.

Option C - Hoarding disorder: Hoarding disorder is a condition characterized by persistent difficulty in parting with or discarding possessions, regardless of their value. It involves excessive accumulation of items, leading to clutter and significant distress or impairment in daily functioning. Hoarding disorder is considered an OCD-related disorder because it shares similarities with OCD in terms of compulsive behaviors and difficulties with decision-making and organization.

Option D - Skin picking (excoriation disorder): Excoriation disorder, also known as skin picking disorder, is a disorder characterized by recurrent picking at one's own skin, resulting in skin lesions. It is considered an OCD-related disorder because it involves repetitive behaviors similar to those seen in OCD. Individuals with excoriation disorder may engage in skin picking to relieve tension, experience a sense of gratification, or attempt to improve the appearance of their skin.

Solution for Question 20:

Correct Option B - Abreaction:

- The correct option for the emotional release and discharge after consciously relieving a painful experience that has been repressed is " Abreaction ."
- Abreaction is a therapeutic process that involves the re-experiencing or reliving of traumatic events under hypnosis or in a deeply relaxed state. It aims to facilitate the release and processing of suppressed emotions, memories, and trauma-related distress.

Incorrect Options:

Option A - Catharsis: Catharsis refers to the process of releasing or purging strong emotions, usually through an artistic or expressive outlet such as writing, painting, or engaging in physical activity. While catharsis involves emotional release, it does not specifically refer to reliving a

repressed painful experience.

Option C - Venting out: "Venting out" generally refers to expressing one's thoughts, feelings, or frustrations openly and often in a cathartic manner. While it involves emotional release, it does not specifically refer to reliving a repressed painful experience.

Option D - Guided relaxation: Guided relaxation is a technique used to induce a state of deep relaxation and reduce stress or anxiety. It typically involves following verbal instructions or audio recordings that guide individuals through relaxation exercises. While it can be beneficial for emotional well-being, it does not specifically involve the conscious reliving of a repressed painful experience.

Solution for Question 21:

Correct Option D - Post- traumatic stress disorder:

- The lady's experience of waking up in the night after nightmares of the car accident suggests the presence of PTSD.
- Post-traumatic stress disorder (PTSD): PTSD is a mental health condition that develops after experiencing or witnessing a traumatic event.
- Symptoms of PTSD include intrusive thoughts, nightmares, flashbacks, avoidance of reminders of the event, emotional distress, and hyperarousal.

Incorrect Options:

Option A - Acute stress reaction: Acute stress reaction refers to an immediate response to a traumatic event. It involves symptoms such as anxiety, dissociation, and emotional distress. However, these symptoms usually resolve within a short period, typically within a few days or weeks. In the given scenario, the symptoms have persisted for two months, suggesting a more chronic condition like PTSD.

Option B - Adjustment disorder: Adjustment disorder is a psychological reaction to a stressful life event, such as a major life change or trauma. It is characterized by emotional and behavioral symptoms that occur within three months of the stressful event. While the lady's symptoms may initially resemble adjustment disorder, the persistent and recurring nature of the nightmares and distress point more towards PTSD.

Option C - Mania: Mania is a symptom associated with bipolar disorder and is characterized by an elevated or irritable mood, increased energy, racing thoughts, and impulsive behavior. The

symptoms described in the scenario, such as nightmares and distress related to the car accident, are not consistent with mania.

Solution for Question 22:

Correct Option D - Suicide attempt:

- Patients with panic attack experience fear of dying but suicide attempt is less likely to be a symptom of panic attack.
- But few studies report that patients experience fear of dying but it could be an independent risk factor for subsequent suicide attempts, especially for those patients with depressive disorders.

Incorrect Options:

Option A,B & C are incorrect.

- Palpitations, fear of dying, and chest discomfort are experienced by patients with a panic attack along with sweating, tremors, and shortness of breath.
- Patients experiencing panic attacks will be having the fear of dying, they might be overwhelmed, and frustrated with the inability to control the symptoms of panic attacks. This might or less likely lead to suicide attempts.

Solution for Question 23:

Correct Option D - Acrophobia:

- Acrophobia is the fear of heights, which is a specific phobia characterized by an intense and irrational fear or anxiety response when exposed to heights or situations involving heights. People with acrophobia may experience symptoms such as dizziness, rapid heartbeat, shortness of breath, sweating, and a strong desire to avoid heights or high places.

Incorrect Options:

Option A - Acarophobia: Acarophobia refers to the fear of mites or small insects.

Option B - Algophobia : Algophobia is the fear of pain. It is not directly related to the fear of heights, as it specifically focuses on the fear of experiencing physical pain.

Option C - Agoraphobia : Agoraphobia is a fear or anxiety about being in situations or places where escape might be difficult or embarrassing, or where help may not be available in the event of a panic attack or other distressing symptoms. Agoraphobia is typically associated with fears of being in open spaces, crowded areas, using public transportation, or being outside the home alone.

Solution for Question 24:

Correct Option D - PTSD:

- Trichotillomania, body dysmorphic disorder, and hypochondriac disorders are classified as OCD as per ICD 11.
- Other disorders include olfactory reference disorder and hoarding disorder, also under this classification.
- PTSD is a separate entity and is not under OCD and is classified under trauma and stress-related disorder.
- It is marked by excessive stress and anxiety following exposure to a traumatic/ stressful event like an accident/ military combat/ assault/ being kidnapped/ sexual abuse.

Incorrect Options:

Option A - Trichotillomania: Trichotillomania is the irresistible and repeated urge to pull hair and the inability to stop. It is classified under obsessive-compulsive disorder as per ICD-11.

Option B - Body dysmorphic disorder: Body dysmorphic disorder is a preoccupation with an imagined defect/slight anomaly in physical appearance characterized by repetitive behavior like mirror checking and mental acts like comparing oneself with others. It is classified as an obsessive-compulsive disorder as per ICD-11.

Option C - Hypochondriac disorder: Hypochondriac disorder is a preoccupation with fear of a serious physical illness despite investigations and medical reassurance. It is classified as OCD as per ICD-11.

Solution for Question 25:

Correct Option B - Exposure and response prevention:

- The patient is exhibiting symptoms of obsessive-compulsive disorder (OCD). Contamination is the most common pattern observed in OCD.
- Patients wash their hands excessively and avoid leaving their homes due to the fear of germs. The emotional response to this condition is anxiety.
- The best treatment is exposure and response prevention by exposing the patient to thoughts that trigger their obsession (Contamination) and resisting the compulsive behavior (Handwashing), thereby hoping to reduce anxiety associated with this.

Incorrect Options:

Option A - DBT: Dialectical behavioral therapy is not considered the first management line in OCD. It is commonly used for individuals diagnosed with borderline personality disorder or mood disorders. The first line for the management of OCD is exposure and response prevention.

Option C - Systematic desensitization : Desensitization, thought-stopping, flooding, and aversive conditioning are also used to treat OCD. However, exposure and response prevention is the best way of managing this condition.

Option D - CBT : Exposure and response prevention is a kind of CBT that helps reduce anxiety and distress associated with repeated exposure to thoughts or objects triggering the condition.

Solution for Question 26:

Correct Option C - Depression is a common comorbidity:

- The only statement true in the following is that depression is the most common comorbidity seen in OCD. Prevalence is around 2-3% in the general population. Contamination is the most common pattern observed in OCD.

Incorrect Options:

Option A - Atypical antipsychotics are the first line of treatment: Typical psychotics are used to treat OCD. However, they are not the first-line treatment of OCD. The best method of treatment is exposure and response prevention.

Option B - Prevalence is 7-10% in the general population: The prevalence of OCD is 2-3% in the general population.

Option D - Contamination is an uncommon obsession: Contamination is the most common obsession, and pathological doubt is the second most common obsession.

Solution for Question 27:

Correct Option D - Panic attack:

- A panic attack is a sudden period of intense fear or apprehension that may last from minutes to hours. In this case, fear of death from COVID-19 infection triggered the panic attack. Patients can feel physical symptoms like choking, palpitations, chest discomfort, nausea or gut discomfort, sweating, etc.

Incorrect Options:

Option A - Post Traumatic Stress Disorder: Post Traumatic Stress Disorder follows a traumatic or stressful event. Symptoms include avoidance, negative mood, and hyperarousal which are present for more than a month.

Option B - Bipolar disorder: Bipolar disorder presents as either manic or depressive episodes. An elevated, expansive, or irritable mood is the hallmark of a manic episode. The depressive episodes are similar to those of depressive disorders.

Option C - Depression: Depression features include a stooped posture, decreased movement, and a downward averted gaze.

Solution for Question 28:

Correct Option A - Panic disorder:

- Panic disorder is a type of anxiety disorder characterized by recurrent and unexpected panic attacks. Panic attacks are sudden episodes of intense fear or discomfort that reach a peak within minutes. They are often accompanied by physical symptoms such as sweating, trembling, heart palpitations, shortness of breath, chest pain, and a feeling of impending doom.
- In the given scenario, the student experiences intermittent episodes of a feeling of impending doom accompanied by intense perspiration, particularly prior to exams. These symptoms are consistent with panic attacks, which commonly occur in situations that provoke anxiety or fear, such as exams or stressful events.

Incorrect Options:

Option B - Conversion disorder: Conversion disorder involves the presence of neurological symptoms that cannot be explained by any underlying medical condition. The symptoms are thought to be related to psychological stress or conflict. While anxiety and stress can contribute to various physical symptoms, the specific symptoms described in the scenario are more indicative of panic attacks, suggesting panic disorder rather than conversion disorder.

Option C - Social anxiety disorder: Social anxiety disorder is characterized by an intense fear of social situations, leading to avoidance or distress in social settings. The symptoms typically revolve around social interactions, while the described symptoms in the scenario are more consistent with panic attacks occurring in the context of exams.

Option D - Generalized anxiety disorder: Generalized anxiety disorder involves excessive worry and anxiety about various aspects of life, persisting for at least six months. While exam-related stress and worry may be present in generalized anxiety disorder, the intense physical symptoms of panic attacks described in the scenario are not specific to generalized anxiety disorder.

Solution for Question 29:

Correct Option B - Post-traumatic stress disorder:

- Based on the provided information, the likely diagnosis for the individual is Post-traumatic stress disorder (PTSD). PTSD is a mental health condition that can develop after experiencing or witnessing a traumatic event. The symptoms described, such as fear of driving, experiencing flashbacks of the accident, and difficulty going to work, are consistent

with the diagnostic criteria for PTSD. It is important for the individual to seek professional help for an accurate diagnosis and appropriate treatment.

Incorrect Options:

Option A - Acute Stress Disorder: Is characterised by symptoms that are similar to PTSD but last for 3 days to 1 month. The case is less likely to have ASD because the symptoms have lasted 6 weeks.

Option C - Panic disorder: Is distinguished by repeated panic attacks and concern about future attacks. While the individual may be anxious due to the accident, the symptoms stated do not match the essential characteristics of panic disorder.

Option D - Major Depressive Disorder: Involves symptoms like persistent sadness, loss of interest, and changes in appetite or sleep patterns. In the given case, the primary focus is on the trauma-related symptoms (flashbacks, avoidance, etc.) rather than the main symptoms of MDD.

Solution for Question 30:

Correct Option D - In children, it occurs more in females than in males:

- The case scenario suggests a conversion disorder. La-belle-indifference is the lack of concern towards the symptoms despite the apparent severity of the disability produced, which is observed in this case scenario. Two different types of disturbances are observed in conversion disorder; one is motor, and the other is sensory. In children, it occurs more often in females. However, it does not mean it never happens in males.

Incorrect Options:

Option A - In children, equally among males and females: There is a slight female preponderance with respect to conversion disorder occurring in children.

Option B - In adults, equally among males and females: It is more commonly seen among women than in men. Women usually outnumber men in such cases.

Option C - In adults, it occurs more in males than in females: In adults, conversion disorder shows a female preponderance.

Substance use and addictive disorders

1. A 30-year-old male patient presents to the emergency room after using opioids. He reports that shortly after taking the drug, he felt an intense euphoria, which was soon followed by a state of deep sedation. What is the term used to describe this state of alternating euphoria and sedation?

- A. Mainlining
- B. Skin popping
- C. Chasing the dragon
- D. Nodding off

2. A 22-year-old male is brought to the emergency department after collapsing at a nightclub. On examination, he is disoriented, has a temperature of 39.5°C, and exhibits dilated pupils, tachycardia, and muscle rigidity. His friends report that he consumed a tablet given to him at the club. Which of the following drugs is the most likely cause of his symptoms?

- A. Methylenedioxymethamphetamine
- B. Cannabis
- C. Psilocybin
- D. Cathinone

3. A 45-year-old male with a 20-year history of smoking presents to the outpatient clinic. He admits that he is aware of the health risks associated with smoking but feels that he is not ready to quit at this time. However, he has started reading articles about smoking cessation. According to the Transtheoretical Model of Change, which stage of behavior change is he most likely in?

- A. Precontemplation
 - B. Contemplation
 - C. Preparation
 - D. Action
-

4. A 35-year-old man reports increasing financial troubles due to frequent horse racing bets. He's attempted to stop multiple times but fails, often betting more than intended. His job performance is suffering, and he's borrowed money from family under false pretenses. Which of the following is the most likely diagnosis?

- A. Impulse control disorder
 - B. Obsessive-compulsive disorder
 - C. Gambling disorder
 - D. Bipolar disorder
-

5. A 45-year-old man presents with mood changes, increased anxiety, and difficulty sleeping, which have progressively worsened over the past week. He has a history of heavy smoking (30 cigarettes/day for 20 years) and quit smoking abruptly 8 days ago due to concerns about his health. Despite wanting to stay off cigarettes, he reports intense cravings, inability to focus, and feeling "on edge" all the time. He has attempted nicotine replacement therapy (NRT) in the past without success. Which of the following medications is most likely to help alleviate both his nicotine cravings and mood symptoms without causing significant weight gain?

- A. Varenicline
 - B. Bupropion
 - C. Nicotine patch
 - D. Nortriptyline
-

6. A 32-year-old male, with a history of frequent cocaine use for the past 6 years, presents with intense fatigue, excessive sleep, increased appetite, and feelings of anhedonia. He states that he stopped using cocaine abruptly 3 days ago. On further questioning, he reports that over the past few months, his cocaine use had escalated to daily use. He also mentions experiencing intense cravings for cocaine and irritability but denies hallucinations or delusions. On physical examination, the patient is lethargic, and his vital signs are stable. Which of the following is the most appropriate management strategy for this patient at this stage?

- A. Immediate initiation of dopamine agonist therapy
 - B. Long-term use of antipsychotic medications
 - C. Supportive care and monitoring
 - D. Administration of benzodiazepines to prevent withdrawal seizures
-

7. A 25-year-old man is brought by his friends after collapsing at a party. He had been using a substance throughout the night that reportedly gave him a surge of euphoria and energy, but soon after, he started experiencing palpitations, intense sweating, and chest pain. On examination, his heart rate is 140 bpm, and his blood pressure is 190/120 mm Hg. The patient appears agitated and confused. His pupils are dilated. An ECG shows signs of acute myocardial ischemia. Which of the following best explains the mechanism of action of the substance this patient likely consumed?

- A. Increased release of gamma-aminobutyric acid (GABA)
 - B. Inhibition of norepinephrine reuptake
 - C. Enhanced serotonin degradation
 - D. Stimulation of opioid receptors
-

8. A 22-year-old college student is brought to the emergency department by his friends after exhibiting bizarre behavior at a party. He appears disoriented, has dilated pupils, and reports seeing vivid colors and patterns. His vital signs show tachycardia and mild hypertension. Which of the following is the most likely cause of his symptoms?

- A. Phencyclidine (PCP) intoxication
 - B. Lysergic acid diethylamide (LSD) intoxication
 - C. Methamphetamine overdose
 - D. Cannabis-induced psychosis
-

9. A 28-year-old man is brought to the emergency department by police after exhibiting bizarre behavior in public. He appears disoriented, has nystagmus, and is expressing paranoid thoughts. His vital signs show hypertension and hyperthermia. Upon questioning, he admits to using a substance he calls "angel dust." Which of the following is the most likely diagnosis?

- A. Alcohol intoxication
 - B. Opioid overdose
 - C. Phencyclidine intoxication
 - D. Ketamine intoxication
-

10. A 23-year-old college student presents to the university health center with complaints of anxiety, paranoid thoughts, and difficulty concentrating. He reports daily cannabis use for the past year. Which of the following is the most likely diagnosis?

- A. Cannabis-induced psychotic disorder
- B. Cannabis-induced anxiety disorder
- C. Amotivational syndrome

D. Cannabis use disorder

11. A 28-year-old man presents to his primary care physician with complaints of irritability, anxiety, and difficulty sleeping. He reports having been a daily cannabis user for the past 5 years but stopped abruptly 3 days ago. Which of the following is the most appropriate first-line approach for managing this patient's symptoms?

- A. Prescribe benzodiazepines for anxiety and sleep
 - B. Start bupropion for mood stabilization
 - C. Provide supportive counseling and psychoeducation
 - D. Initiate medically assisted withdrawal with cannabinoid agonists
-

12. A 20-year-old college student is brought to the emergency department by his roommate. The patient appears disoriented and anxious, with conjunctival injection and tachycardia. His roommate reports that they were experimenting with various forms of cannabis earlier in the evening. Which of the following statements about cannabis use and intoxication is correct?

- A. Oral ingestion of cannabis typically leads to a shorter duration of effects compared to inhalation
 - B. Cannabidiol (CBD) is the primary psychoactive component responsible for the "high" associated with cannabis use
 - C. Cannabinoid receptors are Gq subtype of G-protein coupled receptors
 - D. Tachycardia and conjunctival redness are common features of cannabis intoxication
-

13. A 35-year-old male with opioid use disorder presents with severe muscle aches, diarrhea, abdominal cramps, restlessness, and chills after stopping heroin "cold turkey." He has piloerection, dilated pupils, tachycardia, and mild hypertension. He

refuses opioid for managing the withdrawals. What is the most appropriate initial pharmacological treatment for his opioid withdrawal symptoms?

- A. Methadone
 - B. Buprenorphine + Naloxone combination therapy
 - C. Clonidine
 - D. Naloxone
-

14. A 30-year-old male is found unresponsive on the floor of his apartment by a friend. Upon arrival, emergency medical personnel observe the following: the patient has a weak pulse, slow and shallow respirations, and pinpoint pupils. His skin is cold and clammy, and there are needle track marks on his arms. His blood pressure is significantly low. What is the most appropriate initial treatment for this patient?

- A. Administer intravenous naloxone at a rapid rate
 - B. Ensure an adequate airway and mechanically ventilate the patient
 - C. Give intravenous fluids to manage hypotension
 - D. Perform a CT scan to rule out intracranial hemorrhage
-

15. A 28-year-old woman is brought by her family. They report that over the past year, she has been using increasing amounts of opioids, initially prescribed for chronic back pain. The patient admits to using opioids in larger amounts than intended and has made several unsuccessful attempts to cut down. She has given up her hobbies and often misses work due to opioid use. Recently, she was arrested for stealing prescription pads from her doctor's office. Despite these issues, she continues to use opioids daily. Which of the following statements BEST characterizes this patient's condition based on the information provided?

- A. The patient meets criteria for opioid dependence, with evidence of tolerance and withdrawal

- B. The patient exhibits opioid abuse but does not meet the full criteria for dependence due to the absence of physiological symptoms
 - C. The patient meets the criteria for opioid dependence, with prominent features of impaired control and social impairment
 - D. The patient's behavior suggests opioid misuse rather than dependence, as the initial use was medically sanctioned
-

16. A 45-year-old male patient with a history of alcohol use disorder presents for treatment. He has recently been diagnosed with compensated liver failure. The patient expresses a strong desire to quit drinking and is motivated to maintain abstinence. His medical history includes no liver failure, and he is currently not consuming alcohol. Which pharmacological treatment is the most appropriate for this patient to help manage his alcohol cravings?

- A. Disulfiram
 - B. Acamprosate
 - C. Naltrexone
 - D. Ondansetron
-

17. A 45-year-old man with a history of alcohol use disorder presents after abrupt cessation of drinking. He shows anxiety, tremors, and experiences a seizure. Which medication would be most appropriate for managing his withdrawal symptoms?

- A. Haloperidol
 - B. Carbamazepine
 - C. Thiamine
 - D. Lorazepam
-

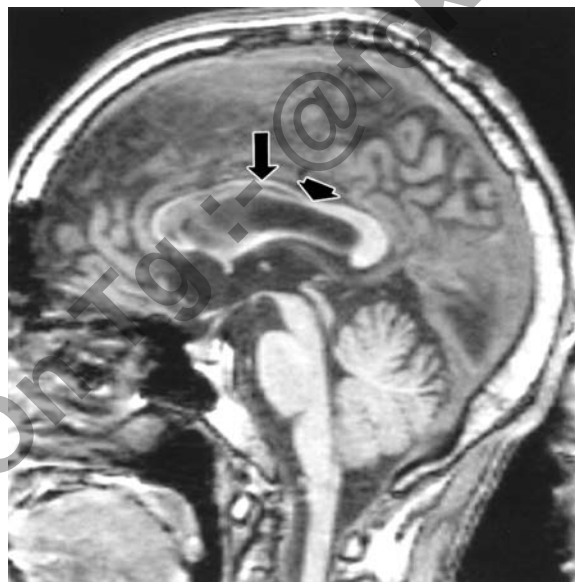
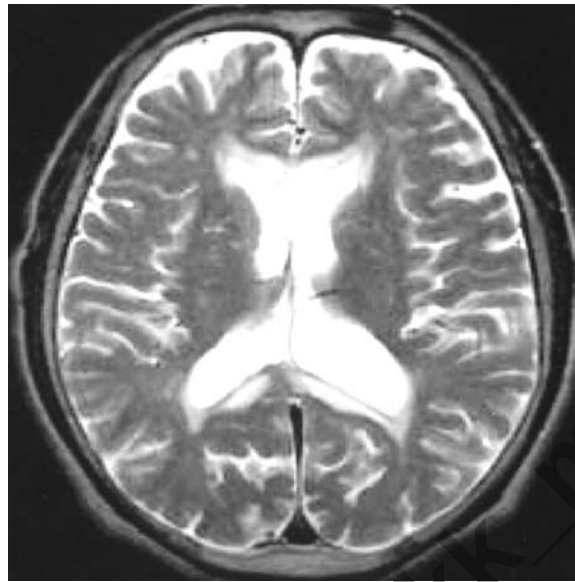
18. A 50-year-old man comes to the clinic for a routine evaluation after his friends expressed concern about his drinking habits. He reports consuming alcohol heavily for several years. The physician orders laboratory tests to assess his alcohol intake. Which test result is most indicative of his heavy alcohol consumption?

- A. Mean corpuscular volume
 - B. Alanine aminotransferase
 - C. γ -glutamyltransferase
 - D. Carbohydrate-deficient transferrin
-

19. A 45-year-old man visits a sleep specialist due to persistent insomnia. He mentions that he often drinks alcohol in the evenings, believing it helps him sleep better. After discussing his sleep patterns, the specialist explains that alcohol has several effects on sleep quality and architecture. Which of the following is not an effect of alcohol on sleep?

- A. Decreases sleep latency
 - B. Increases REM sleep
 - C. Increases sleep fragmentation
 - D. Decreased deep sleep due to its sedative properties
-

20. A 55-year-old man with a history of chronic alcohol use presents to the emergency with sudden onset loss of consciousness and seizures. He also shows signs of confusion, apathy, dementia, visual hallucinations and aggressive behavior. An MRI is performed, revealing the findings shown below. What is the most likely diagnosis?



- A. Alcohol-Induced Dementia
- B. Wernicke's Encephalopathy
- C. Marchiafava-Bignami Syndrome
- D. Alexander Disease

21. A 45-year-old man with a history of chronic alcohol use is referred to a psychiatrist after he exhibits significant memory impairment, particularly anterograde amnesia, and frequently confabulates to fill memory gaps. After initial stabilization, what is the most appropriate next step in the management of this patient?

- A. Increase thiamine dosage to 1g daily
- B. Initiate treatment with Niacin
- C. Initiate thiamine supplementation at 100 mg, 2-3 times/day
- D. Start cognitive rehabilitation therapy

22. A 45-year-old man with a history of chronic alcoholism is brought to the emergency department by his family. They report that he has been confused and unsteady on his feet for the past two days. On examination, the patient is disoriented and has difficulty maintaining his balance. You also notice bilateral nystagmus and weakness in his lateral eye movements. Which of the following is the most appropriate immediate treatment?

- A. Intravenous Glucose
- B. Intravenous thiamine
- C. Benzodiazepines
- D. Haloperidol

23. A 50-year-old man with a 20-year history of daily alcohol use is admitted to the hospital for pneumonia. Two days after admission, he becomes agitated and reports seeing insects crawling on the walls. His vital signs show tachycardia (HR 110 bpm), hypertension (BP 160/95 mmHg), and a fever of 38.2°C (100.8°F). He is disoriented and tremulous. Which of the following is the most accurate statement about this patient's condition?

- A. He is experiencing early symptoms of alcohol withdrawal

- B. He is likely developing Wernicke's encephalopathy
 - C. He is showing signs of delirium tremens
 - D. His symptoms are typical of impending alcohol withdrawal seizures
-

24. A 25-year-old man is brought to the emergency department after being found unconscious at a party. His friends report that he consumed a large amount of alcohol over a short period. On examination, the patient is unresponsive to verbal stimuli, has irregular breathing, and a heart rate of 50 beats per minute. Which range of blood alcohol concentration (in mg/dL) is most likely associated with this patient's condition?

- A. 20-30 mg/dL
 - B. 30-80 mg/dL
 - C. 80-200 mg/dL
 - D. >300 mg/dL
-

25. Which of the following statements about alcohol absorption and metabolism is correct?

- A. The majority of alcohol absorption occurs in the stomach
 - B. Peak Blood Alcohol Concentration (BAC) is typically reached within 5-10 minutes
 - C. Consuming alcohol with food accelerates its absorption
 - D. Only less than 10% is excreted unchanged in breath, sweat, and urine
-

26. A 35-year-old man is brought to the emergency department after heavy alcohol consumption. His blood alcohol concentration (BAC) was recorded at two different times: once when it was rising to 100 mg/dL and later when it was falling to the

same level. The physician noted that the patient appeared more intoxicated during the rising phase of BAC compared to the falling phase, even though the BAC was identical at both points. Which of the following best explains this phenomenon?

- A. Alcohol-Mellanby effect
 - B. Reverse tolerance
 - C. Acute tolerance
 - D. Delirium tremens
-

27. A 28-year-old male presents with a long-standing history of cocaine abuse. Which of the following best explains the etiology and maintenance of his substance use disorder?

- A. Genetic polymorphisms affecting dopamine production
 - B. Alexithymia leading to self-medication with cocaine
 - C. Conditioned learning and neuroplasticity in the brain-reward circuitry
 - D. Modulation of the GABA system resulting in tolerance
-

28. A 25-year-old man presents to the emergency room after consuming a large quantity of alcohol during a party. He mentions that this was the first time he drank so much, and he was urged by friends to continue drinking. He has no history of alcohol use before this event, and he assures that he does not intend to consume alcohol regularly. His physical exam reveals elevated blood pressure, nausea, and mild confusion. His liver function tests are slightly elevated. What is the most appropriate diagnosis?

- A. Alcohol dependence
- B. Alcohol withdrawal
- C. Single episode of harmful use
- D. Delirium tremens

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Correct Answers

Question	Correct Answer
Question 1	4
Question 2	1
Question 3	2
Question 4	3
Question 5	2
Question 6	3
Question 7	2
Question 8	2
Question 9	3
Question 10	2
Question 11	3
Question 12	4
Question 13	3
Question 14	2
Question 15	3
Question 16	2
Question 17	4
Question 18	4

Question 19	2
Question 20	3
Question 21	3
Question 22	2
Question 23	3
Question 24	4
Question 25	4
Question 26	1
Question 27	3
Question 28	3

Solution for Question 1:

Correct Answer: D) Nodding off

Explanation:

- Nodding off refers to the state where initial euphoria is followed by sedation.
- It describes the characteristic behavior of opioid users when they experience a "nod," indicating a state of drowsiness or semi-consciousness.

Route of Administration of opioids:

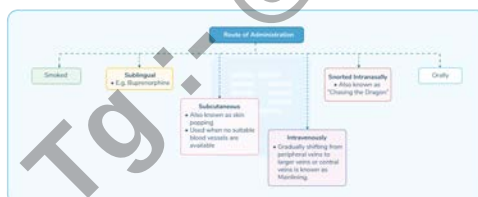
- Oral
- Smoked
- Sublingual- eg. Buprenorphine
- Subcutaneous- Also known as skin popping. Used when no suitable blood vessels are available



- Intravenously- Gradually shifting from peripheral veins to larger veins or central veins is known as Mainlining .



- Shorted intranasally- Also known as "Chasing the Dragon".



Clinical Features

- Produces a feeling of intense Euphoria ("Rush"), especially when taken Intravenously.
- Associated symptoms Feeling of warmth Dry mouth Heaviness of Extremities Facial Flushing
- Feeling of warmth
- Dry mouth
- Heaviness of Extremities
- Facial Flushing
- Initial euphoria may be followed by a period of Sedation, known as Nodding Off .

- First time users may report Dysphoria, Nausea or vomiting.
- Feeling of warmth
- Dry mouth
- Heaviness of Extremities
- Facial Flushing

Solution for Question 2:

Correct Answer: A) Methylenedioxymethamphetamine

Explanation:

The patient has likely consumed methylenedioxymethamphetamine (MDMA), which is an example of club drugs.

Club drugs is a generic term for drugs associated with dance clubs and dance parties (rave parties). Also known as rave drugs . These include:

- Ecstasy (Methylenedioxymethamphetamine - MDMA).
- Flunitrazepam (Ropynol/Roofies).
- Gamma HydroxyButyrate (GHB).
- Ketamine.
- LSD.
- Methamphetamine.

Death Rape Drugs: Consumption of these drugs produce disorientation and sweating effects. After the effect goes away, the person may not remember what happened during that period. They include:

- K - Ketamine
- G - GHB
- F - Flunitrazepam

Inhalant Drugs (Volatile substances/solvents):

- They are volatile hydrocarbons that vaporize into gaseous fumes at room temperature, inhaled through the nose or mouth.

- They are divided into four commercial classes: Solvents for glues and adhesives. Propellants (e.g., for aerosol paint sprays, hair sprays, and shaving cream). Thinners (e.g., for paint products and correction fluids). Fuels (e.g., gasoline, propane).
- Solvents for glues and adhesives.
- Propellants (e.g., for aerosol paint sprays, hair sprays, and shaving cream).
- Thinners (e.g., for paint products and correction fluids).
- Fuels (e.g., gasoline, propane).
- Common methods of use include: Sniffing: Inhaled directly from the neck of the bottle. Huffing: Cloth dipped in solvent and held over nose and mouth. Bagging: Inhaled into a bag filled with solvent. Dusting: Spraying it directly over the nose or mouth.
- Sniffing: Inhaled directly from the neck of the bottle.
- Huffing: Cloth dipped in solvent and held over nose and mouth.
- Bagging: Inhaled into a bag filled with solvent.
- Dusting: Spraying it directly over the nose or mouth.
- Serious adverse effects: Long-term use of these inhalers can cause irreversible liver disease or renal damage, muscle damage, peripheral neuropathy, and brain damage.
- Solvents for glues and adhesives.
- Propellants (e.g., for aerosol paint sprays, hair sprays, and shaving cream).
- Thinners (e.g., for paint products and correction fluids).
- Fuels (e.g., gasoline, propane).
- Sniffing: Inhaled directly from the neck of the bottle.
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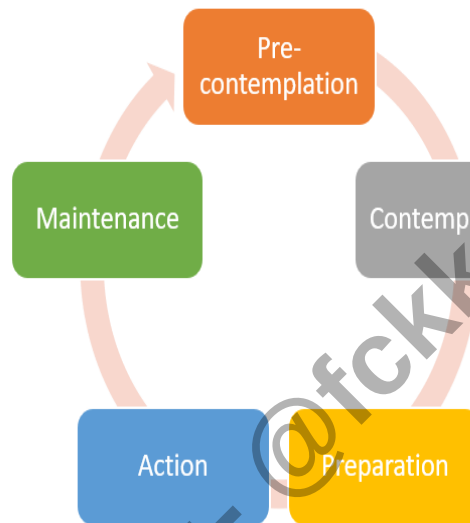
Solution for Question 3:

Correct Answer: B) Contemplation

Explanation:

Contemplation refers to an individual recognizing the problem and considering change, but who is not yet ready to take action based on the Transtheoretical Model of Change.

- The Transtheoretical Model of Change (also known as the Stages of Change Model) is a comprehensive model that describes an individual's journey through different stages of behavior change.
- Developed by Prochaska and DiClemente, this model is relevant in addiction, smoking cessation, and lifestyle changes.
- The model comprises five stages:



Precontemplation	The individual is not considering a change in behavior in the foreseeable future. They may be unaware or in denial about the need for change. In the scenario above, the patient is aware of the risks and is already exploring options, which rules out precontemplation. (Option A)
Contemplation	The individual recognizes the problem and is considering change, but is not yet ready to take action. The patient in this case is aware of the health risks and is actively seeking information, which aligns with contemplation. They are ambivalent about quitting, indicating they are considering, but not committed to, change.
Preparation	The individual is intending to take action in the near future (usually within the next 30 days) May have started taking small steps toward change. This patient has not yet reached this stage, as he is still ambivalent and has not committed to making concrete plans to quit smoking. (Option C)

Precontemplation	The individual is not considering a change in behavior in the foreseeable future. They may be unaware or in denial about the need for change. In the scenario above, the patient is aware of the risks and is already exploring options, which rules out precontemplation. (Option A)
Action	The individual has recently made changes (within the last 6 months) Actively working to maintain those changes. Since the patient has not taken any steps to quit, he is not in the action stage. (Option D)
Maintenance	The individual has sustained their behavior change for more than 6 months Working to prevent relapse.

Solution for Question 4:

Correct Answer: C) Gambling Disorder

Explanation:

Gambling Disorder is an addictive disorder where patients exhibit a persistent and recurring pattern of gambling that leads to clinically significant impairment.

Features include:

- Preoccupation with gambling,
- Needing to gamble with increasing amounts of money,
- Failed attempts to stop,
- Continued gambling despite severe consequences.

This disorder has characteristics similar to substance use disorders, including tolerance, withdrawal symptoms (irritability when unable to gamble), and the use of gambling as a means of coping with stress.

Treatment for Gambling Disorder:

- Cognitive Behavioral Therapy (CBT) : Helps patients identify and change maladaptive thought patterns and behaviors related to gambling.

- Gamblers Anonymous (GA) : A peer-support group based on the 12-step model, offering mutual support for those trying to overcome their addiction.
- Pharmacotherapy : Medications such as SSRIs (e.g., fluoxetine) for underlying depression, mood stabilizers (e.g., lithium, lamotrigine) for mood regulation, and opioid antagonists (e.g., naltrexone) for reducing the urge to gamble.
- Family Therapy : Engaging family members in therapy helps to repair relationships and provides support for the gambler's recovery.
- Hospitalization : In severe cases, temporary hospitalization may be required to remove patients from their gambling environment■.

Other Addictive Behavioural disorder:

Internet Gaming Disorder, as defined in the DSM-5, refers to excessive and compulsive participation in online gaming that significantly impairs personal, social, and occupational functioning. Unlike Gambling Disorder, it typically does not involve financial, legal, or relational harms.

The disorder is characterized by persistent or recurring gaming behavior (whether digital or video games), either online (via the internet) or offline, which manifests through:

- Impaired control over gaming : Difficulty regulating the onset, duration, or stopping of gameplay.
- Prioritization of gaming over daily activities : Gaming takes precedence over essential responsibilities.
- Continuation of gaming despite negative consequences : Gaming persists even when it leads to adverse effects on life.

This behavior results in notable impairment in overall functioning.

Impulse control disorder (Option A): While gambling disorder was previously classified under impulse control disorders, it's now considered a behavioral addiction in the DSM-5. The term "impulse control disorder" is very broad and doesn't specifically capture the addiction-like nature of problematic gambling behavior.

Obsessive-compulsive disorder (OCD) (Option B): Although there can be obsessive thoughts about gambling, OCD is characterized by intrusive, unwanted thoughts and repetitive behaviors aimed at reducing anxiety. The described behavior is more consistent with addiction patterns than OCD rituals.

Bipolar disorder (Option D): While people with bipolar disorder may engage in risky behaviors including gambling during manic episodes, the persistent pattern described, without mention of mood episodes, is more indicative of gambling disorder.

Solution for Question 5:

Correct Answer: B) Bupropion

Explanation:

Withdrawal Symptoms of nicotine:

- When nicotine is abruptly stopped, individuals experience withdrawal symptoms such as irritability, anxiety, depression, difficulty concentrating, increased appetite, and insomnia. These symptoms are due to a sudden drop in dopamine and other neurotransmitters.
- Develops within 2 hours of smoking the last cigarette.
- The peak is achieved in 24-48 hours.
- Withdrawal Symptoms- lasts up to 2-3 weeks.

Nicotine Action and Dependence Pathway:

Nicotine ingestion ↓

Nicotine activates nicotinic acetylcholine receptors (nAChRs) ↓

nAChRs stimulation in the brain ↓

Activation of the Ventral Tegmental Area (VTA) ↓

Release of dopamine in the Nucleus Accumbens ↓

Rewarding & reinforcing effects ↓

Chronic stimulation of nAChRs ↓

Development of dependence

Treatment:

The 5 A's Approach:	
1. Ask	Record tobacco use status at every visit.
2. Advise	Strongly encourage quitting in a personalized way.
3. Assess	Determine if the patient is willing to quit at this time.
4. Assist	Use counseling and pharmacotherapy to help in quitting.

The 5 A's Approach:	
5. Arrange	Schedule follow-up within the first week after quitting.

Psychosocial Therapies:

- Behavioral Therapy : Focus on skills training, relapse prevention, and stimulus control.
- Aversive Therapy : Smoking until nausea associates the behavior with unpleasant sensations.
- Hypnosis : Can help by implanting suggestions that discourage smoking.

Pharmacological Therapies:

Medication	Dosage	Mechanism of Action	Adverse Effects	Additional Info
Nicotine Replacement Therapies (NRTs)				
Nicotine Replacement (NRT) Patch (OTC)	Typically 6-12 weeks of maintenance, follow manufacturer guidelines	Delivers nicotine in controlled doses to reduce withdrawal symptoms and cravings.	Skin irritation (patch), insomnia (24-hour patch).	Provides nicotine without harmful tobacco toxins. Combine well with gum for extra support.
Nicotine Gum/Lozenge Oral (OTC)	2 mg for <25 cigarettes/day, 4 mg for >25 cigarettes/day	Replaces nicotine to ease withdrawal symptoms; absorbed via buccal mucosa.	Sore jaw (gum), heartburn, nausea, hiccups, headache.	Avoid acidic drinks during use as they reduce absorption.
Nicotine Nasal Spray/Inhaler	4 mg per cartridge (Inhaler); Nasal spray delivers rapid nicotine	Rapid delivery is similar to smoking, providing immediate relief from cravings.	Rhinitis, coughing, nasal irritation, watering eyes.	Useful for heavily dependent smokers but can cause irritation (rhinitis, coughing).
Non-Nicotine Medications:				

Medication	Dosage	Mechanism of Action	Adverse Effects	Additional Info
Bupropion	150 mg once daily × 3 days, then 150 mg twice daily for 6-12 weeks	Norepinephrine-dopamine reuptake inhibitor (NDRI) : Reduces cravings and withdrawal by modulating dopamine and norepinephrine.	Insomnia, dry mouth, nausea, risk of seizures (rare).	Antidepressant with dual effects on craving and mood . Contraindicated in seizure disorders and eating disorders.
Varenicline (most effective)	Start with 0.5 mg once daily, increase to 1 mg twice daily	Partial agonist at $\alpha 4\beta 2$ nicotinic receptors: Alleviates withdrawal symptoms while blocking nicotine's rewarding effects.	Nausea, insomnia, vivid dreams, neuropsychiatric symptoms (rare).	Reduces both cravings and the reinforcing effects of nicotine . Monitor for neuropsychiatric or cardiac symptoms.

Combining Therapies:

- Using both nicotine replacement and behavior therapy shows better outcomes compared to using either alone

Varenicline (Option A):

- While varenicline is effective for smoking cessation, its main effect is on nicotine cravings rather than mood stabilization.
- Moreover, varenicline can sometimes be associated with mood changes, including depression, which makes it less suitable for a patient already experiencing mood disturbances.

Nicotine Patch (Option C): The nicotine patch delivers a controlled dose of nicotine and helps reduce cravings. However, it does not address the mood-related aspects of withdrawal and has previously failed in this patient.

Nortriptyline (Option D): Nortriptyline is not recommended as a first-line treatment due to its side effect profile, which includes weight gain, sedation, and anticholinergic effects. It is not as well-tolerated or effective as bupropion for treating both nicotine cravings and withdrawal-related mood disturbances.

Solution for Question 6:

Correct Answer: C) Supportive care and monitoring

Explanation:

This patient is exhibiting classic signs of cocaine withdrawal , which include fatigue, hypersomnia, increased appetite, dysphoria (such as anhedonia), and irritability.

- Cocaine withdrawal is not life-threatening and typically does not require pharmacologic treatment beyond symptomatic care.
- The management at this stage involves supportive care with close monitoring.

Cocaine Withdrawal:

- Cocaine withdrawal symptoms are predominantly psychological and include intense cravings, anhedonia, dysphoria, irritability, hypersomnia, and increased appetite.
- Unlike alcohol or benzodiazepine withdrawal, it is not associated with physical complications like seizures or delirium tremens.
- The acute withdrawal phase (" crash ") lasts for a few days, followed by a more prolonged phase with mood disturbances and drug cravings.

Cocaine-Induced Psychotic Disorder:

- Characterized by paranoid delusions and hallucinations .
- Affects up to 50% of stimulant users .
- Types of Hallucinations: Auditory hallucinations are common. Visual and tactile hallucinations are less common but may occur. Cocaine can cause formication/ Cocaine bugs (sensation of bugs crawling under the skin).
- Auditory hallucinations are common.
- Visual and tactile hallucinations are less common but may occur.
- Cocaine can cause formication/ Cocaine bugs (sensation of bugs crawling under the skin).
- Symptoms depend on dose , duration of use , and individual sensitivity to stimulants.
- More frequent in users of intravenous cocaine or crack cocaine .
- Psychotic symptoms are seen more in men than in women.
- Treatment: The preferred treatment is short-term use of antipsychotics like haloperidol , especially for amphetamine-induced psychosis.
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- The preferred treatment is short-term use of antipsychotics like haloperidol , especially for amphetamine-induced psychosis.

Treatment for Cocaine Withdrawal:

- Supportive care : The mainstay of treatment includes managing symptoms such as fatigue and depression, encouraging sleep, and monitoring for suicidal ideation.
- Behavioral interventions : Cognitive-behavioral therapy (CBT) and psychological support are considered the most effective long-term treatments for cocaine addiction for preventing relapse.

Immediate initiation of dopamine agonist therapy (Option A): Dopamine agonist therapy is not part of the recommended management for cocaine withdrawal.

Long-term use of antipsychotic medications (Option B): Antipsychotics are used in the acute management of cocaine-induced psychosis, which typically presents during intoxication. However, this patient is not experiencing psychotic symptoms like hallucinations or delusions, and routine long-term use of antipsychotic medications is not recommended for managing cocaine withdrawal alone.

Administration of benzodiazepines to prevent withdrawal seizures (Option D): Seizures are not a typical feature of cocaine withdrawal, unlike alcohol or benzodiazepine withdrawal syndromes. Benzodiazepines are used for managing agitation or anxiety during cocaine intoxication, but they are not indicated in uncomplicated cocaine withdrawal.

Solution for Question 7:

Correct Answer: B) Inhibition of norepinephrine reuptake

Explanation:

This patient's symptoms (euphoria, dilated pupils, palpitations, hypertension, agitation, and signs of myocardial ischemia) are classic for cocaine intoxication , a powerful stimulant drug.

- Cocaine works primarily by inhibiting the reuptake of norepinephrine, dopamine, and serotonin in the brain, resulting in increased levels of these neurotransmitters in the synaptic cleft.
- This leads to the intense euphoria and stimulation observed in users, along with sympathomimetic effects such as tachycardia and hypertension.

Cocaine:

- Derived from the leaves of the coca plant (*Erythroxylum coca*).



Mechanism of Action:

Cocaine Enters Synapse



Inhibits Presynaptic Transporters

Blocks Reuptake of Neurotransmitters

(Norepinephrine (NE), Dopamine (DA), Serotonin (5-HT))

Increased Concentration of NE, DA, and 5-HT in Synapse

Enhanced Neurotransmitter Effect at Postsynaptic Receptors

Methods of Use:

- Snorting: Cocaine is finely chopped and inhaled through the nose. Least dangerous compared to other methods, but still harmful.
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- Least dangerous compared to other methods, but still harmful.
- Subcutaneous or Intravenous Injection: Cocaine is dissolved and injected under the skin or directly into the bloodstream. Highly dangerous, and associated with a high risk of overdose, cardiac abnormalities, and cerebrovascular diseases.
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- Highly dangerous, and associated with a high risk of overdose, cardiac abnormalities, and cerebrovascular diseases.
- Smoking (Freebasing or Crack Cocaine): Freebasing: Street cocaine is mixed with chemically extracted pure cocaine alkaloid to increase its effect, and then smoked. Crack Cocaine: A potent, freebase form of cocaine sold as "rocks" that are smoked. One of the most dangerous methods, often leading to intense cravings and addiction. Associated with extreme behavioral changes and health risks like cardiac problems and death. Combination of Cocaine and Heroin is called Speed Balls.
- Freebasing: Street cocaine is mixed with chemically extracted pure cocaine alkaloid to increase its effect, and then smoked.
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- Oral Use: Cocaine can be ingested orally, but this method is rare and considered ineffective by users.
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- Cocaine is finely chopped and inhaled through the nose.
- Least dangerous compared to other methods, but still harmful.
- Cocaine is dissolved and injected under the skin or directly into the bloodstream.
- Highly dangerous, and associated with a high risk of overdose, cardiac abnormalities, and cerebrovascular diseases.
- Freebasing: Street cocaine is mixed with chemically extracted pure cocaine alkaloid to increase its effect, and then smoked.
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- One of the most dangerous methods, often leading to intense cravings and addiction.
- Associated with extreme behavioral changes and health risks like cardiac problems and death.
- Combination of Cocaine and Heroin is called Speed Balls.
- Cocaine can be ingested orally, but this method is rare and considered ineffective by users.

Intoxication Effects: Euphoria, increased energy, hyperactivity, enhanced alertness, agitation, tachycardia, hypertension, dilated pupils, and risk of acute cardiovascular events.

Adverse Effects:

- Hypertension

- Tachycardia
- Myocardial ischemia/infarction
- Seizures
- Arrhythmias
- Hyperthermia.
- Chronic use can lead to anxiety, paranoia, and hallucinations.

Increased release of GABA (Option A):

- This would lead to sedative effects, which are the opposite of the symptoms described in the scenario.
- Cocaine is a stimulant, not a sedative.

Enhanced serotonin degradation (Option C):

- Cocaine inhibits the reuptake of serotonin rather than degrading it.
- Enhanced serotonin degradation would decrease serotonin levels and likely cause depressive symptoms rather than the euphoria seen here.

Stimulation of opioid receptors (Option D):

- This describes the mechanism of opioid drugs like heroin or morphine, which produce sedation and pain relief, not the stimulant effects seen with cocaine.

Solution for Question 8:

Correct Answer: B) Lysergic acid diethylamide (LSD) intoxication

Explanation: The patient's symptoms, including disorientation, dilated pupils, visual hallucinations (seeing vivid colors and patterns), tachycardia, and mild hypertension, are consistent with LSD intoxication.

Lysergic acid diethylamide (LSD)

Chemical nature	Synthetic base derived from lysergic acid nucleus from ergot alkaloids
Common forms	Tablets, liquid, blotter paper ("blotter acid")

Chemical nature	Synthetic base derived from lysergic acid nucleus from ergot alkaloids
Potency	100 µg of LSD ≈ 10-15 mg of psilocybin ≈ 300-400 mg of mescaline
Onset and duration	Onset within an hour, peaks in 2-4 hours, lasts 8-12 hours
Common physiologic symptoms	• Dilated pupils • Increased deep tendon motor reflexes • Muscle tension • Mild motor incoordination • Ataxia
Occasional physiologic symptoms	• Increased heart rate • Increased respiration • Hypertension • Nausea • Decreased appetite • Salivation • Sweating • Blurred vision
Psychological effects	• Visual and perceptual changes • Altered sense of time • Mood changes • Possible anxiety or paranoia • Possible feelings of insight or creativity
Associated phenomenon	• Flashback phenomenon • Badtrip (also seen with cannabis) • Reflex hallucination/ morbid synesthesia (may occur with other hallucinogens)
Tolerance	Develops quickly (2-3 days of repeated use), lost after 4-6 days of abstinence
Potential risks	• Accidents due to impaired judgment • Rarely, a syndrome similar to neuroleptic malignant syndrome • Possible triggering of anxiety or depression in some users

Date Rape Drugs:

Substances used to incapacitate victims, facilitating sexual assault by impairing memory and consciousness.

Common drugs include:

While PCP can cause disorientation and altered perceptions, it typically presents with more severe symptoms such as aggressive behavior, nystagmus, and a more pronounced

hypertension. The vivid visual hallucinations described are more characteristic of LSD (Option A)

Methamphetamine intoxication would typically present with more severe sympathomimetic effects, including significant hypertension, hyperthermia, and agitation. The visual hallucinations described are not typical of methamphetamine use (Option C)

While cannabis can cause anxiety and paranoia, the vivid visual hallucinations and dilated pupils are not typical. Cannabis-induced psychosis would more likely present with paranoid delusions rather than visual distortions (Option D)

Solution for Question 9:

Correct Answer: C) Phencyclidine intoxication

Explanation: The patient's symptoms, including disorientation, nystagmus, paranoid thoughts, hypertension, and hyperthermia, are consistent with Phencyclidine (PCP) intoxication. Additionally, "angel dust" is a common street name for PCP.

Phencyclidine & Ketamine	
Origin	Dissociative anesthetic agents; ketamine is closely related to PCP
Forms	Phencyclidine- Crystalline powder/paste added to cannabis or parsley-containing cigarettes Ketamine- Powder or solution for intranasal, oral, inhalational, or intravenous use
Mechanism	Works at the NMDA receptor (as antagonist), causing hallucinations and dissociated state
Physical effects	Cardiovascular stimulation No respiratory depression Hypertension Tachycardia Increased salivation Nystagmus
Psychological effects	Altered sense of body and reality Dissociative effects

Phencyclidine & Ketamine	
Onset and duration	Phencyclidine- Effects occur in about 5 minutes, plateau in 30 minutes when smoked Ketamine- Onset within seconds when used intravenously, analgesia lasting 40 minutes, dissociative effects for hours
Metabolism	N-demethylated by liver microsomal cytochrome P450 (CYP), especially CYP3A4, into norketamine
Detection	Phencyclidine- Detectable in urine & blood Ketamine- Detectable in urine, with half-lives of 3, 4, and 7 hours for ketamine, norketamine, and dehydronorketamine respectively
Recent developments in usage	Phencyclidine- Not used clinically Ketamine- Growing use in ketamine infusion clinics for depression treatment, FDA approval of esketamine nasal spray for treatment-resistant depression

Drug	Street Names
Cocaine	Coke, Blow, Snow, Nose candy
Heroin	Smack, H, Dope, Junk, Brown
Methamphetamine	Meth, Ice, Crystal, Tina
MDMA (Ecstasy)	Molly, E, XTC, Adam
Marijuana	Weed, Pot, Mary Jane, Bud
LSD	Acid, Lucy, Tabs, Trips
Prescription Opioids	Oxy, Hillbilly Heroin, Percs, Vikes
Benzodiazepines	Benzo, Xanax, Bars, Sleepers
Synthetic cannabinoids	Spice, K2, Black Mamba

While alcohol intoxication can cause disorientation and behavioral changes, it doesn't cause nystagmus, hypertension, or hyperthermia. The paranoid thoughts and the use of the term "angel dust" are not characteristic of alcohol intoxication (Option A)

Signs of opioid overdose include respiratory depression, pinpoint pupils, and decreased level of consciousness or coma. The hypertension, hyperthermia, and paranoid thoughts described in the case are not consistent with opioid overdose. Moreover, the term "angel dust" is not associated with opioids (Option B)

While ketamine is also a dissociative drug and can cause some similar effects to PCP, it typically has a shorter duration of action and is less likely to cause the severe hypertension, hyperthermia, and aggressive behavior associated with PCP. The term "angel dust" is not used for ketamine (Option D)

Solution for Question 10:

Correct Answer: B) Cannabis-induced anxiety disorder

Explanation: The patient's symptoms of anxiety and paranoid thoughts during active cannabis use are most consistent with Cannabis-induced anxiety disorder .

- Cannabis-induced anxiety disorder is common during acute intoxication and is characterized by short-lived anxiety states often provoked by paranoid thoughts

Disorder	Key Features
Cannabis Use Disorder (Option D)	• Prolonged daily use of cannabis for weeks to months • Risk increases with earlier age of first use and longer duration of use • Affects approximately 1 in 10 cannabis users
Cannabis-Induced Psychotic Disorder (Option A)	• Rare occurrence of genuine psychotic process • Transient paranoid ideation is more common • "Florid psychosis" or "hemp insanity" in high potency cannabis use • Often associated with preexisting personality disorders
Cannabis-Induced Anxiety Disorder	• Common during acute intoxication • Short-lived anxiety states often provoked by paranoid thoughts • May induce panic attacks • More common in inexperienced users
Cognitive Impairment	• Long-term use may produce subtle cognitive impairments • Affects memory, attention, and organization • More pronounced with longer periods of heavy use • Unclear if impairments resolve after extended abstinence

Disorder	Key Features
Amotivational Syndrome (Option C)	• Controversial cannabis-related syndrome • Characterized by unwillingness to persist in tasks requiring prolonged attention • Associated with long-term heavy use • Individuals may become apathetic and anergic
Flashbacks	• Recurrence of sensations related to cannabis intoxication • Can occur after short-term effects of the substance have disappeared • Not known whether flashbacks are related to cannabis use alone or concomitant use of other substances

Solution for Question 11:

Correct Answer: C) Provide supportive counseling and psychoeducation

Explanation: Supportive counseling and psychoeducation are the first-line approaches in the management of cannabis withdrawal. This patient is experiencing typical symptoms of cannabis withdrawal, which generally do not require pharmacological intervention as a first step.

Cannabis withdrawal

Aspect	Details
Onset and Duration	• Typically occurs 24-48 hours after cessation • Most symptoms peak at days 2-6 • Can last up to 3 weeks or more in heavy users
Common Symptoms	• Irritability • Anxiety • Disturbed sleep/dreaming • Depressed mood • Loss of appetite • Cannabis cravings • Nervousness • Restlessness

Aspect	Details
Less Common Symptoms	• Headaches • Physical tension • Sweating • Chills • Stomach pain • Tremors
First-line Management	• Supportive counseling • Psychoeducation about withdrawal symptoms • Encouraging healthy coping strategies
Pharmacological Management	• No medications specifically approved for cannabis withdrawal • Symptomatic treatment may be considered (e.g., for anxiety, sleep disturbances) • Some agents (e.g., cannabinoid agonists) used off-label in clinical practice
Special Considerations	• Inpatient management may be indicated for patients with significant comorbid mental health disorders or polysubstance use • Withdrawal symptoms may precipitate relapse to cannabis use

Prescribe benzodiazepines for anxiety and sleep (Option A): While benzodiazepines might help with anxiety and sleep disturbances, they are not the first-line approach for cannabis withdrawal due to their own addiction potential.

Start bupropion for mood stabilization (Option B): Bupropion is not typically used for cannabis withdrawal. It's primarily used for depression and smoking cessation.

Initiate medically assisted withdrawal with cannabinoid agonists (Option D): While cannabinoid agonists have been studied for managing withdrawal symptoms, they are not approved specifically for this purpose and are not considered first-line treatment.

Solution for Question 12:

Correct Answer: D) Tachycardia and conjunctival redness are common features of cannabis intoxication

Explanation: Tachycardia and conjunctival redness are common clinical features which are seen during cannabis intoxication.

Cannabis:

Aspect	Details
Composition	• Over 400 compounds • Includes cannabinoids, flavonoids, and terpenoids • Primary psychoactive component: Tetrahydrocannabinol (THC)
Forms	• Dried plant material (marijuana) • Bhang (Derived from dried leaves) • Ganja (derived from flowering stem) • Hashish (concentrated resin) • Oils and extracts (Hash oil) • Edibles • Topical preparations
Receptors	CB1 and CB2 (G-protein coupled receptors- Gi subtype) (Option C)
Methods of Use	• Smoking (most common) • Vaporizing • Oral ingestion • Sublingual administration • Topical application
Clinical Features of Intoxication	• Tachycardia • Conjunctival injection • Anxiety or euphoria • Slowness of time • Impaired cognition and motor coordination • Dry mouth • Increased appetite • Nystagmus • Postural hypotension • Delirium (especially at high doses)
Onset and Duration of Effects	• Smoking: Rapid onset (minutes), duration 2-3 hours • Oral ingestion: Delayed onset (30-90 minutes), duration 4-12 hours

Option A: Oral ingestion of cannabis typically leads to a longer duration of effects compared to inhalation.

Option B: Tetrahydrocannabinol (THC), not cannabidiol (CBD), is the primary psychoactive component responsible for the "high" associated with cannabis use. CBD is non-psychoactive and is being studied for its potential therapeutic effects.

Solution for Question 13:

Correct Answer: C) Clonidine

Explanation:

Clonidine is an alpha-2 adrenergic agonist that helps reduce the autonomic symptoms of opioid withdrawal, such as sweating, restlessness, tachycardia, and hypertension.

- It works by decreasing sympathetic outflow, which alleviates symptoms like piloerection, tremors, and rhinorrhea.
- Clonidine is particularly useful in patients who are unwilling to use opioid-based treatments as in this patient.
- Methadone (Option A) and Buprenorphine and Naloxone combination therapy (Option B) are opioid agonists used for maintenance therapy, but this patient prefers a non-opioid treatment.

Opioid withdrawal symptoms include

- Muscle and Bone aches, Abdominal cramps
- Diarrhea, Lacrimation, Rhinorrhea, Sweating - resembles flu
- Piloerection/ Gooseflesh ("Cold Turkey" Means Abstinence Syndrome)
- Frequent yawning , Insomnia
- Mydriasis
- Hypertension, Tachycardia
- Temperature Dysregulation (Hypothermia Or Hyperthermia)
- Nausea, vomiting, and restlessness

Treatment of Opioid Withdrawal	
Detoxification	Maintenance

Treatment of Opioid Withdrawal	
Opioid Agents Methadone: Opioid Agonist. Buprenorphine :Partial Opioid Agonist.	Opioid Substitution Therapy (OST) Methadone : Full opioid agonist Improves treatment retention and reduces criminal behavior. 80–120 mg/day. Side effects: constipation, sweating, sexual dysfunction. Buprenorphine : Partial opioid agonist with a ceiling effect (limits overdose risk). Combined with naloxone to prevent abuse. Taken sublingually (tablet/film). Side effects: constipation, nausea, drowsiness. Opioid Antagonists (Naloxone, Naltrexone): Naloxone : Reverses opioid overdose (emergency use). Naltrexone : Blocks opioid effects to prevent euphoria, reduces opioid cravings. Available in oral and long-acting injectable forms. Does not cause dependence or withdrawal but requires patient adherence.
Opioid addicted patient but unwilling to be treated with opioids: Symptomatic agents Clonidine Alpha-2 Adrenergic Agonist. Reduces autonomic withdrawal symptoms- Sweating, Restlessness, Tremor or Rhinorrhea. Anxiety: Benzodiazepines Insomnia: Zolpidem Musculoskeletal Pain: NSAIDs (Ibuprofen) GI Symptoms: Antiemetic Diarrhea (Loperamide)	
Pregnant Women with Opioid Use Disorder: Illicit opioid use during pregnancy increases risks for both mother and baby Methadone or buprenorphine maintenance reduces risks, though neonatal opioid withdrawal may still occur. Associated risks- Neonatal infections (HIV), premature labor, low birth weight, and sudden infant death syndrome (SIDS).	

Treatment of Opioid Withdrawal	
Psychotherapy Modalities/ Non- Pharmacological Treatment: Effective treatments: Individual therapy, CBT, family therapy, support groups (NA), and social skills training. Social skills training is key for patients with poor social skills. Residential programs for people with substance use disorders. Education on HIV transmission and safe sex is essential. Needle-exchange programs, where available, help reduce unsafe needle sharing and associated infections. Narcotics Anonymous (NA): A 12-step self-help group for abstinent drug addicts.	

Naloxone (Option D) is a short-acting opioid antagonist used for reversing opioid overdose, not for treating withdrawal.

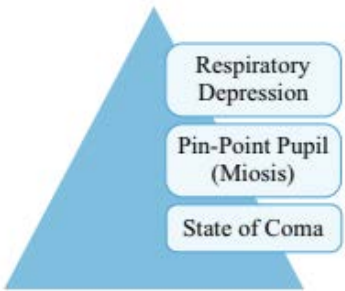
Solution for Question 14:

Correct Answer: B) Ensure an adequate airway and mechanically ventilate the patient.

Explanation:

- The patient is found unresponsive, exhibiting slow and shallow respirations, hypotension, and pinpoint pupils.
- The presence of cold, clammy skin and needle track marks on his arms strongly suggests recent drug use.
- Given these features, the most probable diagnosis is opioid overdose.
- Ensuring an adequate airway and mechanically ventilating the patient is the correct initial action in opioid overdose cases.
- Maintaining airway patency, aspiration of secretions, and providing mechanical ventilation are critical to ensure oxygenation until naloxone can be administered.

Opioid Toxicity	
Clinical features	Treatment

Opioid Toxicity	
 Clinical Triad Opioid overdose often leads to respiratory arrest. Symptoms include marked unresponsiveness, coma, slow respiration, hypothermia, hypotension, and bradycardia. Inspect for needle tracks in arms, legs, ankles, groin, and dorsal vein of the penis.	Ensure an adequate airway; aspirate secretions if needed. Use mechanical ventilation until naloxone is given. Drug of Choice: Intravenous Naloxone Opioid Antagonist which is a short acting. Given at a slow rate of 0.8 mg/70 kg of body weight. (Option A) Signs of improvement Increase respiratory rate Pupillary Dilation If there is no response after the initial dose, repeat naloxone every few minutes. If no response after 4-5 mg, consider nonopioid substances as factors.

Give intravenous fluids to manage hypotension (Option C):

- While managing hypotension is important, it is not the immediate priority in suspected opioid toxicity.
- The main focus should be on airway management and ventilation.

Perform a CT scan to rule out intracranial hemorrhage (Option D): This is not appropriate in the acute management of suspected opioid overdose. The priority is to stabilize the patient's airway and breathing rather than imaging, which can be deferred until the patient is stable.

Solution for Question 15:

Correct Answer: C) The patient meets the criteria for opioid dependence, with prominent features of impaired control and social impairment

Explanation:

The behavioral and clinical features of the patient align with the diagnostic criteria for substance dependence according to DSM-5, as given below:

Dependence:

Dependence is a state where a person exhibits both physical and psychological symptoms related to substance use.

Type of Dependence	Definition
Behavioral Dependence	Focuses on substance-seeking behaviors and the patterns of pathological use.
Physical Dependence	Refers to the physiological effects that result from repeated episodes of substance use.
Psychological Dependence	Also called habituation, characterized by a persistent or occasional craving to avoid dysphoria.

Substance Use Disorder Criteria (DSM-5) :

The criteria for diagnosing substance dependence can be grouped into four main categories:

- Using larger amounts or for longer periods than intended
- Persistent desire or unsuccessful efforts to cut down or control use
- Spending excessive time obtaining, using, or recovering from the substance
- Experiencing intense cravings or urges to use the substance
- Failing to fulfill major role obligations due to substance use
- Continuing use despite persistent social or interpersonal problems
- Giving up important activities because of substance use (Progressive Neglect)
- Recurrent use in physically hazardous situations
- Continued use despite knowledge of physical or psychological problems caused or exacerbated by the substance (Persistence)
- Tolerance: Needing increased amounts to achieve desired effects or experiencing reduced effects with the same amount
- Withdrawal: Experiencing characteristic withdrawal symptoms or using the substance to relieve withdrawal.

(Option A): While the patient likely meets the criteria for opioid dependence, the scenario doesn't provide explicit evidence of tolerance or withdrawal.

(Option B) is incorrect because the patient's presentation goes beyond abuse and meets the criteria for dependence.

(Option D) is incorrect because the origin of use (medically sanctioned) doesn't prevent a diagnosis of dependence. The pattern of use, loss of control, and continued use despite significant problems all point to dependence rather than mere misuse.

Solution for Question 16:

Correct Answer: B) Acamprosate

Explanation:

Acamprosate is effective for managing alcohol cravings and is safe for use in patients with liver dysfunction. It works as an NMDA receptor antagonist and a positive allosteric modulator of GABA-A receptors, making it suitable for this patient.

To maintain the Abstinence, here we have two pillars of maintenance:

- Pharmacological Treatment
- Non-Pharmacological Treatment

Pharmacological Treatment

- Disulfiram (Alcohol Sensitizing Agent).
- Irreversible Inhibitor of Aldehyde Dehydrogenase (ALDH) enzyme.
- This enzyme helps in conversion of acetaldehyde to acetic acid.

Patients on Disulfiram treatment consume alcohol.

↓

Alcohol in the body is converted to acetaldehyde but further metabolism is suppressed.

Increased Acetaldehyde levels

Causes Nausea, Vomiting, burning sensation in face or stomach

Severe reaction can lead to heart failure, MI or respiratory depression

Disulfiram Alcohol Reaction

- Disulfiram should be used in patients with motivation to abstain from alcohol.
- Dose of Disulfiram- 250 mg (daily), given with the consent of the patient.
- Key Points Before Starting Disulfiram to avoid Disulfiram reaction: Wait 12 hours after the last alcohol intake before starting disulfiram. Avoid alcohol for 2 weeks after stopping Disulfiram. Check for alcohol in foods (like vinegar), perfumes, and cough syrups. Monitor liver function tests due to potential risk of hepatitis. (Option A ruled out)
- Wait 12 hours after the last alcohol intake before starting disulfiram.
- Avoid alcohol for 2 weeks after stopping Disulfiram.

- Check for alcohol in foods (like vinegar), perfumes, and cough syrups.
- Monitor liver function tests due to potential risk of hepatitis. (Option A ruled out)
- Wait 12 hours after the last alcohol intake before starting disulfiram.
- Avoid alcohol for 2 weeks after stopping Disulfiram.
- Check for alcohol in foods (like vinegar), perfumes, and cough syrups.
- Monitor liver function tests due to potential risk of hepatitis. (Option A ruled out)
- Both Acamprosate and Naltrexone are FDA approved anti-craving agents for alcohol.

Anticraving Agents	
Acamprosate	Naltrexone
NMDA Antagonist and positive allosteric modulator of GABA-A receptors. Given after stopping the consumption of alcohol. Contraindicated in severe Renal Impairment. In moderate renal impairment (given in reduced doses). FDA approved.	Opioid Antagonist. Can be started when a patient is consuming alcohol or in withdrawal phase. Contraindicated in Liver Failure. (Option C) Check LETs periodically. FDA approved.
Other anti craving Agents are which are not FDA approved but effective: Ondansetron (5HT-3 Antagonist) (Option D) Topiramate (Anti Epileptic agent) Baclofen (GABA-B Agonist) Fluoxetine (SSRI)	

Non- Pharmacological Treatment

- Counseling in the first 3 to 6 months.
- Both individual and group counseling are effective.
- Initially, counseling should occur at least three times a week for the first 2 to 4 weeks, followed by weekly sessions for the next 3 to 6 months.
- Discussions about the importance of a sober peer group, planning alcohol-free social activities, and reestablishing communication with family and friends.
- Relapse prevention
- Family Counseling
- Alcohol Anonymous- Help available 24 hours

Ondansetron (Option D) is an effective anti craving agent but is not FDA approved still. Thus it is not used as a first line anti craving agent.

Solution for Question 17:

Correct Answer: D) Lorazepam

Explanation:

Benzodiazepines (such as Lorazepam) are the drug of choice for managing alcohol withdrawal symptoms.

Alcohol withdrawal Seizures:

- Seizures can occur suddenly; about 1% may experience a single grand mal convulsion, typically on the second day of withdrawal.
- Treatment of choice: Benzodiazepines.
- Anticonvulsants are not more beneficial.

Detoxification

Withdrawal	Treatment
Mild or Moderate Withdrawal	Treatment: Benzodiazepines (e.g., lorazepam, chlordiazepoxide, diazepam) . Short-acting (e.g., lorazepam) or long-acting (e.g., chlordiazepoxide, diazepam). Carbamazepine (daily doses of 800 mg) is also effective and has minimal abuse liability.
Severe withdrawal (Seizures)	Treatment of choice: Benzodiazepines Anticonvulsants are not more beneficial.
Life threatening (Delirium Tremens)	For extreme symptoms (delirium tremens or DTs): Higher doses of Benzodiazepines (e.g., 50-100 mg chlordiazepoxide every 4 hours or lorazepam IV). Supportive care: High-calorie diet, multivitamins. Haloperidol may be used for severe agitation but can lower seizure threshold.
Protracted Withdrawal: Symptoms may last 2 to 6 months.	Acamprosate may help diminish some symptoms. Cognitive and behavioral approaches recommended.

- Treatment of withdrawal in Liver Disease: Lorazepam, Oxazepam, and Temazepam (LOT)
- Lorazepam, Oxazepam, and Temazepam (LOT)
- Lorazepam, Oxazepam, and Temazepam (LOT)
- Treatment of Alcohol Withdrawal-Related Hallucinations: Similar to DTs: Benzodiazepines , adequate nutrition, and fluids. If necessary, add antipsychotics (haloperidol) for long-term cases.

- Similar to DTs: Benzodiazepines , adequate nutrition, and fluids.
- If necessary, add antipsychotics (haloperidol) for long-term cases.
- Similar to DTs: Benzodiazepines , adequate nutrition, and fluids.
- If necessary, add antipsychotics (haloperidol) for long-term cases.
- Treatment of Alcohol-Induced Amnestic Disorders: Wernicke Encephalopathy Treatment: Thiamine 100 mg by mouth 2-3 times daily for 1-2 weeks. If receiving IV glucose, add 100 mg thiamine per liter. Korsakoff Syndrome Treatment: Thiamine 100 mg by mouth 2-3 times daily for 3-12 months. Full recovery is rare; many experience cognitive improvement with thiamine and nutritional support. Alcoholic Pellagra Encephalopathy For patients unresponsive to thiamine: Niacin 50 mg by mouth four times daily or 25 mg parenterally 2-3 times daily.
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- If receiving IV glucose, add 100 mg thiamine per liter.
- Treatment: Thiamine 100 mg by mouth 2-3 times daily for 3-12 months.
- Full recovery is rare; many experience cognitive improvement with thiamine and nutritional support.
- For patients unresponsive to thiamine: Niacin 50 mg by mouth four times daily or 25 mg parenterally 2-3 times daily.
- Treatment of Idiosyncratic Alcohol Intoxication: Protect patients from harm; physical restraint may be required. Consider using Haloperidol to control assaultiveness. Differentiate from other behavioral changes (e.g., complex partial epilepsy).
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- Consider using Haloperidol to control assaultiveness.
- Differentiate from other behavioral changes (e.g., complex partial epilepsy).

Haloperidol (Option A) is used for severe agitation and hallucinations; it does not address withdrawal symptoms or prevent seizures effectively.

Thiamine (Option C) is essential for preventing Wernicke encephalopathy, but does not directly manage acute withdrawal symptoms.

Carbamazepine (Option B) is also effective in managing withdrawals, but benzodiazepines are the drug of choice.

Solution for Question 18:

Correct Answer: D) Carbohydrate-deficient transferrin

Explanation:

A Carbohydrate-deficient transferrin (CDT) level above 3.0% is a well-established and specific biomarker for heavy alcohol intake. It reflects chronic alcohol consumption and is less affected by other conditions compared to the other tests.

Laboratory tests in Alcoholism	
γ -glutamyltransferase (GGT)	>35.0 U/L Chronic alcohol abuse Elevated GGT levels indicate liver disease and are often found in chronic heavy drinkers, but only 30–50% of excessive drinkers show elevated levels (Option C) . Not specific to alcohol use, as other conditions like pancreatitis and prostate disease can also raise its levels.
Carbohydrate-deficient transferrin (CDT)	>3.0% Most important biomarker for heavy alcohol intake. (Option D) Good sensitivity and specificity.
Mean corpuscular volume (MCV)	>91.0 μm^3 Heavy alcohol use MCV levels can remain elevated for several months after someone stops drinking, meaning a person may be abstinent yet still have a high MCV value. (Option A)
Serum glutamic pyruvic transaminase (alanine aminotransferase)- ALT	>45.0 IU/L Chronic alcohol abuse More specific marker of alcohol-induced liver injury since it is mainly located in the liver. (Option B)
Serum glutamic oxaloacetic transaminase (aspartate aminotransferase)- AST	>45.0 IU/L Chronic alcohol abuse The AST to ALT ratio helps confirm heavy alcohol consumption. (Ratio >2 is correlated with alcoholic liver disease)
N-Acetyl- β -Hexosaminidase (Beta-Hex)	Increased beta-Hex subsides to normal levels after only 7 to 10 days of abstinence. Diabetes and hypertension also elevate beta-Hex.

Assessment tools for Alcohol use disorders:

- CAGE Questionnaire (Need to) Cut down on Drinking. Annoyance when asked about your drinking. Guilty about your drinking. Eye openers after drinking first thing in the morning to steady your nerves or get rid of a hangover the morning after heavy drinking. Positive response on 2 or more is suggestive of Alcohol Use Disorders.
- (Need to) Cut down on Drinking.
- Annoyance when asked about your drinking.
- Guilty about your drinking.

- Eye openers after drinking first thing in the morning to steady your nerves or get rid of a hangover the morning after heavy drinking.
- Positive response on 2 or more is suggestive of Alcohol Use Disorders.
- AUDIT - Alcohol Use Disorder Identification Test
- MAST - Michigan Alcoholism Screening Test
- SADQ: Severity of Alcohol Dependence Questionnaire (used to assess severity of dependence).
- (Need to) Cut down on Drinking.
- Annoyance when asked about your drinking.
- Guilty about your drinking.
- Eye openers after drinking first thing in the morning to steady your nerves or get rid of a hangover the morning after heavy drinking.
- Positive response on 2 or more is suggestive of Alcohol Use Disorders.

Solution for Question 19:

Correct Answer: B) Increases REM sleep

Explanation: While alcohol may initially help decrease sleep latency, it actually disrupts overall sleep quality. Specifically, alcohol does not increase REM sleep ; rather, it decreases REM and deep sleep , leading to more fragmented sleep.

Effects of Alcohol on Sleep

- While evening alcohol consumption may make it easier to fall asleep and reduce sleep latency, this effect is largely attributable to its sedative properties. (Option A)
- Alcohol has adverse effects on the sleep architecture: Decrease in REM Sleep and decrease in deep sleep. (Option D) More sleep fragmentations with more and longer episodes of awakening at night. (Option C)
- Decrease in REM Sleep and decrease in deep sleep. (Option D)
- More sleep fragmentations with more and longer episodes of awakening at night. (Option C)
- Chronic use of alcohol leads to tolerance, requiring higher amounts or more frequent dosing to achieve the same sedative effects.

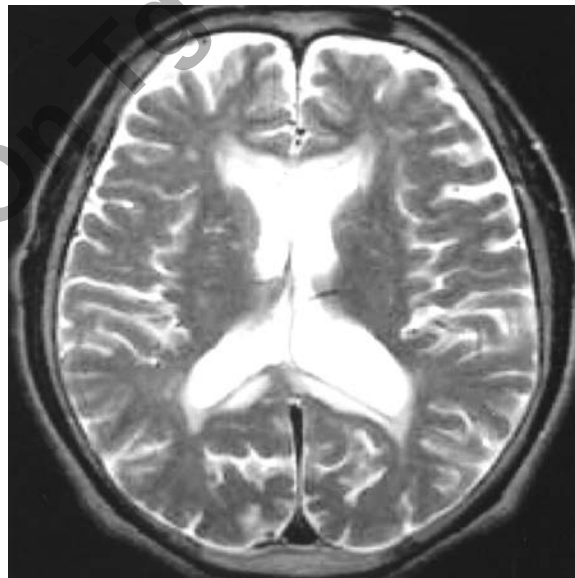
- Rebound Insomnia: Upon withdrawal or after developing tolerance, individuals may experience rebound insomnia, which can be more severe than the original sleep disturbance.
- Idea that drinking improves sleep is a myth. While it may facilitate initial sleep onset, it ultimately disrupts overall sleep quality.
- Decrease in REM Sleep and decrease in deep sleep. (Option D)
- More sleep fragmentations with more and longer episodes of awakening at night. (Option C)

Solution for Question 20:

Correct Answer: C) Marchiafava-Bignami Syndrome

Explanation: The sudden onset of neurological symptoms along with the characteristic MRI findings of high signal intensity in the splenium of the corpus callosum, coupled with the patient's chronic alcohol use, strongly indicates Marchiafava-Bignami Syndrome.

Fig: An axial T2-weighted MR image reveals signal abnormalities in the corpus callosum and periventricular white matter.



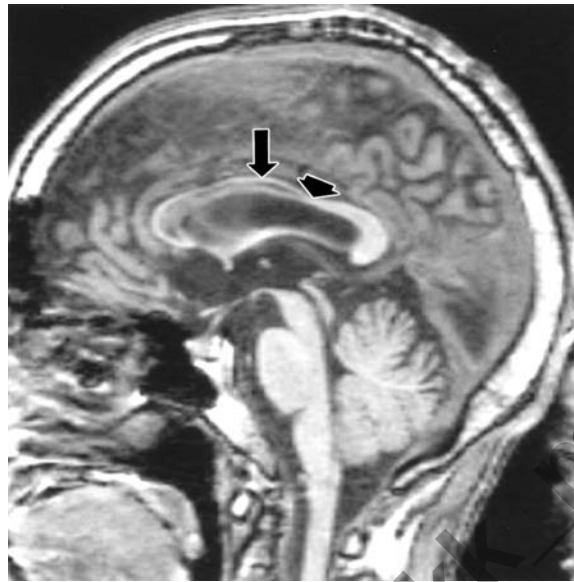


Fig: A sagittal T1-weighted MR image demonstrates corpus callosum atrophy (short arrow). The central layers of the corpus callosum appear hypointense, while the dorsal and ventral layers are spared, creating the characteristic "sandwich sign" (long arrow)

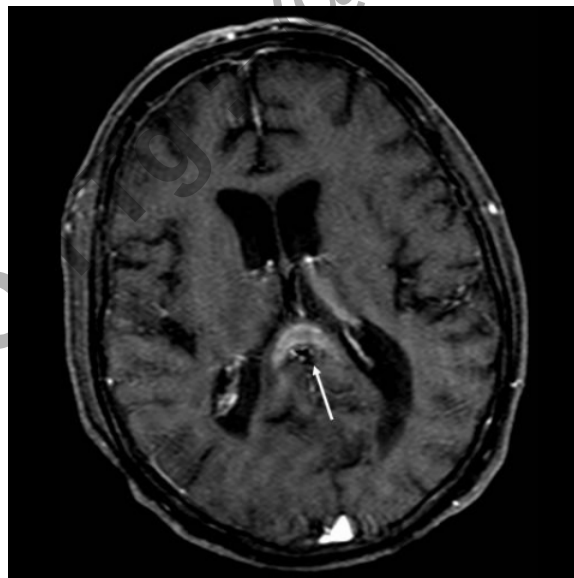


Fig: Axial T1 weighted image after fat saturation and gadolinium injection. Peripheral rim enhancement (arrow) of splenium. Diffuse mild cortical and subcortical atrophy is also present.

Marchiafava Bignami Syndrome

- A rare neurological disorder characterized by demyelination and necrosis of the corpus callosum, the structure connecting the brain's two hemispheres.

- Alcohol-related hypovitaminosis, primarily of vitamin B1, along with oxidative stress, can directly harm the corpus callosum, leading to neurological impairment associated with B1 deficiency.
- Clinical features: Acute- Sudden onset of loss of consciousness and seizures. Other features can be apathy, aggressiveness, confusion, and psychosis. Subacute- Depression, ataxia, apraxia, agraphia, anomia, dysarthria, visual dyslexia. Chronic- Progressive severe global dementia, visual hallucinations, auditory delusions, and behavioral abnormalities.
- Acute- Sudden onset of loss of consciousness and seizures. Other features can be apathy, aggressiveness, confusion, and psychosis.
- Subacute- Depression, ataxia, apraxia, agraphia, anomia, dysarthria, visual dyslexia.
- Chronic- Progressive severe global dementia, visual hallucinations, auditory delusions, and behavioral abnormalities.
- Magnetic Resonance Imaging is the gold standard imaging method. Characteristically, the splenium of the corpus callosum exhibits high signal intensity on T2-weighted images and low signal intensity on T1-weighted images, along with peripheral rim enhancement after gadolinium injection. Also shows adjacent subcortical atrophy.
- Treatment includes high-dose intravenous thiamine (500 mg three times daily) in normal saline or 5% dextrose, oral vitamin B complex, amantadine, and folate along with alcohol withdrawal.
- Acute- Sudden onset of loss of consciousness and seizures. Other features can be apathy, aggressiveness, confusion, and psychosis.
- Subacute- Depression, ataxia, apraxia, agraphia, anomia, dysarthria, visual dyslexia.
- Chronic- Progressive severe global dementia, visual hallucinations, auditory delusions, and behavioral abnormalities.

Alcohol Induced Dementia (Option A)

- Includes recent memory impairment, decreased intellectual functioning and global cognitive impairment (like language).
- About 50 to 70 percent of these patients exhibit enlarged brain ventricles and atrophy of the cerebral sulci. Atrophic changes seem to be reversible after 1 year of abstinence.

Wernicke's Encephalopathy (Option B): Although confusion is present, the classical triad includes ophthalmoplegia, ataxia, and confusion. MRI findings would typically show hyperintensities in the periaqueductal gray matter, mammillary bodies, hypothalamus, and medial thalami, which were not observed in this case.

Alexander Disease (Option D) is a leukodystrophy that affects the brain's white matter, it typically presents in childhood or early adulthood and is characterized by other clinical features such as developmental delays and seizures, which do not align with this patient's presentation.

Solution for Question 21:

Correct Answer: C) Initiate thiamine supplementation at 100 mg, 2-3 times/day

Explanation:

The patient's memory impairment and confabulation, following chronic alcohol use, indicate a diagnosis of Korsakoff syndrome.

- This condition results from thiamine deficiency, which is common in individuals with prolonged alcohol consumption.
- Continuing thiamine supplementation at 100 mg, 2-3 times/day is essential for addressing the underlying deficiency.
- This treatment helps mitigate cognitive decline and supports recovery from memory impairment.

Korsakoff syndrome	
Onset	Chronic progression of Wernicke Encephalopathy
Symptoms	Impaired recent memory Anterograde amnesia: More common, inability to form new memories. Retrograde amnesia: Inability to recall old memories. Confabulation: Making false stories to fill memory gaps. These are unintentional.
Cause	Thiamine (Vitamin B1) deficiency
Neuropathologic Lesions	Symmetrical involvement of: Mammillary bodies Thalamus, Hypothalamus Midbrain Pons Medulla Cerebrum

Korsakoff syndrome	
Treatment	Supplementation of Thiamine 100 mg, 2-3 times/day for 3-12 months. Full recovery is only seen in 20% of patients, others see partial improvement and symptoms may not completely resolve. If patients do not respond to Thiamine: Consider Alcohol Pellagra Encephalopathy and treat with Niacin . (Option B)

Increasing thiamine dosage to 1g daily (Option A) is not standard practice and may not offer additional benefits compared to established dosing guidelines.

Start cognitive rehabilitation therapy (Option D): While cognitive rehabilitation therapy can be beneficial, it should be introduced after ensuring consistent thiamine supplementation, which is essential for the patient's recovery.

Solution for Question 22:

Correct Answer: B) Intravenous thiamine

Explanation: The patient's symptoms are consistent with Wernicke's encephalopathy, which requires immediate treatment with thiamine to prevent progression to Korsakoff syndrome.

- Wernicke's encephalopathy is an acute neurological emergency caused by thiamine (vitamin B1) deficiency, often seen in chronic alcoholics due to poor nutrition and impaired thiamine absorption.
- In this case, the patient presents with the classic triad of Wernicke's encephalopathy:
- Immediate treatment with intravenous thiamine is crucial to prevent progression to the chronic, often irreversible Korsakoff syndrome.

Feature	Wernicke's Encephalopathy	Korsakoff Syndrome
Onset	Acute	Chronic, often follows Wernicke's
Key Symptoms	• Global Confusion • Ocular abnormalities (nystagmus, gaze palsy) • Ataxia (especially gait)	• Severe anterograde amnesia • Retrograde amnesia • Confabulation (in some cases)
Cause	Thiamine deficiency	Prolonged thiamine deficiency
Neuropathological lesions	Symmetrical and paraventricular, involving mammillary bodies, thalamus, hypothalamus, midbrain, pons, medulla, fornix, and cerebellum	
Treatment	Immediate IV thiamine	Long-term thiamine supplementation, supportive care
Prognosis	Potentially reversible with prompt treatment	Often irreversible, only about 20% recover

NOTE:

- In patients with alcohol-related disorders who are receiving IV glucose solution, it is good practice to include 100 mg of thiamine in each liter of the glucose solution .
- In a patient who appears to have Wernicke–Korsakoff syndrome but who does not respond to thiamine treatment , one should consider the diagnosis of alcoholic pellagra encephalopathy due to niacin (nicotinic acid) deficiency . Treatment: 50 mg of niacin by mouth four times daily or 25 mg parenterally two to three times daily.
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- Treatment: 50 mg of niacin by mouth four times daily or 25 mg parenterally two to three times daily.

Solution for Question 23:

Correct Answer: C) He is showing signs of delirium tremens

Explanation: The patient's symptoms and their timing are consistent with delirium tremens (DTs), which is the most severe form of alcohol withdrawal.

Alcohol Withdrawal- Clinical features

Stage	Symptoms	Onset after last drink
Mild withdrawal	• Mild anxiety • Tremulousness • Nausea • Headache • Palpitations	6-8 hours
Moderate (Alcoholic hallucinosis)	• Visual, auditory, or tactile hallucinations • Intact orientation • Normal vital signs	8- 12 hours
Severe (Withdrawal seizures)	• Generalized tonic-clonic seizures • Usually brief and self-limited • Also known as rum fits	12-24 hours
Life threatening (Delirium tremens)	• Severe agitation • Confusion and disorientation • Hallucinations (often visual) • Autonomic hyperactivity (tachycardia, hypertension, fever) • Profound sweating • Tremors • Insomnia	Within 72 hours

Option A: Early symptoms of alcohol withdrawal is incorrect because the patient's symptoms are more severe and occur later than early withdrawal symptoms.

Option B: Wernicke's encephalopathy is not a component of alcohol withdrawal and while it can cause confusion, it typically also presents with ataxia and ocular changes, which are not mentioned in this case.

Option D: Alcohol withdrawal seizures are a serious complication of alcohol withdrawal, they typically occur earlier and do not account for the other symptoms described.

Solution for Question 24:

Correct Answer: D) >300 mg/dL

Explanation:

- The patient's symptoms are consistent with severe alcohol intoxication, which occurs at blood alcohol concentrations above 300 mg/dL .
- At this level, impaired vital signs and possible death can occur, as evidenced by the patient's unresponsiveness, irregular breathing, and low heart rate.

Principle: Blood alcohol concentration is measured using a breathalyzer which is based on the principle that concentration of alcohol is strongly correlated with concentration of alcohol in arterial blood.

Levels of Impairment at Different Blood Alcohol Concentrations

BAC (mg/dL)	Effects
20-30 (Option A)	Slowed motor performance and decreased thinking ability
30-80 (Option B)	Increase in motor and cognitive problems
80-200 (Option C)	Increase in incoordination and judgment errors, mood lability, deterioration in cognition
200-300	Nystagmus, marked slurring of speech, and alcoholic blackouts
>300	Impaired vital signs and possible death

Solution for Question 25:

Correct Answer: D) Only less than 10% is excreted unchanged in breath, sweat, and urine

Explanation: Alcohol is majorly metabolized in the liver (>90%) and only less than 10% is excreted unchanged in breath, sweat, and urine .

Ethanol- Pharmacokinetics

Aspect	Details
Absorption	20% absorbed through stomach 80% absorbed through small intestine (Option A) Faster absorption on empty stomach, slower absorption if consumed after food intake (Option C)
Distribution	Distributed throughout water-containing portions of body Affects organs with high water content (e.g. liver, brain)
Metabolism	Primary site is liver Enzyme alcohol dehydrogenase converts alcohol to acetaldehyde Further, acetaldehyde is converted into acetate by enzyme aldehyde dehydrogenase <10% excreted unchanged in breath, sweat, and urine Remaining ~90% metabolized by liver
Factors Affecting Kinetics	Food in stomach (slows absorption) Body weight and composition Gender (females generally reach higher BAC) Genetics (e.g. ADH enzyme variants) Tolerance in chronic drinkers
Blood Alcohol Curve	Sharp rise during absorption phase Peak at 45-90 minutes after drinking (Option B) Slow decline during distribution phase Faster decline during elimination phase Mellanby effect- more intoxicating effects when BAC is increasing than for the same BAC when decreasing

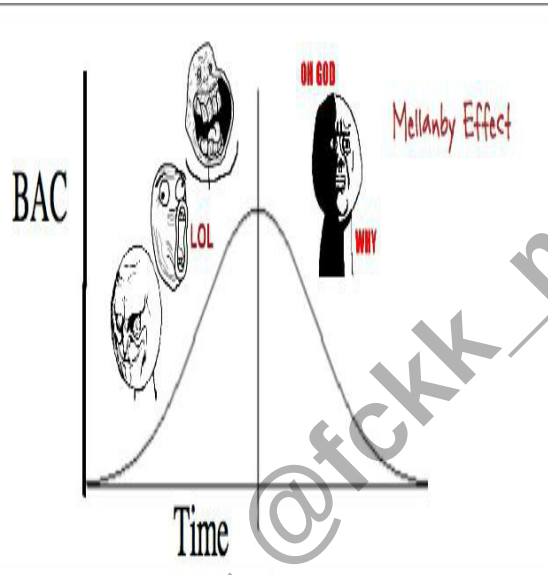
Solution for Question 26:

Correct Answer: A) Alcohol-Mellanby effect

Explanation:

Alcohol-Mellanby effect:

- This effect describes the observation that individuals tend to experience more pronounced intoxication when their blood alcohol concentration (BAC) is rising compared to when it is falling, even when the BAC is at the same level.
- This occurs because the brain's sensitivity to alcohol is higher during the absorption phase (when BAC is increasing) than during the elimination phase (when BAC is decreasing).



Reverse tolerance (Option B): Reverse tolerance refers to the phenomenon where, over time, smaller amounts of alcohol produce the same or heightened effects, usually due to impaired liver function or changes in brain chemistry.

Acute tolerance (Option C):

- Refers to a reduction in the effects of alcohol within a single drinking session, as the brain quickly adapts to alcohol's presence.

Delirium tremens (Option D):

- It is a severe withdrawal syndrome that occurs after sudden cessation of chronic alcohol consumption.
- It is characterized by confusion, agitation, hallucinations, and autonomic hyperactivity.
- This patient's presentation of varying intoxication during rising vs. falling BAC is unrelated to alcohol withdrawal.

Solution for Question 27:

Correct Answer: C) Conditioned learning and neuroplasticity in the brain-reward circuitry

Explanation:

Conditioned learning and neuroplasticity in the brain-reward circuitry: This explanation integrates both psychological and biological aspects.

Repeated drug use leads to conditioned responses to drug-related stimuli and causes changes in the brain's reward system, particularly in the dopaminergic pathways from the ventral tegmental area to the nucleus accumbens. This process explains both the development and maintenance of the disorder.

Etiology of Substance Use Disorders:

Substance use disorders (SUDs) have a multifaceted etiology involving biological, psychological, and environmental factors.

Biological Factors	Genetic Predisposition	
		Studies of twins, adoptees, and siblings indicate a strong genetic component, particularly in alcohol use disorders. Genetic factors may influence neurotransmitter systems, affecting an individual's response to substances.
Neurochemical Changes	SUDs involve alterations in brain neurotransmitter systems, particularly: Dopaminergic system: Critical in the brain's reward circuitry Opioid system: Involved in pain modulation and reward GABA system: Implicated in anxiety reduction and sedation The ventral tegmental area (VTA) and its projections to the nucleus accumbens are key in the brain-reward circuitry.	
Neuroplasticity	Repeated substance use leads to adaptive changes in the brain, affecting: Neurotransmitter release Receptor function Second-messenger systems Gene regulation	

Biological Factors	Genetic Predisposition	Studies of twins, adoptees, and siblings indicate a strong genetic component, particularly in alcohol use disorders. Genetic factors may influence neurotransmitter systems, affecting an individual's response to substances.
Psychological Factors	Learning and Conditioning	Operant conditioning: Substance use is reinforced by its consequences (e.g., euphoria, relief from negative states). Classical conditioning: Environmental cues associated with substance use can trigger craving or withdrawal symptoms.
Cognitive Factors	Beliefs about substance effects Expectations of drug-related outcomes Impaired decision-making processes	
Mental Health	Pre-existing mental health issues may contribute to substance use as a form of self-medication. Some individuals may have difficulty recognizing or expressing emotions (alexithymia), potentially leading to substance use.	
Environmental/Social Factors	Social	Peer pressure and social norms Family dynamics and modeling of substance use behavior Cultural attitudes towards substance use
Availability and Accessibility	Ease of obtaining substances Exposure to drug-using environments	
Stress and Life Events	Chronic stress or traumatic experiences may increase vulnerability to substance use.	

Biological Factors	Genetic Predisposition	Studies of twins, adoptees, and siblings indicate a strong genetic component, particularly in alcohol use disorders. Genetic factors may influence neurotransmitter systems, affecting an individual's response to substances.
Interplay of Factors		Initial use may be influenced by environmental factors (e.g., availability, peer pressure). Continued use is affected by individual biology and psychology (e.g., how drug effects are perceived, and the development of tolerance). Progression to dependence involves neuroadaptive changes, learning processes, and environmental reinforcement.

Genetic polymorphisms affecting dopamine production (Option A) : While genetic factors contribute to vulnerability, they alone don't explain the maintenance of the disorder or account for environmental and learning factors.

Alexithymia leading to self-medication with cocaine (Option B) : This psychological explanation may contribute to some cases but is too narrow to explain the complex etiology of substance use disorders. It doesn't account for the neurobiological aspects or learning processes involved.

Modulation of the GABA system resulting in tolerance (Option D) : While neurotransmitter systems are involved in substance use disorders, this explanation is too specific (more relevant to alcohol use) and doesn't encompass the broader psychological and environmental factors.

Solution for Question 28:

Correct Answer: C) Single episode of harmful use

Explanation:

Single Episode of Harmful Substance Use (ICD-11)

- Diagnosed after an episode where physical or mental health damage occurs, affecting the user or others.
- Allows for early intervention and prevention of further harm.
- ICD-11 Highlight: Explicitly includes harm to others' health.
- Refers to acute effects and may include substance-induced psychological disorders but excludes harm from established patterns of harmful use.

In this case, the patient consumed a large amount of alcohol during one event, resulting in harmful effects such as nausea, elevated blood pressure, and mild confusion.

- There is no history of regular alcohol use, and the patient does not exhibit any signs of dependence or withdrawal.
- This makes the diagnosis of a single episode of harmful use appropriate.

Alcohol dependence (Option A) : Involves a pattern of chronic alcohol use characterized by tolerance, withdrawal symptoms, and an inability to control drinking. This patient does not show these features, and this was a single, isolated event.

Alcohol withdrawal (Option B) : Occurs in individuals who stop or reduce heavy, chronic alcohol use, leading to symptoms such as tremors, anxiety, and seizures. This patient does not have a history of chronic alcohol use, nor is there evidence of withdrawal symptoms.

Delirium tremens (Option D) : It is a severe form of alcohol withdrawal characterized by confusion, agitation, and hallucinations. It occurs in individuals with long-term alcohol dependence who abruptly stop drinking.

Previous Year Questions

1. A 55-year-old man presents to the hospital with auditory hallucinations and inability to recognize his own family members. He reports his last alcohol intake was 48 hours ago. What is the most likely diagnosis?

- A. Delirium tremens
- B. Alcoholic hallucinosis
- C. Alcohol-induced psychosis
- D. Schizophrenia

2. A patient is brought to the hospital following a road traffic accident (RTA). He presents with confusion, irritability, and tremors. The patient has a history of daily alcohol consumption. Which drug is most appropriate to prevent withdrawal symptoms in this patient?

- A. Fomepizole
- B. Naltrexone
- C. Lorazepam
- D. Thiamine

3. A mother brings her daughter to the emergency room, reporting that she ingested an unknown substance. She presents with hypertension, tachycardia, hyperthermia, and mydriasis. Which of the following substances is most likely responsible for these symptoms?

- A. Heroin
- B. Morphine

- C. Cocaine
 - D. Chlorpheniramine
-

4. A 65-year-old chronic alcoholic who began binge drinking ten days ago is brought to the emergency department in an unconscious state. A CT scan of his brain is normal, but his blood glucose level is 45 mg/dL. What is the most appropriate initial management?

- A. Normal saline hydration
 - B. IM Thiamine followed by dextrose
 - C. Dextrose 25%
 - D. Vitamin K with 5% dextrose
-

5. Which drug is used in the maintenance phase of opioid withdrawal?

- A. Buprenorphine
 - B. Disulfiram
 - C. Butorphanol
 - D. Clonidine
-

6. A patient was taking metronidazole for diarrhea. He attended a party where he consumed alcohol. Following this, he developed red rashes, itching all over his body, and a headache. What is the most likely cause of his condition?

- A. Methanol poisoning
- B. Disulfiram like reaction
- C. Cheese reaction

D. None of the above

7. A 50-year-old male with a 20-year history of chronic alcohol consumption, averaging 500 ml daily, presents to the clinic with complaints of sensations of insects crawling on his skin, agitation, and irritation. He reports that his last drink was 3 days ago. What is the most likely diagnosis?

- A. Alcohol intoxication
 - B. Delirium tremens
 - C. Tolerance
 - D. Reaction due to alcohol
-

8. A 62-year-old male with a history of chronic alcohol consumption is admitted to the hospital with features of confusion, vision loss and an unsteady gait. He is unable to remember the most recent incidents. Which vitamin deficiency is responsible for his symptoms?

- A. Ascorbic acid
 - B. Retinol
 - C. Thiamine
 - D. Folate
-

9. Consider the following statements Statement 1-Contemplation - A person who smokes is thinking of quitting smoking Statement 2 - Precontemplation - A person is considering starting morning walks and its benefits to the health Statement 3-Maintenance phase- A person quit smoking one year back and has not smoked since then Statement 4-Action phase- A person with obesity says he can't do anything as it runs deep in his family

- A. Statements 1 and 3 are correct
 - B. Statements 1 and 2 are correct
 - C. Statements 1, 2, 3 are correct
 - D. All are correct
-

10. Jet black pigmentation of the tongue with tactile hallucinations is a feature of:

- A. Heroin
 - B. Opium
 - C. Alcohol
 - D. Cocaine
-

11. What is the probable diagnosis for a 37-year-old male, who is a chronic alcoholic and has recently stopped drinking alcohol, but presents to the emergency department after three days with symptoms of disorientation, tremors, hallucinations, and hypertension?

- A. Alcoholic hallucinosis
 - B. Korsakoff's psychosis
 - C. Delirium tremens
 - D. Marchiafava-Bignami disease
-

12. What is the etiology of delirium tremens in a 48-year-old male who presents to the emergency room with symptoms of confusion and generalized body tremors, and has a known history of heavy alcohol use?

- A. Acute infection

- B. Fatty liver
 - C. Gradual withdrawal of alcohol
 - D. Small doses of regular consumption of alcohol
-

13. What is the probable diagnosis for a 30-year-old male who underwent appendectomy 7 hours ago and is currently experiencing auditory hallucinations and confusion about the source of the voices? The patient maintains orientation, and his vital signs are stable. Upon inquiry, his wife reveals that he occasionally consumes alcohol.

- A. Delirium tremens
 - B. A/c transient psychotic disorder
 - C. Delirium
 - D. Alcoholic hallucinosis
-

14. Which drug among the options provided would assist a chronic smoker in their aim to quit smoking?

- A. Bupropion
 - B. Mirtazapine
 - C. Varenicline
 - D. Both a and c
-

15. What is the probable diagnosis for a 40-year-old male patient who was admitted to the ICU three days ago due to breathing difficulty and is currently experiencing restlessness and widespread tremors? The patient has reported seeing insects crawling on his bed and body and has had one seizure, which was successfully

managed with lorazepam. Further investigation reveals that the patient is a chronic alcoholic.

- A. ICU psychosis
 - B. Alcohol intoxication
 - C. Wernicke-Korsakoff psychosis
 - D. Delirium tremens
-

16. What is the initial indication of alcohol withdrawal?

- A. Tremors
 - B. Delirium tremens
 - C. Seizures
 - D. Hallucinations
-

17. A known alcoholic patient is brought to the emergency in a disoriented state. He complains of seeing snakes and insects crawling on his skin. Which of the following should be given to this patient?

- A. Haloperidol + thiamine
 - B. Lorazepam only
 - C. Lorazepam + thiamine
 - D. Haloperidol only
-

18. Which drug listed below is utilized as an agent for smoking cessation?

- A. Busulfan
- B. Acamprosate

- C. Varenicline
 - D. Gabapentin
-

19. Which of the following statements accurately describes Korsakoff syndrome?

- A. Triad of confusion, ophthalmoplegia, and ataxia
 - B. Loss of long-term memory but intact personality
 - C. Anterograde amnesia with loss of recent memory
 - D. Loss of implicit memory and immediate recall
-

20. What is a true statement about nicotine substitution therapy?

- A. Preferably given by gastrointestinal route.
 - B. Varenicline comes with a black box warning of cardiovascular death
 - C. There should be a 15-minute gap between nicotine gum and coffee/soda/acidic food as they decrease its absorption
 - D. Nicotine chewing gum is better for constant use as it gives 25% higher blood level than lozenges
-

21. Which of the following drug abuses can lead to the symptoms described by a 25-year-old male patient, who complains of a sensation resembling insects crawling beneath his skin, and has presented to the clinic?

- A. Cannabis
- B. Cocaine
- C. Amphetamine
- D. Alcohol

22. A patient stopped alcohol consumption for 3 days and presented with irritability, disorientation, paranoid delusions, agitation, visual hallucinations, and altered sensorium. What is the likely diagnosis in this case?

- A. Delirium tremens
- B. Wernicke's encephalopathy
- C. Korsakoff psychosis
- D. Alcohol-induced psychosis

23. How would you treat a patient who is a long-term alcoholic and is currently experiencing seizures, altered sensorium, auditory hallucinations, and secondary delusions, with his last drink being 2 days ago? The patient has consumed an average of 24 units per day for the past 10 years. Diazepam Thiamine Naltrexone Flupenthixol depot injection

- A. 1, 4
- B. 1, 2, 4
- C. 1, 3, 4
- D. 1, 2

24. What is the probable diagnosis for a chronic alcoholic who arrives at the emergency department presenting confusion, ataxia, painful eye movements, and involvement of the 6th cranial nerve?

- A. Wernicke's encephalopathy
- B. Korsakoff's psychosis
- C. Delirium tremens

D. De Clerambault syndrome

25. What is the predominant manifestation experienced with cannabis consumption?

- A. Anxiety
 - B. Depressed mood
 - C. Dream-like state
 - D. Decreased appetite
-

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Correct Answers

Question	Correct Answer
Question 1	1
Question 2	3
Question 3	3
Question 4	2
Question 5	1
Question 6	2
Question 7	2
Question 8	3
Question 9	1
Question 10	4
Question 11	3
Question 12	3
Question 13	4
Question 14	4
Question 15	4
Question 16	1
Question 17	3
Question 18	3

Question 19	3
Question 20	3
Question 21	2
Question 22	1
Question 23	4
Question 24	1
Question 25	3

Solution for Question 1:

The patient presents with auditory hallucinations and confusion (inability to recognize family members) occurring 48 hours after his last alcohol intake . This strongly suggests delirium tremens (DTs) , a severe form of alcohol withdrawal syndrome that typically occurs 48–96 hours after stopping alcohol. DTs are characterized by altered mental status (confusion, disorientation), hallucinations, autonomic instability (tachycardia, hypertension, fever), and severe agitation .

Alcohol Withdrawal- Clinical features

Stage	Symptoms	Onset after last drink
Minor withdrawal	• Mild anxiety • Tremulousness • Nausea • Headache • Palpitations	6-12 hours
Alcoholic hallucinosis (Option B)	• Visual, auditory, or tactile hallucinations • Intact orientation • Normal vital signs	12-24 hours
Withdrawal seizures	• Generalized tonic-clonic seizures • Usually brief and self-limited • Also known as rum fits	24-48 hours

Stage	Symptoms	Onset after last drink
Delirium tremens	• Severe agitation • Confusion and disorientation • Hallucinations (often visual) • Autonomic hyperactivity (tachycardia, hypertension, fever) • Profound sweating • Tremors • Insomnia	48-96 hours

Other Options:

Alcohol-induced psychosis (Option C): This condition refers to persistent psychotic symptoms (hallucinations, delusions) due to chronic alcohol use, rather than withdrawal. The key difference is that alcohol-induced psychosis is not associated with autonomic instability or withdrawal timing.

Schizophrenia (Option D) is a chronic psychiatric disorder with persistent hallucinations, delusions, and disorganized thought processes. However, the sudden onset following alcohol cessation makes an alcohol-related withdrawal syndrome far more likely.

Solution for Question 2:

Correct Option C - Lorazepam:

- Lorazepam is a short-acting benzodiazepine, which is the drug of choice for managing alcohol withdrawal symptoms.
- Benzodiazepines are effective in reducing the symptoms of alcohol withdrawal, including anxiety, seizures, and delirium tremens (DTs).

Incorrect Options:

Options A, B, and D have no role in treating alcohol withdrawal symptoms.

Solution for Question 3:

Correct Option C – Cocaine:

- Cocaine acts by increasing levels of dopamine, norepinephrine, and serotonin. The symptoms described are all classic signs of cocaine intoxication.

Incorrect Options:

Options A and B – Heroin and morphine:

- It leads to respiratory depression, hypotension, bradycardia, and pinpoint pupils (miosis).

Option D - Chlorpheniramine: It is an antihistamine and is not associated with the given symptoms.

Solution for Question 4:

Correct Option B - IM Thiamine followed by dextrose:

- Chronic alcoholics are often thiamine-deficient. Administering glucose before thiamine can precipitate Wernicke's encephalopathy, a serious neurological disorder because glucose metabolism increases the demand for thiamine. This can lead to acute thiamine deficiency in the brain, exacerbating the condition. Therefore, giving thiamine first helps prevent this complication.

Incorrect Options:

Option A – Normal saline hydration:

- Normal saline alone does not correct hypoglycemia.

Options C and D are incorrect, as prior thiamine supplementation is necessary before glucose administration.

Solution for Question 5:

Correct Option A – Buprenorphine:

- Buprenorphine is used in the maintenance phase of opioid withdrawal.

Incorrect Options:

Option B - Disulfiram:

- It is not used in opioid withdrawal. It is primarily used in the treatment of alcohol dependence.

Option C – Butorphanol:

- It is not used during the maintenance phase of opioid withdrawal.

Option D - Clonidine:

- It is used in the detoxification phase in an opioid-addicted patient who is unwilling to be treated with opioids

Solution for Question 6:

Correct Option B - Disulfiram like reaction:

- Metronidazole inhibit aldehyde dehydrogenase can result in Disulfiram like reaction

Incorrect Options:

Option A - Methanol poisoning: It occurs as a result of drinking contaminated beverages with methanol and present has drowsiness, reduced level of consciousness

Option C - Cheese reaction: Cheese reaction due to Cheese, bear, fish, and red wine are (contains tyramine). Tyramine is sympathomimetic metabolized in GI tract.

Solution for Question 7:

Correct Option B - Delirium tremens:

Within 72 hours (Life threatening)	Delirium tremens • Disturbance of consciousness, • Disoriented (To time, place, person). • Hallucinations, particularly visual, may occur. • Most severe form of withdrawal and medical emergency. • Untreated: Mortality rate of 20%. • Risk of Delirium Tremens increases in presence of: ◦ Hepatic Insufficiency ◦ Pancreatitis ◦ Infections ◦ Heart Failures ◦ Renal Disease • Other symptoms like autonomic Hyperactivity, Tremors may also be seen
------------------------------------	---

Incorrect Options:

Option A - Alcohol intoxication: The excitement that follows drinking is due to the CNS depression

Option C - Tolerance: An Increased amount of substance is required to produce the desired effects which previously were produced by lower amounts

Option D - Reaction due to alcohol: Its not reaction to alcohol because reaction is immediate response in question they have mentioned he is a chronic alcoholic

Solution for Question 8:

Correct option C - Thiamine (Vitamin B1):

- Chronic alcohol consumption can lead to thiamine deficiency due to poor dietary intake and impaired absorption.
- This disorder is known as Wernicke-Korsakoff syndrome and is characterized by confusion, ophthalmoplegia, ataxia, confabulation and amnesia.

Incorrect options:

Option A - Ascorbic acid (Vitamin C): Ascorbic acid deficiency leads to scurvy, characterized by weakness, joint pain, bleeding gums, and skin changes.

Option B - Retinol (Vitamin A): Retinol deficiency can lead to night blindness, dry skin, and vision-related issues.

Option D - Folate (Vitamin B9): Folate deficiency can cause megaloblastic anemia and neural tube defects in pregnant women.

Solution for Question 9:

Correct Option A - Statement 1 and 3 are correct:

- Statement 1 - Contemplation: Contemplation is the stage where a person is aware of the need for change and is seriously considering making a change. In this case, a smoker who is thinking of quitting smoking is in the contemplation stage.
- Statement 3 - Maintenance phase: In this stage, people have sustained their behavior change for more than 6 months and intend to keep moving forward with the change.

Incorrect Options:

Statement 2 & 4 are incorrect

- Statement 2 - Precontemplation: Precontemplation is the stage where a person is not yet considering a change or is unaware of the need for change. In this case, the person considering starting morning walks due to its health benefits is not in the pre-contemplation stage; instead, they are contemplating a positive change.
- Statement 4 - Action phase: The action phase is the stage where a person actively takes steps to make a change. In this case, a person with obesity who says they can't do anything as it runs deep in the family is not in the action phase; rather, they seem to be expressing a belief or barrier that may hinder them from taking action. This person can be considered to be in the phase of precontemplation.

Solution for Question 10:

Correct Option D - Cocaine:

- Among the options provided, cocaine is most commonly associated with black pigmentation of the tongue and tactile hallucinations. The black pigmentation of the tongue is known as "black tongue," and it occurs due to a condition called black hairy tongue. It is characterized by elongation of the filiform papillae on the tongue, which can appear black or dark brown. Tactile hallucinations, also known as formication, are a common symptom of cocaine use and can involve sensations of insects crawling on or under the skin.

Incorrect Options:

Option A - Heroin: Heroin is an opioid drug derived from morphine. Black pigmentation of the tongue is not typically associated with heroin use. Heroin use may cause other physical symptoms, such as constricted pupils, drowsiness, and respiratory depression. Option B - Opium: Opium is another opioid drug derived from the poppy plant. Black pigmentation of the tongue is not a characteristic feature of opium use. Opium use can lead to symptoms such as sedation, euphoria, and constipation. Option C - Alcohol: Alcohol consumption does not typically cause black pigmentation of the tongue. Excessive alcohol intake can have various effects on the body, including impaired judgment, motor coordination, and liver damage.

Solution for Question 11:

Correct Option C - Delirium tremens:

- Description : Delirium tremens (DTs) is a severe form of alcohol withdrawal that occurs typically 48-72 hours after the last drink. It is a medical emergency characterized by severe autonomic hyperactivity and delirium.
- Symptoms : Disorientation, severe tremors, hallucinations (visual, auditory, or tactile), hypertension, hyperthermia, agitation, and diaphoresis.
- The patient's presentation with disorientation, tremors, hallucinations, and hypertension, following a period of alcohol cessation, is classic for delirium tremens. This condition is serious and requires immediate medical attention.

Incorrect options:

Option A - Alcoholic hallucinosis:

- Description : Alcoholic hallucinosis is a condition characterized by the presence of hallucinations, typically auditory, occurring during or after heavy alcohol consumption. The individual is usually aware that the hallucinations are not real and does not experience the

severe autonomic instability seen in delirium tremens.

- Symptoms : Hallucinations (mostly auditory), intact orientation, no severe autonomic symptoms.
- The patient's symptoms of disorientation, tremors, and hypertension, along with hallucinations, suggest a more severe condition than alcoholic hallucinosis.

Option B - Korsakoff's psychosis:

- Description : Korsakoff's psychosis, or Korsakoff's syndrome, is a chronic neuropsychiatric disorder caused by thiamine (vitamin B1) deficiency, often associated with chronic alcoholism. It usually follows an episode of Wernicke's encephalopathy and is characterized by severe memory problems.
- Symptoms : Amnesia, confabulation, lack of insight, and apathy.
- Korsakoff's psychosis is not typically associated with acute symptoms such as tremors, hallucinations, disorientation, and hypertension. It presents with chronic memory issues rather than acute, severe autonomic symptoms.

Option D - Marchiafava-Bignami disease:

- Description : Marchiafava-Bignami disease is a rare condition seen in chronic alcoholics, characterized by the demyelination of the corpus callosum and surrounding white matter. It leads to severe neurological deficits.
- Symptoms : Seizures, dysarthria, spasticity, and progressive cognitive impairment.
- Why Incorrect : The acute presentation of disorientation, tremors, hallucinations, and hypertension is not typical for Marchiafava-Bignami disease. This disease usually presents with more chronic and progressive neurological symptoms.

Solution for Question 12:

Correct Option C - Gradual withdrawal of alcohol:

- Delirium tremens in an alcoholic is caused by the gradual withdrawal of alcohol. Abrupt cessation or significant reduction of alcohol intake leads to hyperexcitability of the brain, resulting in the manifestation of withdrawal symptoms.

Incorrect Options:

Option A - Acute infection: Acute infection is not the cause of delirium tremens in an alcoholic. Delirium tremens occurs due to alcohol withdrawal.

Option B - Fatty liver: Fatty liver is not the direct cause of delirium tremens. It is a consequence of chronic alcohol abuse but does not directly lead to delirium tremens.

Option D - Small doses of regular consumption of alcohol: Small doses of regular alcohol consumption are not the cause of delirium tremens. It occurs with the withdrawal of alcohol rather than regular consumption.

Solution for Question 13:

Correct Option D - Alcoholic hallucinosis:

- Alcoholic hallucinosis is an uncommon intricacy of constant liquor misuse portrayed by dominantly hearable mental trips that happen either during or after a time of weighty liquor utilization. The patient may be confused or agitated, and the hallucinations are typically accusatory or threatening. The condition usually goes away on its own after a few days or a week. The patient, in this instance, is a 30-year-old male who has been drinking for a long time. After having surgery, he is now having auditory hallucinations. The threatening and accusatory hallucinations cause the patient's confusion. Alcoholic hallucinations are consistent with these symptoms. A particular kind of alcohol-induced psychotic disorder known as alcoholic hallucinosis can occur during or shortly after alcohol withdrawal. The way that the patient is a periodic alcoholic further backs this determination, as liquor withdrawal is a typical trigger for the improvement of alcoholic hallucinosis.

Incorrect Options:

Option A - Delirium tremens A more severe withdrawal syndrome known as delirium tremens can affect alcohol users. Tremors, agitation, confusion, and hallucinations are some of its symptoms. Delirium tremens is a medical emergency that must be treated right away. Minor liquor withdrawal is described by the quake, nervousness, queasiness, spewing, and sleep deprivation. Significant liquor withdrawal signs and side effects incorporate visual mind flights and hearable pipedreams, entire body quake, heaving, diaphoresis, and hypertension (hypertension).

Option B - A/c transient psychotic disorder: Psychological stress can lead to the sudden onset of psychotic symptoms in Acute and Transient Psychotic Disorder (ATPD). In this, episodes are present moments, enduring from days to 90 days, trailed by complete abatement. Delusions,

hallucinations, and mood swings are their hallmarks. Most of the time, transient psychotic disorder goes away on its own after a few weeks or a month.

Option C - Delirium: Delirium is a mental state in which you can't think clearly or remember anything, are lost, and are confused. There are many things that can cause delirium, such as an infection, taking medication, or going through a medical procedure. It usually appears out of nowhere. It is frequently brief and treatable.

Solution for Question 14:

Option D - Both A and C:

- Smoking cessation drugs can be an effective tool to help individuals quit smoking. The following drugs are approved by the FDA for smoking cessation: Both bupropion and varenicline are approved by the FDA for smoking cessation and have been shown to be effective in helping individuals quit smoking. They can be used alone or in combination with nicotine replacement therapy.
- Both bupropion and varenicline are approved by the FDA for smoking cessation and have been shown to be effective in helping individuals quit smoking. They can be used alone or in combination with nicotine replacement therapy.
- Both bupropion and varenicline are approved by the FDA for smoking cessation and have been shown to be effective in helping individuals quit smoking. They can be used alone or in combination with nicotine replacement therapy.

Option A - Bupropion: Bupropion is an antidepressant that is also used as a smoking cessation aid. It works by reducing cravings and withdrawal symptoms. It is taken orally and is usually started 1-2 weeks before the quit date.

Option C - Varenicline: Varenicline is a prescription medication that is specifically designed to help individuals quit smoking. It works by reducing cravings and withdrawal symptoms and by blocking the effects of nicotine. It is taken orally and is usually started 1 week before the quit date.

Incorrect Options:

Option B - Mirtazapine: Mirtazapine is not approved by the FDA for smoking cessation and has not been shown to be effective in helping individuals quit smoking.

Solution for Question 15:

Correct Option D - Delirium tremens:

- A person who has been drinking heavily for a long period may have delirium tremens (DTs), a severe type of alcohol withdrawal. DTs can happen if the individual stops drinking suddenly. DTs are a medical emergency that poses a risk to life. Alcohol users may get delirium tremens, a severe withdrawal condition. Some of its symptoms include trembling, anxiety, disorientation, and hallucinations.

Incorrect Options:

Option A - ICU psychosis: ICU psychosis is a mental health condition that can develop in people who are hospitalized in an intensive care unit (ICU). It is often caused by the stress of being in the ICU, as well as the use of sedatives and other medications. ICU psychosis can cause various symptoms, including hallucinations, delusions, and agitation. DTs are caused by alcohol withdrawal, while ICU psychosis is caused by the stress of being in the ICU.

Option B. - Alcohol Intoxication: The patient is not having alcohol intoxication because he has been sober for three days. Alcohol intoxication is a state of intoxication that occurs when a person drinks too much alcohol. Alcohol intoxication can cause a variety of symptoms, including slurred speech, impaired coordination, and loss of consciousness. In severe cases, alcohol intoxication can lead to death.

Option C - Wernicke-Korsakoff psychosis: Wernicke-Korsakoff psychosis is a neurological disorder that is caused by a deficiency of thiamine (vitamin B1). Wernicke-Korsakoff syndrome can cause a variety of symptoms, including confusion, memory loss, and ataxia (loss of coordination). Wernicke's encephalopathy represents the "acute" phase of the disorder, and Korsakoff's amnesic syndrome represents the disorder progressing to a "chronic" or long-lasting stage. The disorder's main features are problems in acquiring new information or establishing new memories and in retrieving previous memories.

Solution for Question 16:

Correct Option A - Tremors:

- Alcohol withdrawal symptoms develop after reducing or stopping alcohol intake. The patient may start to develop certain symptoms:
- After 6-8 hours of stopping alcohol patients develop mild symptoms where the tremor is the classical or the most common sign. The coarse tremors include hand and body shaking. Other early symptoms include GI symptoms (Nausea, Vomiting) , Autonomic Hyperactivity (Anxiety arousal, Sweating, Mydriasis, Tachycardia, Hypertension)

Incorrect Options:

Option B - Delirium tremens : The patient experiences the condition within 72 hours of stopping alcohol. This is a life-threatening symptom.

Option C - Seizures : Occurs after 12-24 hours of stopping alcohol. The withdrawal symptoms are seizures. If the seizures are jerking movements of the body. These seizures are Generalized Tonic-Clonic Seizures (GTCS).

Option D - Hallucinations: Occurs after 8-12 hours of stopping alcohol (Moderate Symptoms) : The withdrawal symptom is alcoholic hallucinosis (Hallucinations, auditory hallucinations more common, visual hallucinations may also be seen). This hallucination occurs in clear consciousness. No impairment of consciousness.

Solution for Question 17:

Correct Option C - Lorazepam + thiamine:

- Here the patient is suffering from chronic alcoholism and shows symptoms of alcohol withdrawal with thiamine deficiency. The patient is in a disoriented state. So the patient is given Lorazepam and thiamine. Lorazepam is a short-acting benzodiazepine that is usually given for alcohol withdrawal. In the treatment of alcohol withdrawal, thiamine is also added with benzodiazepines.
- Many of the time , these patients have a deficiency of Vitamin B1(thiamine) . With thiamine deficiency, the patients exhibit confabulation(from Korsakoff Syndrome) which is making false stories to fill memory gaps. These patients may not be able to remember things, so there are memory gaps. These patients fill those gaps with memories and make false fabricated stories. These are unintentional. Also, because of chronic use of alcohol, there is poor absorption of alcohol from the body.

Incorrect Options:

Option A and D - Haloperidol is indicated only when the symptoms and agitation are severe to control high doses of Benzodiazepines and thiamine may not suffice. In those cases, we can add antipsychotics such as Olanzapine, or Haloperidol.

Option B - Lorazepam only: Here the patient also exhibits symptoms of thiamine deficiency so Lorazepam does not suffice. Thiamine should be given along with lorazepam.

Solution for Question 18:

Correct Option C - Varenicline:

- Varenicline is a medication used as an antismoking agent . It works by targeting nicotine receptors in the brain, reducing the pleasurable effects of smoking and decreasing nicotine withdrawal symptoms. Varenicline is commonly prescribed to individuals who want to quit smoking or tobacco use.

Incorrect Options:

Option A - Busulfan: Busulfan is not used as an antismoking agent. It is a chemotherapy medication primarily used in the treatment of certain types of cancer, such as leukemia.

Option B - Acamprosate : Acamprosate is a medication used in the treatment of alcohol dependence. It helps individuals who have quit drinking alcohol to maintain abstinence by reducing cravings and withdrawal symptoms.

Option D - Gabapentin : Gabapentin is primarily used as an antiepileptic medication, but it can also be prescribed for certain types of chronic pain or neuropathic conditions.

Solution for Question 19:

Correct Option C - Anterograde amnesia with loss of recent memory:

- Korsakoff syndrome is a chronic condition associated with severe thiamine deficiency due to alcohol misuse. Characteristic features are loss of recent memory and anterograde amnesia. They also present with learning difficulties and confabulation.

Incorrect Options:

Option A - Triad of confusion, ophthalmoplegia, and ataxia: This classical triad is seen in Wernicke's encephalopathy, an acute symptom of thiamine deficiency. It is treated with a high dose of parenteral thiamine

Option B - Loss of long-term memory but intact personality: Korsakoff syndrome is characterized by severe anterograde amnesia and Loss of recent memory. Some may present with retrograde amnesia. However, anterograde amnesia is more commonly observed.

Option D - Loss of implicit memory and immediate recall: Korsakoff syndrome affects explicit memories; however, it does not affect implicit memories, which process skills and habits

Solution for Question 20:

Correct Option C - There should be a 15-minute gap between nicotine gum and coffee/soda/acidic food as they decrease its absorption:

- Nicotine substitution therapy treats nicotine addiction and helps reduce withdrawal symptoms associated with smoking cessation.
- Nicotine gums should not be taken with coffee/tea/soda as they are known to reduce absorption. A gap of 15 mins is necessary.

Incorrect Options:

Option A - Preferable given by gastrointestinal route: It is false as nicotine replacement therapy is administered through transdermal patches, lozenges, inhalers, nicotine gums, and nasal sprays.

Option B - Varenicline comes with a black box warning of cardiovascular death: Varenicline helps reduce craving and withdrawal symptoms associated with smoking cessation. It comes with a black box warning of serious neuropsychiatric events like agitation, mood changes, and suicidal thoughts.

Option D - Nicotine chewing gum is better for constant use as it gives a 25% higher blood level than lozenges: It is not true, as nicotine gums and lozenges are equally effective, and one does

not show a 25% higher blood level.

Solution for Question 21:

Correct Option B - Cocaine:

- Cocaine abuse can lead to the development of a condition called "formication," which is the sensation of insects crawling under the skin. This phenomenon is also known as "coke bugs" or "cocaine bugs." Cocaine is a powerful stimulant drug that affects the central nervous system. It increases the levels of dopamine, a neurotransmitter associated with pleasure and reward, in the brain.
- The sensation of formication is a common side effect of cocaine abuse and is believed to be related to the drug's stimulant effects on the nervous system. The exact mechanism of how cocaine causes this sensation is not fully understood, but it is thought to involve alterations in the brain's perception and processing of sensory information.

Incorrect Options:

Option A - Cannabis: (marijuana) Is a different class of drug that primarily acts as a depressant and hallucinogen. It is not typically associated with the symptom of feeling insects crawling under the skin.

Option C - Amphetamine: Is a stimulant drug similar to cocaine, but it is not specifically known to cause the sensation of formication.

Option D - Alcohol abuse: Can lead to various neurological and psychological symptoms, but the feeling of insects crawling under the skin is not a typical manifestation of alcohol abuse.

Solution for Question 22:

Correct Option A - Delirium tremens:

- Delirium tremens is a serious condition that can be extremely dangerous and even fatal. It occurs during alcohol withdrawal. Confusion, intense physical reactions, visual

hallucinations, motor agitation, disorientation, paranoid delusions, and altered sensorium characterize it.

Incorrect Options:

Option B - Wernicke's encephalopathy: Is an acute neurological disorder. It involves symptoms like problems with eye movement (nystagmus), coordination (ataxia) , and confusion . Thiamine (vitamin B1) deficiency is one of its causes.

Option C - Korsakoff psychosis: Results from untreated Wernicke encephalopathy, presents with memory problems, confusion, and behavioural changes, and affects around 80% of patients.

Option D - Alcohol-induced psychosis: Can happen when someone is acutely intoxicated, during alcohol withdrawal, or in individuals with long-term alcohol addiction. The specific term for this type of psychosis is alcohol hallucinosis.

Solution for Question 23:

Correct Option D - 1, 2:

- This patient is probably suffering from an episode of delirium tremens . In such cases, diazepam and thiamine are given.
- They typically present with auditory hallucinations, usually voices that are unstructured. These voices can sound threatening and maligning, while some patients report it sounds pleasant.
- Hallucinations are common in chronic alcohol abusers. Thiamine is ideal for treating alcohol-induced Wernicke's encephalopathy and Korsakoff's psychosis.

Incorrect Options:

Option A - 1, 4: Flupenthixol depot injection is an antipsychotic medication used in treating schizophrenia and is not approved for treating alcohol-induced auditory hallucinations or delusions.

Option B - 1, 2, 4: Flupenthixol is not used to treat patients with delirum tremens.

Option C - 1, 3, 4: Naltrexone is commonly used to treat alcohol dependence. It blocks the effects of opioids in the brain, reducing the desire to consume opioids and alcohol.

Solution for Question 24:

Correct Option A - Wernicke's encephalopathy:

- Wernicke's encephalopathy is an acute , potentially life-threatening, neurological syndrome resulting from thiamine (vitamin B1) deficiency , commonly found in chronic alcoholics .
- The classic triad of symptoms in Wernicke's encephalopathy includes Encephalopathy (confusion), Oculomotor dysfunction (painful eye movements, nystagmus, ophthalmoplegia) Ataxia .
- Encephalopathy (confusion),
- Oculomotor dysfunction (painful eye movements, nystagmus, ophthalmoplegia)
- Ataxia .
- The 6th cranial nerve (which controls lateral movement of the eye) is also involved.
- Encephalopathy (confusion),
- Oculomotor dysfunction (painful eye movements, nystagmus, ophthalmoplegia)
- Ataxia .

Incorrect Options:

Option B - Korsakoff psychosis: Korsakoff psychosis is a chronic condition characterized by memory problems, confabulation, and personality changes.

Option C - Delirium tremens: Delirium tremens is a severe form of alcohol withdrawal that involves sudden and severe mental or nervous system changes. It is characterized by confusion, rapid heartbeat, and fever.

Option D - De Clerambault syndrome: De Clerambault syndrome is a delusional disorder in which a person believes that another person, typically of higher social status, is in love with them.

Solution for Question 25:

Correct Option C - Dream-like state:

- The correct answer is the dream-like state. It refers to feeling in a dream or altered state of consciousness, often described as feeling "stoned" or "high." It's important to note that the effects of cannabis can vary widely depending on the individual, the amount consumed, and the method of consumption (e.g., smoking, edibles, etc.). Some people may experience different or additional symptoms, while others may not experience any noticeable effects at all.

The correct answer is the dream-like state. It refers to feeling in a dream or altered state of consciousness, often described as feeling "stoned" or "high." It's important to note that the effects of cannabis can vary widely depending on the individual, the amount consumed, and the method of consumption (e.g., smoking, edibles, etc.). Some people may experience different or additional symptoms, while others may not experience any noticeable effects at all.

Incorrect Options:

- Option A - Anxiety: While some people may experience anxiety or paranoia after using cannabis, this is not the most common symptom. In fact, some strains of cannabis are known for their calming and anxiety-reducing effects.
- Option B - Depressed mood: Cannabis can have mood-altering effects, but a depressed mood is not the most common symptom. In some cases, cannabis may actually improve mood and alleviate symptoms of depression.
- Option D - Decreased appetite: Cannabis is known for causing "the munchies," or an increase in appetite rather than a decrease. While some strains may have more appetite-suppressing effects, this is not the most common symptom.

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1. Which of the following statements regarding treatments for Autism Spectrum Disorder (ASD) is correct?

- A. Early Start Denver Model (ESDM) primarily focuses on role-play activity for children with ASD.
- B. Social Skills Training involves only individual therapy sessions where children practice social skills face-to-face with a therapist.
- C. Treatment and Education of Autistic and Communication-Related Handicapped Children (TEACCH) employs structured teaching methods to support learning in children with ASD.
- D. Pivotal Response Training emphasizes the use of pharmacological interventions to enhance communication skills in children with ASD.

2. A 10 year old girl is brought to the clinic with concerns about her frequent eye-blinking, shoulder-shrugging and vocalizations that started at age 5. Her symptoms have worsened over the past year, leading to social difficulties and bullying at school. What is the most appropriate initial treatment approach for this child?

- A. Cognitive Behavioral Therapy
- B. Psychoeducation for Family and School
- C. Typical Antipsychotics like Haloperidol
- D. Noradrenergic Agents like Clonidine

3. A 9 year old girl has repetitive shoulder-shrugging and grunting sounds that occur daily. These actions are involuntary but she can suppress them for short

periods. Which of the following best describes her tics?

- A. Simple Motor and Simple Vocal Tics
 - B. Complex Motor and Simple Vocal Tics
 - C. Simple Motor and Complex Vocal Tics
 - D. Complex Motor and Complex Vocal Tics
-

4. A 9 year old boy has frequent temper outbursts at home and school that are out of proportion to the situation, occurring at least three times per week. Between outbursts, he is persistently irritable and has difficulty maintaining friendships. Which of the following is the most likely diagnosis?

- A. Oppositional Defiant Disorder
 - B. Conduct Disorder
 - C. Disruptive Mood Dysregulation Disorder
 - D. Attention Deficit Hyperactivity Disorder
-

5. A 9 year old boy frequently argues with his parents and teachers, often refuses to follow rules, and deliberately annoys his classmates. He seems angry most of the time but has no history of physical aggression or breaking the law. What is the most likely diagnosis?

- A. Conduct Disorder
 - B. Oppositional Defiant Disorder
 - C. Attention-Deficit/Hyperactivity Disorder
 - D. Disruptive Mood Dysregulation Disorder
-

6. A multidisciplinary team is reviewing treatment protocols for a 10 year old child diagnosed with intellectual disability. Which of the following does NOT form part of the treatment protocols for mental retardation?

- A. Screening for PKU and dietary management
 - B. Public education on avoiding alcohol during pregnancy
 - C. Risperidone to enhance cognitive functioning
 - D. Family education and counselling
-

7. A 12 year old boy is brought to a clinic for evaluation. His parents report that he has attended school only up to 2nd class and struggles with basic tasks such as dressing and feeding himself. The child is often unable to communicate effectively and requires constant supervision. He has an IQ of 40 and a mental age of approximately 5 years. What is the most likely diagnosis?

- A. Mild mental retardation
 - B. Moderate mental retardation
 - C. Severe mental retardation
 - D. Profound mental retardation
-

8. A 15 year old boy has been assessed for cognitive abilities, and his mental age is determined to be 9 years. What is the IQ based on the provided information?

- A. 60
 - B. 75
 - C. 90
 - D. 100
-

9. A 10 year old girl has always loved school but is now struggling academically. Her teacher notes that she confuses similar-looking letters when reading and has trouble decoding words. She often asks for help with reading and feels frustrated with writing tasks. Her writing is messy and often lacks punctuation. Although she can do basic math, she struggles to remember mathematical signs and makes careless calculation errors. Which specific learning disorder is she most likely experiencing?

- A. Specific learning disorder with impairment in reading
- B. Specific learning disorder with impairment in written expression
- C. Specific learning disorder with impairment in mathematics
- D. Mixed disorder of scholastic skills

10. A 7 year old boy diagnosed with Autism Spectrum Disorder presents with significant irritability, including frequent temper tantrums and aggressive behaviours towards others. His parents report that these behaviours have led to difficulties at home and school. Which of the following pharmacological treatments would be most appropriate for this child's irritability?

- A. Methylphenidate
- B. Risperidone
- C. Clonidine
- D. Fluoxetine

11. A 6 year old boy is brought to the clinic due to concerns about his inability to stay seated during class. His teacher notes that he often runs around the classroom during lessons, disrupting other students by talking out of turn and blurting out answers. At home, his parents observe that he struggles to wait for his turn when playing games and talks excessively. He does not show signs of forgetfulness or distractibility. What is the most likely diagnosis?

- A. ADHD - Combined Presentation
 - B. ADHD - Predominantly Hyperactive/Impulsive Presentation
 - C. ADHD - Predominantly Inattentive Presentation
 - D. Conduct Disorder
-

12. A 4 year old boy is brought to the clinic because his parents have noticed a significant regression in his abilities over the past year. He previously spoke in full sentences, engaged in social play, and was potty trained but has now lost these skills. He exhibits behaviours resembling those seen in autism. Which of the following conditions is the most likely diagnosis?

- A. Asperger's Syndrome
 - B. Heller's Syndrome
 - C. Tourette Syndrome
 - D. Rett Syndrome
-

13. A 2 year old girl is brought to the paediatrician due to concerns about her development. Initially, she was meeting her developmental milestones but has started to lose skills such as spoken language and purposeful hand movements. O/E, she demonstrates facial grimacing and stereotypical hand-wringing movements. What is the most likely diagnosis?

- A. Autism Spectrum Disorder
 - B. Rett Syndrome
 - C. Cerebral Palsy
 - D. Global Developmental Delay
-

14. Which of the following is NOT a common behavioural symptom associated with Autism Spectrum Disorder?

- A. Repetitive use of toys
 - B. Early development of a social smile
 - C. Delays in language development
 - D. Strong attachment to inanimate object
-

15. A 5 year old child with known Autism Spectrum Disorder (ASD) exhibits difficulties in understanding social cues and rarely participates in group activities at school. The teacher reports that he often plays by himself, lining up toys in a particular order, and shows little interest in playing with others. What core symptom of ASD is most likely causing his difficulties in social interaction?

- A. Impaired Theory of Mind
 - B. Resistance to Change
 - C. Delayed Motor Skills
 - D. Excessive Aggression
-

16. A 10 year old boy with a history of Attention-Deficit/Hyperactivity Disorder presents with tics that have developed over the past few months. The tics are moderate in severity but interfere with his social interactions and academic performance. Which of the following medications is the most appropriate treatment for managing his ADHD while minimizing the risk of exacerbating the tics?

- A. Methylphenidate
 - B. Amphetamine
 - C. Modafinil
 - D. Guanfacine
-

17. A 9 year old boy is brought to the clinic by his parents due to difficulties in school. After a thorough evaluation, the diagnosis of ADHD is confirmed. The parents are concerned about medication and ask about the first line of treatment. What is the most effective first-line treatment for managing this child?

- A. Stimulant medications
 - B. Parent training programs
 - C. Atomoxetine
 - D. Cognitive Behavioral Therapy
-

18. Which of the following is a known contributing factor to the development of ADHD?

- A. Dysfunction of the serotonin system
 - B. Familial genetic links, with a heritability of about 75%
 - C. Increased activity in the prefrontal cortex
 - D. Increased activity of norepinephrine
-

19. A 7 year old boy is referred for evaluation due to behavioural concerns. He frequently interrupts others, struggles to stay seated during class, and exhibits difficulty concentrating on tasks. His teacher completes a standardised ADHD rating scale. Which of the following is most useful in confirming an ADHD diagnosis?

- A. Symptom duration of at least three months
- B. Symptoms persist after the age of 18
- C. Symptoms impair functioning in multiple settings
- D. Symptoms first appear after age 12

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Correct Answers

Question	Correct Answer
Question 1	3
Question 2	2
Question 3	1
Question 4	3
Question 5	2
Question 6	3
Question 7	2
Question 8	1
Question 9	4
Question 10	2
Question 11	2
Question 12	2
Question 13	2
Question 14	2
Question 15	1
Question 16	4
Question 17	1
Question 18	2

Solution for Question 1:

Correct Answer: C) Treatment and Education of Autistic and Communication-Related Handicapped Children (TEACCH) employs structured teaching methods to support learning in children with ASD.

Explanation:

TEACCH uses structured teaching methods, including visual aids, to facilitate learning and social interactions.

ASD Treatment:

- Improve social interaction skills.
- Enhance communication abilities.
- Develop strategies for successful school integration.
- Foster meaningful peer relationships.
- Increase long-term independence in daily living.

Non-Pharmacological Treatment of Autism Spectrum Disorders (ASD):

Psychosocial Interventions:

A) Early Intensive Behavioural and Developmental Interventions:

Model	Description
UCLA/LOVAAS-Based Model	Uses applied behaviour analysis (ABA) through one-on-one sessions focusing on specific skills.
Early Start Denver Model (ESDM)	Emphasises intervention in natural settings, involving parents and educators in daily routines (Option A) .
Parent Training Approaches	Pivotal Response Training: Teaches parents to facilitate social and communication skills through daily activities (Option D) .

B) Social Skills Approaches: Includes social skills training. Therapists lead group sessions to practise social skills and emotion identification (Option B) .

- Behavioural Therapy: Effective in reducing repetitive behaviours; often combined with medication.
- Cognitive Behavioural Therapy (CBT): Helpful for anxiety, depression and OCD.
- TEACCH (Treatment and Education of Autistic and Communication-Related Handicapped Children) involves structured teaching methods using visual aids and controlled environments to promote learning and enhance social skills (Option C) .
- Broad-Based Approaches : Combine various teaching methods, incorporating behavioural analysis and language remediation.

Computer-Based and Virtual Reality Approaches: Computer programs are used to teach language and reading skills, such as the "Let's Face It!" program.

Solution for Question 2:

Correct Answer: B) Psychoeducation for Family and School

Explanation:

The child is likely experiencing Tourette Syndrome, characterised by the presence of both motor (eye-blinking, shoulder-shrugging) and vocal tics that began before age 18. Psychoeducation is an appropriate initial treatment approach as it helps the family and school understand the condition, reducing stress and misinterpretations, which can improve the child's social interactions.

Tourette Syndrome

- Definition : Neuropsychiatric disorder characterized by motor and vocal tics, usually emerging in childhood.
- Onset & Course : Age: Tics begin between 4-6 years, and peak at 10-12 years. Waxing and waning tics, with possible remission periods.
- Age: Tics begin between 4-6 years, and peak at 10-12 years.
- Waxing and waning tics, with possible remission periods.
- Pathophysiology : Basal ganglia dysfunction, particularly dopaminergic transmission in cortico-striatothalamic circuits.
- Diagnosis : Both motor and vocal tics for over 1 year, onset before 18. Motor tics typically start in the face & neck, and progress downward.

- Both motor and vocal tics for over 1 year, onset before 18.
- Motor tics typically start in the face & neck, and progress downward.
- Indications for Treatment : Severe tics causing social or emotional issues, physical discomfort, or academic problems.
- Age: Tics begin between 4-6 years, and peak at 10-12 years.
- Waxing and waning tics, with possible remission periods.
- Both motor and vocal tics for over 1 year, onset before 18.
- Motor tics typically start in the face & neck, and progress downward.

Treatment	
Psychoeducation	Educating families and schools to reduce stress and avoid misinterpretation of tics.
Behavioural Therapy	Habit Reversal Training (HRT): Most effective and currently the first-line behavioural treatment for tic disorders Exposure and Response Prevention (ERP): Prolonged tic suppression.
Pharmacotherapy	Atypical Antipsychotics: Risperidone (most studied), Aripiprazole (fewer side effects). Typical Antipsychotics : Haloperidol, Pimozide (higher risk of side effects). Noradrenergic Agents : Clonidine, Guanfacine (modest efficacy, useful with ADHD). Other: Tetrabenazine, Topiramate, THC (understudy)
Supportive Approaches	Relaxation training, biofeedback, SSRIs for OCD.

Cognitive Behavioral Therapy (CBT) (Option A): While CBT can be helpful, the initial approach should focus on education to manage misconceptions about tics.

Typical Antipsychotics like Haloperidol (Option C): These are generally reserved for cases where tics cause significant distress, making them a secondary option after psychoeducation.

Noradrenergic Agents like Clonidine (Option D) : These agents can be considered later, especially if there is comorbid ADHD, but they are not the first line in this scenario.

Solution for Question 3:

Correct Answer: A) Simple Motor and Simple Vocal Tics

Explanation:

Shoulder-shrugging (simple motor) and grunting (simple vocal) are both rapid, brief, and uncomplicated tics.

Tic Disorders and Types of Tics

- Definition : Tic disorders are neuropsychiatric conditions marked by brief, rapid involuntary motor movements or vocalizations, often triggered by irresistible urges.
- Pathophysiology : Involves dysfunction in the basal ganglia and disrupted dopaminergic transmission within the cortico-striatothalamic circuits.
- Age of Onset : Tics typically emerge between ages 5-6, peaking in severity from ages 10-12.
- Course : Tics may be transient or chronic, exhibiting a waxing and waning pattern.

Types of Tics :

Motor Tics	Involuntary movements
Simple Motor Tics	Rapid contractions of specific muscle groups (e.g., eye-blinking, head-jerking, shoulder-shrugging).
Complex Motor Tics (Options B and D)	More purposeful movements (e.g., grooming behaviours, jumping, touching objects).
Vocal Tics	Involuntary Vocalizations
Simple Vocal Tics	Basic sounds (e.g., throat-clearing, grunting, sniffing), echopraxia (imitation of observed behaviour), and copropraxia (display of obscene gestures).
Complex Vocal Tics (Option C)	Elaborate phrases or behaviours (e.g., coprolalia - involuntary use of obscene words; palilalia - repeating one's own words; echolalia - repeating the last heard words from others).

Solution for Question 4:

Correct Answer: C) Disruptive Mood Dysregulation Disorder

Explanation:

The scenario presents frequent severe outbursts with chronic irritability between episodes, a hallmark of Disruptive Mood Dysregulation Disorder

Disorder	Disruptive Mood Dysregulation Disorder (DMDD)
Definition	Severe, developmentally inappropriate temper outbursts with a persistently irritable or angry mood. Chronic disorder, with a high risk of progression to major depressive disorder, dysthymia, and anxiety.
Diagnostic Criteria (DSM-5)	Duration: ≥ 12 months, with no symptom-free period exceeding 3 months. Outbursts: Occur ≥ 3 times/week, verbal or physical, out of proportion to the situation. Age: Symptoms start before age 10 but diagnosis is not made before age 6 or after 18. Mood Between Outbursts: Persistently irritable/angry, observable by others. Settings: Symptoms present in at least two settings (home, and school).
Epidemiology	More common in males (78%) than females (22%). Mean age of onset: 5 to 11 years.
Clinical Features	Verbal or physical aggression towards people or property. They have persistently irritable moods between outbursts. Impacts social relationships and school performance.
Treatment	Pharmacologic: SSRIs and stimulants (if pathophysiology resembles unipolar depression or anxiety). Atypical antipsychotics and mood stabilizers (if resemble bipolar disorder). Psychosocial Interventions: Cognitive-behavioral therapy (CBT), behavioural psychotherapy.

Oppositional Defiant Disorder (Option A): ODD involves defiance and irritability but lacks the explosive temper outbursts characteristic of DMDD.

Conduct Disorder (Option B): Conduct disorder involves aggression and violation of others' rights, which are not mentioned in this case.

Attention Deficit Hyperactivity Disorder (Option D): ADHD is characterised by inattention, hyperactivity, and impulsivity, but not the frequent, severe outbursts seen in DMDD.

Solution for Question 5:

Correct Answer: B) Oppositional Defiant Disorder

Explanation:

The clinical scenario of frequent arguments, refusal to follow rules, and deliberate annoyance, without severe aggression or rule violations, point to a diagnosis of Oppositional defiant disorder.

Disorder	Oppositional Defiant Disorder (ODD)	Conduct Disorder
Definition	The persistent pattern of negativistic, defiant, disobedient, and hostile behaviour toward authority figures.	A persistent pattern of behaviour involving aggression, violating others' rights, and disregarding social norms.
Diagnostic Criteria (DSM-5)	Duration: ≥ 6 months. Symptoms (≥ 4): Losing temper. Easily annoyed/sensitive. Angry/resentful. Argues with authority figures. Refuses requests/rules. Deliberately annoys others. Blames others for their own mistakes/behaviors. Spiteful/vindictive (at least 2x in 6 months). No severely aggressive or delinquent behaviours.	Duration : ≥ 12 months, with 1 symptom present in the last 6 months. Symptoms (≥ 3) : Physical aggression or threats. Destruction of property. Deceitfulness or theft. Harming animals. Violation of rules (e.g., truancy, running away).
Types (DSM-5)	Angry/Irritable Mood: Frequent temper loss, persistent irritability. Argumentative/Defiant Behavior: Argues, refuses to comply, blames others. Vindictiveness: Spiteful/vindictive actions ≥ 2 x in 6 months.	Childhood-onset : Symptoms before age 10. Adolescent-Onset : Symptoms after age 10. Unspecified-Onset : Age of onset unknown
Severity Specifiers	Mild: Symptoms in 1 setting. Moderate: Symptoms in 2 settings. Severe: Symptoms in ≥ 3 settings.	Mild : Few conduct problems, minor harm. Moderate : Intermediate symptoms. Severe : Many symptoms, significant harm caused.

Disorder	Oppositional Defiant Disorder (ODD)	Conduct Disorder
Treatment	Family Interventions: Parent training in child management, reinforcing positive behaviour. Cognitive Behavioral Therapy (CBT): Parents are taught to discourage the child's oppositional behaviour by not paying much attention to it. Parenting Techniques: Focus on reducing harsh punishment, and increasing positive interactions.	Psychosocial Interventions : Early intervention, CBT, school-based programs. Pharmacologic Interventions : Antipsychotics (risperidone, quetiapine), SSRIs for impulsivity/irritability, sodium valproate.

Conduct Disorder (Option A) : Does not fit, as there is no evidence of severe aggression or serious rule-breaking behaviour.

Attention-Deficit/Hyperactivity Disorder (Option C) : No signs of inattention or hyperactivity are mentioned, which are key in ADHD.

Disruptive Mood Dysregulation Disorder (DMDD) (Option D) : Lacks the severe, recurrent temper outbursts and mood dysregulation typical of DMDD.

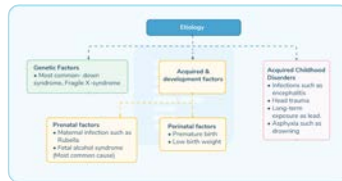
Solution for Question 6:

Correct Answer: C) Risperidone to enhance cognitive functioning

Explanation:

There are no specific medications proven to enhance cognitive functioning in individuals with intellectual disabilities, making this option not part of standard treatment protocols. However, symptomatic treatment for associated features can be given like Risperidone (Antipsychotic) for rage, aggression, and self-injurious behaviours and SSRIs for depression.

Aetiology of Mental retardation:



Treatment:

1. Prevention

- Primary Prevention involves actions to eliminate or reduce conditions that lead to intellectual disability, such as screening for PKU and providing a low phenylalanine diet (Option A) . Public education on avoiding alcohol during pregnancy (Option B) . Improving public health policies. Genetic counselling for families with a history of genetic disorders.
- Public education on avoiding alcohol during pregnancy (Option B) .
- Improving public health policies.
- Genetic counselling for families with a history of genetic disorders.
- Secondary Prevention focuses on promptly addressing medical and psychiatric complications to diminish their impact on individuals with intellectual disabilities.
- Tertiary Prevention aims to minimise the long-term consequences of intellectual disabilities through effective early treatments for hereditary disorders like PKU and hypothyroidism, using dietary control or hormone replacement therapy.
- Public education on avoiding alcohol during pregnancy (Option B) .
- Improving public health policies.
- Genetic counselling for families with a history of genetic disorders.

2. Pharmacotherapy

- For symptomatic treatment.
- Antipsychotics like aripiprazole and risperidone for aggression, rage, irritability, and self-injurious behaviour.
- Depression- Selective serotonin reuptake inhibitors (SSRIs) .

3. Psychosocial Interventions

- Educational Interventions: Comprehensive programs for academics, adaptive, social, and vocational skills. Focus on communication and quality of life.
 - Comprehensive programs for academics, adaptive, social, and vocational skills.
 - Focus on communication and quality of life.
 - Behavioural and Cognitive-Behavioral Interventions: Use of behaviour therapy and positive reinforcement. Cognitive therapy for false beliefs and relaxation. Psychodynamic therapy for family conflicts and anxiety.
 - Use of behaviour therapy and positive reinforcement.
 - Cognitive therapy for false beliefs and relaxation.
 - Psychodynamic therapy for family conflicts and anxiety.
 - Family Education (Option D) : Educate families on enhancing competence and managing expectations. Provide ongoing counselling and support. Share current medical information and resources.
 - Educate families on enhancing competence and managing expectations.
 - Provide ongoing counselling and support.
 - Share current medical information and resources.
 - Social Intervention: Improve social skills and reduce isolation. Programs like the Special Olympics enhance fitness, friendships, and self-esteem.
 - Improve social skills and reduce isolation.
 - Programs like the Special Olympics enhance fitness, friendships, and self-esteem.
 - Comprehensive programs for academics, adaptive, social, and vocational skills.
 - Focus on communication and quality of life.
 - Use of behaviour therapy and positive reinforcement.
 - Cognitive therapy for false beliefs and relaxation.
 - Psychodynamic therapy for family conflicts and anxiety.
 - Educate families on enhancing competence and managing expectations.
 - Provide ongoing counselling and support.
 - Share current medical information and resources.
 - Improve social skills and reduce isolation.
 - Programs like the Special Olympics enhance fitness, friendships, and self-esteem.
-

Solution for Question 7:

Correct Answer: B) Moderate mental retardation

Explanation:

The child has an IQ of 40, which falls within the range of 35-49, indicating moderate mental retardation. He can be trained for semi-skilled or unskilled work but requires significant support.

Classification of mental retardation	
Mild mental retardation	Most common in nearly 85% of cases IQ of 50-69 (Option A) Can study up to 6th class. Educable Maximum mental age that can be achieved - 12 years Can achieve self-sufficiency
Moderate mental retardation	IQ of 35-49 Can study up to 2nd class Trainable Maximum mental age that can be achieved - 9 years Can be trained for semi-skilled or unskilled work
Severe mental retardation	IQ of 20-34 (Option C) No formal education Dependent Maximum mental age that can be achieved - 6 years Can perform simple tasks under close supervision
Profound mental retardation	Least common - 1-2% of cases. IQ < 20 (Option D) No formal education Need support in basic things such as eating, and using the washroom Maximum mental age that can be achieved - 3 years Need continuous supervision

Solution for Question 8:

Correct Answer: A) 60

Explanation:

The child's IQ calculated using the formula is 60 , indicating significant limitations in intellectual functioning.

- Mental Age: 9 years
- Chronological Age: 15 years

Intellectual Disability

- Significant limitations in both: Intellectual functioning –reasoning, learning, problem-solving. Adaptive behaviour – conceptual, social, and practical skills.
- Intellectual functioning –reasoning, learning, problem-solving.
- Adaptive behaviour – conceptual, social, and practical skills.
- Onset before 18 years.
- Intellectual functioning –reasoning, learning, problem-solving.
- Adaptive behaviour – conceptual, social, and practical skills.

Intelligence quotient

- Intelligence is usually measured by calculating IQ.
- IQ is an intelligence quotient.
- $IQ = (Mental\ age \div Chronological\ age) \times 100$ Chronological age - Age calculated from the date of birth. Mental age - The age of one's intellect. Various tests are done to calculate mental age.
- Chronological age - Age calculated from the date of birth.
- Mental age - The age of one's intellect.
- Various tests are done to calculate mental age.
- Normal IQ: 90-109
- Borderline IQ: 80-89
- Mental retardation: $IQ < 70$
- Chronological age - Age calculated from the date of birth.
- Mental age - The age of one's intellect.
- Various tests are done to calculate mental age.

Solution for Question 9:

Correct Answer: D) Mixed disorder of scholastic skills

Explanation:

The child exhibits impairments in reading (difficulty decoding words), written expression (messy writing, omitted punctuation), and mathematics (difficulty with signs and calculations). The combination of these challenges indicates a mixed disorder of scholastic skills.

Specific Learning Disorders

- Neurodevelopmental disorder beginning in early childhood
- Both heritable and environmental factors like lead poisoning, Fetal Alcohol Syndrome, and in-utero drug exposure are responsible.
- Persistent difficulty in learning academic skills such as reading, written expression or mathematics.
- Inconsistent with the intellectual ability of the child.
- The IQ range is usually normal.

Types of Specific Learning Disorders	
Specific learning disorder with impairment in reading (Dyslexia)	Identified in children aged 7-9 years. Difficulty in recognizing words and sounding out words. Slow, inaccurate reading. Poor comprehension. Difficulties with spelling.
Specific learning disorder with impairment in written expression (Specific Spelling Disorder/ Dysgraphia)	Poor writing skills. Poor spelling. Errors in grammar and punctuation. Poor handwriting.
Specific learning disorder with impairment in mathematics (dyscalculia)	Identified in children aged 6-10 years. Difficulty learning numerical. Difficulty remembering signs such as +/ -. Slow, inaccurate calculations. Poor linguistic, perceptual, mathematical, and attention skills.
Mixed disorder of scholastic skills	Combined impairment of reading, written expressions, and mathematics.

Specific learning disorder with impairment in reading (Option A), Specific learning disorder with impairment in written expression (Option B), and Specific learning disorder with impairment in mathematics (Option C) each focus on individual disabilities associated with reading, writing, and math, respectively, while this child exhibits difficulties across all three areas.

Solution for Question 10:

Correct Answer: B) Risperidone

Explanation:

Risperidone is FDA-approved for treating irritability in children with ASD and is commonly prescribed for this purpose.

Psychopharmacological Interventions of Autism Spectrum Disorders (ASD):

1. Medications for Irritability:

Medication	Description
Risperidone	FDA-approved Blocks dopamine and serotonin receptors Side effects: Weight gain, sleepiness.
Aripiprazole	FDA-approved It is a serotonin and dopamine antagonist (5HT _{2A} and D ₂), but a partial D ₂ agonist also. Effective for tantrums and aggression Side effects: Sedation and dizziness.
Olanzapine	Reduces irritability but is associated with significant weight gain.

2. Medications for Hyperactivity, Impulsivity, and Inattention:

Medication	Description
Methylphenidate (Ritalin)	Stimulant for ADHD and ASD; may increase irritability. (Option A)
Atomoxetine (Strattera)	Non-stimulants are effective for hyperactivity and inattention; side effects include sedation.

3. Medications for Repetitive and Stereotypic Behaviour:

Medication	Description
Antidepressants and Mood Stabilisers	Fluoxetine and risperidone may be used in managing repetitive behaviours. (Option D)

Medication	Description
Quetiapine, Clozapine, Ziprasidone	Used for aggression and self-injury; requires monitoring due to side effects.
Lithium	Used clinically for treating aggressive or self-injurious behaviours when antipsychotic medications are ineffective.

Clonidine (Option C) may help with hyperactivity but is not specifically a first-line treatment for irritability in children with ASD.

Solution for Question 11:

Correct answer: B) ADHD - Predominantly Hyperactive/Impulsive Presentation

Explanation:

The child shows significant hyperactivity and impulsivity, such as difficulty sitting still and talking out of turn, with no signs of inattention.

Subtypes of ADHD		
Combined Presentation (Option A)	Predominantly Inattentive Presentation (Option C)	Predominantly Hyperactive/Impulsive Presentation (Option B)
6 or more symptoms of both inattention and Hyperactivity-Impulsivity	6 or more symptoms of Inattention <6 symptoms of Hyperactivity-Impulsivity	6 or more symptoms of Hyperactivity/Impulsive subtype <6 symptoms of Inattention

Clinical Features of ADHD			
Hyperactivity	Attention Deficit	Impulsivity	Other

Clinical Features of ADHD			
Restlessness Difficulty remaining seated Inappropriate running and climbing Excessive talking Interrupting others	Short attention span Distractibility Difficulty following instructions Forgetfulness Disorganisation Procrastination	Acting without thinking Difficulty waiting for their turn Interrupting others Blurting out answers	Emotional lability Explosiveness Irritability Memory and thinking defects Specific learning difficulties Speech and Hearing deficits

Conduct Disorder (Option D): Conduct Disorder is associated with aggressive behaviour and rule violations, which are not observed here.

Solution for Question 12:

Correct Answer: B) Heller's Syndrome

Explanation:

- Heller's Syndrome is characterised by marked regression in multiple areas of development after at least two years of normal development , typically between the ages of 3 and 4.

Heller's Syndrome: Also known as childhood disintegrative disorder or disintegrative psychosis classified under Autism Spectrum Disorder (ASD) in DSM-5.

Characteristics:

- Marked regression in multiple developmental areas after at least two years of normal development.
- It typically occurs between 3 and 4 years of age.

Affected Developmental Domains :

Domain	Description
Language skills	Significant loss of previously acquired language abilities
Social responsiveness	Deterioration in social interactions
Play	Loss of interest or ability to engage in play

Domain	Description
Motor skills	Regression in gross and fine motor skills
Bladder and bowel control	Loss of previously established control

- After regression, children closely resemble those with autistic disorder.

Asperger's Syndrome:

- No longer a separate diagnosis in DSM-5; now classified under ASD.
- Characterised by: Impaired social relatedness. Repetitive and stereotyped behaviours. No significant language delays or abnormalities. Cognitive development is normal.
- Impaired social relatedness.
- Repetitive and stereotyped behaviours.
- No significant language delays or abnormalities.
- Cognitive development is normal.
- Impaired social relatedness.
- Repetitive and stereotyped behaviours.
- No significant language delays or abnormalities.
- Cognitive development is normal.

Asperger's Syndrome (Option A) does not typically involve regression of skills, as it is a form of autism without significant delays in language or cognitive development.

Tourette Syndrome (Option C): This condition involves the presence of motor and vocal tics and does not involve regression of previously acquired developmental skills.

Rett syndrome (Option D) primarily affects girls and usually presents with stereotypical hand-wringing and loss of motor function, not seen in this case.

Solution for Question 13:

Correct Answer: B) Rett Syndrome

Explanation:

Rett Syndrome is characterised by normal early development followed by a regression in skills, particularly communication and motor skills, along with stereotypic hand movements.

Rett Syndrome:

- Rett Syndrome is now classified as a form of Autism Spectrum Disorder (ASD) in the DSM-5.
- A degenerative condition caused by a dominant X-linked gene (hence seen in females > males) due to mutations in the MECP2 gene.

Developmental Progression (Option B):

- Initial development appears normal in infants till around 6 months of age.
- After that, there is a loss of acquired skills (e.g., purposeful hand movements, spoken language) often replaced by unusual behaviours (e.g., stereotypical hand-wringing).

Symptoms of Rett Syndrome:

- Ataxia
- Facial grimacing & Teeth grinding
- Deterioration of verbal communication
- Breathing Issues: eg. hyperventilation, apnea, and breath-holding
- Stereotypical Hand Movements : Eg. hand-wringing and hand-washing behaviours.
- Gait disturbance leading to the development of scoliosis
- Seizures (in 75% of the cases)
- Muscle tone abnormalities: Evolution from hypotonia to spasticity and rigidity
- Cerebral atrophy

Management

- Trofinetide is a tripeptide approved by the FDA for Rett syndrome.

Long-term Prognosis - Generally poor prognosis:

- May live up to 10 years post-symptom onset.
- The developmental level typically stalls at that of a child under one-year-old.
- Usually wheel-chair bound.

Autism Spectrum Disorder (ASD) (Option A) : While Rett Syndrome is now classified under ASD, it specifically involves a distinct pattern of regression and additional symptoms not typical of ASD alone.

Cerebral Palsy (Option C) : This condition primarily involves motor impairment without the specific regression and stereotypical movements seen in Rett Syndrome.

Global Developmental Delay (Option D) refers to delays across multiple developmental domains but doesn't typically include the loss of previously acquired skills.

Solution for Question 14:

Correct Answer: B) Early development of a social smile

Explanation:

Children with ASD often lack a social smile during infancy, which indicates deficits in social communication.

Physical Symptoms of Autism:

- Higher rates of minor physical anomalies (e.g., ear malformations).
- Delayed or lack of handedness and lateralization (prolonged ambidexterity).
- Abnormal dermatoglyphics (unusual fingerprint patterns).
- Macrocephaly in >20% of patients (large head size develops by 5 years of age).

Behavioural Symptoms of Autism:

Symptom Category	Examples
Social Communication and Interaction Deficits	Lack of social smile, poor eye contact (Option B) Atypical attachment behaviour Difficulty with non-verbal communication Challenges in forming and maintaining relationships Awkward or inappropriate social behaviour
Restricted and Repetitive Behaviour Patterns	Limited exploratory play Repetitive toy use (Option A) Rigid, monotonous activities Strong attachment to objects (Option D) Self-stimulatory or self-injurious behaviours
Language Development Disturbances	Delays in language development (Option C) Minimal or absent babbling Echolalia (repeating words or phrases) Pronoun reversals Peculiar voice quality and rhythm Hyperlexia (early reading without comprehension)
Intellectual Disability	Approximately 30% of children with ASD have intellectual disabilities

Symptom Category	Examples
Irritability	Aggression, self-injurious behaviours, severe temper tantrums Sudden, unexplained mood swings
Atypical Response to Sensory Stimuli	Over- or under-responsiveness to sensory input such as sound or pain
Hyperactivity and Inattention	Common in young children with ASD
Precocious Skills	Exceptional abilities in specific areas such as rote memory, calculation, or music

Solution for Question 15:

Correct answer: A) Impaired Theory of Mind

Explanation:

The child in this case struggles with social cues and prefers to play alone, which aligns with the social deficits linked to an impaired theory of mind. This inability to understand how others are feeling or what they are thinking makes it challenging for them to engage in typical social interactions, resulting in poor social reciprocity and difficulty in making friends.

Core Symptoms of Autism Spectrum Disorder (ASD)	
Deficits in Social Communication and Interaction	Restricted and Repetitive Patterns of Behaviour
Reciprocal Social Skills : Difficulty in engaging in back-and-forth social exchanges, may lack typical gestures and eye contact. Attachment Behaviours : Exhibits atypical responses to familiar individuals, showing less emotional expression. Social Cues and Friendships : Difficulty understanding social norms, leading to awkward interactions and trouble making friends. Impaired Theory of Mind: Struggles to recognize or interpret the emotions and intentions of others, hindering empathy and social relationships.	Atypical Play Patterns : Engages in repetitive activities rather than imaginative play, and may show rigidity in how toys are used. Ritualistic Behaviours : Tends to perform repetitive actions (e.g., lining up objects) and may form strong attachments to specific items. Resistance to Change : Experiences distress during changes in routine, which can result in anxiety or temper outbursts. (Option B)

Aetiology of Autism Spectrum Disorder (ASD):

- Genetic Factors : Highly heritable with higher prevalence in families and monozygotic twins. Specific genes implicated (e.g., related to serotonin and synaptic plasticity).
- Highly heritable with higher prevalence in families and monozygotic twins.
- Specific genes implicated (e.g., related to serotonin and synaptic plasticity).
- Environmental Factors : Prenatal and perinatal complications (e.g., advanced parental age, low birth weight). Potential link to hypoxia affecting brain development.
- Prenatal and perinatal complications (e.g., advanced parental age, low birth weight).
- Potential link to hypoxia affecting brain development.
- Other Factors : Possible immunologic factors, though not fully confirmed.
- Highly heritable with higher prevalence in families and monozygotic twins.
- Specific genes implicated (e.g., related to serotonin and synaptic plasticity).
- Prenatal and perinatal complications (e.g., advanced parental age, low birth weight).
- Potential link to hypoxia affecting brain development.

Delayed Motor Skills (Option C): Although some children with ASD may have motor skill delays, such as difficulties with coordination or fine motor tasks, the child's main issue in this case is related to social interaction, not motor development.

Excessive Aggression (Option D): Aggressive behaviour can occasionally be observed in children with ASD, but it is not a core feature of the disorder. The child described in this scenario shows social withdrawal and a preference for solitary play, which are more indicative of impaired social communication rather than aggression

Solution for Question 16:

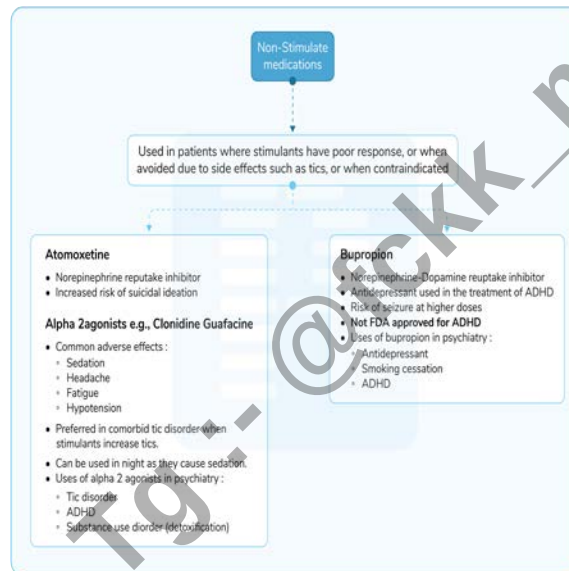
Correct Answer: D) Guanfacine

Explanation:

Alpha 2 agonists, e.g., Clonidine, Guanfacine, are FDA approved for the treatment of ADHD in children and adolescents.

- These are non stimulant medications .
- Their common adverse effects include: Sedation Headache Fatigue Hypotension
- Sedation

- Headache
- Fatigue
- Hypotension
- Preferred in ADHD and comorbid tic disorder when stimulants increase tics.
- Sedation
- Headache
- Fatigue
- Hypotension



Methylphenidate (Option A) is a CNS stimulant medication and is the drug of choice in ADHD.

- However, It can exacerbate tics and hence be avoided in children with comorbid tic disorders.

Amphetamine (Option B) is also a CNS stimulant medication which is the second drug of choice in ADHD.

- They can exacerbate tics, therefore avoided in such conditions.

Modafinil (Option C) is also a CNS stimulant medication used in the treatment of ADHD but it is not FDA approved.

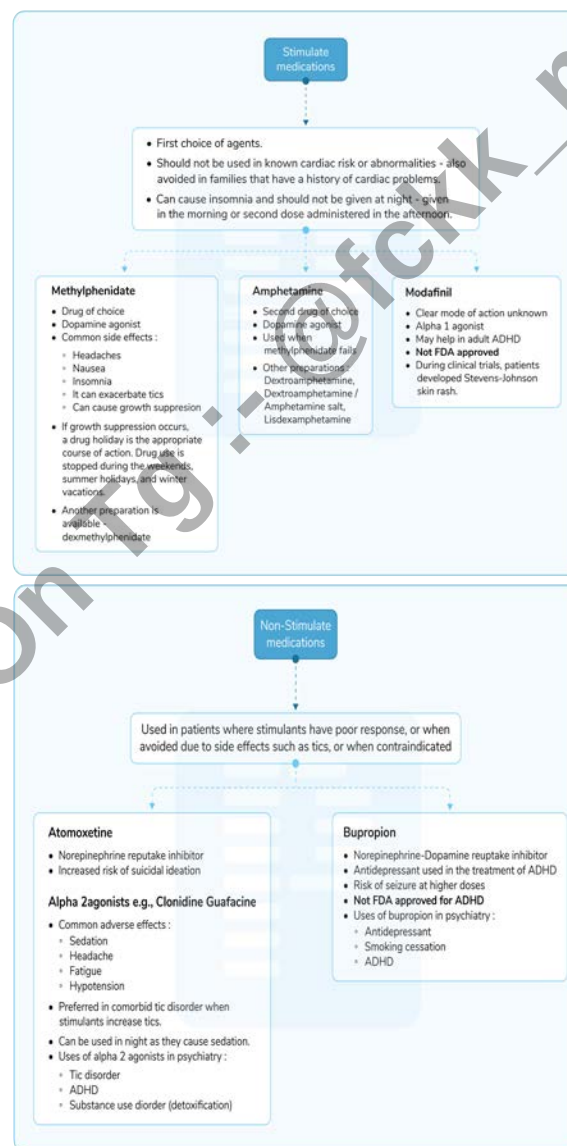
Solution for Question 17:

Correct answer: A) Stimulant medications

Explanation:

Stimulant medications, such as methylphenidate and amphetamines, are considered the first-line treatment for ADHD.

Pharmacologic treatment is considered the first line of treatment for ADHD. CNS stimulants are the first choice of agents in that they have the highest efficacy with generally mild tolerable side effects.



Parent training programs (Option B): While parent training helps manage ADHD symptoms, especially behavioural aspects, it is usually part of a broader treatment plan, not the first-line treatment.

Cognitive Behavioral Therapy (CBT) (Option D): CBT is beneficial for older children and adolescents, helping them develop coping strategies. Psychosocial interventions usually are activities involving paying attention, waiting for their turn, and following instructions. However, it is not the primary treatment in younger children with ADHD.

Solution for Question 18:

Correct answer: B) Familial genetic links, with a heritability of about 75%

Explanation:

Genetic factors play a major role in ADHD, with a heritability of about 75%. Genes like DAT1 and DRD4 are linked to the condition. Previously it was described as “minimal brain damage”.

Aetiology of ADHD			
Genetic Factors (Option B)	Neurochemical Factors	Neurophysiological and Neuroanatomical Factors	Developmental Factors Psychosocial Factors
ADHD has a strong genetic link, with a heritability of about 75%. Children with ADHD are more likely to have parents or siblings with the condition, indicating a familial pattern. DAT1 gene and DRD4 gene, are known to be associated with ADHD.	ADHD is linked to dopamine dysfunction. The noradrenergic system, which influences attention and alertness, is also believed to play a role. ADHD medications typically increase levels of dopamine and norepinephrine.	EEG usually shows theta and beta waves in the frontal brain- indicating inattention and mood swings respectively. Decreased activity in brain areas like the prefrontal cortex, anterior cingulate, and basal ganglia. (Option C)	Certain prenatal and perinatal factors, such as premature birth, maternal infections during pregnancy, and low birth weight, may increase the risk of ADHD. Severe abuse, neglect, or maltreatment can worsen or mimic ADHD symptoms.

Dysfunction of the serotonin system (Option A): The dopamine system is indicated in ADHD, not the serotonin. ADHD medications work to increase the levels of Dopamine in the patient

Increased activity of norepinephrine (Option D): Norepinephrine activity is usually decreased in patients with ADHD.

Solution for Question 19:

Correct answer: C) Symptoms impair functioning in multiple settings

Explanation:

For an ADHD diagnosis to be confirmed, symptoms must be present in at least two different environments, such as both school and home. This ensures that the behaviour is not limited to one specific context and indicates a broader issue with attention and self-regulation.

Symptom duration of at least three months (Option A): The diagnosis of ADHD requires that symptoms of inattention, hyperactivity, or impulsivity persist for at least six months, according to standard diagnostic guidelines.

Symptoms persist after the age of 18 (Option B): While ADHD can continue into adulthood, the diagnosis is based on symptoms that first emerge before the age of 12. Persisting into adulthood is not a requirement for diagnosis, though it can affect long-term management.

Symptoms first appear after age 12 (Option D): According to diagnostic criteria, ADHD symptoms must be observed before age 12. If symptoms emerge after this age, they may be attributable to other factors or conditions.

Previous Year Questions

1. Which of the following is not included in diagnosing autism spectrum disorder?

- A. Abnormalities in social communication
- B. Cognitive dysfunction
- C. Abnormality in socialization
- D. Restricted pattern of behaviour and interest

2. Which of the following is a selective norepinephrine reuptake inhibitor used in the management of ADHD?

- A. Modafinil
- B. Reboxetine
- C. Guanfacine
- D. Methylphenidate

3. A child with untreated ADHD is most likely to develop which of the following disorders during adolescence?

- A. Binge eating
- B. Selective mutism
- C. Separation from parent leading to depression
- D. Conduct disorder

4. Which of the subsequent diseases is solely observed in females?

- A. Rett syndrome
 - B. Asperger syndrome
 - C. Selective mutism
 - D. Atypical autism
-

5. Which of the following is a new drug approved for Rett syndrome?

- A. Trofinetide
 - B. Ambien
 - C. Lunesta
 - D. GHB opioids
-

6. According to the revision of DSM-5 by the American Psychiatric Association, what is the current term used to refer to 'mental retardation'?

- A. Feeble mindedness
 - B. Mental subnormality
 - C. Intellectual disability
 - D. Intellectually challenged
-

7. Which disorder was previously known as Minimal brain dysfunction?

- A. Dyslexia
- B. Attention deficit/hyperactivity disorder
- C. Mental subnormality

D. Oligophrenia

8. The parents of a 13-year-old boy have sought the assistance of a child psychiatrist due to their concerns. They have observed that the child frequently engages in dishonesty both at home and in school, displays disruptive behavior towards his teacher, often intimidates younger children, and has been caught stealing money multiple times. What is the probable diagnosis for this child?

- A. Oppositional defiant disorder
 - B. Conduct disorder
 - C. Antisocial personality disorder
 - D. Disruptive mood dysregulation
-

9. What could be the diagnosis for a child who exhibits below-average mathematical skills despite having normal IQ and age-appropriate learning abilities, and remains unaffected by a year-long private mathematics tutoring?

- A. ADHD
 - B. Mathematics learning disorder
 - C. Aspergers syndrome
 - D. Specific learning disorder
-

10. A 6 year old boy was presented in the psychiatric OPD with complaints of inattention at school, hitting his friends and excess talking, poor academic performance and cannot sit in a place for a long time. What drug is used in the management of this disorder?

- A. Methylphenidate

- B. Modafinil
 - C. Atenolol
 - D. Amitriptyline
-

11. A patient is showing stubbornness, rigidity and frugality. In which stage we can see this kind of behaviour due to fixation according to psychosexual stages of development?

- A. Anal
 - B. Oral
 - C. Phallic
 - D. Genital
-

12. What is the term used to describe the phenomenon where a patient in the ward responds to the question "what is your name?" by repeating the same sentence?

- A. Coprolalia
 - B. Echopraxia
 - C. Copropraxia
 - D. Echolalia
-

13. What is the likely diagnosis for a 10-year-old child experiencing selective mutism?

- A. Childhood depression
- B. Childhood psychosis
- C. Childhood anxiety disorder

D. Hyperkinetic disorder

14. What should be the treatment approach for a 7-year-old child who has been experiencing bed-wetting about twice a week for the past year, following a thorough investigation to exclude any underlying organic cause?

- A. Pharmacotherapy with imipramine
 - B. Psychodynamic psychotherapy
 - C. Bladder training, with reward for delaying micturition during daytime
 - D. Bell and pad based classification conditioning
-

15. The likely diagnosis for a girl who exhibits typical language milestones but primarily focuses on observing her own hands and lacks interaction with others is:

- A. ADHD
 - B. Autism
 - C. Asperger's syndrome
 - D. Rett's syndrome
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	4
Question 4	1
Question 5	1
Question 6	3
Question 7	2
Question 8	2
Question 9	4
Question 10	1
Question 11	1
Question 12	4
Question 13	3
Question 14	4
Question 15	3

Solution for Question 1:

Correct Option B - Cognitive dysfunction:

- Cognitive dysfunction is not a criterion for diagnosing ASD.
- While some individuals with ASD may have cognitive impairments, cognitive dysfunction is not a diagnostic criterion itself.
- ASD diagnosis focuses on social communication difficulties and restricted behaviors, not cognitive abilities.

Incorrect Options:

Option A - Abnormalities in social communication: This is included in diagnosing ASD.

Individuals with ASD typically have difficulties with social communication, such as challenges in understanding and using verbal and non-verbal communication.

Option C - Abnormality in socialization: This is included in diagnosing ASD. Abnormalities in socialization, such as difficulty in developing, maintaining, and understanding relationships, are a key component of ASD diagnosis.

Option D - Restricted pattern of behaviour and interest: This is included in diagnosing ASD. Individuals with ASD often exhibit restricted and repetitive patterns of behavior, interests, or activities. This is a core criterion for ASD diagnosis.

Solution for Question 2:

Correct Option B – Reboxetine:

- Reboxetine is a selective norepinephrine reuptake inhibitor (NRI) used in the management of Attention Deficit Hyperactivity Disorder (ADHD). However, it is not FDA-approved, so it is not used in India and is currently limited to Europe.

Incorrect Options:

Options A, C, and D are used in the management of ADHD but are not selective norepinephrine reuptake inhibitors.

Solution for Question 3:

Correct Option D - Conduct disorder:

- Children with untreated ADHD are at an increased risk of developing conduct disorder during adolescence.

Incorrect Options:

Options A, B, and C are less likely to develop in the adolescence of a child with untreated ADHD.

Solution for Question 4:

Correct Option A - Rett syndrome:

- Rett syndrome is exclusively seen in females and occurs due to a mutation of the MECP2 gene on the X chromosome. It is characterized by normal development for a period of 6 months, followed by stereotyped hand movements, loss of purposeful motions, poor coordination, decreased social engagement, breathing difficulty, loss of speech, and cognitive impairment.

Incorrect Options:

Option B - Asperger syndrome: Asperger's disorder is classified under autistic spectrum disorder seen in both men and women. It is characterized by impairment in social relatedness and repetitive behavior without delay in language development and usage.

Option C - Selective mutism is a social anxiety disorder characterized by a lack of speech in a child, especially in a school setting. The child may whisper but does not speak. However, they may speak when not in a socially anxiety-producing situation. It occurs in children between 4-5 years of age and is seen in both girls and boys; however, girls are more commonly affected.

Option D - Atypical autism: Impaired development becomes evident by 3 years or after 3 years. There are abnormalities in social interaction and restricted, repetitive patterns of behavior or activities. It is common among both men and women. But women with atypical autism exhibit different symptom patterns, making diagnosis extremely challenging.

Solution for Question 5:

Correct Option A - Trofinetide:

- Trofinetide is a medication that has shown promise in the treatment of Rett syndrome. Rett syndrome is a rare genetic disorder that predominantly affects females and is characterized by developmental regression, loss of purposeful hand skills, motor abnormalities, and intellectual disability.

Incorrect Options:

Option B - Ambien: Ambien (generic name: zolpidem) is a sedative medication primarily used for the treatment of insomnia.

Option C - Lunesta: Lunesta (generic name: eszopiclone) is another sedative medication used for the treatment of insomnia. Similar to Ambien,

Option D - GHB opioids: GHB (gamma-hydroxybutyrate) and opioids. GHB is a central nervous system depressant that has been used recreationally and has sedative effects. Opioids, on the other hand, are a class of medications used for pain relief. Neither GHB nor opioids are approved drugs specifically indicated for the treatment of Rett syndrome.

Solution for Question 6:

Correct Option C - Intellectual disability:

- The correct option for what "mental retardation" is now called according to the American Psychiatric Association in its revision of DSM-5 is "Intellectual disability."

Incorrect Options:

Option A - Feeble-mindedness: Feeble-mindedness was a term used in the past to describe individuals with intellectual disabilities. However, this term is outdated and is no longer used in professional diagnostic criteria.

Option B - Mental subnormality: Mental subnormality is another outdated term that was previously used to describe individuals with intellectual disabilities. It is no longer used in current

diagnostic criteria.

Option D - Intellectually challenged: This term is not used in professional diagnostic criteria.

Solution for Question 7:

Correct Option B - Attention deficit/hyperactivity disorder:

- The disorder that was earlier known as "minimal brain dysfunction" is Attention deficit/hyperactivity disorder (ADHD).

Incorrect Options:

Option A - Dyslexia: Dyslexia refers to a specific learning disorder that primarily affects a person's ability to read, write, and spell. It is characterized by difficulties with accurate or fluent word recognition and poor spelling and decoding abilities. .

Option C - Mental subnormality: Mental subnormality is an outdated term that was previously used to describe what is now known as intellectual disability. Intellectual disability refers to significantly below-average intellectual functioning and limitations in adaptive behavior, which manifest during the developmental period.

Option D - Oligophrenia: Oligophrenia is another outdated term that was used to describe what is now known as intellectual disability. It refers to a condition characterized by significantly below-average intellectual functioning and limitations in adaptive behavior.

Solution for Question 8:

Correct Option B - Conduct disorder:

- The above case is that of a conduct disorder.
- Patterns of behavior characterized by aggression and violation of the rights of others, such as: Physical aggression, a threat to harm, and bullying others. Destruction of property, theft, and harming animals. Frequent violation of age-appropriate roles. Staying out at night

despite parent prohibition.

- Physical aggression, a threat to harm, and bullying others.
- Destruction of property, theft, and harming animals.
- Frequent violation of age-appropriate roles.
- Staying out at night despite parent prohibition.
- Physical aggression, a threat to harm, and bullying others.
- Destruction of property, theft, and harming animals.
- Frequent violation of age-appropriate roles.
- Staying out at night despite parent prohibition.

Incorrect Options:

Option A - Oppositional defiant disorder: In oppositional defiant disorder, they do not resort to physically aggressive or disruptive behavior.

Option C - Antisocial personality disorder: In this case, the individual has no regard for others and is violent and aggressive.

Option D - Disruptive mood dysregulation: Severe anger outbursts are not seen therefore, this is not the correct answer.

Solution for Question 9:

Correct Option D - Specific learning disorder:

- The information suggests that the child most likely has a mathematically impaired specific learning disorder (SLD). A neurodevelopmental disorder known as SLD impairs a person's capacity for learning in one or more particular subject areas, such as reading, writing, or mathematics. Individuals with SLD might have ordinary knowledge

Incorrect Options:

Option A - ADHD: ADHD is one of the most widely recognized neurodevelopmental problems of experience growing up.. People with ADHD frequently daydream, forget or lose a lot, squirm or fidget, talk too much, make careless mistakes or take unnecessary risks, struggle to resist temptation, struggle to take turns, and struggle to get along with others are common symptoms.

Option B - Mathematics learning disorder: Dyscalculia is a learning handicap that makes math testing to process and comprehend. Side effects range from trouble with counting and essential mental math to issues with saying what time it is. Despite having adequate intelligence and learning abilities in other areas, people with MLD struggle with various aspects of mathematics, such as number sense, arithmetic operations, understanding mathematical concepts, and applying them to real-world situations

Option C - Syndrome of Asperger's: For the most part, youngsters and adolescents with Asperger's Condition can talk with others and can perform genuinely well in their school work. However, they struggle to comprehend social situations and subtle forms of communication, such as humor, sarcasm, and body language. They might also only want to participate in a limited number of activities or think and talk a lot about a single subject or interest.

Solution for Question 10:

Correct Option A- Methylphenidate:

- The signs and symptoms in the question is suggestive of Attention Deficit Hyperactivity Disorder (ADHD)
- Treatment: Pharmacotherapy and Psychosocial interventions
- Pharmacotherapy Methylphenidate (CNS stimulant) Drug of choice Dopamine agonist Common side effects: headaches, nausea, insomnia, can exacerbate tics, can cause growth suppression
- Methylphenidate (CNS stimulant) Drug of choice Dopamine agonist Common side effects: headaches, nausea, insomnia, can exacerbate tics, can cause growth suppression
- Drug of choice
- Dopamine agonist
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- Dopamine agonist

- Common side effects: headaches, nausea, insomnia, can exacerbate tics, can cause growth suppression
- Drug of choice
- Dopamine agonist
- Common side effects: headaches, nausea, insomnia, can exacerbate tics, can cause growth suppression

Incorrect Options:

Option B - Modafinil:

- Unknown mode of action
- α -1 agonist
- May be helpful in ADHD

Option C- Atenolol:

- It is a beta-blocker, which is used in the treatment of hypertension.

Option D - Amitriptyline:

- This drug belongs to the class of tricyclic antidepressant drugs.

Solution for Question 11:

Correct Option A- Anal:

- In Sigmund Freud's psychosexual stages of development, the anal stage occurs typically around ages 1 to 3 years. During this stage, the child's focus of pleasure and development centers on the anal region. The main conflict of this stage is related to toilet training and the child's ability to control their bowel movements. The child learns to gain satisfaction from retaining and expelling feces.

Incorrect Options:

Option B - Oral Stage: The oral stage occurs during infancy (birth to around 1 year old). Pleasure centers on the mouth and the activities related to it, such as sucking and feeding. Fixation during this stage might result in behaviors like smoking, overeating, or a dependency on oral stimulation for comfort.

Option C - Phallic Stage: The phallic stage emerges around ages 3 to 6 years. During this stage, children become aware of their own gender and develop feelings of attraction towards the opposite-sex parent and rivalry with the same-sex parent (Oedipus and Electra complexes). Fixation during this stage might lead to challenges in forming healthy relationships and issues related to gender identity and self-esteem.

Option D - Genital Stage: The genital stage occurs from puberty onwards. This stage is characterized by the mature development of sexual interests and the establishment of healthy intimate relationships. Fixation during this stage might lead to difficulties in forming lasting relationships and a lack of fulfillment in adult sexual relationships.

Solution for Question 12:

Correct Option D - Echolalia:

- Echolalia is a language and speech disorder characterized by the involuntary repetition of words or phrases spoken by others. This can occur immediately after hearing the words or at a later time. Sometimes, the individual may repeat words or phrases they have heard on television or in their environment. In the given scenario, the patient is seen to repeat the question asked to him involuntarily.
- It is often associated with autism spectrum disorder, but can also occur in other neurological conditions such as dementia, schizophrenia, and traumatic brain injury.

It is often associated with autism spectrum disorder, but can also occur in other neurological conditions such as dementia, schizophrenia, and traumatic brain injury.

Incorrect Options:

Option A - Coprolalia: Coprolalia is a symptom of Tourette's syndrome, where the person involuntarily utters socially inappropriate words or phrases.

Option B - Echopraxia: Echopraxia is a phenomenon where a person imitates the movements of another person without any apparent purpose.

Option C - Copropraxia: Copropraxia is a symptom of Tourette's syndrome, where the person involuntarily makes obscene gestures or movements.

Solution for Question 13:

Correct Option C - Childhood anxiety disorder:

- Childhood anxiety disorder. Selective mutism is a childhood anxiety disorder characterized by consistent failure to speak in certain social situations, despite being capable of speech in other settings. It often occurs in situations where there is an expectation to speak, such as school or public places. The disorder is typically related to anxiety and fear of social evaluation or embarrassment.

Incorrect Options:

Option A - Childhood depression: Childhood depression involves persistent sadness, irritability, and a loss of interest or pleasure in activities. While depression can manifest with various symptoms, selective mutism is not a characteristic symptom of childhood depression.

Option B - Childhood psychosis: Childhood psychosis refers to severe disturbances in thinking, perception, and behavior. Selective mutism is not a symptom commonly associated with childhood psychosis.

Option D - Hyperkinetic disorder: Hyperkinetic disorder, also known as attention-deficit/hyperactivity disorder (ADHD), is characterized by persistent patterns of inattention, hyperactivity, and impulsivity. Selective mutism is not a primary symptom of hyperkinetic disorder.

Solution for Question 14:

Correct Option D - Bell and pad based classification conditioning:

- Bell and pad-based conditioning is a behavioral therapy used in the treatment of enuresis (bed-wetting). It involves using a sensor pad placed on the child's bed that triggers an alarm when it detects moisture. The child is awakened by the alarm and then attempts to go to the bathroom. Over time, this helps to establish a conditioned response of waking up to urinate and gradually reduces bed-wetting episodes.

Incorrect Options:

Option A - Pharmacotherapy with imipramine: Imipramine is an antidepressant medication that is sometimes used off-label for the treatment of enuresis. However, it is not considered the first-line treatment option and is generally reserved for cases that do not respond to behavioral interventions or when there are specific clinical indications.

Option B - Psychodynamic psychotherapy: Psychodynamic psychotherapy focuses on exploring the unconscious processes and unresolved conflicts that may contribute to psychological symptoms. While psychotherapy can be helpful in managing psychological distress associated with enuresis, it is not the primary treatment modality for addressing bed-wetting itself.

Option C - Bladder training with reward for delaying micturition during daytime: Bladder training techniques, which involve scheduled voiding and gradually increasing the time between voids, are primarily used for addressing urinary frequency, urgency, and incontinence. They are not typically the main approach for treating bed-wetting (nocturnal enuresis).

Solution for Question 15:

Correct Option C - Asperger's syndrome:

- Asperger's Syndrome: Asperger's Syndrome is an older term that was once used to describe a subtype of autism characterized by higher cognitive abilities and less severe language impairments. However, in current diagnostic criteria, Asperger's Syndrome is no longer recognized as a separate diagnosis. The symptoms associated with Asperger's Syndrome overlap with those of autism, including social interaction difficulties and repetitive behaviors.

Incorrect Options:

Option A - ADHD (Attention-Deficit/Hyperactivity Disorder): ADHD is a neurodevelopmental disorder characterized by difficulties with attention, impulsivity, and hyperactivity. While individuals with ADHD may display difficulties in social interactions and exhibit certain repetitive behaviors, the specific symptom mentioned in the question, is not typically associated with ADHD. Option B - Autism: This option is the correct answer. Autism, or Autism Spectrum Disorder (ASD), is a neurodevelopmental disorder characterized by persistent deficits in social communication and interaction, as well as restricted and repetitive patterns of behavior, interests, or activities. Individuals with autism often exhibit impairments in social interaction, including difficulties in engaging with others, understanding social cues, and expressing themselves. The described behavior of spending time seeing her own hands and not interacting

with others is consistent with the social interaction deficits observed in autism.

Option D - Rett's Syndrome: Rett's Syndrome is a rare genetic disorder that predominantly affects females. It is characterized by severe cognitive and physical impairments, loss of purposeful hand skills, and the development of repetitive hand movements. While Rett's Syndrome involves significant impairments in social interaction and communication, the specific symptom mentioned in the question, spending time seeing her own hands, is not typically associated with this disorder.

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Somatic Symptom and Related Disorders

1. A 32 y/o woman named Sarah goes missing after a traumatic breakup. Two weeks later, she is found in a different city, identifying herself as "Lily" and claiming she has always lived there. She cannot recall any details of her past or her true identity. What is the most likely diagnosis for Sarah?

- A. Dissociative Amnesia
- B. Possession Trance Disorder
- C. Depersonalization Disorder
- D. Dissociative Fugue

2. A 40 y/o woman visits a cultural healer, stating that she sometimes feels overwhelmed by a spirit that speaks through her. During these episodes, she adopts a distinct persona, telling her family to respect certain traditions as if she were that spirit. After the episodes, she cannot recall what transpired and feels distressed about her lack of control. Based on this description, what is the most likely diagnosis for this patient?

- A. Dissociative Trance Disorder
- B. Possession Trance Disorder
- C. Derealization disorder
- D. Multiple personality disorder

3. A 28-year-old woman reports feeling as if she's living in a dream, with her surroundings appearing foggy and unreal. She recognizes that everything is real but feels detached from her surroundings, and people seem distant or cartoon-like. These episodes last from a few minutes to several hours and have been occurring

for months, causing significant anxiety and interfering with her daily activities. What is the most likely diagnosis?

- A. Depersonalization Disorder
 - B. Derealization Disorder
 - C. Dissociative Identity Disorder
 - D. Dissociative Fugue
-

4. A 25-year-old woman develops severe back pain after a breakup, with no physical cause found. She is unaware of any psychological reason for her pain, but the support from family and friends reinforces her focus on the symptoms. She shows no signs of exaggerating her condition for personal gain. What is the most likely factor contributing to the maintenance of her symptoms?

- A. Tertiary Gain
 - B. Primary Gain
 - C. La Belle Indifférence
 - D. Secondary Gain
-

5. A 30 y/o woman, recently rescued from a religious cult, exhibits severe anxiety and difficulty trusting others, struggling with her identity and beliefs. Which of the following best describes the primary focus of treatment for individuals who have experienced this condition?

- A. Immediate medication to suppress symptoms
 - B. Deprogramming through intensive psychotherapy
 - C. Short-term counselling sessions
 - D. Isolation from social interactions
-

6. A 45 y/o man frequently visits multiple doctors over the past year, expressing intense worry about having various serious illnesses, despite medical tests showing no significant findings. He avoids certain activities due to the fear of becoming ill. Which of the following best describes the diagnosis for this patient?

- A. Somatic Symptom Disorder
 - B. Illness Anxiety Disorder
 - C. Munchausen Syndrome
 - D. Malingering
-

7. A 35-year-old woman frequently brings her 8-year-old son to the emergency department, claiming he has severe and recurrent gastrointestinal symptoms. Despite extensive testing, no medical cause is found. The child's symptoms noticeably improve when he is away from his mother, raising concerns about the mother's role in the situation. What is the most likely diagnosis for this mother's behaviour?

- A. Factitious Disorder
 - B. Somatic Symptom Disorder
 - C. Factitious Disorder Imposed on Another
 - D. Illness Anxiety Disorder
-

8. A 28-year-old man presents to the clinic expressing profound discomfort with his left leg, insisting that it does not belong to him. He reports feeling an overwhelming desire to have it amputated, describing intense distress whenever he sees or uses it. He has started to avoid social situations, fearing others will notice his discomfort. What is the most likely diagnosis for this individual's condition?

- A. Malingering
 - B. Body Integrity Dysphoria (BID)
 - C. Conversion Disorder
 - D. Somatic Symptom Disorder
-

9. A 35-year-old man complains of headaches, irritability, and difficulty concentrating due to increased workload and stress. O/E there is muscle tension and elevated blood pressure. Which stage of General Adaptation Syndrome (GAS) is he most likely experiencing?

- A. Alarm Reaction Stage
 - B. Resistance Stage
 - C. Exhaustion Stage
 - D. Recovery Stage
-

10. A 28 y/o woman visits a therapist. She describes experiencing morning sickness, breast enlargement, and noticeable abdominal distension. Despite multiple home pregnancy tests that have returned negative results, she insists that she feels the baby moving. What is the most likely diagnosis for this patient?

- A. Pseudologia fantastica
 - B. Pseudocyesis
 - C. Nihilism
 - D. Lilliputian hallucination
-

11. A 35 y/o male is disoriented and gives bizarre answers, responding "5" to "What is 2 + 2?" His wife notes significant work-related stress, and he reports

hallucinations. There is no history of substance abuse or brain injury, and his neurological exam is normal.

- A. Korsakoff syndrome
 - B. Dissociative Identity Disorder
 - C. Ganser syndrome
 - D. Schizophrenia
-

12. A 26 y/o male from a rural area in India presents to the clinic with complaints of anxiety, fatigue, and weakness. He is preoccupied with the belief that he is losing semen during urination and nocturnal emissions. He also describes his semen as "milk-like." He is concerned that this will lead to physical weakness and impotence. Which of the following is the most likely diagnosis for this patient?

- A. Koro syndrome
 - B. Dhat syndrome
 - C. Premature ejaculation
 - D. Hypochondriasis
-

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	2
Question 3	2
Question 4	4
Question 5	2
Question 6	2
Question 7	3
Question 8	2
Question 9	2
Question 10	2
Question 11	3
Question 12	2

Solution for Question 1:

Correct Answer: D) Dissociative Fugue

Explanation:

Given the sudden travel and complete memory loss regarding her past identity, Sarah's situation aligns with a dissociative fugue .

Dissociative fugue (Option D)

- Dissociative fugue, a subtype of dissociative amnesia, involves sudden, unexpected travel away from home accompanied by memory loss about one's past and potential identity confusion or loss, lasting from minutes to months, and is also not attributable to substance effects or medical conditions.
- Example: Mr. D from Delhi faced financial loss and stress, leading to his sudden disappearance. A month later, a friend found someone resembling Mr. D as Mr. C in Chennai. When asked to return home, the man claimed to be Mr. C, a shop helper in Chennai, not Mr. D.
- Mr. D from Delhi faced financial loss and stress, leading to his sudden disappearance. A month later, a friend found someone resembling Mr. D as Mr. C in Chennai. When asked to return home, the man claimed to be Mr. C, a shop helper in Chennai, not Mr. D.
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Dissociative amnesia (Option A ruled out)

- Dissociative amnesia is characterized by an inability to recall significant personal information, typically related to trauma or stress, that exceeds ordinary forgetfulness and is not due to physiological causes or substance use or injury.
- Patients may experience symptoms like depression, derealization, and physical manifestations, frequently related to past abuse or trauma.
- Example: A man rescued two days after his kidnapping couldn't recall anything from that period when questioned by police or family. However, he remembered everything before and after the incident. Due to excessive stress, his mind dissociated some parts of his memory from the other functions.
- A man rescued two days after his kidnapping couldn't recall anything from that period when questioned by police or family. However, he remembered everything before and after the incident.
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- A man rescued two days after his kidnapping couldn't recall anything from that period when questioned by police or family. However, he remembered everything before and after the incident.
- Due to excessive stress, his mind dissociated some parts of his memory from the other functions.

Possession Trance Disorder (Option B): This disorder involves trance states in which there is a marked alteration in the individual's state of consciousness and the individual's customary sense

of personal identity is replaced by an external 'possessing' identity and in which the individual's behaviours or movements are experienced as being controlled by the possessing agent. an individual adopting a new identity believed to be influenced by an external entity. Sarah's case does not suggest any belief in possession or behaviour linked to external influences.

Depersonalization Disorder (Option C): Depersonalization involves feelings of detachment from oneself, where individuals feel like they are observing their own actions from the outside. Sarah's scenario is characterized by confusion about her identity and memory loss, not by feelings of detachment.

Solution for Question 2:

Correct Answer: B) Possession Trance Disorder

Explanation:

Possession Trance Disorder is characterized by episodic identity change attributed to a spirit, culturally specific behaviours, and amnesia, fitting the woman's experience perfectly.

Trance Disorder (Option A ruled out)

- Characterized by a temporary and marked alteration in consciousness or loss of personal identity without the emergence of an alternate identity.
- Key Features: Marked change in consciousness. Loss of customary sense of personal identity. No replacement by an alternate identity.
- Marked change in consciousness.
- Loss of customary sense of personal identity.
- No replacement by an alternate identity.
- Example: A teenage girl was brought to the ER after she suddenly entered a trance-like state during a family gathering. She remained unresponsive to external stimuli for several hours, with no recollection of the event and no awareness of her surroundings during the episode
- A teenage girl was brought to the ER after she suddenly entered a trance-like state during a family gathering. She remained unresponsive to external stimuli for several hours, with no recollection of the event and no awareness of her surroundings during the episode
- Marked change in consciousness.

- Loss of customary sense of personal identity.
- No replacement by an alternate identity.
- A teenage girl was brought to the ER after she suddenly entered a trance-like state during a family gathering. She remained unresponsive to external stimuli for several hours, with no recollection of the event and no awareness of her surroundings during the episode

Possession Trance Disorder

- A variant of dissociative trance involves episodes where an individual's customary identity is exchanged for a new identity, often attributed to a spirit, deity, or another person.
- Key Features: Episodic alterations in consciousness. Presence of stereotypical, culturally determined behaviours. Experience of an entity controlling the individual. Partial or full amnesia regarding the episode.
- Episodic alterations in consciousness.
- Presence of stereotypical, culturally determined behaviours.
- Experience of an entity controlling the individual.
- Partial or full amnesia regarding the episode.
- Episodic alterations in consciousness.
- Presence of stereotypical, culturally determined behaviours.
- Experience of an entity controlling the individual.
- Partial or full amnesia regarding the episode.
- Example: A newly married woman claims to be possessed by her deceased grandmother, speaking in a different voice and urging her in-laws to care for her. She later insists she was unaware of her actions.
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- A newly married woman claims to be possessed by her deceased grandmother, speaking in a different voice and urging her in-laws to care for her. She later insists she was unaware of her actions.

Derealization Disorder (Option C) involves feeling detached from surroundings. The woman does not describe this feeling, so it doesn't match her symptoms.

Multiple Personality Disorder (Option D) involves distinct personality states developed within the individual. The woman's experience is attributed to a spirit, not a new personality, making this diagnosis unsuitable.

Solution for Question 3:

Correct Answer: B) Derealization Disorder

Explanation:

- Feelings of detachment from the world and experiences of unreality characterize Derealization Disorder.
- The patient's description of feeling disconnected from his environment appearing foggy and her perception of people as distant or cartoon-like perfectly matches this disorder.

Depersonalization-derealization disorder

Depersonalisation	Derealization
A feeling of detachment from the self	A feeling of detachment from the world
The person may feel like they are watching themselves from outside the body. (Option A ruled out)	The person feels as if the world is unreal.

- The brain uses defensive mechanisms to protect against acute anxiety or trauma, which can lead to recurring feelings of depersonalization or derealization.
- Depersonalization disorder causes individuals to feel as if their experiences are not real and often describe their experiences with "As if", but they maintain intact reality testing, knowing what's real and what's not.
- Differential Diagnosis Seizure Disorders Migraines Substance use like Cannabis and Lysergic Acid Diethylamide (LSD) Head injury.
- Seizure Disorders
- Migraines
- Substance use like Cannabis and Lysergic Acid Diethylamide (LSD)
- Head injury.
- Seizure Disorders
- Migraines
- Substance use like Cannabis and Lysergic Acid Diethylamide (LSD)
- Head injury.

Dissociative Identity Disorder (Multiple Personality Disorder) (Option C ruled out)

- The presence of two or more distinct identities or personality states is called alters .

- Each identity has its own way of perceiving and interacting with the environment and self.
- More than two personalities are possible as well.
- DID is characterized by symptoms such as amnesia, depersonalization, and derealization.
- A key diagnostic feature is the experience of "losing time," which goes beyond ordinary forgetfulness.
- The disorder presents diverse signs and symptoms, complicating diagnosis.
- Only one personality shows at a time, with the second personality usually unaware of the first, or it may know about the primary one.
- This is different from possession trance disorder, where both personalities exist together but one controls the person's behavior.
- Example: Ms P returns home to find photos on her social media with strangers she appears to know, but she has no memory of the experience. Her first identity lived through it, while her second identity remains unaware.
- Ms P returns home to find photos on her social media with strangers she appears to know, but she has no memory of the experience.
- Her first identity lived through it, while her second identity remains unaware.
- Ms P returns home to find photos on her social media with strangers she appears to know, but she has no memory of the experience.
- Her first identity lived through it, while her second identity remains unaware.

Dissociative Fugue (Option D) involves sudden, unexpected travel away from home or customary work locations, often accompanied by an inability to recall one's past.

- The patient's symptoms do not include travel or amnesia for identity.

Solution for Question 4:

Correct Answer: D) Secondary Gain

Explanation:

Secondary gain is the most likely factor contributing to the maintenance of her symptoms, as her chronic pain elicits sympathy and support from others, reinforcing her focus on the condition.

Conversion Disorder(Functional neurological disorder)

- Predominantly, the symptoms of conversion disorder are neurological and are paradoxically experienced as involuntary, or lacking in self-agency.
- Seen in Females > males
- Onset- Late childhood to early adulthood.
- Presentation of conversion disorder: Presents with symptoms suggestive of deficit in motor or sensory functions or both. But without any evidence of any physical disorder. Reason- Too much stress has converted the stress into these neurological symptoms. At times, the symptoms are inconsistent with the anatomy and physiology of the body. E.g. inconsistent tremors.
- Presents with symptoms suggestive of deficit in motor or sensory functions or both.
- But without any evidence of any physical disorder.
- Reason- Too much stress has converted the stress into these neurological symptoms.
- At times, the symptoms are inconsistent with the anatomy and physiology of the body. E.g. inconsistent tremors.
- Presents with symptoms suggestive of deficit in motor or sensory functions or both.
- But without any evidence of any physical disorder.
- Reason- Too much stress has converted the stress into these neurological symptoms.
- At times, the symptoms are inconsistent with the anatomy and physiology of the body. E.g. inconsistent tremors.

Sensory symptoms	Motor symptoms
Paresthesia Anesthesia Deafness Blindness Hemianesthesia	Abnormal movement Paralysis, paresis Gait disturbances Astasia-abasia gait Pseudo seizures

- Various gains in conversion disorder:

Gains	Features	Examples
Primary Gains	Achieved by keeping the internal conflicts outside the awareness. Done unconsciously by producing the symptoms that reduce the internal conflicts (internal gains). (Option B ruled out)	A person facing exam stress develops blindness, which helps keep his anxiety out of awareness. He becomes more focused on his blindness than on studying or preparing for the exam.
Secondary Gains	Advantages or benefits a person gets as a result of getting sick (external gains) . (Option A ruled out)	The person is excused from studying due to sickness, receives family support, and is relieved of duties and responsibilities.

Gains	Features	Examples
Tertiary Gains	Sometimes the whole family will get the benefit because of the illness.	The parents receive money from relatives for treatment and get relief from their jobs to care for the patient.

- Example: A seven-year-old girl overhears her parents arguing about divorce and suddenly reports blindness. After a thorough examination, she is found to be normal, indicating classical conversion disorder. Due to high stress, the mind converts that stress into neurological symptoms, such as blindness.
- A seven-year-old girl overhears her parents arguing about divorce and suddenly reports blindness.
- After a thorough examination, she is found to be normal, indicating classical conversion disorder.
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- A seven-year-old girl overhears her parents arguing about divorce and suddenly reports blindness.
- After a thorough examination, she is found to be normal, indicating classical conversion disorder.
- Due to high stress, the mind converts that stress into neurological symptoms, such as blindness.
- La Belle Indifference (Option C ruled out) La belle indifférence refers to a patient's inappropriate lack of concern regarding severe symptoms, which can appear in conditions like conversion disorder. While it may also be seen in seriously ill patients who develop a stoic attitude, it is not exclusive to conversion disorder. For instance, the girl who is not able to see; may be sitting very calmly.
- La belle indifférence refers to a patient's inappropriate lack of concern regarding severe symptoms, which can appear in conditions like conversion disorder.
- While it may also be seen in seriously ill patients who develop a stoic attitude, it is not exclusive to conversion disorder.
- For instance, the girl who is not able to see; may be sitting very calmly.
- La belle indifférence refers to a patient's inappropriate lack of concern regarding severe symptoms, which can appear in conditions like conversion disorder.
- While it may also be seen in seriously ill patients who develop a stoic attitude, it is not exclusive to conversion disorder.
- For instance, the girl who is not able to see; may be sitting very calmly.

Solution for Question 5:

Correct Answer: B) Deprogramming through intensive psychotherapy

Explanation:

Deprogramming through intensive psychotherapy is the correct answer as the treatment primarily involves reversing brainwashing effects through intensive psychotherapy, which is essential in addressing the psychological manipulation experienced.

Brainwashing

- Definition : Brainwashing is the deliberate psychological manipulation involving isolation, intimidation, and coercion, causing identity disturbance and culture shock.
- It aims to break down coping skills, leading to significant personality and belief changes.

Treatment of Brainwashing

- Deprogramming : Involves reversing the effects of brainwashing through intensive psychotherapy and confrontation, particularly effective with religious cult members.
- Psychotherapy : Focuses on deconstructing harmful beliefs and restoring individual autonomy
- Reeducation : This involves guiding the individual through a gradual process of learning and accepting new, healthier thought patterns, challenging the beliefs that were forcibly imposed during brainwashing.
- Long-Term : Treatment is usually long-term, focusing on rebuilding healthy self-esteem and coping skills, especially in those with PTSD and dissociative symptoms.

Deprogramming : Involves reversing the effects of brainwashing through intensive psychotherapy and confrontation, particularly effective with religious cult members.

Psychotherapy : Focuses on deconstructing harmful beliefs and restoring individual autonomy

Reeducation : This involves guiding the individual through a gradual process of learning and accepting new, healthier thought patterns, challenging the beliefs that were forcibly imposed during brainwashing.

Long-Term : Treatment is usually long-term, focusing on rebuilding healthy self-esteem and coping skills, especially in those with PTSD and dissociative symptoms.

Immediate medication to suppress symptoms (Option A): While medication may be used to manage specific symptoms, it is not the primary focus of treatment for brainwashing.

Short-term counselling sessions (Option C): Brainwashing treatment is generally long-term, focusing on reeducation and ongoing support rather than short-term interventions.

Isolation from social interactions (Option D): Isolation can be a tactic used in brainwashing but is not a treatment strategy; rather, the treatment aims to re-engage individuals with supportive social interactions.

Solution for Question 6:

Correct Answer: B) Illness Anxiety Disorder

Explanation:

The patient has persistent anxiety about having a serious illness with minimal physical symptoms, aligning with the characteristics of Illness anxiety disorder.

Somatic symptom disorder (SSD)	Illness anxiety disorder (IAD)
Definition : Persistent excessive anxiety and preoccupation with one or more somatic symptoms, often triggered by a history of stressors, despite medical evidence showing no significant findings. Symptoms : Anxiety, distress, and excessive focus on symptoms or health, with significant time and energy devoted to them. Duration : Symptoms persist for ≥ 6 months (DSM-5) or ≥ 2 years (ICD-10). Psychosocial Impact : Causes significant distress or functional impairment in social, interpersonal, or familial areas. Severity Specifiers (DSM-5): Mild: 1 symptom Moderate: ≥ 2 symptoms Severe: ≥ 2 symptoms with multiple or severe complaints Symptom Specifier : Predominant pain Exclusions : Symptoms are not due to malingering or identifiable medical causes.	Definition : Persistent preoccupation with having or developing a serious illness, despite minimal or no physical symptoms and evidence to the contrary. Symptoms : Preoccupation with health and fear of illness. Excessive health-related behaviours (e.g., frequent medical visits) or avoidance of medical care. Anxiety and distress related to the belief. Duration : Symptoms persist for ≥ 6 months (DSM-5). Specifiers : Care-seeking type : Frequent reassurance from medical providers. Care-avoidant type : Avoids medical appointments due to fear. Key Difference : Unlike somatic symptom disorder, patients have minimal physical symptoms but are fixated on potential illness Differential Diagnosis: Delusional disorder (Somatic type) Hypochondriasis : The ICD-10 equivalent of illness anxiety disorder, characterized by a persistent fear of having a serious illness despite little to no physical symptoms.

Somatic Symptom Disorder (Option A) : This disorder involves a persistent belief in having a serious undetected disease with significant somatic symptoms, which is not the case here.

Munchausen Syndrome (Option C) is incorrect because the patient is not deliberately fabricating or inducing symptoms to assume the sick role, as seen in Munchausen Syndrome.

Malingering (Option D) implies the intentional fabrication of symptoms for secondary gain, which is not suggested in the scenario.

Solution for Question 7:

Correct Answer: C) Factitious Disorder Imposed on Another

Explanation:

The mother is inducing or fabricating symptoms in her child to seek medical attention and assumes the sick role indirectly, which aligns with this diagnosis.

Disorder	Factitious disorder	Factitious Disorder Imposed on Another
Definition	Feigning or inducing illness for medical care without external rewards (formerly Munchausen syndrome).	A caregiver (often a parent) induces or fabricates illness in a dependent (often a child).
Motivation	Desire to assume the sick role without external incentives.	The caregiver assumes the sick role by proxy or seeks to relieve caretaking duties.
Symptoms	Fabrication of symptoms Self-harm Manipulation of lab tests Demand for invasive treatments.	Faked illness Altered medical records Unnecessary treatments The child's condition improves away from the caregiver.
Clues to Diagnosis	Multiple hospital visits with inconsistent symptoms Eager for invasive procedures Noncompliance with treatment History in healthcare Disputed claims by lab tests.	Repeated hospital visits Altered samples Unexplained injuries Improvement in the child when separated from the caregiver.

Factitious Disorder (Option A) : This disorder involves the individual feigning illness for themselves, not for another person.

Somatic Symptom Disorder (Option B) : This diagnosis is characterized by excessive focus on somatic symptoms without feigning or fabricating illnesses.

Illness Anxiety Disorder (Option D) : This disorder involves excessive worry about having a serious illness, but the focus here is on the mother's manipulation of her child's health.

Solution for Question 8:

Correct Answer: B) Body Integrity Dysphoria (BID)

Explanation:

The man's intense discomfort with his healthy limb and desire for amputation align with BIID, characterized by feelings that certain body parts do not belong.

Body Integrity Identity Dysphoria (BID)

- Definition : A condition where individuals feel that a healthy body part does not belong, leading to desires for amputation or paralysis.
- Symptoms : Strong desire to be physically disabled or without a body part. Discomfort with healthy body parts. Preoccupation disrupting social activities. Behaviors mimicking disability despite being healthy. Self-harm or self-mutilation of unwanted body parts.
- Strong desire to be physically disabled or without a body part.
- Discomfort with healthy body parts.
- Preoccupation disrupting social activities.
- Behaviors mimicking disability despite being healthy.
- Self-harm or self-mutilation of unwanted body parts.
- Commonly Affected Body Parts : Most commonly legs (especially the left leg), also arms, fingers, toes, eyes, ears, and teeth.
- Most commonly legs (especially the left leg), also arms, fingers, toes, eyes, ears, and teeth.
- Causes : Potentially linked to brain abnormalities affecting body perception (somatosensory cortex, parietal lobe, insula).
- Complications : Risks from self-harm include bleeding, infection, nerve damage, severe pain, and phantom limb pain.
- Diagnosis : Conducted by psychiatrists via discussions on body perception with the patient; recognized in ICD-11 but not in DSM-5.

- Treatment : Cognitive behavioural therapy and medications for depression/anxiety. Use of prosthetics or mobility aids. Virtual reality simulations to explore life without a body part. No single treatment exists; focus on reducing distress and harmful behaviours.
- Cognitive behavioural therapy and medications for depression/anxiety.
- Use of prosthetics or mobility aids.
- Virtual reality simulations to explore life without a body part.
- No single treatment exists; focus on reducing distress and harmful behaviours.
- Strong desire to be physically disabled or without a body part.
- Discomfort with healthy body parts.
- Preoccupation disrupting social activities.
- Behaviors mimicking disability despite being healthy.
- Self-harm or self-mutilation of unwanted body parts.
- Most commonly legs (especially the left leg), also arms, fingers, toes, eyes, ears, and teeth.
- Cognitive behavioural therapy and medications for depression/anxiety.
- Use of prosthetics or mobility aids.
- Virtual reality simulations to explore life without a body part.
- No single treatment exists; focus on reducing distress and harmful behaviours.

Malingering (Option A ruled out)

- Definition : Deliberate falsification of symptoms for secondary gain (e.g., avoiding military duty, financial compensation, evading prosecution).
- Contexts for Malingering : Criminal Context : Pretending incompetence or insanity to avoid trial. Military Context : Feigning illness to evade conscription or hazardous duties. Financial Gain : Seeking undeserved benefits eg. disability insurance. Social Responsibility : Escaping unpleasant situations.
- Criminal Context : Pretending incompetence or insanity to avoid trial.
- Military Context : Feigning illness to evade conscription or hazardous duties.
- Financial Gain : Seeking undeserved benefits eg. disability insurance.
- Social Responsibility : Escaping unpleasant situations.
- Epidemiology : Higher prevalence in military, prison, and litigious populations, particularly among men.
- Associated Disorders : Conduct and anxiety disorders in children; antisocial, borderline, and narcissistic personality disorders in adults.
- Course and Prognosis : Malingering persists as long as rewards are anticipated; symptoms fade once goals are met. Ignoring behaviour in structured environments may reduce it.

- Treatment : Maintain neutrality; conduct differential diagnosis. Tactfully confront suspected malingering and explore motivations. Assess for coexisting psychiatric disorders. Engagement is essential unless the patient is manipulative.
- Maintain neutrality; conduct differential diagnosis.
- Tactfully confront suspected malingering and explore motivations.
- Assess for coexisting psychiatric disorders.
- Engagement is essential unless the patient is manipulative.
- Criminal Context : Pretending incompetence or insanity to avoid trial.
- Military Context : Feigning illness to evade conscription or hazardous duties.
- Financial Gain : Seeking undeserved benefits eg. disability insurance.
- Social Responsibility : Escaping unpleasant situations.
- Maintain neutrality; conduct differential diagnosis.
- Tactfully confront suspected malingering and explore motivations.
- Assess for coexisting psychiatric disorders.
- Engagement is essential unless the patient is manipulative.

Conversion Disorder (Option C) : This disorder presents with neurological symptoms that are inconsistent with known medical conditions, not a desire for amputation.

Somatic Symptom Disorder (Option D) : This involves excessive focus on physical symptoms rather than the specific desire to alter or remove a healthy body part.

Solution for Question 9:

Correct Answer: B) Resistance Stage

Explanation:

The patient's symptoms, including irritability, concentration difficulties, and muscle tension, suggest he is in the resistance stage, where the body is attempting to cope with stress but is showing signs of strain.

Stress and General Adaptation Syndrome (GAS): Stress is a physiological and psychological response to challenging situations; GAS describes the body's response to stress.

1. Alarm Reaction Stage	Immediate fight-or-flight response. (Option A ruled out) Increased heart rate and release of adrenaline and cortisol.
2. Resistance Stage	The body recovers itself while remaining on high alert. Signs include irritability, frustration, and poor concentration.
3. Exhaustion Stage	Prolonged stress depletes resources. Symptoms include fatigue, burnout, anxiety, and a weakened immune system. (Option C ruled out)

Cardiovascular	Acute stress: Increases heart rate and blood pressure. Chronic stress: Risk factors for hypertension, heart attack, and stroke.
Respiratory	Stress: Causes shortness of breath; worsens asthma. Management: Breathing techniques can alleviate symptoms.
Gastrointestinal	Stress: Impacts gut health, alters appetite; can exacerbate IBS.
Musculoskeletal	Effects: Muscle tension leads to headaches and chronic pain. Relief: Relaxation techniques reduce tension.
Endocrine	Activation: The HPA axis leads to increased cortisol. Consequences: Chronic high cortisol impairs immune function.
Nervous system	Response: Activates SNS for "fight or flight." Long-term: Chronic activation leads to body wear and tear.
Reproductive	Male: Lowers testosterone, impacting libido and sperm quality. Female: Disrupts menstrual cycles, and fertility; increases anxiety in pregnancy and menopause.

Recovery Stage (Option D) : This stage implies the patient is starting to recover from stress, which is not the case given his current symptoms.

Solution for Question 10:

Correct Answer: B) Pseudocyesis

Explanation:

Pseudocyesis involves the belief that one is pregnant, often accompanied by real physical symptoms such as morning sickness, breast enlargement, and abdominal distension, despite negative pregnancy tests.

Pseudocyesis/ Phantom pregnancy

- It is a rare condition where a nonpregnant individual exhibits signs and symptoms of pregnancy.
- The symptoms are real and can mimic actual pregnancy, but there is no fetal development.
- Symptoms: Abdominal distension Breast enlargement Skin pigmentation changes Cessation of menses Morning sickness
- Abdominal distension
- Breast enlargement
- Skin pigmentation changes
- Cessation of menses
- Morning sickness
- Causes: Strong desire to become pregnant Multiple miscarriages Loss of a child Infertility Sexual abuse
- Strong desire to become pregnant
- Multiple miscarriages
- Loss of a child
- Infertility
- Sexual abuse
- Treatments for phantom pregnancy include: Clinical tests that disprove pregnancy- The first step in treating pseudocyesis is to reassure the individual they are not pregnant, typically using ultrasound to show the absence of a fetus. Psychodynamic and Supportive Psychotherapy Medications such as antidepressants or antipsychotics Hormonal therapy- Helps the person get their menstrual period. Uterine dilation and curettage (D&C;)
- Clinical tests that disprove pregnancy- The first step in treating pseudocyesis is to reassure the individual they are not pregnant, typically using ultrasound to show the absence of a fetus.
- Psychodynamic and Supportive Psychotherapy
- Medications such as antidepressants or antipsychotics
- Hormonal therapy- Helps the person get their menstrual period.
- Uterine dilation and curettage (D&C;)

- Abdominal distension
- Breast enlargement
- Skin pigmentation changes
- Cessation of menses
- Morning sickness
- Strong desire to become pregnant
- Multiple miscarriages
- Loss of a child
- Infertility
- Sexual abuse
- Clinical tests that disprove pregnancy- The first step in treating pseudocyesis is to reassure the individual they are not pregnant, typically using ultrasound to show the absence of a fetus.
- Psychodynamic and Supportive Psychotherapy
- Medications such as antidepressants or antipsychotics
- Hormonal therapy- Helps the person get their menstrual period.
- Uterine dilation and curettage (D&C;)

Pseudologia fantastica (Option A) involves uncontrollable lying and the creation of elaborate fantasies.

- The patient's symptoms focus on the belief of being pregnant, not on lying or creating fantasies

Nihilism (Option C) involves a delusion of the nonexistence of self or a part of self or an extreme rejection of established values.

- The patient does not exhibit these beliefs; her focus is on her pregnancy symptoms.

Lilliputian hallucination (Option D) refers to visual sensations where objects or people appear smaller than they are, also called micropsia.

- The patient's symptoms do not involve visual distortions, making this diagnosis irrelevant.

Solution for Question 11:

Correct Answer: C) Ganser syndrome

Explanation: Ganser syndrome is characterized by giving approximate answers (paralogia), which the patient demonstrates by responding with incorrect but related answers (e.g., $2 + 2 = 5$). The presence of disorientation, hallucinations, and the recent stressor fits with this diagnosis.

Ganser Syndrome

- Definition : A condition characterized by giving approximate answers (paralogia) alongside clouded consciousness, hallucinations, and dissociative or conversion symptoms.
- Epidemiology : More common in men.
- Precipitating Factors : Stress, personal conflicts, brain injury, childhood adversity
- Clinical Features : Paralogia : Incorrect but related answers (e.g., $2 + 2 = 5$). Clouded consciousness : Disorientation, amnesia, impaired reality testing. Hallucinations : Visual/auditory in about 50% of cases. Dissociative symptoms : Amnesia, conversion, trance-like states. Hysterical stigmata : Non-neurologic analgesia, shifting hyperalgesia.
- Paralogia : Incorrect but related answers (e.g., $2 + 2 = 5$).
- Clouded consciousness : Disorientation, amnesia, impaired reality testing.
- Hallucinations : Visual/auditory in about 50% of cases.
- Dissociative symptoms : Amnesia, conversion, trance-like states.
- Hysterical stigmata : Non-neurologic analgesia, shifting hyperalgesia.
- Differential Diagnosis : Includes organic brain syndromes (seizures, trauma, dementia), Korsakoff syndrome, and dissociative identity disorder.
- Treatment : Supportive care in a hospital setting. Low-dose antipsychotics may help. Exploration of stressors (hypnosis), avoid confronting incorrect answers.
- Supportive care in a hospital setting.
- Low-dose antipsychotics may help.
- Exploration of stressors (hypnosis), avoid confronting incorrect answers.
- Prognosis : Recovery is usually within days, up to a month.
- Paralogia : Incorrect but related answers (e.g., $2 + 2 = 5$).
- Clouded consciousness : Disorientation, amnesia, impaired reality testing.
- Hallucinations : Visual/auditory in about 50% of cases.
- Dissociative symptoms : Amnesia, conversion, trance-like states.
- Hysterical stigmata : Non-neurologic analgesia, shifting hyperalgesia.
- Supportive care in a hospital setting.
- Low-dose antipsychotics may help.
- Exploration of stressors (hypnosis), avoid confronting incorrect answers.

Korsakoff syndrome (Option A) involves amnesia and confabulation, typically associated with chronic alcohol use and thiamine deficiency, which is not present in this case.

Dissociative Identity Disorder (Option B) involves the presence of two or more distinct identities or personalities, which is not described in the scenario.

Schizophrenia (Option D) could present with hallucinations, but it does not typically involve giving approximate answers (paralogia), and the rapid onset of symptoms makes Ganser syndrome more likely.

Solution for Question 12:

Correct Answer: B) Dhat syndrome

Explanation:

- Dhat syndrome is characterized by the fear of semen loss through nocturnal emission, urination, or other means, leading to anxiety, somatic complaints, and weakness, which fits the patient's presentation.
- His cultural background and the description of his semen as "milk-like" align with the typical presentation of Dhat syndrome.

Culture bound syndromes: patterned behaviors of distress or illness whose symptoms appear distinct from psychiatric categories and considered unique to particular cultural settings.

- Such as Dhat syndrome and Koro syndrome.

Dhat syndrome	Koro syndrome
Definition : A psychiatric condition marked by fear of semen loss through ejaculation, nocturnal emission, or other means. Cultural Context : Leading culture-bound syndrome in the Indian subcontinent, tied to Ayurvedic beliefs that semen is vital for health. Demographics : Affects young, unmarried males from lower socioeconomic and rural backgrounds. Age : The average onset is around 26 years. This can lead to depression and sexual dysfunction Clinical Presentation Symptoms : Anxiety, fatigue, weakness, and impotence from the fear of semen loss. Beliefs : Semen loss through nocturnal emissions, urination, defecation, or arousal. Somatic Complaints : Physical symptoms tied to weakness, infertility, and fear of death. Semen Description : Often described as "milk-like" or "pus-like." Treatment Psychiatric Care : Treated similarly to depression, with psychotherapy (sex education to dispel the myths) and antidepressants. Diagnosis ICD-10 : Concern over semen loss (seen as whitish discharge in urine) with associated anxiety and physical symptoms. Female Dhat syndrome , presenting with nonpathological vaginal discharge ("safedpaani") seen as a vital fluid, is triggered by warm foods, stress, and dietary imbalances, causing somatic symptoms and linked to Ayurvedic beliefs about bodily balance.	Definition : A psychiatric disorder characterized by an intense fear of genital retraction into the body, believed to result in death (in males, the penis; in females, the vulva or breasts). Epidemiology : Epidemic Koro : Common in parts of East and Southeast Asia (e.g., China, India, Malaysia), linked to cultural beliefs and myths. Sporadic Koro : Occurs globally, unrelated to culture, often secondary to psychiatric disorders (e.g., schizophrenia, depression) or substance use. Clinical Presentation : Overwhelming anxiety with belief of genital retraction and impending death. Symptoms: Panic attacks, physical attempts to prevent retraction, measuring genitals, insomnia, and fear of death Treatment : Psychotherapy : Sex education and counselling to dispel myths. Medications : Anxiolytics, antidepressants, antipsychotics, and sedatives (based on psychiatric comorbidities).

Koro syndrome (Option A) involves the fear of genital retraction rather than semen loss and is prevalent in different regions of Asia.

Premature ejaculation (Option C) is a sexual dysfunction, but the focus in this case is on semen loss and its perceived effects, not on ejaculation timing.

Hypochondriasis (Option D) involves excessive health anxiety, but Dhat syndrome specifically involves culturally bound fears of semen loss rather than general health anxiety.

Previous Year Questions

1. Which of the following is not considered a possible factor contributing to SIADH?

- A. Demeclocycline
- B. Vincristine
- C. Carbamazepine
- D. Nicotine

2. Which mental functions are examined in the Mental State Examination (100-7)?

- A. Judgment
- B. Language
- C. Orientation
- D. Concentration

3. Confusion assessment test is used in:

- A. Delirium
- B. Dementia
- C. Schizophrenia
- D. Depression

4. Which of the following characteristics provides stronger evidence in support of delirium?

- A. Occurs gradually over a period of time
 - B. Fluctuating course
 - C. Preserved consciousness
 - D. Commonly associated with auditory hallucinations
-

5. What is the purpose of utilizing the confusion assessment test?

- A. Dementia
 - B. Depression
 - C. Delirium
 - D. Schizophrenia
-

6. Which of the following statements is false regarding Alzheimer's disease?

- A. Intracellular neurofibrillary tangles are seen
 - B. Neuritic plaques made of beta amyloid are found
 - C. Nucleus of Meynert is not affected
 - D. Short term memory is affected
-

7. In the physiology viva, the examiner inquires about the individual who initially recorded the first EEG. You respond by admitting that you are unaware of the answer. Sympathetic to your lack of knowledge, the examiner provides a hint indicating that the name resembles a common item found at McDonald's. With this clue, you are able to recall the name and provide the correct response. Which form of memory is involved in this situation?

- A. Semantic memory

- B. Implicit memory
 - C. Associative learning
 - D. Priming
-

8. Which one of the options below is not classified as a form of subcortical dementia?

- A. Pick's disease
 - B. Parkinson's disease
 - C. Huntington's disease
 - D. Wilson's disease
-

9. What are the distinguishing characteristics between delirium and dementia in Alzheimer's disease?

- A. Acuity of onset and agitation
 - B. Acuity of onset and loss of consciousness
 - C. Visual hallucinations and impaired memory
 - D. Agitation and irritation
-

10. A patient of Alzheimer's disease refused to take oral drugs. Which of the following drugs can be prescribed to this patient?

- A. Memantine
- B. Donepezil
- C. Rivastigmine
- D. Galantamine

11. A 65-year-old patient presents with a unique psychological phenomenon in which they believe that a close family member has been replaced by an imposter with an uncanny resemblance. This is commonly seen in which type of dementia?

- A. Alzheimer's disease
 - B. Lewy body dementia
 - C. Vascular dementia
 - D. Frontotemporal dementia
-

12. What deficiency is responsible for cognitive impairment in the elderly?

- A. Vitamin C
 - B. Vitamin B12
 - C. Vitamin B6
 - D. Vitamin B1
-

13. What is the psychiatric disorder most frequently observed in patients with medical or surgical conditions?

- A. Depression
 - B. Schizophrenia
 - C. Delirium
 - D. Dementia
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	4
Question 3	1
Question 4	2
Question 5	3
Question 6	3
Question 7	4
Question 8	1
Question 9	2
Question 10	3
Question 11	2
Question 12	2
Question 13	3

Solution for Question 1:

Correct choice: A

Demeclocycline is not a cause of SIADH, It is used in the treatment of SIADH Other Options: B. Vincristine: Vincristine is a chemotherapy medication commonly used in the treatment of cancer.

While it primarily affects cell division, it has been associated with the development of SIADH as a potential side effect. C. Carbamazepine: Carbamazepine is an anticonvulsant medication used to treat seizures and various neurological disorders. It can also cause SIADH, although the exact mechanism is not well understood. It is believed to involve the alteration of ADH release or response. D. Nicotine: Nicotine, the primary addictive component of tobacco, is not known to directly cause SIADH. However, it is important to note that smoking and nicotine use can have various effects on the body, including influencing hormonal regulation. There are certain case reports suggesting an association between smoking and SIADH, but it is not considered a well-established or common cause.

Solution for Question 2:

Correct Option D - Concentration:

- The Mental State Examination (MSE), 100-7 is a commonly used test to assess concentration. This test involves asking the patient to subtract 7 from 100, then continue subtracting 7 from the resulting number, and so on. The test is stopped after a minute, and the number of correct answers is noted.
- Concentration refers to a person's ability to maintain focus and attention on a task or topic. Concentration can be assessed in the MSE by asking the patient to perform tasks such as counting backward or reciting the months of the year in reverse order.

Incorrect Options:

Option A - Judgment: Judgment refers to a person's ability to make sound decisions and choices. It involves considering multiple options, weighing the pros and cons, and deciding based on reason and logic. Judgment can be assessed in the MSE by asking the patient to solve a hypothetical problem, such as what they would do if they found a stamped envelope on the street.

Option B - Language: Language refers to a person's ability to communicate effectively using spoken or written words. Language can be assessed in the MSE by asking the patient to name objects, repeat phrases, describe a picture, or write a sentence.

Option C - Orientation: Orientation refers to people's awareness of their surroundings and place in time and space. Orientation can be assessed in the MSE by asking the patient to state their name, the date, location, and the reason for their hospitalization.

Solution for Question 3:

Correct Option A - Delirium:

- Confusion Assessment Test (CAM) is primarily used for the assessment and diagnosis of delirium. Delirium is a state of acute confusion and changes in cognition that often occurs in older adults, particularly those who are hospitalized or have underlying medical conditions. The CAM is a validated screening tool that helps healthcare professionals identify the presence of delirium and distinguish it from other cognitive disorders or psychiatric conditions.

Incorrect Options:

Option B - Dementia : While both delirium and dementia involve changes in cognition, they are distinct conditions. Dementia is a chronic and progressive disorder characterized by a decline in memory, thinking, and reasoning skills. The CAM is not typically used for diagnosing dementia.

Option C - Schizophrenia: Schizophrenia is a psychiatric disorder characterized by disturbances in perception, thought processes, and emotions. It does not typically present with acute confusion or cognitive impairment, which are hallmarks of delirium. The CAM is not used for assessing schizophrenia.

Option D - Depression : Depression is a mood disorder that primarily affects a person's emotional state, such as persistent sadness, loss of interest, and changes in appetite or sleep patterns. While depression can sometimes lead to cognitive impairments, the CAM is not specifically designed for assessing depression or its cognitive symptoms.

Solution for Question 4:

Correct Option B - Fluctuating course:

- Fluctuating course is a hallmark feature of delirium.
- It refers to changes in the intensity and severity of symptoms, including attention, cognition, and consciousness, that can vary throughout the day.

- Delirium is characterized by an altered state of consciousness that fluctuates, with periods of lucidity and confusion .

Incorrect Options:

Option A - Occurs gradually over a period of time: Delirium typically has an acute onset, meaning it develops rapidly, often within hours or days, rather than occurring gradually over a period of time. This symptom is more consistent with a chronic cognitive disorder rather than delirium.

Option C - Preserved consciousness: Delirium is associated with an impaired level of consciousness . However, it is important to note that consciousness in delirium can vary, ranging from hyperactive (agitated and restless) to hypoactive (lethargic and withdrawn). While consciousness is impaired, it is not preserved, making this option incorrect.

Option D - Commonly associated with auditory hallucinations: Delirium is more commonly associated with perceptual disturbances, such as visual hallucinations or illusions.

Solution for Question 5:

Correct Option C - Delirium:

- Delirium is an acute state of confusion and cognitive impairment that often occurs due to underlying medical conditions or medication side effects. The CAT is commonly used as a quick and reliable tool to assess for the presence of delirium. It helps identify cognitive deficits, attention disturbances, and other symptoms associated with delirium.

Incorrect Options:

Option A - Dementia- Dementia is a chronic cognitive disorder characterized by progressive decline in memory, thinking, and behavior. The CAT is not specifically designed to assess dementia. Instead, there are other specific assessment tools, such as the Mini-Mental State Examination (MMSE), that are commonly used to evaluate cognitive impairment in dementia.

Option B - Depression- Depression is a mood disorder characterized by persistent feelings of sadness, loss of interest, and changes in appetite, sleep, and concentration. The CAT is not primarily used for assessing depression. Diagnostic criteria for depression involve a comprehensive evaluation of mood symptoms, psychological and physical symptoms, and impairment in functioning.

Option D - Schizophrenia- Schizophrenia is a chronic mental disorder characterized by hallucinations, delusions, disorganized thinking, and impaired social functioning. The CAT is not specifically designed to assess schizophrenia. The diagnosis of schizophrenia involves a comprehensive evaluation of symptoms, duration, and functional impairment.

Solution for Question 6:

Correct Option C - Nucleus of Meynert is not affected:

- The nucleus of Meynert, also known as the basal nucleus of Meynert, is a group of neurons located in the basal forebrain that produces and releases the neurotransmitter acetylcholine.
- In Alzheimer's disease, there is a significant degeneration of the nucleus of Meynert, which contributes to the cholinergic dysfunction observed in the disease.
- The loss of cholinergic neurons and acetylcholine depletion is associated with cognitive impairments.

Incorrect Options:

Option A - Intracellular neurofibrillary tangles are seen: One of the pathological hallmarks of Alzheimer's disease is the presence of intracellular neurofibrillary tangles. These tangles are made up of abnormal tau protein that has accumulated inside neurons. Neurofibrillary tangles contribute to the disruption of neuronal function and eventual cell death.

Option B - Neuritic plaques made of beta-amyloid are found: Neuritic plaques, also known as senile plaques, are another characteristic feature of Alzheimer's disease. These plaques consist of aggregates of beta-amyloid protein that accumulate between neurons in the brain. The presence of neuritic plaques is associated with inflammation and neuronal damage.

Option D - Short-term memory is affected: Short-term memory impairment is a hallmark symptom of Alzheimer's disease. As the disease progresses, individuals may have difficulty retaining and recalling new information, experiencing deficits in short-term memory. This is often one of the initial cognitive symptoms noticed in Alzheimer's disease.

Solution for Question 7:

Correct Option D - Priming:

- Priming: In this scenario, the clue provided by the examiner ("the name sounds like something you get commonly in McDonalds") triggers the retrieval of the memory associated with the first EEG recording. This is an example of priming, where exposure to a stimulus (the clue) influences subsequent processing or recognition of related stimuli.

Incorrect Options:

Option A - Semantic memory: Semantic memory refers to general knowledge and facts about the world, not specific memories or retrieval processes.

Option B - Implicit memory: Implicit memory refers to the unconscious recall of information or skills without conscious awareness. It does not involve the retrieval of specific memories based on cues or clues.

Option C - Associative learning: Associative learning refers to the process of forming associations between stimuli and responses. While priming can be considered a form of associative learning, it is more specific to the activation of related memories through cues or stimuli.

Solution for Question 8:

Correct Option A - Pick's disease:

- Pick's disease. Pick's disease is a type of frontotemporal dementia characterized by progressive degeneration of the frontal and temporal lobes of the brain. It is associated with the accumulation of abnormal protein aggregates in the brain called Pick bodies. Symptoms of Pick's disease include changes in personality, behavior, and language difficulties.

Incorrect Options:

Option B - Parkinson's disease: Parkinson's disease is a neurodegenerative disorder primarily affecting the motor system. It is characterized by tremors, rigidity, bradykinesia (slowness of movement), and postural instability. While cognitive impairment can occur in advanced stages, it is not classified as a subcortical dementia.

Option C - Huntington's disease: Huntington's disease is a genetic disorder characterized by progressive degeneration of brain cells, primarily affecting the basal ganglia. It leads to motor symptoms such as chorea (involuntary jerking movements), as well as cognitive and psychiatric symptoms. It is not classified as a subcortical dementia.

Option D - Wilson's disease: Wilson's disease is a rare inherited disorder that causes copper to accumulate in various organs, including the liver, brain, and other tissues. It primarily affects the liver, but neurological symptoms can also occur. While it can lead to cognitive impairment, it is not classified as a subcortical dementia.

Solution for Question 9:

Correct Option B - Acuity of onset and loss of consciousness:

- Delirium and dementia are both cognitive disorders, but they have distinct features that differentiate them. Delirium is characterized by a rapid onset of symptoms, often within hours or days, and it is accompanied by fluctuations in attention, confusion, disorientation, and sometimes a reduced level of consciousness. On the other hand, dementia is a progressive condition with a gradual decline in cognitive function, primarily affecting memory, language, and problem-solving abilities. In dementia, there is usually no significant impairment of consciousness.

Incorrect Options:

Option A - Acuity of onset and agitation: While delirium can involve agitation, it is not a specific feature that distinguishes it from dementia. Agitation can be present in both conditions.

Option C - Visual hallucinations and impaired memory: Visual hallucinations can occur in certain types of dementia, such as dementia with Lewy bodies, but they are not exclusive to dementia and may also be present in delirium.

Option D - Agitation and irritation: Agitation and irritation can be seen in both delirium and dementia. These symptoms are not specific enough to differentiate between the two conditions.

Solution for Question 10:

Correct Option C - Rivastigmine :

- Rivastigmine is a reversible anticholinesterase used in the treatment of Alzheimer's disease.
- Rivastigmine is also available in a transdermal patch formulation.
- The patch is applied to the skin, and it releases the medication slowly over a 24-hour period, providing a consistent and continuous therapeutic effect.

Incorrect Options:

Option A - Memantine:

- Memantine is an NMDA antagonist, used in moderate-severe disease.
- It works by regulating the activity of glutamate, a neurotransmitter involved in learning and memory.
- It is administered orally.

Option B - Donepezil:

- Donepezil is a reversible anticholinesterase used in the treatment of mild to moderate Alzheimer's disease.
- It is typically administered orally.

Option D - Galantamine:

- Galantamine is a reversible anticholinesterase used in the treatment of Alzheimer's disease.
- It is typically administered orally.

Solution for Question 11:

Correct Option B - Lewy body dementia:

- Capgras syndrome is commonly seen in Lewy body dementia.
- Lewy body dementia is characterized by cognitive fluctuations, visual hallucinations and parkinsonism symptoms.
- Capgras syndrome involves a delusion that a familiar person has been replaced by an imposter.

Incorrect Options:

- Option A, C & D are incorrect because Capgras syndrome is not associated with option A, C and D

Solution for Question 12:

Correct Option B - Vitamin B12:

- Low vitamin B12 levels can cause megaloblastic anemia, inhibition of cell division, and dysfunctional myelination.
- Elevated methylmalonic acid and serum homocysteine are the specific markers of vitamin B12 deficiency. These are also associated with poor myelination. These could be underlying factors that lead to cognitive impairment in old age and can lead to Reversible dementia in old age.

Low vitamin B12 levels can cause megaloblastic anemia, inhibition of cell division, and dysfunctional myelination.

Elevated methylmalonic acid and serum homocysteine are the specific markers of vitamin B12 deficiency. These are also associated with poor myelination. These could be underlying factors that lead to cognitive impairment in old age and can lead to Reversible dementia in old age.

Incorrect Options:

Option A - Vitamin C: Deficiency in old age is characterized by general weakness, fatigue, irritability, and joint pain.

Option C - Vitamin B6 : Deficiency is characterized by a swollen and reddened tongue, dermatitis, and irritability.

Option D - Vitamin B1: Deficiency is characterized by nausea, vomiting, delirium, and blurry vision.

Solution for Question 13:

Correct Option C - Delirium:

- Delirium is a common psychiatric condition seen in medico-surgical patients. It is an acute and fluctuating disturbance of consciousness and cognitive function that often occurs in the context of a medical illness, surgery, or medication use. Delirium is characterized by a rapid onset and can cause impairments in attention, memory, perception, and other cognitive domains. It is commonly associated with underlying medical conditions, such as infections, metabolic imbalances, drug intoxication, or withdrawal. Delirium is considered a medical emergency that requires prompt evaluation and treatment.

Incorrect Options:

Option A - Depression: While depression can occur in medico-surgical patients, it is not the most commonly seen psychiatric illness in this population. Depression is a mood disorder characterized by persistent feelings of sadness, loss of interest or pleasure, changes in appetite or sleep patterns, and other symptoms. While depression can be prevalent in various clinical settings, including medical and surgical patients, delirium is more frequently observed in this context.

Option B - Schizophrenia: Schizophrenia is a chronic psychiatric disorder characterized by disturbances in perception, thinking, emotions, and behavior. It is typically diagnosed during early adulthood and often requires ongoing treatment and support. While schizophrenia can coexist with medical or surgical conditions, it is not the most commonly observed psychiatric illness in medico-surgical patients.

Option D - Dementia: Dementia is a chronic and progressive neurocognitive disorder characterized by a decline in memory, thinking, behavior, and the ability to perform daily activities. While dementia can be seen in medico-surgical patients, it is not the most commonly encountered psychiatric illness in this population. Delirium, with its acute onset and fluctuating course, is more frequently observed in the hospital setting compared to chronic conditions like dementia.

Sleep Disorders

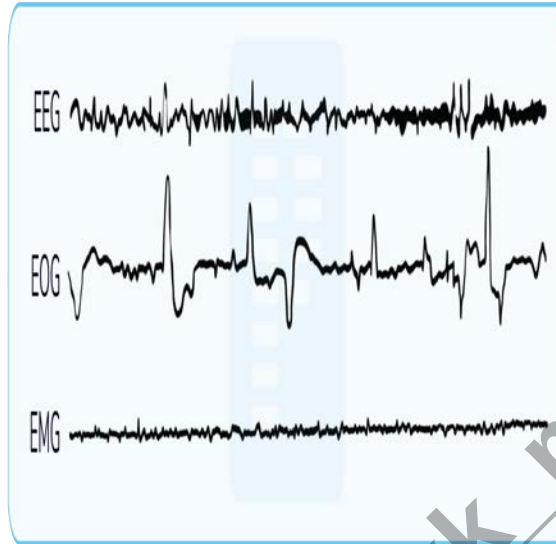
1. Which of the following EEG wave patterns is typically associated with specific states of consciousness or neurological activity in humans, and what does it correctly indicate?

- A. Alpha waves (8-12 Hz) - Indicate deep sleep and unconsciousness.
- B. Theta waves (4-7 Hz) - Often seen in drowsiness and can indicate potential seizures.
- C. Beta waves (13-30 Hz) - Commonly associated with relaxed wakefulness and mental activity.
- D. Delta waves (0.5-4 Hz) - Associated with REM sleep and alertness.

2. Which stage of sleep is primarily associated with the deepest sleep, characterized by the lowest levels of brain activity and the greatest difficulty in awakening?

- A. N1
- B. N2
- C. N3
- D. REM

3. Examine the given EEG, EOG, and EMG recordings. Which of the following statements best describes the physiological characteristics based on these recordings? {{caption_text}}



- A. EEG shows high amplitude, low-frequency waves; EOG shows minimal eye movement; EMG shows high muscle tone.
 - B. EEG shows low amplitude, mixed frequency waves; EOG shows rapid eye movements; EMG shows atonia (reduced muscle tone).
 - C. EEG shows slow waves; EOG shows no eye movement; EMG shows regular muscle contractions.
 - D. EEG shows spikes and waves; EOG shows slow rolling eye movements; EMG shows normal muscle tone
-

4. What does a sleep stage histogram typically represent in sleep studies, and which of the following best describes its key components?

- A. The frequency of waking episodes throughout the night and their duration.
- B. The distribution of sleep stages (NREM and REM) over the course of the night and the amount of time spent in each stage.
- C. The average heart rate during each hour of sleep shows changes in cardiac activity.
- D. The brainwave activity is measured in Hertz (Hz) across different sleep stages, illustrating neurological patterns.

5. A sleep study reveals that a patient experiences increased muscle tone and vivid dreaming. Based on this finding, which of the following statements best differentiates REM sleep from NREM sleep?

- A. REM sleep is associated with increased muscle tone, while NREM sleep features muscle atonia.
 - B. REM sleep is associated with muscle atonia, whereas NREM sleep features increased muscle tone.
 - C. REM sleep occurs only in the early hours of the night, while NREM sleep predominates throughout the night .
 - D. REM sleep is marked by delta waves, while NREM sleep is marked by alpha waves.
-

6. A 45-year-old man presents with difficulty falling asleep and frequent awakenings for the past six months. He reports feeling fatigued during the day but denies any history of substance use, psychiatric illness, or significant stressors. His physical examination and laboratory tests are unremarkable. He has tried over-the-counter sleep aids with little effect. Which of the following is the next best step in the management of this patient?

- A. Cognitive Behavioral Therapy for Insomnia (CBT-I)
 - B. Melatonin supplements
 - C. Benzodiazepines
 - D. Antihistamines
-

7. A 45-year-old woman presents with complaints of difficulty falling asleep and frequent nocturnal awakenings. She describes an uncomfortable sensation in her legs, often accompanied by an irresistible urge to move them, particularly in the evening and at night. These symptoms subside temporarily after moving her legs but recur shortly after. She denies any history of substance use, and her medical

history is unremarkable. What is the most likely cause of her insomnia?

- A. Periodic Limb Movement Disorder
 - B. Obstructive Sleep Apnea
 - C. Restless Leg Syndrome
 - D. Generalized Anxiety Disorder
-

8. A 25-year-old male presents with excessive daytime sleepiness for the past six months. He describes episodes where he suddenly loses muscle tone and falls to the ground after laughing. These episodes increase in situations when he becomes very emotional. He also reports experiencing vivid, dream-like hallucinations upon waking and occasional paralysis on waking from sleep. Based on the clinical picture, what is the most likely diagnosis?

- A. Narcolepsy Type 1
 - B. Obstructive Sleep Apnea
 - C. Idiopathic Hypersomnia
 - D. Narcolepsy Type 2
-

9. A 30-year-old woman is diagnosed with narcolepsy after presenting with excessive daytime sleepiness and episodes of cataplexy triggered by laughter. She also experiences sleep paralysis and hypnagogic hallucinations. Which of the following is the most appropriate initial treatment for her excessive daytime sleepiness?

- A. Modafinil
 - B. Amphetamines
 - C. Zolpidem
 - D. Fluoxetine
-

10. A 19-year-old male presents with recurrent episodes of excessive sleep lasting for several days to weeks. His parents report that during these episodes, he sleeps for 18-20 hours a day, is difficult to wake and when awake, he exhibits irritability, hyperphagia, and hypersexual behavior. Between episodes, he returns to his normal baseline functioning without any neurological deficits. What is the most likely diagnosis?

- A. Narcolepsy
- B. Kleine-Levin Syndrome
- C. Major Depressive Disorder with Atypical Features
- D. Bipolar Disorder - Manic Episode

11. A 32-year-old woman presents to a sleep clinic with episodes of walking around her house at night while appearing to be asleep. She performs complex tasks, sometimes injuring herself, and has no recollection of these activities in the morning. Her doctor notes she experiences heightened stress and occasional sleep deprivation. Which of the following is NOT classified as an NREM-related parasomnia?

- A. Sleepwalking (Somnambulism)
- B. Night Terrors
- C. Confusional Arousal
- D. Nightmares

12. A 28-year-old man is brought to the clinic by his partner, who reports that he frequently acts out his dreams during sleep. She describes him kicking, punching, and even shouting while asleep, often appearing very distressed. He has no recollection of these events upon waking and has no history of sleepwalking. His partner expresses concern about the risk of injury to both of them during these

episodes. On polysomnography, REM sleep without atonia was seen. Which condition is this patient most likely experiencing?

- A. Nightmare Disorder
 - B. REM Sleep Behavior Disorder
 - C. Recurrent Isolated Sleep Paralysis
 - D. Sleep Terrors
-

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Correct Answers

Question	Correct Answer
Question 1	2
Question 2	3
Question 3	2
Question 4	2
Question 5	2
Question 6	1
Question 7	3
Question 8	1
Question 9	1
Question 10	2
Question 11	4
Question 12	2

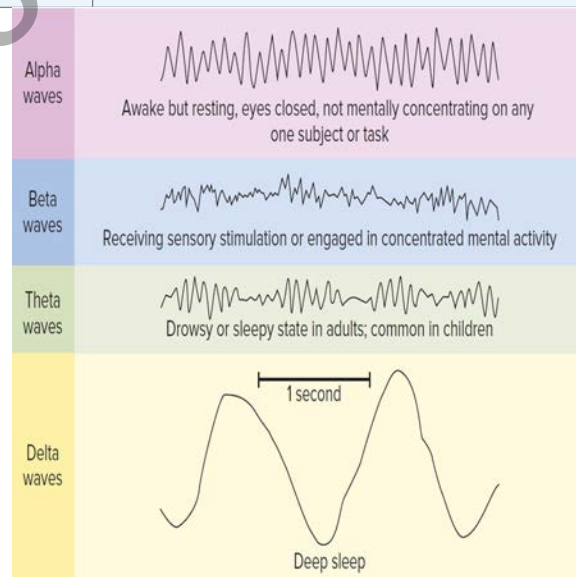
Solution for Question 1:

Correct Answer: B) Theta waves (4-7 Hz) - Often seen in drowsiness and can indicate potential seizures.

Explanation:

- EEG waves are patterns of electrical activity in the brain that are categorized by their frequency.
- Each type of wave is associated with different states of consciousness or neurological activity:

Delta Wave	Seen in slow wave deep sleep or delta sleep. (Option D) Also seen in infants. Frequency: 0.5-3 Hz. Amplitude or the micro voltage- High, ≥ 75 V.
Theta waves	The transition from wakefulness to sleep. 1st wave that is encountered in sleep. Also seen in children. Frequency: 4-7 Hz. Low-voltage waves. Prominently seen in the temporal and parietal regions.
Alpha Waves	Seen at rest - When the person is awake but the eyes are closed. (Option A) Frequency: 8-13Hz. Low-voltage waves. Seen in occipital and parieto-occipital areas.
Beta Wave	Seen in a fully awake person (eyes open). (Option C) Frequency: Higher, 14-30 Hz. Low-voltage waves. Present in the frontal and parietal areas.
Gamma wave	High-frequency brain activity. Frequencies between 30-100 Hz. Prominent during REM sleep. Contribute to the consolidation of memories and processing of information.



Solution for Question 2:

Correct Answer: C) N3 sleep

Explanation:

- N3 sleep is associated with the lowest levels of brain activity and the most difficulty in awakening.

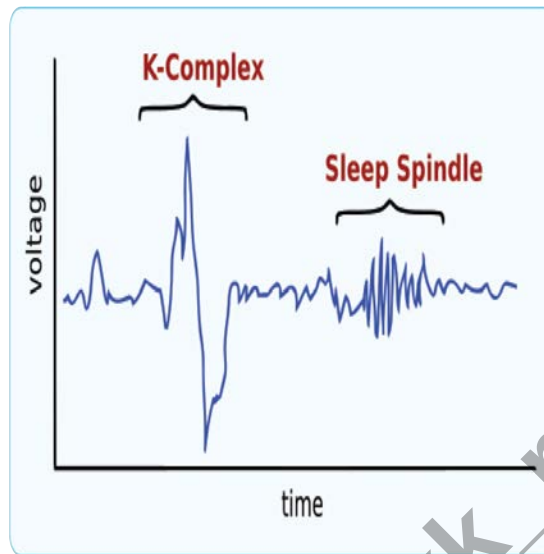
Sleep is divided into 2 stages:

NREM:

NREM is divided into 3 stages

- Stage 1: N1 stage
- Stage 2: N2 stage
- Stage 3 and Stage 4: Stages 3 and 4 have been combined into one stage known as the N3 stage.

Stage	EEG	EOG	EMG
Awake (Stage W)	Alpha waves (8-13 Hz) with eyes closed	Eye movements and blinks present	High tonic activity, voluntary movements
N1 Stage (Light Sleep) (Option A)	Loss of alpha waves, theta waves (4-7 Hz)	Slow eye movements	Tonic activity decreases
N2 Stage (Moderate Sleep) (Option B)	Sleep spindles (12-14 Hz), K complexes (sharp negative followed by positive waves)	None	Very low tonic activity
N3 Stage (Deep Sleep)	Delta waves (slow, high amplitude)	None	Low tonic activity



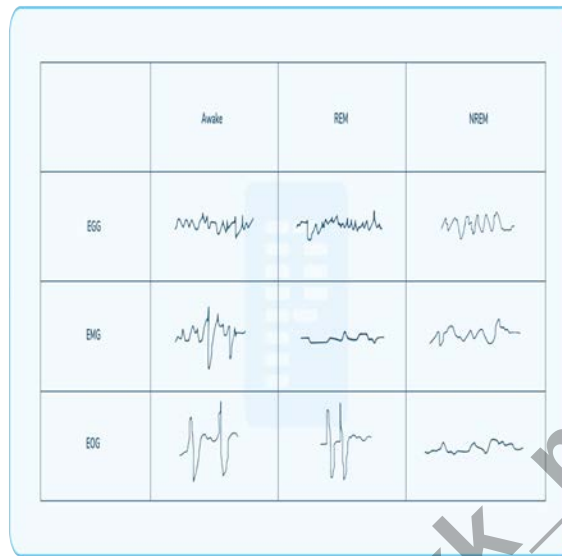
Stage 4 NREM (Option D) is sometimes differentiated but largely overlaps with Stage 3 in terms of characteristics. Predominantly delta waves (low-frequency, high-amplitude brain waves), indicate very deep sleep.

Solution for Question 3:

Correct Answer: B) EEG shows low amplitude, mixed frequency waves; EOG shows rapid eye movements; EMG shows atonia (reduced muscle tone).

Explanation:

- EEG in REM sleep shows low amplitude, mixed frequency waves, similar to an awake state.
- EOG shows rapid eye movements, characteristic of this sleep stage.
- EMG shows atonia, meaning there is a near-complete loss of muscle tone, preventing movement during vivid dreaming.



REM sleep:

EEG findings:

- EEG activity is increased similar to the awake state (beta activity) along with the return of alpha activity (Option A)
- The typical appearance of these waves which are known as Sawtooth waves or Sawtooth appearance.
- Paradoxical sleep: Even though the person is sleeping the EEG shows activity similar to the awake state.
- PGO (Ponto Geniculo Occipital) spikes are also seen - Large phasic potentials that originate from Pons (the cholinergic neurons) and go towards the lateral geniculate body and to the occipital spikes.

EOG findings: Rapid eye movements are seen (Option A)

EMG findings: Tonic atonia (as if the muscle is in paralysis) with phasic twitches (Option D)

EEG shows slow waves; EOG shows no eye movement; EMG shows regular muscle contractions (Option C): Slow waves are seen in deep NREM sleep (N3), not REM sleep, and there are no muscle contractions in REM due to atonia.

Solution for Question 4:

- A sleep stage histogram is a graphical representation that shows the distribution of different sleep stages (such as NREM and REM sleep) throughout the sleep cycle.
- It typically displays the amount of time spent in each stage over the course of the night, providing insights into sleep architecture.
- This helps clinicians evaluate sleep quality and diagnose sleep disorders.

It typically reflects:

- The sequence and duration of various sleep stages (N1, N2, N3, and REM) throughout the night.
- The cyclic nature of sleep shows transitions between stages over time and the relative amount of time spent in each stage.

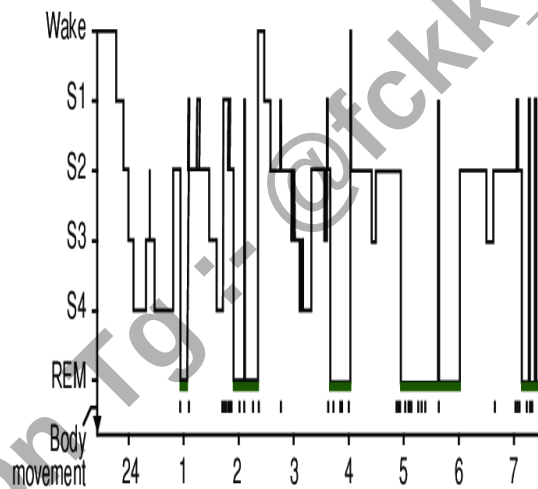


Fig: Sleep stage histogram

The frequency of waking episodes throughout the night and their duration (Option A): The sleep stage histogram does not specifically focus on waking episodes or their durations; its primary function is to map the sleep architecture.

The average heart rate during each hour of sleep shows changes in cardiac activity (Option C): This option refers to cardiac activity, which is not depicted in a sleep stage histogram. Cardiac data may be part of a polysomnographic study but is represented separately.

The brainwave activity is measured in Hertz (Hz) across different sleep stages, illustrating neurological patterns (Option D): Brainwave activity is measured by EEG but is not typically represented in a sleep stage histogram. The hypnogram instead emphasizes sleep stages, not detailed EEG frequencies.

Solution for Question 5:

Correct Answer: B) REM sleep is associated with muscle atonia, whereas NREM sleep features increased muscle tone.

Explanation:

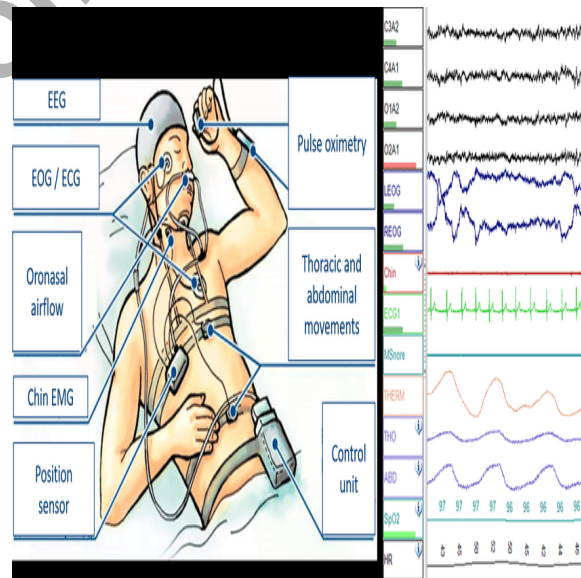
- REM sleep is characterized by muscle atonia (nearly complete paralysis of skeletal muscles), except for the muscles responsible for eye movement and breathing.
- This atonia helps prevent the body from acting out dreams.
- NREM sleep, on the other hand, has increased muscle tone and does not involve vivid dreaming as seen in REM sleep.

REM	NREM
Paradoxical sleep - asleep but the EEG is showing the activity of the active day. Brain activity and physiological activity area increased or same as in the awake state except for the muscle tone (the muscles are in the state of paralysis). Known as desynchronized sleep : Lots of variables in REM Body movements are absent. Penile erection is present. Dreams are remembered. Disorders: Narcolepsy Nightmares REM sleep behavior disorder Sleep apnea	Known as orthodox or conventional sleep Brain activity & physiological activity are lower than in the awake state. Known as synchronized- minimal variations are there in parameters such as pulse rate, respiratory rate, and BP. Pulsatile release of Gonadotropins and Growth hormones. Body movements are present. Penile erection is absent. Dreams are not remembered. Disorders: Somnambulism (sleepwalking) Night terror Bruxism Nocturnal enuresis Somniloquy (sleep talking)

Polysomnography:

- It is the continuous, attended, comprehensive recording of the physiologic changes that occur during sleep.
- A polysomnogram is typically recorded at night and lasts between 6 and 8 hours.
- The following are assessed during sleep: Brain wave activity (EEG) Eye movements (EOG) Submental is electromyography activity (EMG) Nasal–oral airflow Nasal pressure Respiratory effort Oxyhemoglobin saturation Heart rhythm Leg movements Body position Snoring sounds
- Brain wave activity (EEG)

- Eye movements (EOG)
- Submentalisan electromyography activity (EMG)
- Nasal–oral airflow
- Nasal pressure
- Respiratory effort
- Oxyhemoglobin saturation
- Heart rhythm
- Leg movements
- Body position
- Snoring sounds
- Brain wave activity (EEG)
- Eye movements (EOG)
- Submentalisan electromyography activity (EMG)
- Nasal–oral airflow
- Nasal pressure
- Respiratory effort
- Oxyhemoglobin saturation
- Heart rhythm
- Leg movements
- Body position
- Snoring sounds



REM sleep is associated with increased muscle tone, while NREM sleep features muscle atonia (Option A):

- This option incorrectly switches the characteristics of muscle tone in REM and NREM sleep.
- In REM sleep, there is muscle atonia (loss of muscle tone), whereas in NREM sleep, muscle tone is generally increased.

REM sleep occurs only in the early hours of the night, while NREM sleep predominates throughout the night (Option C):

- REM sleep does not occur only in the early hours; it increases in duration as the night goes on, particularly in the latter part of sleep.
- NREM sleep does dominate the early part of the night but is interspersed with REM periods throughout the sleep cycles

REM sleep is marked by delta waves, while NREM sleep is marked by alpha waves (Option D):

- NREM sleep (Non-Rapid Eye Movement sleep) consists of several stages, including light sleep (Stage 1), moderate sleep (Stage 2), and deep sleep (Stage 3), which are characterized by slower brain waves, including delta waves in deep sleep .
- Muscle tone is generally more preserved during NREM sleep compared to REM.

Solution for Question 6:

Correct Answer: A) Cognitive Behavioral Therapy for Insomnia (CBT-I)

Explanation:

- Cognitive Behavioral Therapy for Insomnia (CBT-I) is the first-line treatment for chronic insomnia.
- It addresses maladaptive thoughts and behaviors related to sleep and has long-term benefits with minimal side effects.
- Sleep restriction therapy and stimulus control are components of CBT-I, where patients gradually reduce their time in bed to increase sleep efficiency.

Insomnia is characterized by dissatisfaction with the amount or quality of sleep, leading to significant distress or difficulties in social, occupational, or other important areas of functioning.

Types of Insomnia:

As per ICSD-3 (International Classification of Sleep Disorder)

- Acute or Short-term insomnia disorder: <3 months
- Chronic insomnia disorder: ≥ 3 months

Previous guidelines classify insomnia into:

- Primary - without a specific cause Psychophysiologic Idiopathic Paradoxical
- Psychophysiologic
- Idiopathic
- Paradoxical
- Secondary- due to some other disorder
- Psychophysiologic
- Idiopathic
- Paradoxical

DSM-5 classifies as:

- Episodic (lasting 1–3 mo)
- Persistent (lasting ≥ 3 mo)
- Recurrent (two or more episodes occurring within 1 y)

Insomnia in general may present as:

- Difficulty falling asleep - Initial Insomnia
- Difficulty maintaining sleep - Middle Insomnia
- Early morning awakenings - Late Insomnia

Management of Insomnia:

1. Cognitive Behavioral Therapy for Insomnia (CBT-I)

CBT-I is the first-line treatment for chronic insomnia, conducted over a 6- to 8-week period. It is highly effective and focuses on changing behaviors and thoughts that affect sleep. It involves several components:

- Cognitive Therapy
- Relaxation or Meditation Therapy
- Sleep Education
- Sleep Restriction Therapy
- Stimulus Control Therapy

2. Medications for Insomnia

- Benzodiazepine Receptor Agonists - zolpidem, zaleplon, and eszopiclone

- Melatonin Receptor Agonists - Ramelteon
- Orexin Receptor Antagonists - Suvorexant
- Benzodiazepines

Melatonin supplements (Option B):

- Melatonin is often used for circadian rhythm sleep disorders, such as delayed sleep phase syndrome, and may have some benefit in insomnia.
- However, it is not considered a first-line treatment for chronic insomnia.

Benzodiazepines (Option C):

- While effective in inducing sleep, benzodiazepines are not first-line treatments for chronic insomnia due to their potential for tolerance, dependence, and withdrawal.
- They may be considered for short-term use in acute insomnia but should be avoided for long-term management.

Antihistamines (Option D):

- While easily available, antihistamines are not effective long-term solutions for insomnia and can cause side effects like drowsiness and confusion, especially in older adults.

Solution for Question 7:

Correct Answer: C) Restless Leg Syndrome

Explanation:

Restless Leg Syndrome (RLS) also known as Willis-Ekbom Disease is characterized by an uncomfortable sensation in the legs, especially in the evening or at night, which leads to an irresistible urge to move them.

- The movement temporarily relieves the discomfort, but the symptoms soon return, causing difficulty falling asleep or maintaining sleep.
- This leads to insomnia, which is a common complaint in patients with RLS.

Features of RLS include:

- Sensations that begin after rest
- Relief of discomfort with movement
- Worsening of symptoms at night with a distinct symptom-free period in the early morning

Treatment of Restless Legs Syndrome (RLS)

- No definite cure, but symptoms can be managed.
- Movement of affected limbs may provide temporary relief.
- Treat underlying conditions like peripheral neuropathy, diabetes, or iron deficiency anemia to control symptoms.

Medications:

- Iron supplements : For RLS linked to iron deficiency.
- Anti-seizure drugs (first-line treatment): FDA-approved: Gabapentin enacarbil , others like pregabalin .
- FDA-approved: Gabapentin enacarbil , others like pregabalin .
- Dopaminergic agents : FDA-approved: Ropinirole, pramipexole, rotigotine . Levodopa + carbidopa are used intermittently.
- FDA-approved: Ropinirole, pramipexole, rotigotine .
- Levodopa + carbidopa are used intermittently.
- Opioids : For severe cases unresponsive to other drugs (e.g., methadone, oxycodone).
- Benzodiazepines : For better sleep (clonazepam, lorazepam).
- FDA-approved: Gabapentin enacarbil , others like pregabalin .
- FDA-approved: Ropinirole, pramipexole, rotigotine .
- Levodopa + carbidopa are used intermittently.

Lifestyle Modifications:

- Avoid alcohol, nicotine, and caffeine.
- Maintain a regular sleep pattern.
- Engage in moderate exercise.
- Use leg massage, warm baths, heating pads, or foot wraps for relief.

Periodic Limb Movement Disorder (PLMD) (Option A):

- PLMD involves repetitive, involuntary jerking or movements of the limbs during sleep, which may cause brief arousals but does not necessarily involve the urge to move the legs as seen in RLS.
- While both PLMD and RLS can cause sleep disturbances, RLS symptoms occur primarily before sleep onset, while PLMD typically occurs during sleep.

Obstructive Sleep Apnea (OSA) (Option B):

- OSA is characterized by repetitive episodes of partial or complete obstruction of the airway during sleep, leading to fragmented sleep and daytime fatigue.

- While OSA can cause insomnia, the hallmark of OSA is loud snoring and witnessed apneas, neither of which is mentioned in this case.

Generalized Anxiety Disorder (GAD) (Option D):

- GAD can cause insomnia due to excessive worry and hyperarousal, but the physical symptoms of RLS, such as leg discomfort and the urge to move, are not characteristic of anxiety disorders.
- Anxiety-related insomnia typically involves difficulty falling asleep due to racing thoughts rather than discomfort in the legs.

Solution for Question 8:

Correct Answer: A) Narcolepsy Type 1

Explanation:

Narcolepsy Type 1 is a REM disorder characterized by excessive daytime sleepiness, reduced REM latency, cataplexy (sudden loss of muscle tone triggered by emotions like laughter), hypnagogic or hypnopompic hallucinations (vivid hallucinations occurring while falling asleep or waking up), and sleep paralysis (temporary inability to move upon awakening).

- These are classic symptoms described in this scenario.

Types of Narcolepsy:

As per ICSD 3:

Type 1:

Type 2:

- Cataplexy and/or hypocretin deficiency not present
- Other features of type 1 are present.

According to DSM 5:

- Narcolepsy without cataplexy but with hypocretin deficiency
- Narcolepsy with cataplexy but without hypocretin deficiency
- Autosomal dominant cerebellar ataxia, deafness, and narcolepsy (due to genetic mutation and characterized by late onset in the 30s and 40s)

- Autosomal dominant narcolepsy, obesity, and type 2 diabetes (associated with mutation in myelin oligodendrocyte glycoprotein gene)
- Narcolepsy secondary to another medical condition

Severity of Narcolepsy as per DSM 5:

- Mild : <1/week, 1–2 naps daily mildly, or intermittently disturbed sleep.
- Moderate : cataplexy 1/day. Disturbed sleep, need for multiple naps.
- Severe : near constant. Multiple attacks/day. Disturbed sleep. Resistant to medications.

Treatment of Narcolepsy:

- Scheduled naps (15-20 minutes) throughout the day.
- Maintain a proper nighttime sleep schedule.
- Emotional support, mental health counselling, and education on avoiding alcohol/illicit drugs.
- Exercise is also recommended.
- For Excessive Daytime Sleepiness: First-line: Modafinil (twice daily) or Armodafinil (once daily). Second-line: Amphetamines .
- First-line: Modafinil (twice daily) or Armodafinil (once daily).
- Second-line: Amphetamines .
- For Cataplexy: First-line: Sodium Oxybate (taken in bed, second dose after 2.5-4 hours). Sodium oxybate is restricted and should not be mixed with CNS depressants or alcohol. Alternatives: Tricyclic antidepressants (protriptyline, clomipramine), SNRIs/SSRIs (venlafaxine, fluoxetine).
- First-line: Sodium Oxybate (taken in bed, second dose after 2.5-4 hours). Sodium oxybate is restricted and should not be mixed with CNS depressants or alcohol.
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- First-line: Modafinil (twice daily) or Armodafinil (once daily).
- Second-line: Amphetamines .
- First-line: Sodium Oxybate (taken in bed, second dose after 2.5-4 hours). Sodium oxybate is restricted and should not be mixed with CNS depressants or alcohol.
- Alternatives: Tricyclic antidepressants (protriptyline, clomipramine), SNRIs/SSRIs (venlafaxine, fluoxetine).

Obstructive Sleep Apnea (Option B) presents with daytime sleepiness due to disrupted nighttime sleep but does not feature cataplexy, hypnagogic hallucinations, or sleep paralysis.

- It is caused by intermittent airway obstruction during sleep.

Idiopathic Hypersomnia (Option C) also leads to excessive daytime sleepiness but lacks the hallmark features of cataplexy, hypnagogic hallucinations, and sleep paralysis seen in narcolepsy.

Narcolepsy Type 2 (Option D) has similar symptoms to Type 1 but lacks cataplexy.

- The presence of cataplexy in this patient favors Narcolepsy Type 1.

Solution for Question 9:

Correct Answer: A) Modafinil

Explanation:

Modafinil is a first-line treatment for excessive daytime sleepiness in narcolepsy.

- It promotes wakefulness with a lower potential for abuse compared to traditional stimulants like amphetamines.
- It is well-tolerated and improves the patient's quality of life by reducing daytime sleepiness.
- The management of narcolepsy involves a combination of behavioral modifications and pharmacological interventions to address both excessive daytime sleepiness and cataplexy.

1. Behavioral Modification:

- Patients are encouraged to take 15 to 20-minute naps at strategic times during the day and to maintain good sleep hygiene at night. These naps help to alleviate excessive daytime sleepiness.
- Exercise is also recommended.

2. Pharmacological Treatment:

- For Excessive Daytime Sleepiness (EDS): The first-line treatment is Modafinil (administered twice daily) or Armodafinil (once daily), both of which promote wakefulness. In cases where these are ineffective, amphetamines may be used as second-line treatment.
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- In cases where these are ineffective, amphetamines may be used as second-line treatment.
- For Cataplexy: The first-line treatment for cataplexy is Sodium Oxybate, which induces sleep quickly and is administered in two doses—once while in bed and again 2.5 to 4 hours later. Tricyclic antidepressants (e.g., protriptyline, clomipramine) and SSRIs/SNRIs (e.g.,

venlafaxine, fluoxetine) are also used effectively for cataplexy management.

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- The first-line treatment for cataplexy is Sodium Oxybate , which induces sleep quickly and is administered in two doses—once while in bed and again 2.5 to 4 hours later.
- Tricyclic antidepressants (e.g., protriptyline, clomipramine) and SSRIs/SNRIs (e.g., venlafaxine, fluoxetine) are also used effectively for cataplexy management.

Clonazepam (Option B) is primarily used to treat parasomnias such as REM sleep behavior disorder, but it is not effective for managing daytime sleepiness or cataplexy in narcolepsy.

Zolpidem (Option C) is a sedative-hypnotic used for the short-term management of insomnia.

- It would not address excessive daytime sleepiness or cataplexy in narcolepsy patients.

Fluoxetine (Option D) is a selective serotonin reuptake inhibitor (SSRI) that can be used to treat cataplexy by reducing the frequency of episodes, but it is not the initial treatment for excessive daytime sleepiness.

- It may be considered an adjunct therapy for cataplexy management.

Solution for Question 10:

Kleine-Levin Syndrome (KLS) is a rare, recurrent disorder characterized by episodes of hypersomnia, hyperphagia, hypersexuality, and behavioral disturbances such as irritability and confusion.

- The episodes typically last days to weeks and are separated by periods of normal behavior and cognitive function.
- The hallmark of the syndrome is the cyclical pattern of symptoms with complete recovery between episodes .

Kleine-Levin Syndrome (KLS):

- Periods of excessive sleep (18-20 hours/day) lasting for days to weeks, followed by normal functioning. (Recurrent Hypersomnia)
- Typically affects adolescent males, though females and older adults can also be affected. Most cases begin between ages 10 and 21.
- Symptoms During Episodes : Extreme and prolonged sleep Hyperphagia (excessive eating), hypersexuality, irritability, social withdrawal, and disinhibition (e.g., aggression).
- Extreme and prolonged sleep
- Hyperphagia (excessive eating), hypersexuality, irritability, social withdrawal, and disinhibition (e.g., aggression).
- Episodes may occur 1 to 10 times per year.
- Some patients exhibit only hypersomnia without other behavioral symptoms.
- The condition is usually sporadic, though familial cases have been reported
- Extreme and prolonged sleep
- Hyperphagia (excessive eating), hypersexuality, irritability, social withdrawal, and disinhibition (e.g., aggression).

Narcolepsy (Option A) is characterized by excessive daytime sleepiness, sleep attacks, and often cataplexy, hypnagogic hallucinations, and sleep paralysis.

- It does not involve prolonged episodes of hypersomnia, hyperphagia, or hypersexuality.
- Narcolepsy is more continuous, rather than episodic like Kleine-Levin Syndrome.

Major Depressive Disorder with Atypical Features (Option C) may involve hypersomnia and hyperphagia, but it lacks the episodic nature and additional symptoms of hypersexuality and irritability seen in Kleine-Levin Syndrome.

- Patients with atypical depression typically experience mood reactivity, feelings of heaviness in limbs, and sensitivity to rejection, which are absent in KLS.

Bipolar Disorder - Manic Episode (Option D) may present with hypersexuality and irritability, but it is typically associated with a reduced need for sleep rather than hypersomnia.

- Additionally, mania includes symptoms like grandiosity, pressured speech, and increased goal-directed activity, which are not features of Kleine-Levin Syndrome.

Solution for Question 11:

Sleepwalking , Sleep Terrors , and Confusional Arousals are all NREM-related parasomnias, while nightmares occur during REM sleep

NREM related Parasomnias	
Sleepwalking (Somnambulism)	Involves walking or engaging in complex behaviors while in a state of unconsciousness, which may lead to injuries or dangerous situations. Typically occurs during slow-wave sleep, starting in the first or second slow-wave episodes. Can be triggered by sleep deprivation and disruption of slow-wave sleep.. Difficult to awaken; they often appear confused when roused. More common in children , with a peak prevalence between ages 4 and 8, and usually resolves by adolescence. Can occur as a primary disorder or secondary to another sleep issue, like sleep apnea.
Night Terrors	Night terrors involve sudden arousal with intense fear, often starting with a scream. Accompanied by signs of severe anxiety and confusion; individuals are unresponsive to stimuli. Typically occurs during slow-wave sleep and may be triggered by fever or CNS depressant withdrawal. Amnesia for the episode is common. Unlike nightmares, sleep terrors may lack detailed imagery.
Confusional Arousals	Defined as a milder form of NREM sleep parasomnia, often seen in children. Involves partial awakening, confusion, and then returning to sleep. Believed to be on a continuum with sleepwalking and sleep terrors.
Sleep-Related Eating Disorder	Classified under NREM sleep arousal disorders, specifically as "sleepwalking with sleep-related eating." Involves episodes of eating during sleep, often with no awareness of the behavior. Can lead to weight gain and health issues, including injuries while preparing food.

NREM related Parasomnias	
Sleep Talking (Somniloquy)	Speaking during sleep without awareness is often incoherent or nonsensical. Occurs during stages N1 and N2, but can also happen in REM. Can vary from rare occurrences to frequent episodes, lasting for a few seconds or minutes. Can occur independently or alongside other parasomnias like sleepwalking or night terrors. Stress, sleep deprivation, alcohol, or fever may trigger episodes.

Solution for Question 12:

Correct Answer: B) REM Sleep Behavior Disorder

Explanation:

REM Sleep Behavior Disorder (RBD) is characterized by acting out dreams due to the absence of muscle paralysis during REM sleep, matching the patient's symptoms of kicking, punching, and shouting.

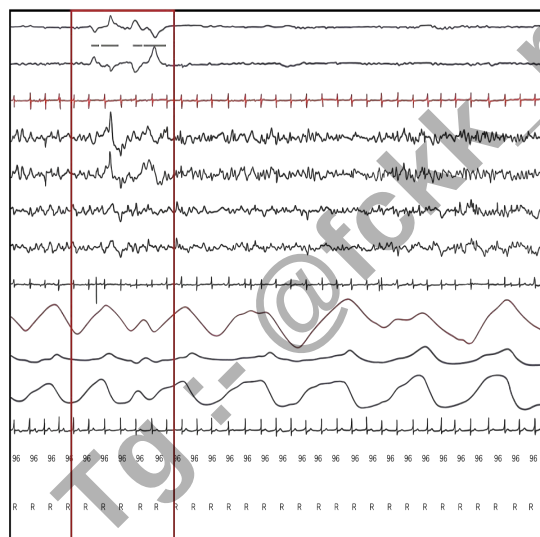
REM related Parasomnias	
Nightmare Disorder (Option A ruled out)	Nightmares are frightening dreams that wake the person sleeping. They are often remembered. Common in children (ages 3-6) but rare in adults. Can lead to insomnia due to fear of sleeping. Associated with PTSD and traumatic events. Certain medications and substance abuse can trigger nightmares. Treatment: Behavioral techniques like desensitization, exposure therapy, image rehearsal therapy, and cognitive therapy can be effective. Medications such as prazosin, nitrazepam, and triazolam may also help, especially for PTSD-related nightmares.

REM related Parasomnias	
REM Sleep Behavior Disorder (RBD)	RBD involves acting out dreams due to lack of sleep paralysis. Individuals may punch or kick while dreaming. They are unaware of their surroundings during episodes. Actions relate directly to dream content. There is a risk of injury to themselves or others. Various drugs and conditions can worsen RBD. On polysomnography, REM sleep without atonia may be seen. May be associated with DLB, Parkinson's disease.
Recurrent Isolated Sleep Paralysis (Option C ruled out)	Sleep paralysis is the inability to move during sleep onset or awakening. It can be distressing and may include hallucinations. It can occur independently of narcolepsy. Paralysis can last from one to several minutes. Risk factors include irregular sleep, stress, and sleep deprivation. Treatment: Improved sleep hygiene and ensuring sufficient sleep are primary treatments. Rapid eye movements or being touched may help terminate episodes.

Sleep Terrors (Option D) : These typically occur during non-REM sleep, involve screaming or intense fear, and are usually not remembered by the individual upon waking.

Previous Year Questions

1. On performing polysomnography in a patient, the waves of EOG, EEG, and EMG, from above downwards, are seen below. Which stage of sleep do the marked areas represent? EOG: Electrooculography EEG: Electroencephalography EMG: Electromyography

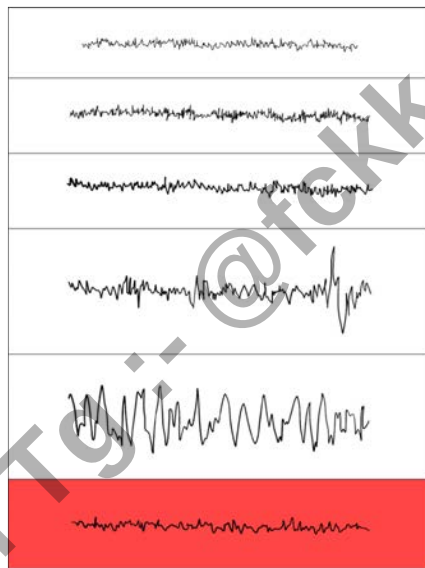


- A. REM sleep
- B. NREM I sleep
- C. NREM II sleep
- D. NREM II sleep

2. What medication is typically prescribed to treat the condition experienced by a 25-year-old woman who complains of a sensation resembling creepy crawling in her legs? This symptom is more pronounced at night and hinders her ability to sleep. However, she finds relief by either engaging in walking or moving her legs.

- A. Gabapentin
- B. Pramipexole
- C. Vitamin B12
- D. Iron tablets

3. Please determine the sleep stage based on the red-highlighted EEG findings.



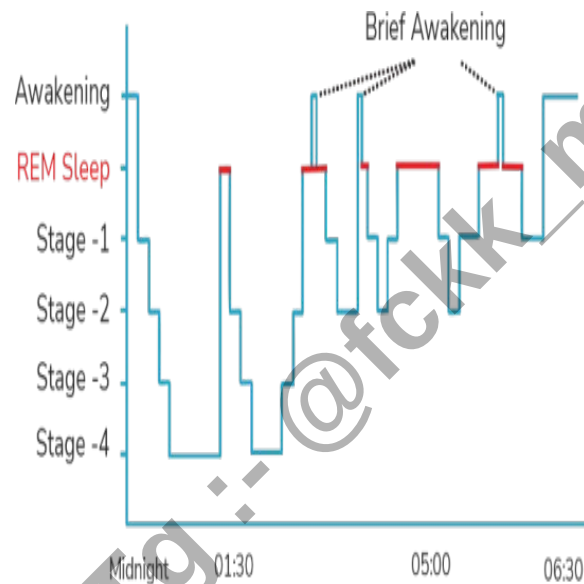
- A. Stage 1
- B. Stage 2
- C. Stage 3
- D. Stage 4

4. Which of the following statements about somnambulism is incorrect?

- A. Disorder of sleep arousal
- B. Person will have full consciousness

- C. Usually, terminates in awakening followed by confusion
- D. There may be neurological condition associated

5. What does the given image represent?



- A. Hypnogram
- B. Electroencephalogram
- C. Polysomnogram
- D. Epworth sleep assessment

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	1
Question 3	4
Question 4	2
Question 5	1

Solution for Question 1:

Correct Option A - REM sleep:

- The given polysomnography shows REM sleep. Rapid eye movements in REM sleep are observed in the following graph. During REM sleep, there will be temporary muscle atonia.
- The brain is quite active during this stage and resembles a state of wakefulness. There will be fluctuations in the heart rate and rate of breathing. Vivid dreams usually occur during REM sleep.

Incorrect Options:

Option B - NREM I: It occurs before wakefulness and sleep. Heart rate and respiration are reduced during this stage. There is a reduction in muscle tone as well as core body temperature. Alpha and theta waves are typically observed during the NREM I stage of sleep.

Option C - NREM II: K complexes and sleep spindles are observed. In this graph, K complexes or sleep spindles are not visualized.

Option D - NREM III: Characterized by low-frequency delta waves. Heart rate and respiratory rate are slow.

Solution for Question 2:

Correct Option A - Gabapentin:

- The symptoms described by the patient are consistent with restless legs syndrome (RLS), a neurological disorder characterized by uncomfortable sensations in the legs, usually accompanied by an irresistible urge to move them. The symptoms are typically worse during periods of rest or inactivity, such as when sitting or lying down, and are relieved by movement, such as walking or stretching.
- Gabapentin is one of the medications commonly used to treat restless legs syndrome. It is an anticonvulsant and analgesic drug that can help alleviate sensory symptoms and improve sleep quality in individuals with RLS. It works by affecting the transmission of certain neurotransmitters in the brain, reducing abnormal electrical activity, and calming overactive nerve signals.

Incorrect Options:

Option B - Pramipexole: Pramipexole is a dopamine agonist that is also used in the treatment of restless legs syndrome. It helps regulate dopamine levels in the brain, which can help alleviate the symptoms of RLS. Pramipexole is a valid treatment option, but in this case, Gabapentin is a more appropriate choice.

Option C - Vitamin B12 : Vitamin B12 deficiency can cause various neurological symptoms, including neuropathy, but it is not the primary treatment for restless legs syndrome. However, if the underlying cause of RLS is determined to be vitamin B12 deficiency, supplementation may be necessary.

Option D - Iron tablets : Iron deficiency anemia is a potential cause of restless legs syndrome, especially in individuals with low iron stores. In such cases, iron supplementation may be prescribed to correct the deficiency and alleviate the symptoms of RLS. However, iron tablets alone are not the treatment for all cases of RLS.

Solution for Question 3:

Correct Option D - Stage 4:

- The EEG findings of sawtooth waves indicate the presence of slow-wave sleep (SWS), which is typically observed in Stage 4 of the sleep cycle. Stage 4 is characterized by the presence of deep, restorative sleep with synchronized slow-wave activity on the EEG.

Incorrect Options:

Option A - Stage 1: In Stage 1 of sleep, the EEG typically shows low-amplitude, high-frequency waves, known as theta waves. It is a transitional stage between wakefulness and sleep and is not characterized by sawtooth waves.

Option B - Stage 2: In Stage 2 of sleep, the EEG shows characteristic features such as sleep spindles (bursts of high-frequency waves) and K-complexes (sharp waveforms). It does not typically exhibit sawtooth waves.

Option C - Stage 3: In Stage 3 of sleep, also known as slow-wave sleep (SWS), the EEG shows a mix of slower delta waves and higher-frequency waves. However, it does not specifically show sawtooth waves.

Solution for Question 4:

Correct Option B - Person will have full consciousness:

- Somnambulism, also known as sleepwalking, is a parasomnia disorder characterized by complex motor behaviors and actions during sleep. During an episode of somnambulism, individuals are in a state of partial arousal from sleep, and their consciousness is typically impaired or altered.

Incorrect Options:

Option A - Disorder of sleep arousal : Somnambulism is classified as a disorder of sleep arousal because it involves partial awakenings from sleep with the occurrence of complex behaviors.

Option C - Usually terminates in awakening followed by confusion: Somnambulism episodes often end with the sleepwalker waking up, and upon awakening, they may experience confusion and disorientation about their surroundings or actions during the episode.

Option B - There may be a neurological condition associated : While somnambulism can occur in otherwise healthy individuals, there can also be underlying neurological conditions or factors that contribute to its occurrence. These may include conditions such as epilepsy, sleep disorders, or

certain medications.

Solution for Question 5:

Correct Option A - Hypnogram:

- A hypnogram is a graphical representation of sleep stages over a period of time. It typically shows the different stages of sleep, such as wakefulness, rapid eye movement (REM) sleep, and various stages of non-REM sleep. It provides a visual depiction of sleep architecture.

Incorrect Options:

Option B - Electroencephalogram (EEG): An electroencephalogram is a test that measures the electrical activity of the brain using electrodes placed on the scalp. It is used to record and analyze brain wave patterns and can provide information about brain function and abnormalities.

Option C - Polysomnogram: A polysomnogram is a comprehensive sleep study that involves monitoring various physiological parameters during sleep. It typically includes the recording of brain activity (EEG), eye movements (electrooculogram), muscle activity (electromyogram), heart rate and rhythm (electrocardiogram), respiratory airflow and effort, and oxygen saturation levels. A polysomnogram is used to diagnose sleep disorders and evaluate sleep quality.

Option D - Epworth sleep assessment: The Epworth Sleepiness Scale is a questionnaire used to assess an individual's daytime sleepiness or propensity to fall asleep in various situations. It does not involve monitoring physiological parameters or providing a visual representation like the other options mentioned.

Sexual Disorders

1. A therapist discusses the sexual response cycle with a patient. Which of the following statements is false about the sexual response cycle?

- A. The excitement stage involves increased blood pressure and heart rate.
- B. Males experience swelling of the glans and testes during the plateau stage.
- C. Females experience contractions of the vagina during orgasm.
- D. Both genders have a refractory period after orgasm.

2. A 50-year-old man reports to his doctor that he is having difficulty achieving an erection during intercourse. He mentions this issue has been ongoing for the past year, despite having previously enjoyed a healthy sex life. Which statement about his condition is TRUE?

- A. Erectile dysfunction is common in all ageing men.
- B. Phosphodiesterase-5 inhibitors can be safely used with nitrates.
- C. Situational erectile disorder means he can achieve erections in certain situations.
- D. Organic Erectile Dysfunction occurs in men with relationship issues.

3. A 30-year-old man visits his doctor expressing frustration with his sexual performance. He reports that he ejaculates within 30 seconds of vaginal penetration, which has caused distress in his relationship. Which of the following treatments is not typically used for premature ejaculation?

- A. Squeeze technique
- B. Stop-start technique
- C. SSRI

D. Sildenafil

4. A 32-year-old woman presents to her gynecologist with concerns about her inability to achieve orgasm during sexual activity with her partner. She reports that this has been an ongoing issue for the past 8 months and is causing distress in her relationship. She has normal sexual desire but is unable to reach orgasm despite adequate stimulation. She has no history of medical conditions or medication use that might explain these symptoms. Which of the following is the most likely diagnosis based on the DSM-5 criteria?

- A. Hypoactive sexual desire disorder
 - B. Female sexual interest/arousal disorder
 - C. Female orgasmic disorder
 - D. Sexual aversion disorder
-

5. A 7-year-old child assigned female at birth has been consistently expressing a strong desire to be a boy for the past 2.5 years. The child prefers traditionally masculine clothing, toys, and activities, and becomes upset when referred to as a girl. What is the most appropriate diagnosis according to the ICD-11 criteria?

- A. Transsexualism
 - B. Dual-role transvestism
 - C. Gender identity disorder of adolescence
 - D. Gender incongruence in childhood
-

6. A 28-year-old male presents to the psychiatric clinic with distressing thoughts and behaviors involving intense sexual arousal from observing unsuspecting individuals in private settings. He reports that he has experienced these urges for

several years and has acted on them multiple times, which has caused legal and interpersonal problems. Which of the following paraphilias best fits his symptoms?

- A. Voyeurism
 - B. Exhibitionism
 - C. Fetishism
 - D. Masochism
-

7. Which of the following paraphilias involves sexual arousal from interaction with corpses?

- A. Necrophilia
 - B. Coprophilia
 - C. Klismaphilia
 - D. Nymphomania
-

8. Which of the following sexual pain disorders involves painful intercourse due to involuntary muscle spasms in the pelvic floor?

- A. Sexual aversion disorder
 - B. Dyspareunia
 - C. Vaginismus
 - D. Vulvodynia
-

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	3
Question 3	4
Question 4	3
Question 5	4
Question 6	1
Question 7	1
Question 8	3

Solution for Question 1:

Correct Answer: D) Both genders have a refractory period after orgasm.

Explanation:

- Males experience a refractory period after ejaculation, during which they cannot become aroused again for a period of time.
- Females do not have a refractory period and can often become aroused again quickly after orgasm.

Phases of the sexual response cycle	
Desire phase	Consists of sex fantasies and the desire to have sex Associated disorder: Hypoactive sexual desire disorder (HSDD) FDA drug approved to treat HSDD in premenopausal women Flibanserin- Partial 5HT1A agonist, oral drug Bremelanotide- Melanocortin receptor agonist, subcutaneous injection)
Excitement Stage	Muscles tense, heart rate and blood pressure increase. (Option A ruled out) Males: Penis becomes erect. Females: Vaginal walls become moist and wider; clitoris enlarges. Associated disorder: Male erectile dysfunction, Female Sexual arousal disorder
Plateau Stage	Breathing becomes faster; muscles continue to tense. Males: Glans swell; testes enlarge. (Option B ruled out) Females: Outer vagina contracts; clitoris retracts.
Orgasm Stage	Neuromuscular tension releases. Males: Penis contracts, ejaculating sperm and semen. Females: Vagina contracts rhythmically. (Option C ruled out) Associated disorder: Female orgasmic disorder, premature ejaculation, delayed ejaculation.
Resolution Stage	The body gradually returns to its resting state. Males: Penis shrinks; includes a refractory period where further arousal is not possible. Females: No refractory period; can become aroused again quickly.

Solution for Question 2:

Correct Answer: C) Situational erectile disorder means he can achieve erections in certain situations.

Explanation:

Situational erectile disorder allows for erections in specific circumstances.

Erectile Dysfunction

- Incidence increases with age; reported in 2-8% of young adults and is more common in older men.
- Not universal in aging; factors like having a partner and consistent sexual activity are protective. (Option A)
- Episodes of erectile disorder can increase anxiety in future sexual encounters.

Causes:

- Organic Causes: Due to medical conditions or medications. Medical Conditions: Cardiovascular diseases (hypertension, atherosclerosis, coronary artery disease), Diabetes mellitus, Neurological disorders (multiple sclerosis, Parkinson's disease, spinal cord injuries), Endocrine disorders (hypogonadism, hyperprolactinemia, thyroid dysfunction), Chronic kidney disease, Liver cirrhosis, Pelvic or prostate surgeries, Sleep disorders (obstructive sleep apnea). Medications: Antihypertensives (beta-blockers, thiazide diuretics), Antidepressants (SSRIs, tricyclic antidepressants), Antipsychotics (typical and atypical), Anti-androgens (finasteride, spironolactone), Opioids, Chemotherapy agents, Anticonvulsants, H2-receptor antagonists (cimetidine).
- Medical Conditions: Cardiovascular diseases (hypertension, atherosclerosis, coronary artery disease), Diabetes mellitus, Neurological disorders (multiple sclerosis, Parkinson's disease, spinal cord injuries), Endocrine disorders (hypogonadism, hyperprolactinemia, thyroid dysfunction), Chronic kidney disease, Liver cirrhosis, Pelvic or prostate surgeries, Sleep disorders (obstructive sleep apnea).
- Medications: Antihypertensives (beta-blockers, thiazide diuretics), Antidepressants (SSRIs, tricyclic antidepressants), Antipsychotics (typical and atypical), Anti-androgens (finasteride, spironolactone), Opioids, Chemotherapy agents, Anticonvulsants, H2-receptor antagonists (cimetidine).
- Psychogenic Causes: More common in young and middle-aged men; often related to psychological factors such as fear, anxiety, or relationship issues. (Option D)
- Medical Conditions: Cardiovascular diseases (hypertension, atherosclerosis, coronary artery disease), Diabetes mellitus, Neurological disorders (multiple sclerosis, Parkinson's disease, spinal cord injuries), Endocrine disorders (hypogonadism, hyperprolactinemia, thyroid dysfunction), Chronic kidney disease, Liver cirrhosis, Pelvic or prostate surgeries, Sleep disorders (obstructive sleep apnea).
- Medications: Antihypertensives (beta-blockers, thiazide diuretics), Antidepressants (SSRIs, tricyclic antidepressants), Antipsychotics (typical and atypical), Anti-androgens (finasteride, spironolactone), Opioids, Chemotherapy agents, Anticonvulsants, H2-receptor antagonists (cimetidine).

Types of Male Erectile Disorder:

- Never able to achieve an erection sufficient for insertion. (Option B)
- Rare, occurring in about 1% of men under 35.
- Previously able to achieve penetration but later unable to do so.
- Affects 10-20% of all men.
- Able to have erections in certain situations (e.g., with a sex worker) but not with a partner.

Investigations used to record nocturnal erections are:

- Nocturnal penile tumescence
- Penile plethysmography (which measures blood pressure in the penis).

These are used to differentiate between organic and psychogenic erectile dysfunction.

Treatment of Erectile Dysfunction	
Pharmacotherapy	Phosphodiesterase-5 inhibitors (PDE-5 inhibitors) Nitric oxide enhancers- facilitates blood flow into the penis and enhances erection. Drugs- Sildenafil, tadalafil, vardenafil and Avanafil. Not used along with organic nitrates- Can cause sudden fall in blood pressure and can be fatal. (Option B)
Other medications: Oral phentolamine : May reduce sympathetic tone and relax muscles of corpora cavernosa and subsequently improve the erection. Alprostadil contains naturally occurring prostaglandin E (vasodilating agent). Alprostadil- Directly injected into corpora cavernosa or given intraurethral through a cannula as a pellet.	
Psychotherapy	Dual sex therapy: It was derived by Masters and Johnson. It is also effective for other sexual disorders as well. It means it treats the couple (not individual). Both the partners are involved in the treatment. Couples are taught: To improve their communication and Exercises to increase sensory awareness.

Treatment of Erectile Dysfunction	
Sensate focus exercises: It has two stages: Non-genital sensate focus: Couples should touch, rub, or kiss each other body parts excluding breasts, and genitals Genital sensate focus: Stage II Involves only breasts and genitals. Subsequently, the couple can engage in complete intercourse. Purpose of exercises is to make the couple aware that pleasure can be given and received by methods other than sexual intercourse (penetration).	
Others: Behavioral therapy Hypnotherapy Psychoanalysis	

Solution for Question 3:

Correct Answer: D) Sildenafil

Explanation:

- Sildenafil is used to treat erectile dysfunction, not premature ejaculation.
- It works by increasing blood flow to the penis but does not address the timing of ejaculation.

Premature Ejaculation (PME)

- A pattern of persistent or recurrent ejaculation with minimal sexual stimulation before or immediately after vaginal penetration.
- DSM-5- Ejaculation occurs within <1 minute following vaginal penetration
- The cause of PME is usually psychogenic.

Ejaculation (PME) treatment" data-author="" data-hash="1174" data-license="" data-source="" data-tags="March2025" src="https://image.prepladder.com/notes/FEpnBa0ExDBZVS8Yf0SO1742541734.png" />

Solution for Question 4:

Correct Answer: C) Female orgasmic disorder

Explanation:

- The patient's symptoms of persistent inability to achieve orgasm during sexual activity, lasting for at least 6 months and causing distress, align with the diagnostic criteria for female orgasmic disorder, according to DSM-5.

Female orgasmic disorder (DSM-5)

Definition	Recurrent or persistent inhibition of female orgasm, as manifested by delay in, or absence of orgasm after a normal sexual excitement phase
Duration	≥6 months
Symptoms	Difficulties with orgasms Delayed/reduced frequency of orgasms Reduced intensity of orgasms
Diagnostic Criteria	≥1 symptom during most sexual encounters
Treatment	Psychotherapy

Hypoactive sexual desire disorder (Option A): Hypoactive sexual desire disorder involves a reduced sexual desire, which is not mentioned in the patient's presentation.

Female sexual interest/arousal disorder (Option B): Female sexual interest/arousal disorder involves a lack of sexual interest or arousal, which is not the primary complaint here.

Sexual aversion disorder (Option D): Sexual aversion disorder is not a DSM-5 diagnosis and involves an aversion to sexual contact, which is not described in this case.

Solution for Question 5:

Correct Answer: D) Gender incongruence in childhood

Explanation:

This case meets the criteria for gender incongruence of childhood as defined by ICD-11:

- There is a marked incongruence between the child's experienced/expressed gender and assigned sex.
- The child shows a strong desire to be a different gender than the assigned sex.
- The child exhibits preferences for clothes, toys, and activities typical of the experienced gender.
- The duration of symptoms (2.5 years) exceeds the requirement of 2 years.
- The child shows distress when gender identity is not acknowledged (upset when referred to as a girl).
- The child is prepubertal.

According to ICD-11 , gender dysphoria has been classified into:

- Gender incongruence of adolescence or adulthood (Option C) : A marked incongruence between an individual's experienced/expressed gender and assigned sex, lasting at least 6 months, manifesting in adolescence or adulthood.
- Gender incongruence of childhood: A marked incongruence between a child's experienced/expressed gender and assigned sex, lasting at least 6 months, manifesting before puberty onset.
- Gender incongruence, unspecified: A condition where an individual experiences a mismatch between their gender identity and assigned sex, without specifying age of onset or meeting full diagnostic criteria.

Previously in ICD-10 , gender dysphoria consisted of:

- Transsexualism (Option A): Desire to live as the opposite sex, often seeking surgery and hormone therapy.
- Dual-role transvestism (Option B): Cross-dressing without a desire for permanent sex change.
- Gender identity disorder of childhood : Persistent discomfort with one's gender starting in childhood.

Solution for Question 6:

Voyeurism, or scopophilia, is a persistent focus on fantasies and behaviors involving the observation of unaware individuals who are nude or involved in sexual activities.

Disorders of Sexual Preference (Paraphilias/Perversions):

Fetishism (Option C)	In fetishism, sexual interest centers on objects (such as shoes, gloves, or stockings) closely linked to the human body, or on non-genital body parts. The latter focus is sometimes called partialism. Exclusively found in men.
Exhibitionism (Option B)	Exhibitionism is the recurrent urge to expose the genitals to a stranger or an unsuspecting person. Almost exclusively found in men.
Transvestism/ Eonism	Previously known as transvestic fetishism. Trans means opposite and vest means clothes Persons dressing in opposite gender clothes for sexual pleasure.
Frotteurism	Characterized by a man rubbing his penis against the buttocks or other body parts of a fully clothed woman to achieve orgasm.
Voyeurism	Persistent focus on fantasies and behaviors involving the observation of unaware individuals who are nude or involved in sexual activities Also known as scopophilia
Masochism (Option D)	Recurrent sexual arousal from being humiliated, beaten, bound, or made to suffer.
Sadism	Recurrent sexual arousal from the physical or psychological suffering of another person.
Pedophilia	Pedophilia involves recurrent intense sexual urges toward, or arousal by, children. As per DSM-5, the child must be under 14 years old, and the patient must be at least 16 years old and at least five years older than the victims.
Other paraphilic disorders	Necrophilia: Obtaining sexual gratification from corpses/cadavers. Zoophilia: Sexual arousal from animals. Telephone and computer scatologia: Sexual arousal involving obscene phone calls or obscene messages/images to unsuspecting persons. Coprophilia: Sexual pleasure associated with desire to defecate on a partner/to be defecated on/eat feces. Urophilia: Sexual pleasure associated with desire to urinate on a partner/to be urinated on. Klismaphilia: Use of enemas for sexual stimulation.

Treatment: Psychotherapy - Cognitive Behavior Therapy (CBT)

Solution for Question 7:

Correct Answer: A) Necrophilia

Explanation:

Necrophilia is a type of paraphilic disorder that refers to obtaining sexual gratification from corpses/cadavers.

- Necrophilia (Option A): Obtaining sexual gratification from corpses/cadavers.
- Zoophilia: Sexual arousal from animals.
- Telephone and computer scatologia: Sexual arousal involving obscene phone calls or obscene messages/images to unsuspecting persons.
- Coprophilia (Option B): Sexual pleasure associated with desire to defecate on a partner/to be defecated on/eat feces.
- Urophilia: Sexual pleasure associated with desire to urinate on a partner/to be urinated on.
- Klismaphilia (Option C): Use of enemas for sexual stimulation.

Nymphomania (Option D) is a type of sexual addiction which includes abnormal, excessive, insatiable desire in a woman for sexual intercourse.

- In a man, a similar condition is called satyriasis, which involves excessive sexual desire in males.

Solution for Question 8:

Correct Answer: C) Vaginismus

Explanation:

- Vaginismus is a constriction of the outer third of the vagina due to involuntary pelvic floor muscle tightening or spasm.
- It interferes with penile insertion and intercourse.

Genito-Pelvic Pain/Penetration Disorder:

In DSM-5, this disorder refers to one or more of the following complaints, of which any two or more may occur together:

- Difficulty having intercourse
- Genito-pelvic pain
- Fear of pain or penetration
- Tension of the pelvic floor muscles

These pain disorders are diagnosed as dyspareunia or vaginismus:

- Dyspareunia (Option B): Recurrent or persistent genital pain occurring before, during, or after intercourse. Although they frequently coexist, it does not specifically involve involuntary muscle spasms making vaginismus a better answer here.
- Recurrent or persistent genital pain occurring before, during, or after intercourse.
- Although they frequently coexist, it does not specifically involve involuntary muscle spasms making vaginismus a better answer here.
- Vaginismus: ■■■■■■ It is the constriction of the outer third of the vagina due to involuntary pelvic floor muscle tightening or spasm. It interferes with penile insertion and intercourse.
- ■■■■■■ It is the constriction of the outer third of the vagina due to involuntary pelvic floor muscle tightening or spasm.
- It interferes with penile insertion and intercourse.
- Recurrent or persistent genital pain occurring before, during, or after intercourse.
- Although they frequently coexist, it does not specifically involve involuntary muscle spasms making vaginismus a better answer here.
- ■■■■■■ It is the constriction of the outer third of the vagina due to involuntary pelvic floor muscle tightening or spasm.
- It interferes with penile insertion and intercourse.

Treatment:

- Dilating vagina (or) opening with fingers or with dilators.
- Dilators may also help in dyspareunia.

Sexual aversion disorder (Option A) is characterized by a strong aversion to sexual activity, rather than physical pain during penetration.

Vulvodynia (Option D) refers to chronic vulvar pain that is not associated with an identifiable cause, which can occur during sexual activity, but it is not primarily caused by muscle spasms.

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Previous Year Questions

1. A man experiences sexual arousal from wearing female undergarments and high heels during sexual activity with women, but he is not attracted to men. What is the most likely diagnosis?

- A. Gender Dysphoria
- B. Transvestic Fetishism
- C. Hermaphroditism
- D. Gay Identity Disorder

2. Match the following: Eonism Exhibitionism Frotteurism Necrophilia
A. Exposure of one's genitals to an unsuspecting stranger
B. Sexual gratification by intercourse with dead bodies
C. Sexual pleasure by wearing clothes of the opposite sex
D. Sexual gratification by rubbing private parts against another person

Eonism
Exhibitionism
Frotteurism
Necrophilia

- A. Exposure of one's genitals to an unsuspecting stranger
- B. Sexual gratification by intercourse with dead bodies
- C. Sexual pleasure by wearing clothes of the opposite sex
- D. Sexual gratification by rubbing private parts against another person

- A. 1-B, 2-A, 3-D, 4-C
- B. 1-C, 2-A, 3-D, 4-B
- C. 1-C, 2-D, 3-A, 4-B
- D. 1-C, 2-B, 3-D, 4-A

3. What diagnosis would you give to a 30-year-old woman who is brought to the psychiatric OPD by her friend, where she claims that her boss is in love with her and stalks her, but her friend denies these allegations?

- A. Megalomania
 - B. Delusion of reference
 - C. Delusion of control
 - D. Erotomania
-

4. Premature ejaculation is a disorder of which phase of the normal sexual cycle?

- A. Desire
 - B. Arousal
 - C. Orgasm
 - D. Pain
-

5. For which of the following conditions is the squeeze technique utilized?

- A. Retrograde ejaculation
 - B. Erectile dysfunction
 - C. Premature ejaculation
 - D. Painful ejaculation
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	4
Question 4	3
Question 5	3

Solution for Question 1:

Correct Option B - Transvestic Fetishism:

- Transvestic fetishism (also known as transvestism or eonism) is characterized by intense sexual arousal from cross-dressing, typically wearing clothing associated with the opposite gender, such as female undergarments and high heels, in this case.

Incorrect Options:

Option A - Gender dysphoria:

- Gender dysphoria refers to the psychological distress that occurs when a person's gender identity does not align with their biological sex.

Option C - Hermaphroditism:

- Hermaphroditism occurs when an individual is born with ambiguous genitalia or both male and female reproductive organs.

Option D - Gay identity disorder:

- Gay identity disorder refers to homosexuality, in which a person is attracted to individuals of the same sex.

Solution for Question 2:

Correct Option B - 1-C, 2-A, 3-D, 4-B:

- Eonism is sexual pleasures attained by wearing clothes of the opposite sex.
- Exhibitionism is the exposure of one's genitals to a stranger.
- Frotteurism is sexual gratification by rubbing private parts against another person.
- Necrophilia is the sexual gratification of having intercourse with dead bodies.

Incorrect Options:

- Option A, C & D are incorrect.
-

Solution for Question 3:

Correct Option D - Erotomania:

- Erotomaniac type of delusion: Also known as "psychosis passionata". This type of delusion involves a belief that a person usually of higher status is in love with the patient. These patients are usually socially withdrawn, dependent, and sexually inhibited with a poor level of social and/or occupational functioning. "Paradoxical conduct" is an important characteristic wherein all denials of affection are rationalized as affirmations. Males with this type of delusion can be more aggressive.

Incorrect Options:

Option A - Megalomania or delusion of grandeur : The patient believes that he has some exceptional identity or power.

Option B - Delusion of reference: The patient believes that events happening around him are somehow related to him. For example, a patient claimed that the tube light of his apartment was flickering as there was a camera fitted inside through which his movements are being recorded.

Option C - Delusion of control: the patient believes others are trying to control their thoughts or behaviors. For example: "My doctor is trying to plant thoughts inside my head."

Solution for Question 4:

Correct Option C - Orgasm:

- Premature ejaculation is a disorder that occurs during the orgasm phase of the normal sexual cycle. It is characterized by the persistent or recurrent ejaculation that happens before or shortly after the initiation of sexual activity, often with minimal stimulation. This results in an inability to delay ejaculation and causes distress or interpersonal difficulties.

Incorrect Options :

Option A - Desire : The desire phase refers to the initial stage of sexual response where an individual experiences sexual interest or desire.

Option B - Arousal: The arousal phase involves physical and psychological changes that prepare the body for sexual activity, such as increased heart rate, genital vasocongestion, and subjective sexual arousal.

Option D - Pain: The pain phase refers to the presence of pain or discomfort during sexual activity, which can be caused by various factors such as medical conditions or psychological issues.

Solution for Question 5:

Correct Option C - Premature ejaculation:

- The Masters and Johnson's squeeze technique is indeed used for Premature ejaculation. It involves the application of pressure or squeezing on the head or base of the penis to delay ejaculation.

Incorrect Options:

Option A - Retrograde ejaculation: Retrograde ejaculation is a condition in which semen flows backward into the bladder instead of being expelled through the penis during ejaculation. It can

be caused by certain medical conditions, medications, or surgery.

Option B - Erectile dysfunction: Erectile dysfunction (ED) refers to the inability to achieve or maintain an erection sufficient for sexual intercourse. It can have various causes, including physical, psychological, or a combination of both.

Option D - Painful ejaculation: Painful ejaculation, also known as dysorgasmia, refers to discomfort or pain during ejaculation. It can be caused by various factors, including infections, inflammation, structural abnormalities, or psychological factors. The approach to managing painful ejaculation involves identifying and addressing the underlying cause.

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Eating disorders

1. A 21-year-old college student presents with feelings of anxiety. She explains that since starting college, she has become increasingly insecure about her weight, particularly after gaining a few kgs. To cope with the stress of her coursework and social pressures, she often binge-eats high-calorie snacks during late-night study sessions. After these episodes, she feels intense guilt and frequently induces vomiting to avoid weight gain. On physical examination, her BMI is measured at 16.5 kg/m and her heart rate is 70 beats per minute. Which of the following is the most likely diagnosis?

- A. Anorexia nervosa
- B. Bulimia nervosa
- C. Binge eating disorder
- D. Orthorexia nervosa

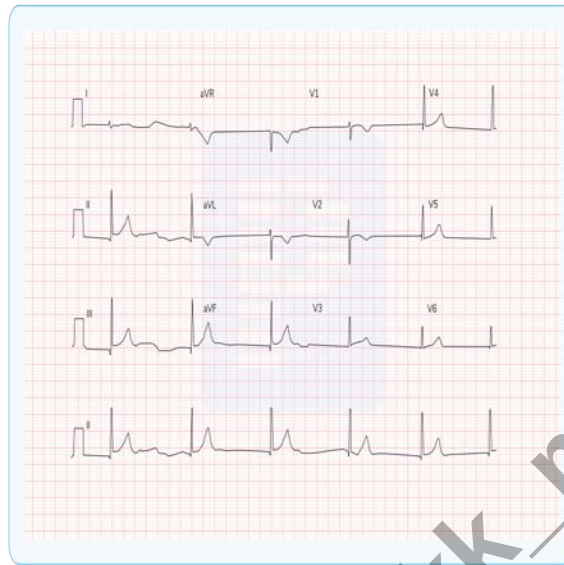
2. A 22-year-old female who is an aspiring model presents with complaints of gastrointestinal discomfort and concerns about her weight. Since beginning her modeling career, she has become increasingly anxious about her body image, feeling pressure to maintain a slim figure to secure modeling opportunities. Recently, she has also started binge eating high-calorie foods during stressful periods, followed by feelings of guilt. To counteract this, she regularly uses over-the-counter laxatives, believing they help her feel thinner. Her BMI is measured at 17 kg/m² and her heart rate is 70 beats per minute. What is the most likely diagnosis?

- A. Anorexia with dietary restriction
- B. Anorexia with purging
- C. Bulimia with purging
- D. Binge eating disorder

3. A 19-year-old female presents with significant anxiety about gaining weight. She has a BMI of 16 kg/m² and she engages in excessive exercise. During the consultation, she reports feeling isolated from her friends and family, who are worried about her health. What is the most appropriate next best step for this patient?

- A. Bupropion for anxiety and weight gain
 - B. Supervised feeding with individual psychotherapy
 - C. Inpatient care for severe malnutrition
 - D. Nutritional counseling with a focus on portion control
-

4. A 15-year-old boy presents to the hospital with lethargy and giddiness. He has a history of being bullied at school due to his physical appearance, which led him to become increasingly interested in various forms of exercise. Over the past three months, he has lost 24 kilograms due to exercise. An ECG performed upon admission is shown below. He appears cachectic and pale. He is assessed by a dietitian and started on nutritional supplements. Two days later, the patient exhibits involuntary muscle contractions, followed by periods of unresponsiveness and confusion. Which electrolyte abnormality would most likely be seen in this patient?



- A. Hypophosphatemia
- B. Hyperkalemia
- C. Hypermagnesemia
- D. Hyponatremia

5. A 22-year-old female medical student presents with a preoccupation with her body image that has persisted since age 13. She describes engaging in constant mirror checking and spending significant time scrutinizing her reflection. However, she acknowledges that her friends and family do not notice any flaws in her appearance and often compliment her. She reports episodes of binge eating followed by engaging in excessive exercise twice a week, for the past 6 months. She expresses significant emotional distress, stating, "I feel like I can't stop myself from overeating, and then I panic about gaining weight". What is the most likely diagnosis?

- A. Bulimia nervosa
- B. Binge-eating Disorder
- C. Body dysmorphic Disorder
- D. Obsessive-compulsive Disorder

6. A 30-year-old woman presents with symptoms of anxiety that have persisted for the past six months. She is seeking medication to help manage her symptoms. Her medical history is unremarkable; she is non-diabetic, non-hypertensive, and has no family history of medical illnesses except for a recent history of bulimia nervosa. Given her background, which of the following medications would pose the highest risk of side effects if prescribed?

- A. Escitalopram
- B. Duloxetine
- C. Venlafaxine
- D. Bupropion

7. A 32-year-old woman with a BMI of 32 kg/m² reports frequent episodes of overeating, feeling disgusted afterwards, and eating even when not hungry. She denies compensatory behaviors like vomiting. What is the most appropriate first-line treatment?

- A. Lisdexamfetamine
- B. Cognitive Behavioral Therapy (CBT)
- C. Interpersonal Psychotherapy (IPT)
- D. Topiramate

8. A 10-year-old boy is brought to the pediatrician by his parents due to concerns about his eating habits. They report that he has a limited variety of foods he is willing to eat and has lost interest in mealtime, often refusing to eat altogether. The boy does not express any fear of weight gain and has not experienced significant weight loss. Which of the following characteristics most likely indicates that he is

suffering from Avoidant Restrictive Food Intake Disorder (ARFID)?

- A. Intense fear of gaining weight leading to severe caloric restriction
 - B. Disturbance in eating patterns due to sensory sensitivities or aversive reactions to food
 - C. Compulsive exercise to maintain a specific body weight
 - D. Engagement in binge-eating episodes followed by purging behavior
-

9. Which of the following best differentiates night-eating syndrome from sleep-related eating disorder?

- A. Higher prevalence of comorbid bulimia nervosa and binge-eating disorder
 - B. Consumption of non-edible foods or substances during eating episodes
 - C. Lack of concern about body image or weight
 - D. Increased risk of developing obesity and metabolic syndrome
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	2
Question 3	2
Question 4	1
Question 5	1
Question 6	4
Question 7	2
Question 8	2
Question 9	2

Solution for Question 1:

Correct Answer: A) Anorexia nervosa

Explanation:

- The correct diagnosis is Anorexia nervosa due to: The patient's significantly low BMI of 16.5 kg/m². Having an intense fear of weight gain and a distorted body image, which are evident in the patient's anxiety about her weight after gaining a few pounds. The combination of binge eating followed by purging behaviors, such as self-induced vomiting, aligns with the criteria for the binge/purge subtype of anorexia nervosa.
- The patient's significantly low BMI of 16.5 kg/m².

- Having an intense fear of weight gain and a distorted body image, which are evident in the patient's anxiety about her weight after gaining a few pounds.
- The combination of binge eating followed by purging behaviors, such as self-induced vomiting, aligns with the criteria for the binge/purge subtype of anorexia nervosa.
- The patient's significantly low BMI of 16.5 kg/m².
- Having an intense fear of weight gain and a distorted body image, which are evident in the patient's anxiety about her weight after gaining a few pounds.
- The combination of binge eating followed by purging behaviors, such as self-induced vomiting, aligns with the criteria for the binge/purge subtype of anorexia nervosa.

Anorexia Nervosa - Clinical Features:	
Restriction of Energy Intake	Low body weight (Most characteristic feature) BMI <18.5kg/m ² in adults BMI-for-age under 5th percentile in children/adolescents
Intense Fear of Weight Gain	Persistent fear of gaining weight or becoming obese
Body Image Distortion	Undue influence of body weight/shape on self-evaluation Inaccurate perception of being normal or overweight
Amenorrhea	Previously required but no longer necessary for diagnosis in DSM-5 or ICD-11
Peculiar Food Behaviors	Hiding food Disposing of food in napkins Rearranging food on plate Collecting recipes and cooking for others
Delayed sexual development	Have poor sexual adjustment Adults may have decreased interest in sex.
Gender prevalence	More common in females than males

Anorexia Nervosa - Medical Complications	
Vital signs	Bradycardia, hypotension, hypothermia, poikilothermia
General	Muscle atrophy, loss of body fat
Central nervous system	Generalized brain atrophy with enlarged ventricles
Cardiovascular	Peripheral (starvation) edema, superior mesenteric artery syndrome
Hematologic	Anemia of starvation, leukopenia

Anorexia Nervosa - Medical Complications	
Gastrointestinal	Decreased intestinal lipase and lactase
Metabolic	Hypercholesterolemia, nonsymptomatic hypoglycemia, elevated liver enzymes, decreased bone mineral density
Endocrine	Decrease: LH, FSH, Estrogen or Testosterone, T 3 , T 4 Increase: Reverse T3, Cortisol, GH, prolactin Normal: TSH

Bulimia nervosa (Option B): Inappropriate compensatory behaviors, like purging, can also be seen in Bulimia Nervosa but bulimics typically maintain a normal or slightly elevated weight, which contrasts with this patient's underweight status.

Binge eating disorder (Option C): Does not involve purging behaviors and is characterized by episodes of eating large quantities of food without compensatory actions.

Orthorexia nervosa (Option D): An unhealthy preoccupation with healthy eating, does not fit the symptoms presented here.

Solution for Question 2:

Correct Answer: B) Anorexia with purging

Explanation:

- This patient is best diagnosed as anorexia (purging type). She exhibits significant weight control behaviors, including binge eating followed by laxative use, while also maintaining a low BMI. This aligns with the criteria for anorexia nervosa, particularly the binge-eating/purging type, where she engages in binge eating and compensatory behaviors such as purging (laxative use here) to manage her weight.
- She exhibits significant weight control behaviors, including binge eating followed by laxative use, while also maintaining a low BMI.
- This aligns with the criteria for anorexia nervosa, particularly the binge-eating/purging type, where she engages in binge eating and compensatory behaviors such as purging (laxative use here) to manage her weight.

- She exhibits significant weight control behaviors, including binge eating followed by laxative use, while also maintaining a low BMI.
- This aligns with the criteria for anorexia nervosa, particularly the binge-eating/purging type, where she engages in binge eating and compensatory behaviors such as purging (laxative use here) to manage her weight.

Type of Anorexia Nervosa	Characteristics
Binge-Eating/Purging type	Recurring purging behaviors (e.g., laxative, diuretic abuse, self-induced vomiting) Episodes of binge eating Occurs at least once a week over the last 3 months
Restricting type	Primary behaviors include dieting, fasting, and/or excessive exercise No recurring purging behaviors or binge eating over the last 3 months

Anorexia with dietary restriction (Option A): While this option describes a subtype of anorexia, it specifically refers to individuals who do not engage in binge eating or purging behaviors.

Bulimia with purging (Option C): Bulimia nervosa is characterized by recurrent episodes of binge eating followed by inappropriate compensatory behaviors (like purging) but typically occurs in individuals with a normal or higher weight.

Binge Eating Disorder (Option D): This is defined by recurrent episodes of binge eating without compensatory behaviors. This patient does engage in binge eating, but she also uses laxatives to prevent weight gain.

Solution for Question 3:

Correct Answer: B) Supervised feeding with individual psychotherapy

Explanation:

- The patient exhibits symptoms consistent with Anorexia nervosa, characterized by significant anxiety about weight gain, a low BMI (16 kg/m^2), and excessive exercise.
- A supervised feeding program ensures the patient receives adequate nutrition, while individual psychotherapy addresses her psychological issues and promotes healthier eating behaviors.

Indications for Inpatient Care: (Option C)

- Weight for height- 20% below normal
- Temperature <35.5°C
- Bradycardia (heart rate <50 beats per minute)
- Blood pressure <80/50 mmHg
- Postural hypotension (>20 mmHg drop)
- Arrhythmias
- Postural tachycardia (>20 beats per minute increase)
- Electrolyte abnormalities (e.g., hypokalemia, hyponatremia)
- Hypoglycemia
- Neutropenia (low white blood cell count)

Management:

Management Type	Details
Non-Pharmacological Management	
Individual Psychotherapy	Focus on behavior modification and improving body image.
Supervised Feeding	Structured meal plans in a monitored setting to ensure nutritional intake.
Nutritional Rehabilitation	Educating patients about nutrition and healthy eating patterns.
Counseling to address maladaptive thoughts and attitudes toward food and body image.	
Family Support	Involvement of family members to provide emotional support and understanding.
Treat Life-Threatening Complications	Address any medical complications (e.g., electrolyte imbalances, dehydration) as necessary.
Pharmacological Treatment	
SSRIs	Can be prescribed to help induce weight gain and manage underlying anxiety or depression.
Olanzapine	May be considered if there is no adequate response to psychotherapy or SSRIs.
Bupropion (Option A)	Contraindicated in patients with anorexia or bulimia due to the increased risk of seizures.

Nutritional counseling with a focus on portion control (Option D): is inadequate because it does not sufficiently address her psychological issues, which are crucial in her treatment.

Solution for Question 4:

Correct Answer: A) Hypophosphatemia

Explanation:

- This is a case of Anorexia nervosa, as evidenced by the 15-year-old boy's significant weight loss, cachectic appearance, and bradycardia observed on his ECG.
- He began a nutritional rehabilitation program to address his malnutrition which led to refeeding syndrome as evidenced by the complication of seizure (involuntary muscle contractions, followed by periods of unresponsiveness and confusion)

Refeeding Syndrome

- Refeeding syndrome is a potentially life-threatening condition that can occur in significantly malnourished patients when they are reintroduced to calories, particularly carbohydrates, after a period of starvation or very low caloric intake.
- The syndrome is characterized by a series of metabolic and electrolyte disturbances due to the sudden increase in insulin secretion in response to increased calorie intake.

Category	Details
Risk Factors	Severe malnutrition (e.g., anorexia nervosa, prolonged fasting) Patients who have experienced significant weight loss or are undergoing aggressive refeeding
Metabolic Changes (Option B, C)	Increased insulin levels lead to shifts in electrolytes, particularly: Decreased phosphate (PO_4^{3-}) : Essential for energy production and cellular function. Most common electrolyte abnormality seen in this condition; it is accompanied by prolonged QT interval and bradycardia. Decreased potassium (K^+) : Important for cardiac and muscular function. Decreased magnesium (Mg^{2+}): Critical for many biochemical reactions and muscle function.

Category	Details
Clinical Complications	Cardiac complications: Arrhythmias due to electrolyte imbalances. Rhabdomyolysis: Muscle breakdown caused by electrolyte disturbances and increased workload. Seizures: Can occur due to severe electrolyte imbalances.
Management	To prevent refeeding syndrome, it is crucial to: Start feeding at a reduced caloric rate and gradually increase. Monitor electrolytes closely, particularly phosphate, potassium, and magnesium. Provide thiamine supplementation before correcting glucose levels.

Hyponatremia (Option D): This does not align with the rapid weight loss and subsequent neurological disturbances observed in this case.

Solution for Question 5:

Correct Answer: A) Bulimia nervosa

Explanation:

- The most likely diagnosis is Bulimia Nervosa. She exhibits episodes of binge eating, characterized by consuming large amounts of food followed by feelings of guilt, which she attempts to counteract with excessive exercise.
- She exhibits episodes of binge eating, characterized by consuming large amounts of food followed by feelings of guilt, which she attempts to counteract with excessive exercise.
- She exhibits episodes of binge eating, characterized by consuming large amounts of food followed by feelings of guilt, which she attempts to counteract with excessive exercise.

Binge Eating Episodes	Recurring episodes of binge eating, occurring at least weekly for the last 3 months.
Compensatory Behaviors	Purging behaviors (self-induced vomiting, laxative abuse, diuretics), excessive exercise, fasting.
BMI	Often normal or slightly overweight (compared to anorexia nervosa).

Binge Eating Episodes	Recurring episodes of binge eating, occurring at least weekly for the last 3 months.
Gender prevalence	More common in females than males
Physical Signs	Parotid gland hypertrophy (may see increased serum amylase) Enamel erosion (from vomiting) Mallory-Weiss syndrome (esophageal tears) These signs appear due to repeated purging.
Electrolyte Disturbances	Commonly hypokalemia and hyponatremia.
Metabolic Changes	Metabolic alkalosis due to vomiting.
Physical Findings	 Dorsal hand calluses from induced vomiting (Russell sign).

Binge-eating Disorder (Option B): This would involve binge eating without compensatory behaviors (excessive exercise), which she does exhibit.

Body dysmorphic Disorder (Option C): Her concerns are more related to her eating behavior rather than a specific perceived defect.

Obsessive-compulsive Disorder (Option D): Although her constant mirror checking could suggest OCD, the primary focus on binge eating and compensatory behaviors aligns more with Bulimia nervosa.

Solution for Question 6:

Correct Answer: D) Bupropion

Explanation:

- Bupropion is contraindicated in individuals with a history of eating disorders, including bulimia nervosa, due to an increased risk of seizures.
- This risk arises from the potential for electrolyte imbalances caused by purging behaviors, which can be exacerbated by bupropion's mechanism of action.

Treatment for Bulimia Nervosa

- This is the first-line treatment, focusing on modifying distorted thoughts and behaviors related to eating and body image.
- Aimed at restoring healthy eating patterns and addressing nutritional deficiencies.
- Fluoxetine: The most effective medication for reducing binge-eating episodes and associated symptoms.
- Topiramate: This can be effective in suppressing binge eating, but not as commonly used as fluoxetine.
- Bupropion: Contraindicated due to an increased risk of seizures, especially in patients with eating disorders.

Evidence indicates that CBT and medications (particularly fluoxetine) are the most effective combination.

Escitalopram, Duloxetine, and Venlafaxine (Option A, B, and C): are generally considered safer options for treating anxiety in patients with a history of eating disorders.

Solution for Question 7:

Correct Answer: B) Cognitive Behavioral Therapy (CBT)

Explanation:

- Cognitive Behavioral Therapy (CBT) is the first-line treatment for Binge Eating Disorder (BED), helping to reduce binge eating episodes and associated psychological problems like depression, although it may not consistently lead to weight loss.

Binge Eating Disorder (BED)

- BED involves frequent binges, often without physical hunger.
- Binges occur in private, typically involving high-calorie foods.

- Individuals feel a lack of control during these episodes.
- It is more common in females than males.
- It is the most common eating disorder.

Etiology:

- Less is known about the causes compared to anorexia and bulimia.
- Risk factors: Childhood obesity Mood disorders Negative family dynamics
- Childhood obesity
- Mood disorders
- Negative family dynamics
- Dietary restriction may trigger binge eating, but some experience binges without restriction, often triggered by emotional stress .
- Childhood obesity
- Mood disorders
- Negative family dynamics

Diagnostic Criteria

- Recurrent binge episodes at least once a week for three months .
- At least three of the following behaviors: Eating faster than usual Eating until feeling overly full Eating when not hungry Eating alone to avoid embarrassment Feelings of guilt or disgust after eating
- Eating faster than usual
- Eating until feeling overly full
- Eating when not hungry
- Eating alone to avoid embarrassment
- Feelings of guilt or disgust after eating
- BED differs from bulimia as there are no compensatory behaviors (e.g., purging).
- BED patients typically have average to obese body weight, unlike anorexia.
- Eating faster than usual
- Eating until feeling overly full
- Eating when not hungry
- Eating alone to avoid embarrassment
- Feelings of guilt or disgust after eating

Comorbidity

Comorbid Conditions	Additional Associations
Mood disorders	Unstable weight (frequent weight cycling)
Anxiety disorders	Insomnia
Substance use disorders	Early menarche
Obesity (nearly half)	Chronic pain in neck, shoulders, or back
Risk of obesity-related issues	Metabolic disorders

Treatment

Psychotherapy

- Cognitive Behavioral Therapy (CBT) : Most effective, first-line treatment. Reduces binge eating and depression but may not lead to weight loss. Enhanced with SSRIs or exercise.
- Interpersonal Psychotherapy (IPT) : Addresses interpersonal issues contributing to BED. (Option C)
- Dialectical Behavior Therapy (DBT) : Some evidence supports its use for BED.

Psychopharmacotherapy

Medication	Effectiveness
Lisdexamfetamine (Option A)	Effective for weight loss and reducing binges
Antidepressants	Improve binge episodes but don't cause sustained weight loss
Topiramate & Zonisamide (Option D)	Moderate weight loss and reduced binge episodes
Combined CBT + Medication	More effective for weight loss than medication or CBT alone

Solution for Question 8:

Correct Answer: B) Disturbance in eating patterns due to sensory sensitivities or aversive reactions to food

Explanation:

- ARFID is characterized by limited food intake due to sensory sensitivities or lack of interest in food, which aligns with the symptoms presented.

Avoidant Restrictive Food Intake Disorder (ARFID)

- ARFID is a recent diagnosis in eating disorders, marked by: Persistent failure to meet nutritional and/or energy needs. Lack of fear of weight gain or drive for thinness, differentiating it from anorexia nervosa (AN). Limited food variety leading to nutritional deficiencies and psychosocial impairment.
- Persistent failure to meet nutritional and/or energy needs.
- Lack of fear of weight gain or drive for thinness, differentiating it from anorexia nervosa (AN).
- Limited food variety leading to nutritional deficiencies and psychosocial impairment.
- Persistent failure to meet nutritional and/or energy needs.
- Lack of fear of weight gain or drive for thinness, differentiating it from anorexia nervosa (AN).
- Limited food variety leading to nutritional deficiencies and psychosocial impairment.

Clinical Features and Diagnosis

Feature	Description
Eating Patterns	Symptoms arise from sensory sensitivities, lack of interest, or aversive reactions to certain foods, not body image concerns.
Demographics	Patients are generally younger with less weight loss before treatment; often depend on enteral nutrition and may have longer hospital stays.
Comorbidities	Overlap with other eating disorders, especially in adolescents; some develop traditional eating disorder psychopathology alongside ARFID symptoms.

Treatment Approaches

Treatment	Description
Cognitive Behavioral Therapy (CBT)	Incorporates systematic desensitization and cognitive restructuring, showing improvement in food acceptance and reduced anxiety related to eating.
Family-Based Therapy	Involves family members in treatment, aiding in maintaining progress at home.
Pharmacological Treatments	Some medications are being researched to alleviate associated anxiety or comorbid conditions.

Intense fear of gaining weight leading to severe caloric restriction (Option A) ARFID does not involve fear of gaining weight or a drive for thinness, which are characteristics of anorexia nervosa.

Compulsive exercise to maintain a specific body weight (Option C) is incorrect because compulsive exercise is more associated with anorexia nervosa or bulimia nervosa.

Engagement in binge-eating episodes followed by purging behavior (Option D) Binge-eating followed by purging is indicative of bulimia nervosa, not ARFID.

Solution for Question 9:

Correct answer: B) Consumption of non-edible foods or substances during eating episodes

Explanation:

- A key distinguishing feature of sleep-related eating disorder (SRED) is the involuntary consumption of non-edible foods or substances and potentially dangerous behaviors while seeking food, which typically occur during a state of partial or full unconsciousness.
- In contrast, night-eating syndrome (NES) involves conscious, delayed eating patterns, and patients are usually not asleep during these episodes.

Night-Eating Syndrome (NES)

- Common among the patients with bulimia nervosa and binge-eating disorder, but is a characteristic sign of NES.
- Food consumption is typically lower during episodes compared to bulimia nervosa or binge-eating disorder.
- Patients are generally not overly concerned about body image or weight.
- Recurrent nocturnal eating episodes, but patients are awake during the eating episodes.
- Low appetite in the morning.
- Difficulty falling asleep, but no sleepwalking or unconscious eating.
- Higher risk for obesity and metabolic syndrome.

Sleep-Related Eating Disorder (SRED)

- Involuntary eating episodes occurring during the night, often while unconscious or asleep.

- Potential ingestion of non-edible substances, dangerous behaviors while searching for or preparing food, and sleep-related injuries.
- Frequently associated with sleepwalking, restless legs syndrome, and obstructive sleep apnea.
- Medication triggers: Episodes have been reported following the use of medications like zolpidem, triazolam, olanzapine, and risperidone.

Option A, C and D does not help to differentiate night-eating syndrome from sleep-related eating disorder

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Previous Year Questions

1. A mother presented her daughter with complaints of eating so much and induce vomiting after food. She was conscious about her weight and her BMI is 25. What is the diagnosis?

- A. Anorexia nervosa
- B. Binge eating disorder
- C. Bulimia nervosa
- D. OCD

2. A young female has episodes of binge eating followed by self-induced vomiting. Which of the following acid- base disturbance is she likely to have?

- A. Respiratory acidosis
- B. Respiratory alkalosis
- C. Metabolic acidosis
- D. Metabolic alkalosis

3. Match the following: 1. Kleptomania A. Intense desire to drink alcoholic drinks 2. Pyromania B. Intense desire to mutilate 3. Mutilomania C. Intense desire to steal items of trivial value 4. Dipsomania D. Intense desire to burn things

1. Kleptomania	A. Intense desire to drink alcoholic drinks
2. Pyromania	B. Intense desire to mutilate
3. Mutilomania	C. Intense desire to steal items of trivial value
4. Dipsomania	D. Intense desire to burn things

- A. 1-D, 2-B, 3-A, 4-C
 - B. 1-C, 2-B, 3-D, 4-A
 - C. 1-C, 2-D, 3-B, 4-A
 - D. 1-D, 2-C, 3-B, 4-A
-

4. What will be your next course of action upon noticing multiple cigarette burns on the forearm of a 14-year-old girl who has presented to the emergency room with a fractured forearm and claims to have tripped and fallen?

- A. Conduct a thorough examination
 - B. Seek a social worker's help
 - C. Inform higher authorities immediately
 - D. Inform your colleagues
-

5. What is a true statement regarding depersonalization disorder?

- A. More common after life threatening trauma
 - B. Seen in seizure and migraine patients
 - C. More common in females
 - D. Mean age at onset is 25 years
-

6. A patient presented with alopecia of unknown origin and abdominal discomfort. What could be the most likely diagnosis based on the image below, which was retrieved from the stomach following a surgery?



- A. Trichobezoar
- B. Splenic rupture
- C. Pancreatic necrosis
- D. Carcinoma stomach

7. What could be the likely diagnosis for a 16-year-old girl who experiences strong food cravings and engages in excessive eating followed by self-induced vomiting?

- A. Anorexia nervosa
- B. Bulimia nervosa
- C. Atypical depression
- D. Binge eating disorder

8. What should be done if a patient with anorexia nervosa follows the recommended diet chart but shows no weight change after one week?

- A. Observe the patient for 2 hours post meal
 - B. Ask the psychiatrist to prescribe anti-anxiety medication
 - C. Increase the calories from 1500 to 2000 in her diet
 - D. Increase water intake
-

9. SCOFF criteria is used as a screening tool for which of the following disorder?

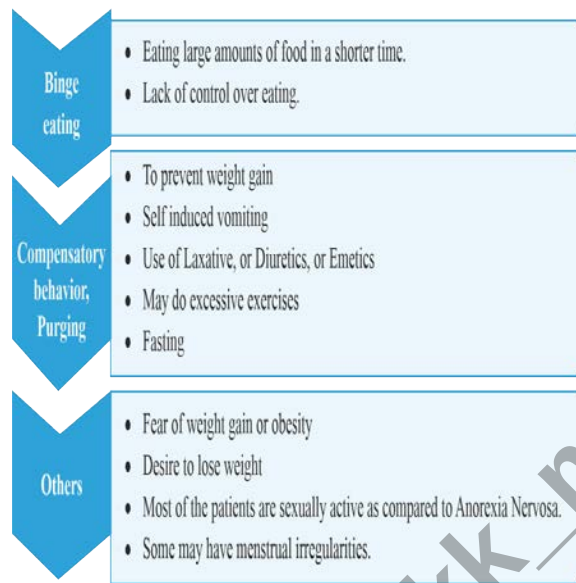
- A. Eating disorder
 - B. Sexual disorder
 - C. Alcohol use disorder
 - D. Schizophrenia
-

Correct Answers

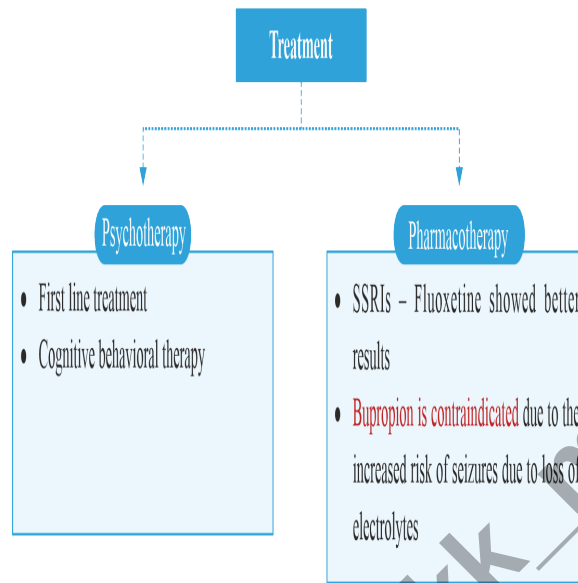
Question	Correct Answer
Question 1	3
Question 2	4
Question 3	3
Question 4	1
Question 5	1
Question 6	1
Question 7	2
Question 8	1
Question 9	1

Solution for Question 1:

Correct Option C - Bulimia nervosa:



- Age of Onset Late adolescence and young adulthood. 8-21 years.
- Late adolescence and young adulthood.
- 8-21 years.
- Episodes of binge eating combined with inappropriate ways of preventing weight gain.
- Occurrence: 1/week for 3 months.
- Weight: Normal or increased
- Features Secondary to Purging Enamel erosion Dental caries Parotitis or salivary gland swelling Russell's Sign: Callus on knuckles- Due to self-induced vomiting, the knuckles get injured by the teeth.
- Enamel erosion
- Dental caries
- Parotitis or salivary gland swelling
- Russell's Sign: Callus on knuckles- Due to self-induced vomiting, the knuckles get injured by the teeth.
- Late adolescence and young adulthood.
- 8-21 years.
- Enamel erosion
- Dental caries
- Parotitis or salivary gland swelling
- Russell's Sign: Callus on knuckles- Due to self-induced vomiting, the knuckles get injured by the teeth.



Incorrect Options:

Option A - In anorexia nervosa: BMI should be decreased

Option B - Binge eating disorder: Alternate attempts at rigorous dieting along with intermittent binge eating and purging. &purging;

Option D - OCD: It's not an eating disorder

Solution for Question 2:

The patient exhibits binge eating followed by self-induced vomiting, which is characteristic of bulimia nervosa. Recurrent vomiting leads to loss of gastric acid (HCl), resulting in hypochloremic metabolic alkalosis. This occurs because:

Binge Eating Episodes	Recurring episodes of binge eating, occurring at least weekly for the last 3 months.
Compensatory Behaviors	Purging behaviors (self-induced vomiting, laxative abuse, diuretics), excessive exercise.
BMI	Often normal or slightly overweight (compared to anorexia nervosa).

Binge Eating Episodes	Recurring episodes of binge eating, occurring at least weekly for the last 3 months.
Physical Signs	Parotid gland hypertrophy (may see increased serum amylase) Enamel erosion (from vomiting) Mallory-Weiss syndrome (esophageal tears) These signs appear due to repeated purging.
Electrolyte Disturbances	Commonly hypokalemia and hypochloremia.
Metabolic Changes	Metabolic alkalosis due to vomiting.
Physical Findings	 Dorsal hand calluses from induced vomiting (Russell sign).

Other Options:

Respiratory acidosis (Option A): This occurs due to hypoventilation (e.g., in chronic lung disease), leading to CO_2 retention and a drop in pH. Vomiting does not cause respiratory acidosis.

Respiratory alkalosis (Option B) : Caused by hyperventilation (e.g., anxiety, panic attacks) leading to excess CO_2 loss, but vomiting does not affect breathing.

Metabolic acidosis (Option C): Seen in conditions such as diarrhea, diabetic ketoacidosis (DKA), or renal failure, where there is an excess loss of bicarbonate (HCO_3^-) or acid buildup. Vomiting removes acid rather than increasing it, making metabolic acidosis unlikely.

Solution for Question 3:

Correct Option C - 1-C, 2-D, 3-B, 4-A:

- Kleptomania is an intense desire to steal items of trivial value.
- Pyromania is an intense desire to burn things
- mutilomania is the intense desire to mutilate,
- dipsomania is an intense desire to drink alcoholic drinks.

These conditions are classified under disruptive, impulse control, and conduct disorders.

Incorrect Options:

- Option A, B & D are incorrect

Solution for Question 4:

Correct Option A - Conduct a thorough examination:

- Conduct a thorough examination: This option is the most appropriate next step. As a healthcare professional, it is crucial to conduct a thorough examination of the patient to assess the extent of their injuries, including the fractured forearm and the presence of multiple cigarette burns. This examination will help determine the severity of the injuries, provide necessary medical treatment, and identify any additional signs of abuse or neglect.

Incorrect Options:

Option B - Seek a social worker's help: Seeking the assistance of a social worker is an important step in cases involving potential child abuse or neglect. Social workers are trained professionals who specialize in addressing such situations. They can provide support, resources, and interventions to ensure the safety and well-being of the child. In many healthcare settings, there are designated professionals who can be called upon to assist in cases involving suspected abuse or neglect. Option C - Inform higher authorities immediately: While it is important to address the situation promptly, informing higher authorities immediately might not be the first step. The immediate focus should be on providing medical care and conducting a thorough examination of the patient. However, if during the examination there are clear signs of abuse or neglect, it is essential to report the situation to the appropriate authorities in accordance with the established protocols and legal requirements in your jurisdiction. This may include contacting child protective services or law enforcement. Option D - Inform your colleagues: While it can be helpful to discuss the case with colleagues, especially if they have experience or expertise in child protection, it should not be the first step in this scenario. The priority is to ensure the immediate safety and medical care of the patient through a thorough examination and, if

necessary, involving a social worker or appropriate authorities.

Solution for Question 5:

Correct Option A - More common after life threatening trauma:

- Depersonalization disorder is a dissociative disorder characterized by persistent or recurrent episodes of depersonalization, where individuals feel detached from themselves and their experiences. It is often associated with significant distress and impairment in daily functioning. While the exact cause of depersonalization disorder is unclear, it is commonly observed that it can be triggered or exacerbated by traumatic experiences, including life-threatening events.

Incorrect Options:

Options B - Seen in seizure and migraine patients: Depersonalization disorder is not specifically associated with seizure or migraine patients. It is a distinct psychiatric condition unrelated to these neurological disorders.

Options C - More common in females: Depersonalization disorder does not show a significant gender bias. It can affect both males and females equally.

Options D - Mean age at onset is 25 years: The age of onset for depersonalization disorder can vary, and there is no specific mean age at onset. It can occur at any age, but it often manifests during adolescence or early adulthood.

Solution for Question 6:

Correct Option A- Trichobezoar:

- Trichobezoar is a mass of hair that forms within the stomach, occurring in individuals with a condition known as trichotillomania.
- Trichotillomania is a compulsive habit of pulling out their own hair and sometimes ingesting it.

- The image shows a mass that could potentially be a trichobezoar, which explains the patient's symptoms of abdominal discomfort.
- The patient also has alopecia of unknown origin, which further suggests a connection to trichotillomania.

Incorrect Options:

- Option B, C and D are incorrect. Refer to the explanation of correct answer.

Solution for Question 7:

Correct Option B - Bulimia nervosa:

- Bulimia nervosa is characterized by recurrent episodes of binge eating, followed by compensatory behaviors such as self-induced vomiting, laxative use, or excessive exercise.
- Individuals with bulimia nervosa often have a normal body weight or maybe slightly overweight, and they tend to have a preoccupation with their body shape and weight.
- The key features are: Strong food cravings Episodes of excessive eating (binge eating) Engaging in behaviors to prevent weight gain (e.g., self-induced vomiting)
- Strong food cravings
- Episodes of excessive eating (binge eating)
- Engaging in behaviors to prevent weight gain (e.g., self-induced vomiting)

The key features are:

- Strong food cravings
- Episodes of excessive eating (binge eating)
- Engaging in behaviors to prevent weight gain (e.g., self-induced vomiting)

Incorrect Options :

Option A - Anorexia nervosa: It is characterized by a restriction of food intake leading to significantly low body weight, a fear of gaining weight, and a distorted body image. While binge eating can occur in anorexia nervosa, purging is not a primary symptom, as individuals with anorexia nervosa tend to be more restrictive in their eating habits.

Option C - Atypical depression: It is a subtype of depression characterized by mood reactivity (i.e., mood improves in response to positive events), increased appetite or weight gain, hypersomnia, and a feeling of being weighed down.

Option D - Binge eating disorder: It is characterized by recurrent episodes of binge eating but without compensatory behaviors such as purging or excessive exercise. Individuals with binge eating disorder may be overweight or obese, and they tend to experience feelings of guilt, shame, or distress after binge eating.

Solution for Question 8:

Correct Option A - Observe the patient for 2 hours post meal:

- In the case of a patient with anorexia nervosa who is following a prescribed diet chart but not experiencing weight changes after one week, it is important to monitor the patient closely for a period after meals. This observation allows healthcare providers to assess the patient's behaviour, including any signs of purging or excessive exercise that may be affecting their weight.

Incorrect Options:

Option B - Asking the psychiatrist to prescribe anti-anxiety medication: While anxiety may be a common comorbidity in anorexia nervosa, it is not the appropriate first step in addressing the lack of weight change. Anxiety medication alone does not directly address the issue of weight gain, and it is essential to rule out other factors before considering medication options.

Option C - Increasing the calories from 1500 to 2000 in her diet: Increasing calorie intake without understanding the underlying cause of the lack of weight change may not be the most appropriate step. It is important to assess the patient's behaviour and rule out any compensatory behaviours or medical factors that may be affecting weight before making changes to the diet plan.

Option D - Increasing water intake: While adequate hydration is important for overall health, increasing water intake alone is unlikely to address the lack of weight change in this scenario. It is crucial to evaluate the patient comprehensively and focus on potential behavioural factors or medical conditions that may be impacting weight.

Solution for Question 9:

Correct Option A - Eating Disorder:

- The SCOFF criteria are primarily used as a screening tool for eating disorders, including anorexia nervosa, bulimia nervosa, and binge eating disorder. The five questions in the SCOFF criteria are related to key symptoms and behaviors commonly seen in individuals with eating disorders, such as a distorted body image, a preoccupation with weight, and loss of control over eating.
- The SCOFF questionnaire comprises the following inquiries: Do you induce vomiting due to feeling uncomfortably full? Do you fear losing control over your eating habits? Have you experienced a weight loss exceeding one stone (14 lb) within a three-month duration? Do you perceive yourself as fat despite others commenting that you are too thin? Would you describe food as dominating your life?
- Do you induce vomiting due to feeling uncomfortably full?
- Do you fear losing control over your eating habits?
- Have you experienced a weight loss exceeding one stone (14 lb) within a three-month duration?
- Do you perceive yourself as fat despite others commenting that you are too thin?
- Would you describe food as dominating your life?
- Do you induce vomiting due to feeling uncomfortably full?
- Do you fear losing control over your eating habits?
- Have you experienced a weight loss exceeding one stone (14 lb) within a three-month duration?
- Do you perceive yourself as fat despite others commenting that you are too thin?
- Would you describe food as dominating your life?

Incorrect Options:

Option B - Sexual disorder: Sexual disorders encompass a range of conditions that affect a person's sexual thoughts, desires, or behaviors. The SCOFF criteria are not designed to assess or evaluate these types of disorders.

Option C - Alcohol use disorder: Alcohol use disorder refers to a problematic pattern of alcohol consumption leading to significant impairment or distress. While screening tools do exist for alcohol use disorder, including the AUDIT (Alcohol Use Disorders Identification Test) and the CAGE questionnaire but the SCOFF criteria are not among them.

Option D - Schizophrenia: Schizophrenia is a complex mental disorder characterized by symptoms such as hallucinations, delusions, disorganized thinking, and impaired social functioning. The SCOFF criteria are specific to screening for eating disorders and do not assess symptoms related to schizophrenia.

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Personality Disorders

1. A 32-year-old male presents with a lifelong pattern of social withdrawal. He avoids social interactions, prefers to be alone, and shows no interest in forming close relationships. He appears emotionally distant and indifferent to praise or criticism. On examination, there are no signs of psychosis, delusions, or hallucinations. What is the most likely diagnosis?

- A. Schizotypal Personality Disorder
- B. Schizoid Personality Disorder
- C. Paranoid Personality Disorder
- D. Avoidant Personality Disorder

2. A 40-year-old man is brought to therapy by his partner, who is frustrated with his constant need for admiration and his belief that he deserves special treatment. He frequently exaggerates his achievements and often feels that others are envious of him. Which type of personality disorder is he most likely exhibiting?

- A. Antisocial Personality Disorder
- B. Borderline Personality Disorder
- C. Histrionic Personality Disorder
- D. Narcissistic Personality Disorder

3. A 29-year-old woman with a history of self-harm, chronic emptiness, and unstable relationships often alternates between idealizing and devaluing others. What defense mechanism is most commonly observed in this disorder?

- A. Sublimation

- B. Splitting
 - C. Rationalization
 - D. Displacement
-

4. A 28-year-old woman presents with intense fear of rejection, leading her to avoid social interactions despite desiring close relationships. She often misinterprets others' comments as negative and feels timid and lacking in self-confidence. Which of the following is the most likely diagnosis?

- A. Dependent Personality Disorder
 - B. Avoidant Personality Disorder
 - C. Obsessive-Compulsive Personality Disorder
 - D. Social Anxiety Disorder
-

5. A 30-year-old male presents with episodes of sudden aggression, leading to property destruction and physical assaults. He reports feeling intense tension before these outbursts and experiences significant guilt and remorse afterward. Which of the following impulse control disorders is most likely affecting him?

- A. Kleptomania
 - B. Pyromania
 - C. Intermittent Explosive Disorder
 - D. Compulsive Hoarding
-

6. A 42-year-old man presents to the clinic, expressing concern over his frequent and uncontrollable urges to engage in sexual activities. He reports spending several hours per day on sexual pursuits, such as browsing explicit material and

seeking multiple sexual partners. Despite attempts to limit his behavior, he continues to act out sexually, leading to significant distress and disruption in his work and personal relationships. He has no history of substance abuse or mood disorders. Which of the following is the most appropriate diagnosis?

- A. Compulsive sexual behavior disorder
 - B. Bipolar disorder with hypersexuality
 - C. Borderline personality disorder
 - D. Sexual aversion syndrome
-

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Correct Answers

Question	Correct Answer
Question 1	2
Question 2	4
Question 3	2
Question 4	2
Question 5	3
Question 6	1

Solution for Question 1:

Answer: B) Schizoid Personality Disorder

Explanation:

Schizoid Personality Disorder : Characterized by lifelong social withdrawal, emotional coldness, and lack of desire for social relationships. This patient fits the classic presentation with no psychosis or interest in forming close relationships.

Personality disorder: An enduring pattern of inner experience and behavior that deviates markedly from the expectations of the individual's culture

- Pervasive and inflexible
- Onset in adolescence or early adult- hood
- Stable over time
- Ego-syntonic (i.e., acceptable to the ego)
- Leads to distress or impairment

DSM-5 divides the disorders into three categories, or clusters: A, B, and C.

Cluster A Personality Disorders

Cluster A personality disorders are characterized by odd and aloof behavior, prone to develop psychosis. They include:

- Paranoid Personality Disorder
- Schizoid Personality Disorder
- Schizotypal Personality Disorder

Disorder	Key Features	Diagnosis	Treatment
Paranoid Personality Disorder (Option C ruled out)	Pervasive distrust and suspiciousness of others Hostility, irritability, anger Difficulty with trust and intimacy Cynical view of world	No fixed delusions (vs. delusional disorder) No hallucinations or formal thought disorder (vs. schizophrenia) No tumultuous relationships (vs. borderline personality disorder) No history of antisocial behavior (vs. antisocial personality disorder)	Psychotherapy : Straightforward and honest therapist, avoid warmth or deep interpretation Group therapy : Helpful for some, but poorly tolerated Behavior therapy : Social skills training, may feel intrusive
Schizoid Personality Disorder (Option B)	Lifelong social withdrawal Eccentric, isolated, and lonely Introverted with a bland affect	No psychotic symptoms (vs. schizophrenia, delusional disorder) More withdrawn and aloof (vs. paranoid personality disorder) No dysphoric loneliness Fewer past relationships, possible autistic reverie No desire for social interaction (vs. avoidant personality disorder)	Psychotherapy : Build trust, encourage expression Group therapy : May benefit, initial silence likely Pharmacotherapy : Limited evidence, medications for specific symptoms (e.g., emotional detachment)
Schizotypal Personality Disorder (Option A ruled out)	Social difficulties Eccentric behavior and thinking ("magical thinking") Peculiar speech, superstitious beliefs, perceptual illusions	Odd behavior, thinking, and perception (vs. schizoid and avoidant personality disorders) Differentiated from autism by the severity of language disorder Lacks psychosis, or it's brief (vs. schizophrenia) No odd behavior (vs. paranoid personality disorder)	Psychotherapy : Sensitive approach to avoid ridicule Antipsychotics : Manage illusions and odd ideas Antidepressants : Used for depressive symptoms

Avoidant Personality Disorder (Option D) : Involves social withdrawal due to fear of rejection, but those individuals still desire social connections, unlike schizoid personality disorder (Cluster C Personality Disorder).

Solution for Question 2:

Correct Answer: D) Narcissistic Personality Disorder

Explanation:

- The patient exhibits a pervasive pattern of grandiosity, a strong need for admiration, and a sense of entitlement.
- These traits include exaggerating achievements and believing that others are envious, which are hallmark features of Narcissistic Personality Disorder.

Cluster B Personality Disorders

Cluster B disorders are characterized by dramatic, impulsive, exploitative, and erratic behaviors.

The disorders include:

- Antisocial Personality Disorder (ASPD)
- Borderline Personality Disorder (BPD)
- Histrionic Personality Disorder (HPD)
- Narcissistic Personality Disorder (NPD)

Disorder	Definition	Key Characteristics	Treatment Options
Antisocial Personality Disorder (ASPD) (Option A ruled out)	Disregard for the rights of others since age 15 (<15 years- "Conduct Disorder")	Disregard legal and social rules, history of lying, manipulation, lack of remorse, criminality, impulsivity, irritable/aggressive Charming demeanor H/o substance use	Psychotherapy: Contingency management. Pharmacotherapy: Anticonvulsants, β -blockers, lithium, antipsychotics.
Borderline Personality Disorder (BPD) (Option B ruled out)	Instability in relationships, unstable self-image, and emotions; marked impulsivity.	Tumultuous relationships, identity diffusion, mood swings, impulsivity, Fear of abandonment. Feeling empty inside h/o suicidal attempts. M/C in females.	Psychotherapy: DBT, MBT, TFP* Pharmacotherapy: Antipsychotics, antidepressants. (avoid benzodiazepines)

Disorder	Definition	Key Characteristics	Treatment Options
Histrionic Personality Disorder (HPD) (Option C ruled out)	Excessive emotionality and attention-seeking behavior.	Strong need for attention, exaggerated emotions, shallow relationships.	Psychotherapy: Psychoanalytically oriented therapy. Pharmacotherapy: Antidepressants, anti-anxiety agents, antipsychotics.
Narcissistic Personality Disorder (NPD)	Pattern of grandiosity, need for admiration, and lack of empathy	Exaggerated self-importance, craving admiration, lack of empathy, sense of entitlement, poor tolerance to criticism, fantasies of unlimited power/ success	Psychotherapy: Individual and group therapy. Pharmacotherapy: Lithium for mood swings, antidepressants

*DBT- Dialectical behavior therapy

*MDT- Mentalization-based therapy

*TFP- Transference-focussed psychotherapy

Solution for Question 3:

Correct answer: B) Splitting

Explanation:

- Splitting is a common defense mechanism in borderline personality disorder, where individuals view people or situations as either all good or all bad, contributing to their unstable relationships.

BORDERLINE PERSONALITY DISORDER:

General Characteristics

- More common in females than in males.
- Persons with BPD are often in a state of crisis, with frequent mood swings, shifting from being argumentative to depressed, and later feeling numb.
- They can experience short-lived psychotic episodes and their behavior is highly unpredictable,

Self-Destructive Behaviors

- Repetitive self-destructive acts such as wrist slashing or self-mutilation, often to elicit help from others, express anger, or numb overwhelming emotions.

Interpersonal Relationships

- Individuals with BPD have tumultuous interpersonal relationships, feeling both dependent on others and hostile toward them when frustrated.
- They cannot tolerate being alone and prefer companionship, even in unhealthy relationships.

Identity and Emotional Experience

- Chronic feelings of emptiness, boredom, and a lack of a consistent sense of identity (identity diffusion) are common, with some complaining of feeling depressed despite fluctuating moods.

Distorted Relationships

- Relationships are distorted, with others seen as either "all good" or "all bad" due to splitting, leading to idealization or devaluation of people.

Defense Mechanism

- Projective identification is a common defense mechanism, where patients project unacceptable aspects of themselves onto others, inducing them to play the projected role.

Treatment

- Dialectical behaviour therapy.
- Mentalization based therapy.
- Transference: Focussed Psychotherapy.
- Pharmacotherapy: Antipsychotics, antidepressants, mood stabilizers like carbamazepine.

Sublimation (Option A) is a mature defense mechanism where individuals redirect unacceptable impulses or emotions into socially acceptable behaviors or activities.

Rationalization (Option C) involves explaining away or justifying behaviors or feelings to make them seem more acceptable or less distressing.

Displacement (Option D) involves shifting emotions from a threatening target to a safer one, such as expressing anger toward a friend instead of an authority figure.

Solution for Question 4:

Answer: B) Avoidant Personality Disorder

Explanation:

- Avoidant Personality Disorder is characterized by hypersensitivity to rejection, a desire for close relationships, and avoidance due to fear of criticism or negative evaluation. This aligns with the patient's symptoms.

Cluster C Personality Disorders

- Characteristics : Cluster C personality disorders are characterized by anxious and fearful features and include: Avoidant Personality Disorder Dependent Personality Disorder Obsessive-Compulsive Personality Disorder
- Avoidant Personality Disorder
- Dependent Personality Disorder
- Obsessive-Compulsive Personality Disorder
- Avoidant Personality Disorder
- Dependent Personality Disorder
- Obsessive-Compulsive Personality Disorder

Cluster C personality disorders:

Cluster C Personality Disorders	Characteristics	Diagnosis	Treatment
Avoidant/ Anxious Personality Disorder	Hypersensitivity to rejection Desires close relationships but avoids them due to fear of rejection or criticism Timid and lacks self-confidence Misinterprets others' comments negatively	Differentiate from: Social Anxiety Disorder: Avoidance of specific situations Panic Disorder with Agoraphobia: Avoidance develops after panic attacks Schizoid Personality Disorder: Desires social interaction unlike those with schizoid disorder.	Psychotherapy: Build a strong therapeutic alliance Encourage gradual risk-taking in social situations Group therapy for understanding social fears Assertiveness training for better self-expression and self-esteem Pharmacotherapy: Beta-adrenergic antagonists like atenolol Serotonergic agents like SSRI

Cluster C Personality Disorders	Characteristics	Diagnosis	Treatment
Dependent Personality Disorder (Option A ruled out)	Pervasive dependent and submissive behavior Difficulty making decisions without reassurance Avoids responsibility Fears being alone; may tolerate abusive relationships for connection	Differentiate from: Histrionic Personality Disorder: More manipulative and often involved in short-term relationships Borderline Personality Disorder: More unstable relationships.	Psychotherapy: Insight oriented therapy, behavior therapy, assertiveness training, family therapy and group therapy are also found useful. Pharmacotherapy: Symptomatic management such as antidepressants.
Obsessive-Compulsive Personality Disorder (Option C ruled out)	Preoccupation with orderliness, perfectionism, and control Rigid and inflexible Difficulty relaxing Highly devoted to work, often neglecting relationships	Differentiate from: Obsessive-Compulsive Disorder (OCD): Symptoms are less severe and time-consuming in OCPD. Ego-syntonic in OCPD vs ego-dystonic in OCD Diagnosis is reserved for those with significant occupational or social impairment.	Psychotherapy: Cognitive therapy or cognitive behavioral therapy (CBT) Interpersonal psychotherapy may help with depressive symptoms Pharmacotherapy: Limited evidence; some studies suggest benefits from fluvoxamine and carbamazepine. Citalopram may help with comorbid depression.

Social Anxiety Disorder (Option D) primarily involves anxiety in specific social situations, not a general avoidance of social contact as seen in avoidant personality disorder.

Solution for Question 5:

Correct Answer: C) Intermittent Explosive Disorder

Explanation:

- Intermittent Explosive Disorder is marked by episodes of uncontrolled aggression with significant tension before the outbursts and feelings of guilt afterward.

Impulse Control Disorders

- Definition : Conditions marked by an inability to resist strong urges to engage in harmful actions toward oneself or others.
- Characteristics : Tension and arousal before the act. Gratification and relief post-act. Followed by remorse, guilt, and secrecy, often delaying treatment.
- Tension and arousal before the act.
- Gratification and relief post-act.
- Followed by remorse, guilt, and secrecy, often delaying treatment.
- Tension and arousal before the act.
- Gratification and relief post-act.
- Followed by remorse, guilt, and secrecy, often delaying treatment.

Examples of Impulse Control Disorders:

Disorder	Description
Intermittent Explosive Disorder (Option C)	Episodes of uncontrolled aggression leading to assaults or property damage. Aggression is disproportionate to triggers; episodes are followed by regret. Important to rule out other causes.
Kleptomania (Option A ruled out)	Recurring urge to steal unnecessary items, with gratification derived from the act. Guilt or anxiety may follow, but not anger or revenge. Most shoplifters do not have kleptomania.
Pyromania (Option B ruled out)	Repeated deliberate fire-setting, accompanied by an urge and a sense of relief. Individuals may show no remorse or concern for consequences.

Comorbidity of Impulse Control Disorders:

Disorder	Common Comorbid Conditions
Intermittent Explosive Disorder	Often co-occurs with substance use disorders, mood disorders, anxiety disorders, personality disorders, PTSD, and eating disorders. Increased risk of self-harm.
Kleptomania	Frequently comorbid with major depressive disorder, anxiety disorders, other impulse control disorders, eating disorders, personality disorders, OCD, and substance use disorders. Increased suicide risk.

Disorder	Common Comorbid Conditions
Pyromania	Often associated with substance use disorders, mood disorders (depressive and bipolar), and other impulse control disorders. Increased suicidal thoughts reported.

Treatment Approaches

Disorder	Recommended Treatments
Intermittent Explosive Disorder	Focus on recognizing warning signs and developing coping mechanisms. Effective therapies include CBT and family therapy. Medications: anticonvulsants, lithium, SSRIs. Best chances of success: combination of medication and psychotherapy.
Kleptomania	Insight-oriented psychotherapy, psychoanalysis, CBT, and behavioral therapies (e.g., systematic desensitization). Medications: SSRIs, tricyclic antidepressants, lithium, naltrexone.
Pyromania	Treatment is challenging; CBT is the most promising approach. Family therapy is essential for youth. Limited medication options; potential benefits from SSRIs, lithium, naltrexone.

Impulsive vs. Compulsive Disorders

- Impulsive Behaviors : Driven by an urge, associated with pleasure or gratification.
- Compulsive Behaviors : Driven by compulsion to relieve anxiety or distress; may not be enjoyable. (Option D ruled out)

Examples :

- Kleptomania : Stealing for gratification.
- Hoarding Disorder : Acquiring possessions due to anxiety about discarding them.

Role of Temperament in Impulse Control Disorders

- Novelty Seeking : Impulsive and easily bored; may engage in risky behaviors for excitement (Eg. Likely to develop gambling disorder.).
- Harm Avoidance : Anxious and sensitive to punishment; may act impulsively to escape negative emotions (Eg. May develop kleptomania to relieve anxiety).

Solution for Question 6:

Correct Answer: A) Compulsive sexual behavior disorder

Explanation:

- The patient's report of spending several hours per day on sexual pursuits, despite efforts to limit these behaviours, fits the diagnosis of Compulsive sexual behavior disorder.

Compulsive sexual behavior disorder:

- Included under impulse control disorders in ICD-11.
- Characterised by persistent failure to control intense, repetitive, sexual impulses resulting in repetitive sexual behaviours.
- The repetitive sexual activity becomes the focus of life and results in adverse life consequences.

Bipolar disorder with hypersexuality (Option B):

- While hypersexuality can occur during manic episodes of bipolar disorder, this patient's symptoms are not clearly linked to mood fluctuations.
- Additionally, the patient's behavior persists consistently, not only during periods of mania.

Borderline personality disorder (Option C):

- The patient's symptoms in this case are more consistent with an isolated compulsive behavior rather than the broader emotional and relational instability characteristic of BPD.

Sexual aversion disorder (Option D):

- It involves the avoidance of sexual contact and a lack of interest related to sexual activity.
 - In contrast, compulsive sexual behavior disorder involves uncontrollable sexual urges or behaviors that cause distress and impairment, as seen in this case.
-

Forensic Psychiatry, Electroconvulsive Therapy (Ect) And Bereavement

1. Match the following: 1. Durham's rule a. A person is not guilty because of insanity if they were unaware of the nature, quality, and consequences of their acts or could not realise they were wrong. 2. Curren's rule b. An accused is not criminally responsible if their unlawful act is the product of mental disease or defect. 3. McNaughten's rule c. An accused is not criminally responsible if, at the time of the act, they couldn't regulate their conduct due to mental disease or defect.

1. Durham's rule	a. A person is not guilty because of insanity if they were unaware of the nature, quality, and consequences of their acts or could not realise they were wrong.
2. Curren's rule	b. An accused is not criminally responsible if their unlawful act is the product of mental disease or defect.
3. McNaughten's rule	c. An accused is not criminally responsible if, at the time of the act, they couldn't regulate their conduct due to mental disease or defect.

- A. 1-b, 2-a, 3-c
- B. 1-a, 2-c, 3-b
- C. 1-b, 2-c, 3-a
- D. 1-c, 2-b, 3-a

2. A 29-year-old woman with severe depression and anxiety relapses due to job loss and a breakup. She expresses suicidal thoughts, prompting her therapist to suggest a treatment plan with advanced directives. Which of the following statements is included in advance directives under the Mental Health Act 2017? Every person, including minors, has the right to make a written advance directive. The individual can specify how they wish to be treated for a mental illness. The individual can specify how they wish not to be treated for a mental illness. Advance

directives are applicable in emergency treatment situations.

- A. 1,2,3
 - B. 2,3
 - C. 3,4
 - D. 1,2,3,4
-

3. According to the Mental Health Care Act (MHCA) 2017, which section deals with the admission of minors to a mental health establishment?

- A. Section 86
 - B. Section 87
 - C. Section 89
 - D. Section 90
-

4. Which of the following statements is not true regarding the decriminalisation of suicide in India as per the Mental Healthcare Act, 2017?

- A. Suicide attempts were decriminalised as per IPC Section 309
 - B. The act provides treatment instead of punishment for suicide attempts.
 - C. Individuals who attempt suicide are presumed to have severe stress under the Act.
 - D. The Act encourages individuals to seek help without fear of legal consequences.
-

5. A 12-year-old girl presents with social withdrawal, anxiety, and bruises. She discloses inappropriate behaviour by her neighbour, and examination reveals signs of possible sexual abuse. What should the healthcare provider do immediately?

- A. Refer the patient to a psychiatrist for evaluation.
 - B. Document the findings and schedule a follow-up appointment.
 - C. Report the suspected abuse to the appropriate authorities.
 - D. Discuss the situation with the patient's parents before taking any action
-

6. Under the Protection of Children from Sexual Offences (POCSO) Act, which of the following scenarios would not be classified as an "aggravated" offence?

- A. Security guard exploits his position of trust to abuse a child under his care.
 - B. Child becoming mentally ill
 - C. Parent inflicts grievous hurt on their child
 - D. Peer shares explicit photos of a 15-year-old child
-

7. A 62-year-old man recently lost his spouse and has been experiencing a profound sense of loss, sadness, and withdrawal from social interactions. This emotional response occurs following the death of a loved one and includes a natural period of adjustment. What term best describes his condition?

- A. Mourning
 - B. Adjustment disorder
 - C. Depression
 - D. Bereavement
-

8. A 62-year-old man is navigating the emotional process following the death of his spouse. Which of the following sequences best represents the stages of grief according to Kübler-Ross's model?

- A. Shock and Denial → Anger → Bargaining → Depression → Acceptance

- B. Shock and Denial → Anger → Depression → Bargaining → Acceptance
 - C. Shock and Denial → Depression → Bargaining → Anger → Acceptance
 - D. Shock and Denial → Depression → Anger → Bargaining → Acceptance
-

9. Which of the following is least aligned with the SPIKES protocol?

- A. Addressing the patient's emotions with empathetic responses
 - B. Use complex medical jargon to explain the situation
 - C. Creating a private setting for the discussion
 - D. Start by asking about the patient's understanding of the condition
-

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	2
Question 3	2
Question 4	1
Question 5	3
Question 6	4
Question 7	4
Question 8	1
Question 9	2

Solution for Question 1:

Correct Answer: C) 1-b, 2-c, 3-a

Explanation:

Rules related to Criminal responsibility:

Durham's rule	An accused person is not criminally responsible if his unlawful act is the product of a mental disease or mental defect. Later, this rule was discarded in the USA.
Curren's rule	An accused person is not criminally responsible if, at the time of committing the act, he did not have the capacity to regulate his conduct to the requirements of the law as a result of mental disease or defect.
McNaughten's rule (Section 84, IPC, Section 22, BNS)	A person is not criminally responsible if, due to a mental disease, they were unaware of the nature and consequences of their actions or unable to recognise that the act was wrong . It has the following requirements: There must be evidence of a mental disorder. The mental disorder must be present at the time of the crime. The disorder must impair the individual's ability to understand the wrongfulness or illegality of the act.
Section 85 IPC (Section 23 BNS)	A person who is intoxicated may not be criminally responsible if they were so impaired at the time of the act that they could not understand the nature of their actions or recognize that their actions were wrong or illegal. This applies only if the intoxication was caused against their will or without their knowledge.
Section 86 IPC (Section 24 BNS)	A voluntarily intoxicated person is still considered criminally responsible if they had the intention or knowledge to commit a crime, despite their intoxication.

Solution for Question 2:

Correct Answer: B) 2,3

Explanation:

Advance directives of the Mental Health Care Act, 2017:

- Every person who is not a minor can create an advance directive. (Statement 1 ruled out)
- Individuals can specify how they wish to be treated for a mental illness. (Statement 2)

- Individuals can specify how they do not want to be treated for a mental illness. (Statement 3)
- Individuals can appoint a nominated representative.
- The nominated representative's role is activated only when the individual loses the capacity to make decisions.
- The psychiatrist or medical officer in charge of treatment is responsible for ensuring that care aligns with the patient's advance directive.
- The patient, caregiver, or nominated representative is responsible for providing access to the advance directive for the psychiatrist or medical officer in charge.
- The doctor cannot be held liable if unforeseen consequences arise from following a valid advance directive.
- Advance directives are not applicable in emergency treatment situations. (Statement 4 ruled out) .

Solution for Question 3:

Correct Answer: B) Section 87

Explanation:

- According to the Mental Health Care Act (MHCA) 2017 , Section 87 pertains to the admission of minors to a mental health establishment.

Admission of patients with mental illness:

- Independent Admission: This is when an individual voluntarily seeks admission for treatment.

Section 86 (Option A)	Any person who is not a minor and believes they have a mental illness may request admission as an independent patient to a mental health establishment. This provision emphasizes the right of individuals to seek treatment voluntarily.
Section 87 (Option B)	Addresses the admission of minors to mental health establishments, ensuring that their rights and needs are specifically considered.

Supported Admission:

- An individual who requires admission due to threats or attempts of bodily harm to themselves or others, or who is unable to care for themselves and has lost the capacity to make decisions regarding their mental healthcare or treatment.
- A nominated representative can provide consent for the admission.

Section 89 (Option C)	Admission and treatment for individuals with mental illness requiring high levels of support can be provided for up to 30 days.
Section 90 (Option D)	Admission and treatment for individuals with mental illness requiring high levels of support may extend beyond 30 days.

Solution for Question 4:

Correct Answer: A) Suicide attempts were decriminalised as per IPC Section 309

Explanation:

- Prior to the Mental Healthcare Act, 2017
- Indian Penal Code (IPC) Section 309: Attempting suicide was historically a criminal offence in India, punishable by imprisonment, fine, or both. This criminalisation treated suicide attempts as violations of law rather than issues of mental health.
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- This criminalisation treated suicide attempts as violations of law rather than issues of mental health.

Change with the Mental Healthcare Act, 2017

- Section 115 explicitly states that individuals who attempt suicide shall not be punished under IPC Section 309. (Option B ruled out)

- Instead, they are presumed to be experiencing severe stress linked to mental illness (Option C ruled out) and should receive care, treatment, and rehabilitation to prevent recurrence.
- This shift promotes a compassionate, health-centric approach, emphasizing the need for psychiatric help rather than imprisonment.
- Mental healthcare professionals are now mandated to provide appropriate psychosocial support and treatment for underlying mental health conditions.
- This act also helps reduce the stigma associated with mental illness, encouraging individuals to seek help without fear of legal repercussions. (Option D ruled out)

Solution for Question 5:

Correct Answer: C) Report the suspected abuse to the appropriate authorities.

Explanation:

When a healthcare provider suspects sexual abuse in a child, the immediate action is to report the suspected abuse to the appropriate authorities.

- Under the Protection of Children from Sexual Offences (POCSO) Act, 2012, individuals aware of an incident of child sexual abuse are legally required to inform the local police or the Special Juvenile Police Unit.
- Failure to do so is considered a punishable offence.
- This Act aims to strengthen the protection of children below 18 years from sexual abuse, encompassing both male and female victims.
- The following offences are covered: Penetrative Sexual Assault Sexual Assault Sexual Harassment Use of a Child in Pornography
- Penetrative Sexual Assault
- Sexual Assault
- Sexual Harassment
- Use of a Child in Pornography
- While psychiatric evaluation (Option A) is important for the child's mental health, it should occur after reporting.
- Documentation and follow-up (Option B) are also necessary but must be secondary to ensuring the child's safety by reporting the abuse.

- Discussing the situation with the parents (Option D) before reporting can compromise the investigation and may put the child at further risk.
- Penetrative Sexual Assault
- Sexual Assault
- Sexual Harassment
- Use of a Child in Pornography

Solution for Question 6:

Correct Answer: D) Peer shares explicit photos of a 15-year-old child

Explanation:

- Sharing explicit photos does not meet the criteria for an aggravated offence under POCSO, as it involves sharing content without direct harm or abuse.

Aggravated Offence in POCSO:

- An offence is considered "aggravated" under the following circumstances: Perpetrated by an individual in a position of trust or authority over the child (e.g., security personnel, police officers, public servants, jail staff). Involves a gang. Involves the use of deadly weapons. Causes grievous hurt. Inflicts life-threatening diseases (e.g., HIV). Involves attempted murder. It is resulting in the child becoming mentally ill. Causes pregnancy in a female child. When it is committed repeatedly. Involves a child under 12 years of age. Occurs during communal or sectarian violence.
- Perpetrated by an individual in a position of trust or authority over the child (e.g., security personnel, police officers, public servants, jail staff).
- Involves a gang.
- Involves the use of deadly weapons.
- Causes grievous hurt.
- Inflicts life-threatening diseases (e.g., HIV).
- Involves attempted murder.
- It is resulting in the child becoming mentally ill.
- Causes pregnancy in a female child.
- When it is committed repeatedly.

- Involves a child under 12 years of age.
- Occurs during communal or sectarian violence.
- Perpetrated by an individual in a position of trust or authority over the child (e.g., security personnel, police officers, public servants, jail staff).
- Involves a gang.
- Involves the use of deadly weapons.
- Causes grievous hurt.
- Inflicts life-threatening diseases (e.g., HIV).
- Involves attempted murder.
- It is resulting in the child becoming mentally ill.
- Causes pregnancy in a female child.
- When it is committed repeatedly.
- Involves a child under 12 years of age.
- Occurs during communal or sectarian violence.

Solution for Question 7:

Correct Answer: D) Bereavement

Explanation:

Bereavement refers to the state of having lost someone close, typically following a death. It encompasses the emotional and psychological responses that occur during the process of adapting to the loss.

Term	Definition	Characteristics
Bereavement	State of having lost someone close, typically due to death.	Involves emotional and psychological responses; natural adjustment period; mild, transient functional impairment. Rarely involves severe symptoms like guilt or suicidal ideation.

Term	Definition	Characteristics
Mourning (Option A)	The outward expression and rituals associated with grief after a loss.	Includes cultural and personal expressions of grief; may involve ceremonies, rituals, and social mourning practices.
Adjustment Disorder (Option B)	A condition characterized by difficulty coping with a significant life stressor.	Emotional or behavioral symptoms occur within three months of the event; can lead to distress and impaired functioning but not a normal grieving process.
Depression (Option C)	A mental health disorder marked by persistent sadness and loss of interest in activities.	Symptoms include severe sadness, worthlessness, suicidal thoughts, and lack of interest; distinct from normal bereavement processes.

Solution for Question 8:

Correct Answer: A) Shock and Denial → Anger → Bargaining → Depression → Acceptance

Explanation:

- Grief is described as a psychological response to loss, typically involving feelings of sadness, despair, and longing for the deceased.
- It may also include a range of emotions and behaviours influenced by cultural and personal factors.

Stages of Grief/Death and Dying: Kübler-Ross identified five common stages of emotional responses to impending death/Grief.

- Stage 1- Shock and Denial: Patients initially react with disbelief and may seek multiple opinions to validate their denial. Effective communication from healthcare providers is essential to reassure and inform patients and families.
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- Effective communication from healthcare providers is essential to reassure and inform patients and families.
- Stage 2- Anger: Patients often feel frustrated and angry as reality sets in, questioning their fate and sometimes blaming others. Providers should approach this anger with empathy, recognising it as a natural response.
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- Providers should approach this anger with empathy, recognising it as a natural response.
- Stage 3- Bargaining: ■■■■■■■■ Patients may negotiate with healthcare providers or a higher power, promising to change behaviours for a better outcome. Reassuring them of comprehensive care, regardless of their actions, is crucial.
- ■■■■■■■■ Patients may negotiate with healthcare providers or a higher power, promising to change behaviours for a better outcome.
- Reassuring them of comprehensive care, regardless of their actions, is crucial.
- ■■■■■■■■ Patients may negotiate with healthcare providers or a higher power, promising to change behaviours for a better outcome.
- Reassuring them of comprehensive care, regardless of their actions, is crucial.
- Stage 4- Depression: ■■■■■■■■ Signs of clinical depression can emerge, including withdrawal and hopelessness. While some sadness is normal, significant depressive symptoms should be treated to maintain quality of life and hope.
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- While some sadness is normal, significant depressive symptoms should be treated to maintain quality of life and hope.
- ■■■■■■■■ Signs of clinical depression can emerge, including withdrawal and hopelessness.
- While some sadness is normal, significant depressive symptoms should be treated to maintain quality of life and hope.
- Stage 5- Acceptance: ■■■■■■■■ In this stage, patients accept their mortality, & realise death is inevitable. Open discussions about death can occur, and spiritual beliefs may provide comfort.
- ■■■■■■■■ In this stage, patients accept their mortality, & realise death is inevitable.
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- ■■■■■■ In this stage, patients accept their mortality, & realise death is inevitable.
- Open discussions about death can occur, and spiritual beliefs may provide comfort.

Solution for Question 9:

Correct Answer: B) Use complex medical jargon to explain the situation

Explanation:

- Using complex medical jargon contradicts the SPIKES protocol, emphasising clear and understandable communication.

SPIKES Protocol:

- A Structured approach for delivering bad news to patients
- It focuses on 6 Steps:

S—Setting Up the Interview	Preparation : Rehearse and anticipate emotional responses. Privacy : Ensure a distraction-free environment (Option C) Involve Others : Let the patient choose participants. Relaxing Atmosphere & Connect with the patient
P—Patient's Perception	Understanding : Ask about the patient's grasp of their condition (Option D) Correct Misunderstandings : Clarify misconceptions. Identify Denial : Look for signs of denial or unrealistic expectations
I—Patient's Invitation	Information preferences: Inquire about the extent of information the patient wishes to receive regarding their diagnosis and prognosis & respect their preference
K—Knowledge & Information	Warning Before Bad News : Indicate difficult news is coming. Simple Language : Use clear, non-technical terms. (Option B) Gradual Delivery : Provide information in small chunks.

S-Setting Up the Interview	Preparation : Rehearse and anticipate emotional responses. Privacy : Ensure a distraction-free environment (Option C) Involve Others : Let the patient choose participants. Relaxing Atmosphere & Connect with the patient
E-Patient's Emotions	Observe Emotions : Acknowledge emotional responses. Acknowledge Feelings : Use empathetic statements (Option A) Encourage Expression : Allow time for feelings to be expressed. Validate Responses : Reassure that reactions are normal
S-Strategy and Summary	Future Plans : Discuss readiness for treatment options. Collaborative Decision-Making : Involve the patient in planning. Clarify Understanding : Summarize to ensure comprehension

1. A 20-year-old male who was consulting a 30-year-old male therapist said to the therapist that he cares just like his brother. Which of the following statements is true regarding this scenario?

- A. Called as countertransference
- B. Can interfere with the treatment of this patient
- C. Such feelings are always positive
- D. The patient is always consciously aware of such feelings

2. All of the following are components of the topographical model of mind described by Sigmund Freud except

- A. Conscious mind
- B. Subconscious mind
- C. Preconscious mind
- D. Unconscious mind

3. Analyze the following statements regarding the structural model of mind and choose the correct set of true statements from these 1) Ego lies in all three-conscious, preconscious & unconscious parts of the mind 2) Instinctual drives are a part of Id. 3) The superego is mostly present in the conscious part of the mind.

- A. 1,2
- B. 1,3
- C. 2,3
- D. 1,2,3

4. Karthik says 'Siddharth hates me', but in reality Karthik hates Siddharth. What kind of defense mechanism is this?

- A. Projection
 - B. Denial
 - C. Regression
 - D. Distortion
-

5. All of the following are neurotic defenses except

- A. Displacement
 - B. Regression
 - C. Repression
 - D. Undoing
-

6. Leena, the sister of Garv disciplines her brother like her mother would when she isn't home. What type of defense is being followed here?

- A. Regression
 - B. Projective identification
 - C. Somatisation
 - D. Introjection
-

7. A medical student feeling overwhelmed by academic pressure starts writing short stories in her free time. Through creative writing, she transforms her anxiety into a productive outlet. What kind of defense mechanism is this?

- A. Affiliation
 - B. Altruism
 - C. Self-assertion
 - D. Sublimation
-

8. Given are various stages of psychosexual development. Which stage of development is a 4-year-old in?

- A. Stage I
 - B. Stage II
 - C. Stage III
 - D. Stage IV
-

9. A child accidentally trips over a toy car and scolds it, saying, "Bad car! You made me fall!". The child believes the toy can cause harm intentionally. Which stage of cognitive development is the child in?

- A. Sensorimotor
 - B. Preoperational
 - C. Concrete operational
 - D. Formal operational
-

10. Which of the following findings in Pavlov's experiment explains extinction?

- A. Dogs salivate on seeing food
 - B. Dogs stopped salivating on seeing food
 - C. Dogs started salivating at sounds associated with food
 - D. Dogs stopped salivating at sounds associated with food
-

11. Which of the following describes negative reinforcement?

- A. Adding a favorable stimulus
 - B. Removing an unfavorable stimulus
 - C. Adding an unfavorable stimulus
 - D. Removing a favorable stimulus
-

12. Which of the following is the most widely used projective personality test?

- A. Rorschach Inkblot Test
 - B. Thematic Apperception Test
 - C. Holtzman Inkblot Technique
 - D. Figure Drawing
-

13. A 45-year-old man has recently experienced several setbacks, including the loss of his job and difficulties in his marriage. He expresses a sense of failure and a fear that his life lacks purpose, and he feels disconnected from his community and his family. According to Erikson's stages of psychosocial development, which stage is he most likely struggling with?

- A. Industry vs. Inferiority
- B. Initiative vs. Guilt

C. Generativity vs. Stagnation

D. Integrity vs. Despair

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Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	1
Question 4	1
Question 5	2
Question 6	4
Question 7	4
Question 8	3
Question 9	2
Question 10	4
Question 11	2
Question 12	1
Question 13	3

Solution for Question 1:

Correct answer: B) Can interfere with the treatment of this patient

Explanation:

- A patient developing feelings (like his brother) for the therapist is suggestive of Transference (not Countertransference - Option A) .

Transference and countertransference:

Transference	Countertransference
Feelings of a patient towards the therapist	Feelings of a therapist towards patient
Can be positive or negative (Option C)	
Can be both conscious and unconscious (predominantly unconscious) (Option D)	
Interferes with treatment of the patient (Option B) (as there could be exaggerated/diminished reactions due to this)	

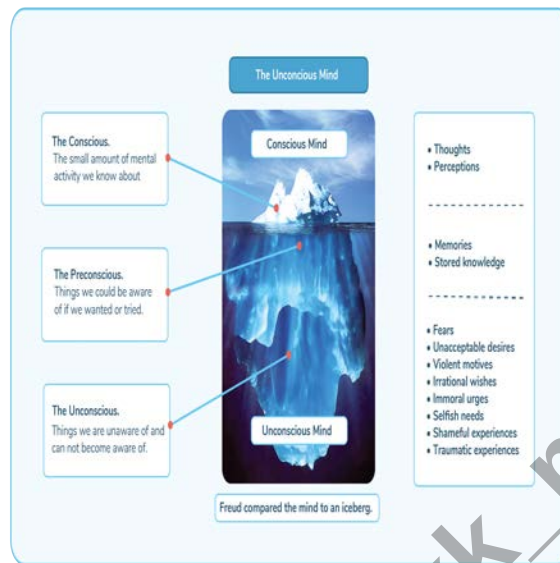
Solution for Question 2:

Correct answer: B) Subconscious mind

Explanation:

- The subconscious mind is not a component of the Topographical model described by Sigmund Freud.

Topographical model of mind (Sigmund Freud):



The unconscious mind can be reached by,

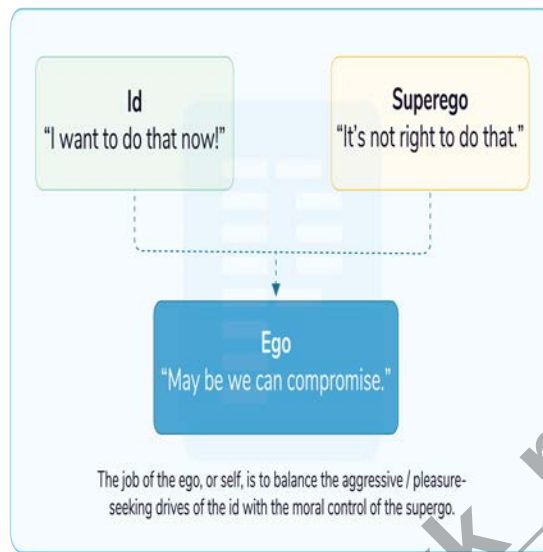
- Psychoanalysis
- Dream analysis

Psychoanalysis	Dream analysis
Techniques to bring out the direct unconscious mind (crossing the repression barrier).	The process involved in the conversion of a Latent dream (unconscious mind) to a Manifest dream (conscious mind) is known as Dreamwork.
Psychoanalysis techniques: Hypnosis Free association (patient is allowed to talk uninterrupted) Parapraxis (slip tongue) Abreaction (recollection of repressed memories with the release of emotions by methods like the use of thiopentone - the same principle used in Narco analysis)	Dreamwork: Condensation Displacement Symbolic representation Secondary revision

Solution for Question 3:

Correct answer: A) 1,2

Explanation:



Structural model of mind (by Sigmund Freud):

Id:	Instinctive drives Based on the pleasure principle (immediate gratification) Operates under primary processes Lies in the unconscious mind
Ego:	The executive organ of the mind Maintains balance between Id and Superego Works on the reality principle Lies in all three - conscious, preconscious and unconscious parts of the mind
Superego:	Moral compass insisting on socially acceptable behavior Develops after 5 years (after resolution of oedipal complex) Predominantly present in the unconscious part of the mind

Solution for Question 4:

Correct answer: A) Projection

Explanation:

- Here the person projects his own thoughts onto someone else, this type of defense mechanism is known as Projection - a part of narcissistic defenses.

Narcissistic defenses:

Defense	Features	Example
Projection (Option A)	A person projects his own thoughts onto someone else. Seen in psychosis.	Mr.P is having an affair but is convinced that his wife has an affair.
Denial (Option B)	Refuse to accept reality Usually used to avoid some painful stimuli	Refusal to admit to having a serious illness
Distortion (Option D)	Reshaping external reality in a person's favor to suit his/her inner needs	Mr.X believed that his boss was rude to him because in fact he likes X more than anyone else
Splitting	Viewing people/things in the world as either good/bad without considering the whole range. Seen in Borderline personality disorder.	Seeing all people without moustache as feminine.

Regression (Option C): Type of immature defense. A person returns to an earlier stage of development (acts childish).

Solution for Question 5:

Correct answer: B) Regression

Explanation:

- Regression is a type of immature defense mechanism, not neurotic.
- Regression is the return to an earlier stage of development (act childish/ younger).
- Example: Husband returns after a stressful day, and converses with wife in baby talks.

Neurotic defenses:

Defense	Features	Example
Inhibition	Unconsciously limiting an aim and accepting partial fulfilment of desire.	If a person is unable to clear the entrance exam for MBBS, he becomes a veterinary doctor.
Intellectualization	Excessive use of intellect to escape painful emotions.	After getting diagnosed with a metastatic tumor, a person starts discussing a great deal about its technical aspects.
Isolation of affect	Separation of emotions related to a stressful event, but reality is accepted.	A man comes home and tells his family of his lung cancer without any emotions.
Displacement (Option A)	Unconscious shifting of impulses from one object to another, to solve a conflict.	A man gets angry with his boss and shouts at his wife, the wife shouts at her son, and the son punches his toy.
Dissociation	Separation of one or more mental functions from remaining functions to avoid distress.	A person was kidnapped and tortured, he says it was as if he was floating on the ceiling and watching it.
Undoing (Option D)	An act to nullify a previous act.	Performing compulsions in OCD, apologizing.
Rationalization	Giving logical explanations for unacceptable behaviors. Seen in Substance use disorders.	A person says he drinks alcohol because it kills all the germs in his body.
Reaction formation	Doing the reverse of your unacceptable emotions.	Over Caring for the person you don't like.
Repression (Option C)	Unconscious forgetting Such memories (stored in the preconscious mind) can be brought out by techniques like Hypnosis/free association.	A child has been abused by his mother, and now has no memory of that abuse.
Externalization	Perceiving one's personality (mood/impulses/etc.) within the external world. Similar to projection.	An argumentative type of person believes others are arguing while the personal self is blameless.

Defense	Features	Example
Retroflexion	Unconscious projection of another's personality to one's hostility.	Mr.X (arrogant and short-tempered in reality) hates me because I am not worth enough.

Solution for Question 6:

Correct answer: D) Introjection

Explanation:

- Attributing to oneself the qualities of another is called Introjection - a type of immature defense mechanism.

Immature defenses:

Defense	Features	Example
Blocking	Inhibition of thinking (temporary) is associated with a component of tension (inability to remember).	A student unable to recall the answer in the exam (temporary), but recalls as he comes out of the exam.
Acting out	Unconscious wishes are enacted in behavior to avoid being aware of them. Seen in impulse control disorders.	A student is angry with his friend, Rather than telling him he throws a book at him (not intentionally).
Regression	Return to an earlier stage of development (act childish/ younger).	Husband returns after a stressful day, and converses with wife in baby talks.
Introjection (Option D)	Opposite to projection. Attributing to oneself the qualities of another. Stockholm syndrome: Hostages get sympathetic towards their captors.	People often treat their children the same way their parents treat them.
Projective identification	A person projects some aspect of himself onto someone else. Tries to coerce another person into identifying that he/she is projected. Seen in Borderline personality disorder.	The wife is aggressive toward the husband, then the husband also shows his aggressiveness.

Defense	Features	Example
Somatization	Unconscious moments are converted to bodily/ physical moments.	Getting a headache while taking exams.
Hypochondriasis	Unconscious transforming of unacceptable impulses to unrelated somatic complaints of pain, and illness.	Exaggerating a physical symptom (e.g. headache) for the purpose of regression.
Passive aggression	Indirectly expressing aggression towards others.	Refusing to take medications after a family quarrel.

Solution for Question 7:

Correct Answer: D) Sublimation

Explanation:

Since the subject is channelling emotional conflicts into creative or physical activities, this is an example of sublimation

Mature defences:

Affiliation	The subject seeks help or advice not to have the problem solved by others, but to find a solution they can implement themselves, and learn from emotionally meaningful interactions. (Option A ruled out)	Going to therapy/supportive group/spiritual/etc.
Altruism	The subject helps others with problems they cannot handle alone, finding personal meaning and gratification in helping, often tied to similar past experiences. (Option B ruled out)	After the death of the husband, due to smoking, the wife starts a 'No smoking' campaign.
Anticipation	The subject prepares emotionally for upcoming stressful situations by thinking ahead, imagining outcomes, and planning how to cope.	After failing the exam, the student plans to argue while going home.

Affiliation	The subject seeks help or advice not to have the problem solved by others, but to find a solution they can implement themselves, and learn from emotionally meaningful interactions. (Option A ruled out)	Going to therapy/supportive group/spiritual/etc.
Humor	The subject uses humor to diffuse tension in difficult or embarrassing situations, often making self-deprecatory or ironic remarks to lighten the mood without harm.	A man comes out and laughs after getting scolded by the boss.
Self-assertion	The subject confidently expresses thoughts, wishes, or feelings, deals with their own needs directly, and stands up for themselves in emotionally difficult situations. (Option C ruled out)	The wife tells her alcoholic husband to choose between her (or) alcohol and leaves the house.
Sublimation	The subject channels emotional conflicts into socially valued ones creative or physical activities, finding satisfaction or mastery in the process. (Option D)	A child gets angry at being scolded at home and scores 100 runs in a school cricket match.
Suppression	The subject consciously sets aside desires or emotions that may lead to bad consequences, postponing emotional responses to handle immediate, urgent tasks effectively.	A student decides to forget about the upcoming exam and take a day's break.

Solution for Question 8:

Correct Answer: C) Stage III

Explanation:

- 3-5-year-old child is in stage III-phallic stage of psychosexual development

Freud's 5 stages of psychosexual maturation:

Stage	Age Range	Erogenous Zone (Center of pleasure)	Explanation
Stage I: Oral	0-1 .5 year (Option A ruled out)	Mouth	Example- feeding (gratification for baby's oral needs) Fixation: Schizophrenia, substance dependence
Stage II: Anal	1-3 years (Option B ruled out)	Bowel	Toilet training is central Control over sphincters especially anal sphincter Fixation: OCD, OCPD
Stage III: Phallic	3- 5 years (Option C)	Genitalia	Development of the Oedipus (or) Electra complex occurs.
Stage IV: Latency	5- 6 to 11-13 years (Option D ruled out)	Dormant (no specific zone)	Focus on school, sports, and social relationships. Fixation: Neurotic disorders
Stage V: Genital	11-13 years until young adulthood	Genital	The ego is fully developed and seeks independence and meaningful relationships. Fixation: Neurotic disorders

Solution for Question 9:

Correct Answer: B) Preoperational

Explanation:

The child is giving the car life-like qualities. This is animistic thinking seen in the preoperational stage of cognitive development.

Piaget's stages of Cognitive Development:

Cognitive Stage	Key Achievements
Sensorimotor (Option A)	Birth to 2 years Infants "think" with their senses (eyes, ears, and touch). Object permanence develops. Improved memory and make-believe play. Mental representation (symbolization) by 18 months
Preoperational (Option B)	2 to 7 years Egocentric thinking dominates. Animistic thinking: Objects are given life-like qualities. Use of symbols in play. Development of language.
Concrete Operations (Option C)	7 to 11 years Logical thinking develops. Understanding of conservation (weight, volume, matter). Ability to classify objects hierarchically. Reversibility (can mentally reverse actions).
Formal Operations (Option D)	11 years through the end of adolescence Hypothetical-deductive reasoning develops. Imaginary audience- everyone is looking at them Personal fable- inflated opinion of themselves Propositional or abstract thinking

Solution for Question 10:

Correct Answer: D) Dogs stopped salivating at sounds associated with food

Explanation:

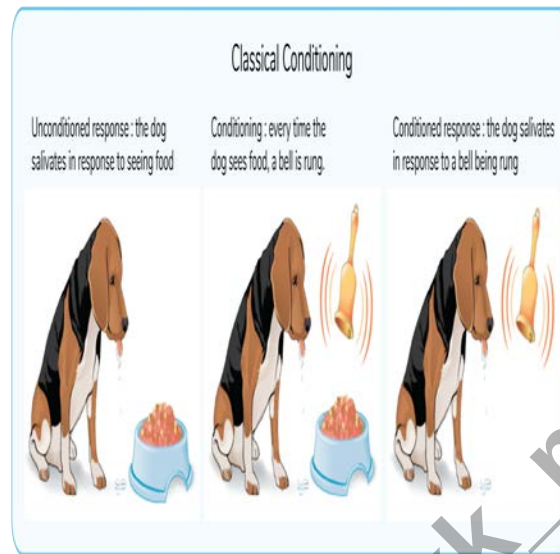
- Extinction is when the conditioned response (salivating at sounds associated with food arriving) weakens and disappears.

Classical conditioning:

- Unconscious learning method in which an automatic, conditioned response is paired with specific stimuli.
- Pavlov's Experiment with Dogs- At first, the dogs would only salivate when the food was placed in front of them. However, later they salivated slightly before their food arrived at the noises related to food arrival; for example, the sound of an approaching food cart. Here, Unconditioned stimulus- food Unconditioned response- salivation on seeing the food

Conditioned stimulus- sound associated with food
Unconditioned stimulus- food
Conditioned response- salivation
Unconditioned response- salivation on seeing the food

- At first, the dogs would only salivate when the food was placed in front of them.
- However, later they salivated slightly before their food arrived at the noises related to food arrival; for example, the sound of an approaching food cart. Here, Unconditioned stimulus- food
Unconditioned response- salivation on seeing the food
Conditioned stimulus- sound associated with food
Conditioned response- salivation on sounds associated with food
- Unconditioned stimulus- food
- Unconditioned response- salivation on seeing the food
- Conditioned stimulus- sound associated with food
- Conditioned response- salivation on sounds associated with food
- At first, the dogs would only salivate when the food was placed in front of them.
- However, later they salivated slightly before their food arrived at the noises related to food arrival; for example, the sound of an approaching food cart. Here, Unconditioned stimulus- food
Unconditioned response- salivation on seeing the food
Conditioned stimulus- sound associated with food
Conditioned response- salivation on sounds associated with food
- Unconditioned stimulus- food
- Unconditioned response- salivation on seeing the food
- Conditioned stimulus- sound associated with food
- Conditioned response- salivation on sounds associated with food
- Unconditioned stimulus- food
- Unconditioned response- salivation on seeing the food
- Conditioned stimulus- sound associated with food
- Conditioned response- salivation on sounds associated with food



	Extinction	Stimulus Generalization
Definition	The conditioned response weakens and eventually disappears due to the conditioned stimulus being presented without the unconditioned stimulus.	A conditioned response is elicited by stimuli that are similar to the conditioned stimulus.
Example	If Pavlov rang the bell (conditioned stimulus) without presenting food (unconditioned stimulus), the dogs would eventually stop salivating at the sound of the bell (conditioned response)	If Pavlov's dogs salivated in response to a sound similar to the bell, such as a cell phone ringing, this would illustrate stimulus generalization.

Solution for Question 11:

Correct Answer: B) Removing an unfavorable stimulus

Explanation:

- Negative reinforcement is removing an unfavorable stimulus (e.g., restrictions or unpleasant tasks) to increase behavior.

Operant Conditioning - deals with how environmental events control voluntary behavior

Positive reinforcement	Adding a favorable stimulus (e.g., money or praise) to increase behavior . (Option A ruled out)
Negative reinforcement	Removing an unfavorable stimulus (e.g., restrictions or unpleasant tasks) to increase behavior. (Option B)
Positive punishment	Adding an unfavorable stimulus (e.g., a physical blow or loud noise) to decrease behavior. (Option C ruled out)
Negative punishment	Removing a favorable stimulus (e.g., money or privileges) to decrease behavior. (Option D ruled out)
Absence of reinforcement/punishment	No consequence, leading to a likely decrease in behavior.

Solution for Question 12:

Correct Answer: A) Rorschach Inkblot Test

Explanation:

The most widely used projective personality test is the Rorschach Test

Projective personality tests - psychological assessments that use ambiguous stimuli, such as images or words, to elicit responses that reflect the individual's thoughts, feelings, and experiences.

Name of the test	Description
Rorschach Inkblot Test (Option A)	 10 stimulus cards of inkblots are shown and the examiner records verbatim what the patient says. The most widely used projective device

Name of the test	Description
Thematic Apperception Test (TAT) (Option B)	 20 stimulus cards depicting various scenes of ambiguity Patients create stories based on the pictures
Sentence Completion Test	Patients receive a series of incomplete sentences to complete with their first response.
Holtzman Inkblot Technique (HIT) (Option C)	Two parallel forms of inkblot cards with 45 cards per form Only one response is allowed per card
Figure Drawing (Option D)	Typically involves human forms, houses, or other figures
Make-a-Picture Story (MAPS)	Similar to TAT; stimuli can be manipulated by the patient

Solution for Question 13:

Correct Answer: C) Generativity vs. Stagnation

Explanation:

- Erikson's stage of Generativity vs. Stagnation, typically encountered during middle adulthood (40–65 years), involves a focus on contributing to society and helping the next generation.
- Failure to achieve this leads to feelings of stagnation and a lack of purpose, as seen in the patient's symptoms.

Erik Erikson's theory of psychosocial development identifies eight stages of development:

Stage 1	Trust vs. Mistrust (Birth to 18 months)	Infants learn to trust or mistrust the world based on the consistency of care from their caregivers.
Stage 2	Autonomy vs. Shame and Doubt (18 months to 3 years)	Children develop a sense of personal control and independence or feel shame and doubt about their abilities.
Stage 3	Initiative vs. Guilt (3 years to 5 years)	Children assert power and control over their environment, leading to a sense of initiative or feelings of guilt. (Option B)
Stage 4	Industry vs. Inferiority (5 years to 13 years)	Children develop self-confidence in abilities when competent or a sense of inferiority when not. (Option A)
Stage 5	Identity vs. Role Confusion (13 years to 21 years)	Adolescents explore their personal identity and values or become confused about their role in life.
Stage 6	Intimacy vs. Isolation (21 years to 40 years)	Young adults form deep, meaningful relationships, or experience loneliness and isolation.
Stage 7	Generativity vs. Stagnation (40 years to 60 years)	Adults seek to contribute to society and future generations, or feel stagnant and disconnected from broader purposes.
Stage 8	Integrity vs. Despair (60 years to death)	Older adults reflect on their life, either feeling a sense of fulfilment and integrity, or regret and despair. (Option D)

Neurocognitive Disorders

1. A 72-year-old male presents with progressive memory decline and difficulty performing daily tasks. Neuroimaging shows significant atrophy in the medial temporal lobe, and cerebrospinal fluid analysis reveals high levels of certain neurotoxic oligomers. Which of the following statements regarding the pathology and pathophysiology of this patient's condition is false?

- A. Soluble neurotoxic oligomers lead to synapse loss.
- B. Neurofibrillary tangles contain hyperphosphorylated tau protein.
- C. Plaques indicate polymerization of the neurotoxic proteins.
- D. Increased acetylcholine, norepinephrine and serotonin.

2. A 45-year-old man is brought to the clinic by his wife due to recent personality changes. She reports that he has become impulsive, shows poor judgment in social situations, and has difficulty planning and organizing tasks. His medical history is significant for a car accident 6 months ago. On examination, he shows emotional lability and perseveration. Which area of the brain is most likely affected in this patient?

- A. Temporal lobe
- B. Parietal lobe
- C. Occipital lobe
- D. Ventromedial orbitofrontal cortex

3. A 65-year-old man with a history of chronic alcohol use is brought to the emergency department by his family due to confusion, difficulty walking, and memory problems. On examination, he is disoriented to time and place, has

difficulty recalling recent events, and shows signs of confabulation. Which of the following is the most likely cause of his symptoms?

- A. Alzheimer's disease
 - B. Transient global amnesia
 - C. Korsakoff syndrome
 - D. Multiple sclerosis
-

4. A 72-year-old woman is brought to the clinic by her daughter due to progressive memory loss and confusion over the past year. She has been diagnosed with Alzheimer's disease. Which of the following is the most appropriate initial pharmacological intervention for this patient?

- A. Memantine
 - B. Donepezil
 - C. Risperidone
 - D. Sertraline
-

5. A 70-year-old woman is brought to the clinic by her son due to concerns about her memory. The physician decides to administer the Mini-Mental State Examination (MMSE). Which of the following statements about the MMSE is correct?

- A. It was originally designed to identify early stages of dementia
 - B. It takes approximately 30 minutes to administer
 - C. A score of 23/24 is traditionally used as a cut-off for suspected cognitive impairment
 - D. The test protocol and variables are the same for different socio-geographic populations
-

6. A 45-year-old man presents with progressive cognitive decline over the past 6 months. He has difficulty with memory, confusion, and occasional myoclonic jerks. His medical history is unremarkable, but his wife mentions he was an avid hunter who often consumed venison. Which of the following is the most likely diagnosis?

- A. Neurocognitive Disorder due to Traumatic Brain Injury
 - B. Neurocognitive Disorder due to HIV Infection
 - C. Neurocognitive Disorder due to Prion Disease
 - D. Alzheimer's Disease
-

7. A 68-year-old female presents to the clinic with noticeable cognitive decline. She is diagnosed with Dementia with Lewy Bodies. Which of the following is not a core feature of the given condition?

- A. Fluctuating levels of attention
 - B. Recurrent visual hallucinations
 - C. Systematized delusions
 - D. Parkinsonian features
-

8. A 65-year-old male presents to the clinic with noticeable personality changes, including increased impulsivity and a lack of empathy towards his family. He has difficulty regulating his emotions and exhibits hyperorality, often eating unusual food combinations. His cognitive functions related to memory and perception appear preserved. A family history of similar symptoms is noted. Based on the presented symptoms, which of the following is the most likely diagnosis?

- A. Alzheimer's Disease
- B. Multi-Infarct Dementia
- C. Lewy Body Dementia
- D. Pick's Disease

9. A 75-year-old male with a history of diabetes and hypertension is evaluated for cognitive decline. His family reports that he has had several episodes of confusion and difficulty with problem-solving. Neurological examination reveals evidence of white matter changes on MRI and he has a history of transient ischemic attacks. Which of the following features is MOST characteristic of vascular dementia in this patient?

- A. Continuous and gradual decline in cognitive function.
 - B. Cognitive changes following a stroke only.
 - C. Stepwise deterioration in cognitive abilities over time.
 - D. Memory loss as the only symptom.
-

10. Which statement best describes the key difference between twilight state and oneiroid state?

- A. The twilight state is characterized by disturbed consciousness with unexpected violent acts, while the oneiroid state involves a more dream-like, less interactive experience.
 - B. The oneiroid state features disorientation and confusion, whereas the twilight state is marked by clarity and coherence.
 - C. In the twilight state, individuals are typically fully unconscious, while those in the oneiroid state exhibit fluctuating consciousness.
 - D. The oneiroid state is primarily associated with external stimuli, while the twilight state is largely independent of environmental factors.
-

11. A 50-year-old female presents to the clinic with complaints of progressive memory loss over the past year, along with increasing difficulty in planning and organizing daily tasks. Her family reports that she often misplaces items and

struggles to follow conversations. Which of the following genes does NOT cause early-onset of this condition?

- A. APP gene
 - B. Presenilin-1
 - C. APOE gene
 - D. Presenilin-2
-

12. A 68-year-old male presents with progressive memory loss, difficulty concentrating, and irritability over the past several months. His family reports recent weight loss and changes in appetite. On examination, he appears pale, and neurological examination shows no focal deficits. Basic laboratory investigations reveal low hemoglobin and hypersegmented neutrophils on peripheral blood smear. Which of the following is the most likely cause of his cognitive decline?

- A. Alzheimer's disease
 - B. Vitamin B12 deficiency
 - C. Vascular dementia
 - D. Creutzfeldt-Jakob disease
-

13. A 75-year-old woman is brought in due to progressive memory loss over two years. She frequently forgets recent events, misplaces items, and has become socially withdrawn. Neurological exam shows intact motor function but poor recall after 5 minutes and disorientation to time. Blood work and imaging are unremarkable. Which of the following is not a typical feature of this condition?

- A. Gradual decline in cognitive function
- B. Memory impairment
- C. Rapid onset and fluctuation of symptoms
- D. Personality changes and apathy

14. A 65-year-old female is brought by her daughter, who reports that her mother has been progressively forgetting names, events, and sometimes even familiar faces over the last year. Recently, she misplaced household items and became suspicious of neighbors, accusing them of stealing. The patient is oriented to time and place but cannot recall what she had for breakfast. Her mood appears stable, and she does not report hearing voices or seeing things. There is no history of psychiatric illness. What is the most likely diagnosis?

- A. Dementia
 - B. Schizophrenia
 - C. Delirium
 - D. Major Depressive Disorder
-

15. A 75-year-old male with a history of hypertension and type 2 diabetes is admitted to the hospital for pneumonia. Three days into the hospital stay, he becomes acutely confused, disoriented to time and place, and exhibits fluctuating levels of consciousness. The patient is agitated, pulling at his IV lines, and experiencing visual hallucinations. His family reports he was mentally alert prior to admission. The patient's vital signs show fever and tachycardia. His medication list includes antibiotics, antihypertensives, and metformin. What is the next best step in management?

- A. Administer antipsychotic medication immediately
 - B. Discontinue all medications
 - C. Rule out metabolic and infectious causes
 - D. Start long-term benzodiazepine therapy
-

16. A 75-year-old male with a history of hypertension and diabetes is admitted to the hospital for pneumonia. On the second day of hospitalization, he becomes confused, agitated, and disoriented to time and place. His family reports that he was cognitively intact before admission. He is unable to focus on simple tasks and frequently hallucinates. His symptoms fluctuate throughout the day, worsening at night. Which of the following is the most likely diagnosis?

- A. Dementia
 - B. Schizophrenia
 - C. Delirium
 - D. Major depressive disorder with psychotic features
-

17. A 45-year-old man came to OPD with complaints of impaired consciousness, confusion, inattention, disorganized thinking, and visual hallucination, with symptoms more at night. Which clinical feature is most characteristic of this psychiatric condition?

- A. Gradual onset of cognitive decline
 - B. Fluctuating levels of consciousness
 - C. Stable and coherent thought processes
 - D. Fixed and focused attention
-

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	4
Question 3	3
Question 4	2
Question 5	3
Question 6	3
Question 7	3
Question 8	4
Question 9	3
Question 10	1
Question 11	3
Question 12	2
Question 13	3
Question 14	1
Question 15	3
Question 16	3
Question 17	2

Solution for Question 1:

Correct Answer: D) Increased acetylcholine, norepinephrine and serotonin.

Explanation:

In Alzheimer's disease, there is a reduction in acetylcholine due to the degeneration of cholinergic neurons.

- Additionally, norepinephrine and serotonin levels are also typically decreased due to degeneration in the respective areas (locus coeruleus and dorsal raphe).

Pathophysiology of Alzheimer's Disease:

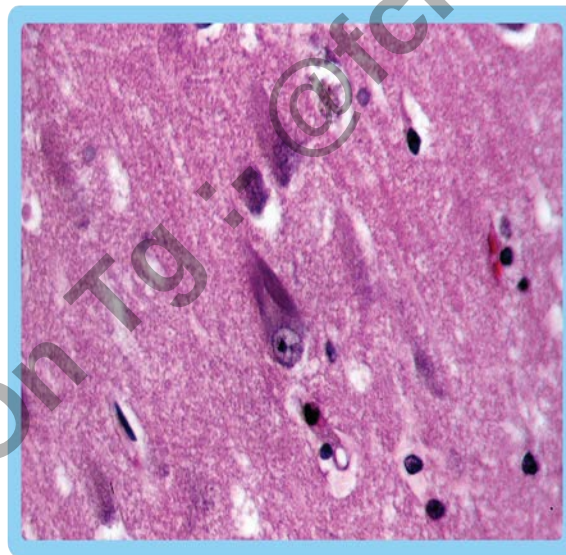
- Self-aggregation of amyloid-beta ($A\beta$) into soluble low-n oligomers causes synaptotoxicity in Alzheimer's disease. (Option A ruled out)
- Soluble $A\beta$ is strongly correlated with cortical synapse loss, an early pathological event.
- Cognitive impairment is a direct consequence of synaptic damage.
- Further polymerization of $A\beta$ leads to fibril formation and the development of amyloid plaques. (Option C ruled out)
- Mutations in the APP gene increase metabolism of APP through β -secretase activity.
- Mutations in presenilin 1 (PS1) and presenilin 2 (PS2) are linked to γ -secretase activity.
- $A\beta$ also accumulates in cerebral arterioles, leading to cerebral amyloid angiopathy.
- $A\beta$ contributes to the phosphorylation of tau protein, resulting in neurofibrillary tangles.

Pathology:

Macroscopic Pathology	Earliest degeneration: Seen in the medial temporal lobe (entorhinal/perirhinal cortex and hippocampus), inferolateral temporal cortex, and nucleus basalis of Meynert. Amyloid plaques: Formed by amyloid beta ($A\beta$), proteoglycans, ApoE, α -antichymotrypsin, and other proteins. Amyloid angiopathy: Accumulation of $A\beta$ in cerebral arterioles.
Microscopic Pathology	Neuritic plaques: Central core of amyloid surrounded by dystrophic neurites and activated microglia. Neurofibrillary tangles (NFTs): Composed of abnormally hyperphosphorylated tau protein, which forms paired helical filaments. (Option B ruled out) Tau pathology: Hyperphosphorylated tau compromises microtubule function and leads to neuronal dysfunction. Coexisting pathologies: Often seen with DLB, TDP-43, or vascular pathology.

Macroscopic Pathology	Earliest degeneration: Seen in the medial temporal lobe (entorhinal/perirhinal cortex and hippocampus), inferolateral temporal cortex, and nucleus basalis of Meynert. Amyloid plaques: Formed by amyloid beta ($A\beta$), proteoglycans, ApoE, α-antichymotrypsin, and other proteins. Amyloid angiopathy: Accumulation of $A\beta$ in cerebral arterioles.
Neurotransmitters	Acetylcholine reduction: Due to degeneration of cholinergic neurons in the nucleus basalis of Meynert. Noradrenergic and serotonergic depletion: From degeneration in the locus coeruleus (norepinephrine) and dorsal raphe (serotonin). Cognitive deficits: Linked to the reduction of acetylcholine and dysfunction in the cholinergic system.

Neurofibrillary tangles



Solution for Question 2:

Correct Answer: D) Ventromedial orbitofrontal cortex

Explanation:

- The patient's symptoms, including impulsivity, poor social judgment, difficulty with planning, emotional lability, and perseveration, are classic signs of frontal lobe syndrome, specifically involving the ventromedial orbitofrontal cortex.

Frontal Lobe Syndrome

Area	Functions	Symptoms when Damaged
Ventromedial Orbitofrontal Cortex	Social behavior Impulse control Decision making Emotional regulation	Impulsivity Poor social judgment Emotional lability Disinhibition
Dorsolateral Prefrontal Cortex	Working memory Planning Attention Problem-solving	Difficulty with planning and organization Poor working memory Reduced motivation (abulia) Perseveration
Anterior Cingulate Cortex	Motivation Error detection Emotional processing	Apathy Reduced initiative Difficulty with emotional regulation
Broca's Area (Left Frontal Lobe)	Speech production	Expressive aphasia (non-fluent)

- While temporal lobe damage can cause memory problems and emotional changes, it doesn't typically cause the impulsivity and poor judgment seen in this patient. (Option A)
- Parietal lobe damage is more associated with spatial awareness problems and sensory integration issues, not the behavioral changes described. (Option B)
- The occipital lobe is primarily involved in visual processing, and damage here would cause visual deficits, not behavioral changes. (Option C)

Solution for Question 3:

Correct Answer: C) Korsakoff syndrome

Explanation: The patient's history of chronic alcohol use, combined with symptoms of confusion, memory problems (particularly for recent events), and confabulation, is highly suggestive of Korsakoff syndrome.

- This condition is caused by thiamine (vitamin B1) deficiency, often associated with chronic alcoholism.

- Korsakoff syndrome frequently follows Wernicke's encephalopathy , which is characterized by the classic triad of ophthalmoplegia, ataxia, and confusion .
- In this case, the presence of confabulation, memory deficits, and alcohol history makes Korsakoff syndrome the most likely diagnosis.

Cause of Confabulation:

Confabulation occurs in several neurological conditions where memory is compromised, including:

- Wernicke-Korsakoff syndrome (thiamine deficiency)
- Anton syndrome (denial of visual impairment)
- Traumatic brain injury (TBI)

Amnestic Disorders- Causes

Amnestic Disorder	Cause	Key Features
Korsakoff Syndrome	Thiamine deficiency, often due to chronic alcoholism	• Anterograde and retrograde amnesia • Confabulation • Personality changes • Often follows Wernicke encephalopathy
Transient Global Amnesia	Unknown, possibly related to temporal lobe ischemia	• Sudden onset of anterograde amnesia • Episodes last 6-24 hours • Complete recovery is typical
Head Injury-Related Amnesia	Traumatic brain injury	• Retrograde amnesia for events leading up to injury • Anterograde amnesia post-injury • Severity correlates with injury severity
Cerebrovascular Disease-Related Amnesia	Infarctions affecting hippocampus or thalamus	• Memory impairment • Often accompanied by other focal neurological signs
Alcoholic Blackouts	Acute alcohol intoxication	• Inability to remember events during intoxication • Awareness of memory gap upon sobering
ECT-Related Amnesia	Electroconvulsive therapy	• Retrograde amnesia for minutes before treatment • Anterograde amnesia after treatment • Usually resolves within hours to months

While Alzheimer's disease can cause memory problems, it typically has a more gradual onset and doesn't present with the acute confusion and confabulation described in this case. (Option A)

This condition is characterized by sudden onset of anterograde amnesia that typically resolves within 24 hours. It doesn't explain the chronic nature of this patient's symptoms or the history of alcohol use. (Option B)

While multiple sclerosis can cause cognitive symptoms, including memory problems, it typically presents with other neurological symptoms and is not associated with chronic alcohol use. (Option D)

Solution for Question 4:

Correct Answer: B) Donepezil

Explanation:

- For patients with mild to moderate Alzheimer's disease, cholinesterase inhibitors like donepezil are the first-line pharmacological treatment .
- They can help improve cognitive function and activities of daily living.

Management of Dementia and Alzheimer's

Management Strategy	Details
Pharmacological	Cholinesterase inhibitors (e.g., donepezil, rivastigmine - used as a transdermal patch, galantamine) for mild to moderate Alzheimer's disease Memantine - blocks NMDA glutamate receptors, for moderate to severe Alzheimer's disease (Option A) Antidepressants for comorbid depression Antipsychotics (with caution) for severe behavioral disturbances
Non-pharmacological	Cognitive stimulation therapy Reality orientation Reminiscence therapy Behavioral management techniques Physical exercise programs

Management Strategy	Details
Environmental Modifications	Simplify the living environment Use memory aids (e.g., calendars, to-do lists) Ensure adequate lighting Remove potential hazards
Caregiver Support	Education about the disease Respite care Support groups Counseling
Newly approved drugs for treatment of early Alzheimer's disease.	Aducanumab (Aduhelm) - Monoclonal antibody targeting amyloid-beta plaques. Lecanemab (Leqembi) - Monoclonal antibody targeting soluble amyloid-beta protofibrils, facilitating plaque clearance. Donanemab - Monoclonal antibody targeting amyloid plaques; selectively binds to a modified form of amyloid-beta.

- While memantine is used in dementia treatment, it's typically reserved for moderate to severe Alzheimer's disease, often in combination with a cholinesterase inhibitor. (Option A)
- Antipsychotics like risperidone are only used cautiously in dementia for severe behavioral disturbances, not as a first-line treatment for cognitive symptoms. (Option C)
- Antidepressants may be used in dementia patients with comorbid depression, but they're not the primary treatment for cognitive symptoms. (Option D)

Solution for Question 5:

Correct Answer: C) A score of 23/24 is traditionally used as a cut-off for suspected cognitive impairment

Explanation: The Mini-Mental State Examination (MMSE), also known as the Folstein test, is a 30-question assessment of cognitive function. A score of 23 or 24 out of 30 is traditionally used as a cut-off for suspected cognitive impairment including dementia.

Mini Mental State Examination (MMSE)

Max. Score	Score	Orientation
5	-----	What is the (year) (season) (day of the week) (month) (date) ?
5	-----	Where are we : (state) (country) (town) (hospital) (floor) ?
3	-----	Registration Name 3 unrelated objects, allow 1 second to say each. Then ask the patient to repeat all 3. Give 1 point for each correct answer. Repeat them until he learns all 3. Count trials and record. Trials : -----
5	-----	Attention and Calculation Ask patient to count backwards from 100 by sevens. 1 point for each correct answer. Stop after 5 answers.
3	-----	Recall Ask patient to recall the 3 objects previously stated. Give 1 point for each correct answer.
9	-----	Language - Show patient a wristwatch : ask patient what it is. Repeat for a pencil (2 points). - Ask patient to repeat the following : "No ifs, ands, or buts" (1 point). - Ask patient to "take a piece of paper in your right hand, fold it in half, and put it on the floor" (3 points). - Ask patient to read and obey the following sentence which you have written on a piece of paper : "Close your eyes" (1 point). - Ask patient to write a 1 sentence (1 point). - Ask patient to copy a design (1 point).
Total Score -----		

Suggested guideline for determining the severity of cognitive impairment :

Mild : MMSE > 21
Moderate : MMSE 10-20
Severe : MMSE < 9

Expected decline in MMSE scores in untreated mild to moderate Alzheimer's patient is 2 to 4 points per year.

Key points about the MMSE include:

- Orientation
- Registration
- Recall
- Attention and concentration
- Language.

Solution for Question 6:

Correct Answer: C) Neurocognitive Disorder due to Prion Disease

Explanation:

- The patient's presentation of progressive cognitive decline, memory issues, confusion, and myoclonic jerks, along with a history of consuming venison (which can potentially be contaminated with prions causing Chronic Wasting Disease), is most consistent with a prion disease such as Creutzfeldt-Jakob Disease .

Causes of Neurocognitive Decline

Characteristic	Due to Traumatic Brain Injury	Due to HIV Infection	Due to Prion Disease
History	H/O traumatic brain injury, with associated noncognitive symptoms	HIV diagnosis, Associated cognitive symptoms	Cognitive impairment: Subtle onset, Rapid progression
Key Symptoms	Loss of consciousness Amnesia following trauma Confusion/ disorientation Neuroimaging evidence of injury/ Neurologic symptoms	Cognitive symptoms associated with HIV infection May include memory issues, attention deficits, and slowed processing speed	Rapid cognitive decline Motor symptoms (e.g., myoclonus/ ataxia) Biochemical marker on testing
Onset	Acute, following traumatic event	Gradual, as HIV progresses	Subtle onset, but rapid progression
Course	Can improve, stabilize, or worsen over time	Often fluctuates with HIV treatment and progression	Progressive rapid decline

EEG findings in Prion Disease:

- Triphasic Waves - Repetitive, sharp waves followed by a slower wave. Frequency: 1-2 Hz.
- Generalized slowing (theta and delta waves). Decreased amplitude of brain waves.
- Periodic sharp wave complexes (PSWCs) may be present.
- Changes are bilateral and symmetric .
- Alpha rhythm may be attenuated or absent

Neurocognitive Disorder due to Traumatic Brain Injury (Option A) is incorrect as there is no history of head trauma mentioned, and the onset is described as progressive rather than acute.

Neurocognitive Disorder due to HIV Infection (Option B) is incorrect because there's no mention of HIV diagnosis or risk factors, and the progression is typically not as rapid as described in this case.

Alzheimer's Disease (Option C) is incorrect because the progression is too rapid for typical Alzheimer's, and the presence of myoclonic jerks is more suggestive of prion disease.

Solution for Question 7:

Correct Answer: C) Systematized delusions

Explanation:

While delusions can occur in DLB, systematized delusions (fixed false beliefs that are well organized) are not considered a core feature of the condition, rather are considered a supporting feature.

- DLB primarily features hallucinations, particularly visual ones, rather than structured delusions.

Dementia with Lewy Bodies

- Dementia with Lewy Bodies (DLB) is a neurodegenerative disorder characterized by cognitive decline sufficient to interfere with social or occupational functioning.
- Early symptoms may include difficulties with attention, fronto-subcortical skills, and visuospatial ability, rather than prominent memory issues.

Clinical Features:

Probable DLB requires two or more core symptoms, while possible DLB requires at least one.

Core Features	Supporting Features
Fluctuating levels of attention and alertness (Option A ruled out) Recurrent visual hallucinations (Option B ruled out) Parkinsonian features (e.g., cogwheeling, bradykinesia, resting tremor) (Option D ruled out) REM Sleep behavior disorder	Repeated falls Syncope Sensitivity to neuroleptics Systematized delusions (Option C) Hallucinations in other modalities (e.g., auditory, tactile)

Differences in features of LBD and PD:

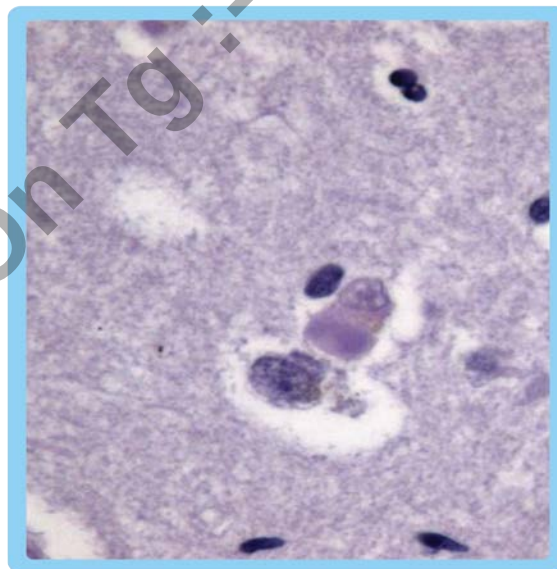
Feature	Lewy Body Dementia (LBD)	Parkinson's Disease (PD)
Onset of Symptoms	Cognitive and psychiatric symptoms appear early; motor symptoms develop later.	Primarily motor symptoms appear first (e.g., bradykinesia, rigidity, tremors).
Visual Hallucinations	Common and can occur early.	Less common initially; may occur later, often due to treatment.
Fluctuations in Attention	Fluctuations in alertness and attention are characteristic.	Fluctuations are primarily related to medication cycles.
Response to Medications	Poor response to dopaminergic medications; risk of worsening psychiatric symptoms.	Generally improves with dopaminergic medications (e.g., levodopa).

Feature	Lewy Body Dementia (LBD)	Parkinson's Disease (PD)
Sleep Disturbances	REM sleep behavior disorder is common and can precede cognitive symptoms.	REM sleep behavior disorder may occur but is primarily linked to motor symptoms.
Progression	Rapid progression of cognitive and behavioral symptoms.	Gradual progression of motor symptoms, with cognitive decline developing later.

Pathology:

- Lewy body disease presents clinically similarly to Alzheimer's disease but is marked by hallucinations, parkinsonian features, and extrapyramidal signs.
- Lewy bodies, which are abnormal accumulations of α -synuclein aggregates, are found in the cerebral cortex and substantia nigra.
- In cases where Lewy body pathology coexists with Alzheimer's, the amygdala is often the only affected region.
- Patients with Lewy body dementia may exhibit Capgras syndrome (reduplicative paramnesia), highlighting the complex nature of their symptoms.

Lewy bodies



Solution for Question 8:

Correct Answer: D) Pick's Disease

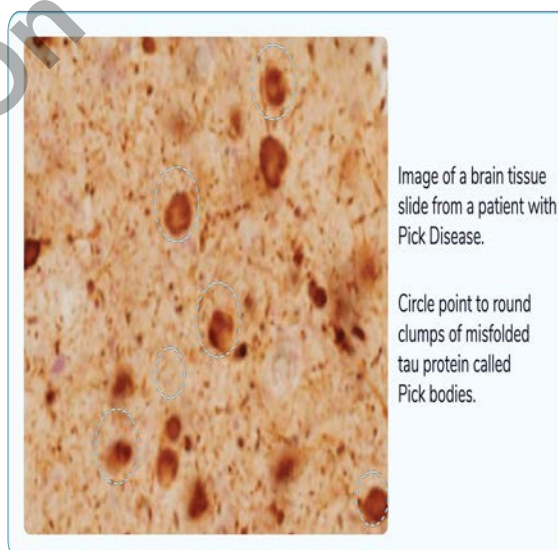
Explanation:

Pick's Disease is characterized by early personality and behavioral changes, including disinhibition, lack of empathy, and hyperorality, while other cognitive functions remain intact initially.

- The patient's age and family history of similar symptoms further support this diagnosis.
- The behavioral variant of frontotemporal dementia aligns closely with the symptoms presented.

Frontotemporal Dementia (Pick Disease)

- Affects frontotemporal regions with atrophy.
- Involves neuronal loss, gliosis, and Pick bodies (not needed for diagnosis).
- Accounts for about 5% of irreversible dementias.
- More common in men, especially with a family history.
- Early symptoms include personality and behavioral changes; other cognitive functions are preserved.
- Typically begins before age 75; familial cases may start earlier.
- Features like hypersexuality and hyperorality are more common than in Alzheimer's disease.
- Difficult to distinguish from Alzheimer's in early stages.



Aspect	Behavioral Variant	Language Variant
Affected	Social cognition/executive function	Language (word finding, naming, grammar, less spontaneous speech)
Spared	Learning/memory/perceptual-motor function	Learning/memory/perceptual-motor function
Associated Symptoms (≥3)	Disinhibition Apathy Lack of sympathy/empathy Perseveration/stereotypes Hyperorality/dietary changes	N/A
Probable	Evidence of a genetic cause (family history/genetic testing) Neuroimaging evidence of frontal/temporal pathology	
Possible	Some but not all the probable criteria met	

Cause of Dementia	Clinical Features
Alzheimer's Disease	Gradual memory loss. Memory impairment is usually the first symptom Difficulty with problem-solving and disorientation Presence of amyloid plaques and neurofibrillary tangles on brain imaging or autopsy Irreversible; progressive neurodegeneration
Vascular Dementia	Cognitive decline following a stroke (CT/MRI showing infarcts) Fluctuating abilities Impaired judgment Potentially reversible if acute vascular events are treated promptly

Cause of Dementia	Clinical Features
Lewy Body Dementia	Hallucinations occurring early - Visual hallucinations Parkinsonism (tremors, rigidity) Fluctuating cognition REM sleep behavior disorder Severe sensitivity to antipsychotics Irreversible; symptomatic treatments may help manage symptoms
Frontotemporal Dementia	Early onset (often under 65 years) Language difficulties and disinhibition Significant changes in social behavior or personality Irreversible; progressive condition
Alcohol-Related Dementia	Memory problems and history of alcohol abuse Difficulty with planning and organization Social withdrawal Nutritional deficiencies (e.g., Wernicke-Korsakoff syndrome) Potentially reversible with abstinence and nutritional support
Creutzfeldt-Jakob Disease	Rapidly progressive dementia Myoclonus Visual disturbances Diagnosis confirmed via MRI (increased signal in basal ganglia) and EEG findings Rare, irreversible; rapidly progressive and fatal
Pick's Disease	Marked personality changes Language difficulties (e.g., aphasia) Emotional blunting MRI/CT shows frontal and temporal lobe atrophy Behavioral changes prominent early Irreversible; part of the frontotemporal dementia spectrum

Alzheimer's Disease (Option A): Primarily involves memory impairment and gradual cognitive decline, not prominent personality changes as seen here.

Multi-Infarct Dementia (Option B): Characterized by stepwise deterioration due to vascular issues, not marked by the significant behavioral symptoms present in this patient.

Lewy Body Dementia (Option C): Features cognitive fluctuations and visual hallucinations, with motor symptoms; the patient's main issues are behavioral, not motor.

Solution for Question 9:

Correct Answer: C) Stepwise deterioration in cognitive abilities over time.

Explanation:

The stepwise deterioration in cognitive abilities is a hallmark of vascular dementia, particularly as seen in patients with a history of TIAs and small vessel disease, which this patient exhibits.

Vascular Dementia

- Vascular dementia, previously known as multi-infarct dementia, is caused by multiple areas of cerebral vascular disease.
 - Most common in men, especially those with hypertension or cardiovascular risk factors.
 - Affects small- and medium-sized cerebral vessels, leading to infarctions and parenchymal lesions.
 - Causes of infarctions include arteriosclerotic plaques or thromboemboli from distant sources (e.g., heart valves). (Option B ruled out)
 - Types: Acute onset after strokes Multi-infarct dementia from transient ischemic attacks Subcortical dementia associated with hypertension (white matter injury) Mixed vascular dementia.
 - Acute onset after strokes
 - Multi-infarct dementia from transient ischemic attacks
 - Subcortical dementia associated with hypertension (white matter injury)
 - Mixed vascular dementia.
 - Typically shows stepwise deterioration , unlike the gradual decline seen in Alzheimer's disease. (Option A ruled out)
 - Symptoms may include carotid bruits, fundoscopic abnormalities, or enlarged cardiac chambers. (Option D ruled out)
 - Diagnosis requires clinical or laboratory evidence of a vascular cause.
 - Acute onset after strokes
 - Multi-infarct dementia from transient ischemic attacks
 - Subcortical dementia associated with hypertension (white matter injury)
 - Mixed vascular dementia.
-

Solution for Question 10:

Correct Answer: A) The twilight state is characterized by disturbed consciousness with unexpected violent acts, while the oneiroid state involves a more dream-like, less interactive experience.

Explanation:

- The twilight state is characterized by disturbed consciousness with unexpected violent acts, while the oneiroid state involves a more dream-like, less interactive experience.

Twilight state:

- Disturbed consciousness with hallucinations.
- Onset/end- Abrupt
- Episodes of unexpected violent acts/ emotional outbursts.
- Seen in cases of epilepsy, brain tumors, and head injury.

Dream-like (oneiroid) state:

- Impairment of consciousness
- Dream-like experiences : Vivid, surreal perceptions resembling a dream, often with hallucinations or delusions.
- Disorientation : Difficulty distinguishing reality from the dream-like mental state.
- Fluctuating consciousness : Periods of clarity mixed with deep confusion.
- Detachment : Reduced interaction with the environment, as the individual is absorbed in internal, fantasy-like experiences.

Level of Consciousness	Features
Alertness	Normal cognitive function Fully awake and aware of the environment Appropriate responses to stimuli Ability to maintain attention.
Lethargy (Somnolence)	Drowsy but can be awakened Decreased alertness Slow responses to questions May fall asleep easily Requires external stimulation to maintain wakefulness.
Obtundation	Reduced awareness and responsiveness Difficult to arouse Responses are slowed The individual may appear confused or disoriented.

Level of Consciousness	Features
Stupor (Semicoma)	Near-unresponsive state; Can only be aroused with vigorous or painful stimuli. With no awareness of the environment.
Coma	Unresponsive state Cannot be aroused even with strong stimuli. No voluntary responses Brain activity may still be present at a basic level (reflexes may occur).

Other terms related to disturbance with consciousness:

Torpor	A state of physical and mental inactivity, often due to fatigue or metabolic disturbances. Reduced responsiveness The individual may appear lethargic and is difficult to arouse without significant stimuli.
Twilight	A transitional state between alertness and lethargy, characterized by confusion and disorientation. Responses may be variable Onset/end- Abrupt Episodes of unexpected violent acts/ emotional outbursts. The individual may exhibit impaired judgment and perception of reality.
Oneiroid State	A dream-like state with vivid hallucinations The individual may be unaware of their surroundings. Experiences of hallucinations or delusions, leading to a distorted perception of reality Unresponsive to external stimuli.

The oneiroid state features disorientation and confusion, whereas the twilight state is marked by clarity and coherence (Option B): This is incorrect. Both states can involve confusion, but the clarity typically associated with the twilight state is not accurate.

In the twilight state, individuals are typically fully unconscious, while those in the oneiric state exhibit fluctuating consciousness (Option C): The twilight state does not necessarily imply full unconsciousness; rather, individuals may be partially aware but still confused or hallucinatory.

The oneiric state is primarily associated with external stimuli, while the twilight state is largely independent of environmental factors (Option D): This is not accurate. Both states can be influenced by external stimuli, but the experience of the oneiric state often resembles a dream in response to both internal and external cues.

Solution for Question 11:

Correct Answer: C) APOE gene

Explanation:

The patient's clinical features—progressive memory loss, difficulties in planning and organizing tasks, and issues with conversation—strongly point to a diagnosis of Alzheimer's disease.

- Given her age and the nature of her symptoms, this suggests early-onset Alzheimer's disease (EOAD).
- The APOE gene, particularly the $\epsilon 4$ allele, is a key risk factor for late-onset Alzheimer's disease (LOAD) rather than early-onset.

Alzheimer's Disease

- Cognitive Changes : Starts with memory impairment, progressing to executive function, language, and visuospatial deficits.
- Recognizable memory loss, with about 50% progressing to Alzheimer's within four years.
- Biomarkers can indicate amyloid pathology up to 20 years before symptoms.
- Late Stage Symptoms : Wandering, loss of judgment, delusions, sleep-wake pattern changes, and parkinsonian features.
- Typical course lasts 8-10 years, but can range from 1 to 25 years; memory impairment is a key feature.
- Epidemiology: Seen in females more than males.

Genetics:

Early-Onset Alzheimer's Disease (EOAD)	Late-Onset Alzheimer's Disease (LOAD)
APP gene (chromosome 21): Associated with early-onset familial Alzheimer's disease (FAD) and Down's syndrome. (Option A ruled out) Presenilin-1 (PSEN-1) : Most common cause (40-70%), mean onset at 45 years, rapid progression (6-7 years) (Option B ruled out) . Presenilin-2 (PSEN-2) : Less common, mean onset at 53 years, slower progression (11 years). (Option D ruled out) Pathogenesis involves mutations that increase Aβ42 levels and amyloid accumulation.	APOE gene (chromosome 19) is a key risk factor. ApoE $\epsilon 4$ allele : Increases risk and lowers age of onset (about 10 years per allele). ApoE $\epsilon 2$ allele : May reduce risk. Risk varies by ethnicity; higher in East Asians, lower in Blacks and Hispanics. Polygenic Risk : Additional genes (e.g., CLU, PICALM, CR1) involved, with smaller individual effects. Variants in TREM2 significantly increase AD risk (odds ratios 3-4).

Solution for Question 12:

Correct Answer: B) Vitamin B12 deficiency

Explanation:

Vitamin B12 deficiency is one of the common reversible causes of dementia.

- In this case, the patient's cognitive decline, weight loss, and pale appearance, along with findings of macrocytic anemia (e.g., hypersegmented neutrophils), strongly suggest vitamin B12 deficiency.
- This deficiency can lead to neuropsychiatric symptoms, including memory loss, irritability, and cognitive impairment.
- The pathophysiology involves the disruption of myelin synthesis and subsequent demyelination, which affects both the central and peripheral nervous systems.

Types of Dementia:

Degenerative dementias	Alzheimer disease Frontotemporal dementias (e.g., Pick disease) Parkinson disease Lewy body dementia Idiopathic cerebral ferrocalsinosis (Fahr disease) Progressive supranuclear palsy Limbic predominant age-related TDP-43 encephalopathy (LATE)
Miscellaneous	Huntington disease Wilson disease Metachromatic leukodystrophy Neuroacanthocytosis
Psychiatric	Pseudodementia of depression Cognitive decline in late-life schizophrenia
Demyelinating diseases	Multiple sclerosis
Cardiac, vascular, and anoxia	Infarction (single or multiple or strategic lacunar) Binswanger disease (subcortical arteriosclerotic encephalopathy) Hemodynamic insufficiency (e.g., hypoperfusion or hypoxia)

Degenerative dementias	Alzheimer disease Frontotemporal dementias (e.g., Pick disease) Parkinson disease Lewy body dementia Idiopathic cerebral ferrocalsinosis (Fahr disease) Progressive supranuclear palsy Limbic predominant age-related TDP-43 encephalopathy (LATE)
Physiologic	Normal pressure hydrocephalus
Metabolic	Vitamin deficiencies (e.g., vitamin B12, folate) Endocrinopathies (e.g., hypothyroidism) Chronic metabolic disturbances (e.g., uremia)
Tumor	Primary or metastatic (e.g., meningioma or metastatic breast or lung cancer)
Traumatic	Dementia pugilistica, posttraumatic dementia Subdural hematoma
Infection	Prion diseases (e.g., Creutzfeldt–Jakob disease, bovine spongiform encephalitis, Gerstmann–Sträussler syndrome) Acquired immune deficiency syndrome (AIDS) Syphilis
Drugs and toxins	Alcohol Heavy metals Irradiation Cognitive dysfunction due to medications (e.g., anticholinergics) Carbon monoxide

Some types of dementia can be attributable to reversible causes, such as metabolic abnormalities (e.g., hypothyroidism), nutritional deficiencies (e.g., vitamin B12 or folate deficiencies), normal pressure hydrocephalus or dementia syndrome caused by depression.

Dementia is classified based on the brain structures involved into following 2 types:

- Involves the cerebral cortex , leading to impairments in memory, language, and higher cognitive functions.
- Examples : Alzheimer's disease, Frontotemporal dementia).
- Affects deeper brain structures (basal ganglia, thalamus), causing psychomotor slowing, mood changes, and movement disorders.
- Examples : Parkinson's disease dementia, Huntington's disease.

Important causes of dementia and their clinical features:

Cause of Dementia	Clinical Features
Alzheimer's Disease	Gradual memory loss. Memory impairment is usually the first symptom Difficulty with problem-solving and disorientation Presence of amyloid plaques and neurofibrillary tangles on brain imaging or autopsy Irreversible; progressive neurodegeneration
Vascular Dementia	Cognitive decline following a stroke (CT/MRI showing infarcts) Fluctuating abilities Impaired judgment Potentially reversible if acute vascular events are treated promptly
Lewy Body Dementia	Hallucinations occurring early - Visual hallucinations Parkinsonism (tremors, rigidity) Fluctuating cognition Severe sensitivity to antipsychotics REM Sleep behavior disorder Irreversible; symptomatic treatments may help manage symptoms
Frontotemporal Dementia	Early onset (often under 65 years) Language difficulties and disinhibition Significant changes in social behavior or personality Irreversible; progressive condition
Alcohol-Related Dementia	Memory problems and history of alcohol abuse Difficulty with planning and organization Social withdrawal Nutritional deficiencies (e.g., Wernicke-Korsakoff syndrome) Potentially reversible with abstinence and nutritional support
Creutzfeldt-Jakob Disease	Rapidly progressive dementia Myoclonus Visual disturbances Diagnosis confirmed via MRI (increased signal in basal ganglia) and EEG findings Rare, irreversible; rapidly progressive and fatal
Pick's Disease	Marked personality changes Language difficulties (e.g., aphasia) Emotional blunting MRI/CT shows frontal and temporal lobe atrophy Behavioral changes prominent early Irreversible; part of the frontotemporal dementia spectrum

Alzheimer's disease (Option A):

- This is the most common cause of dementia but is not reversible.
- It is characterized by progressive memory impairment, language difficulties, and visuospatial deficits.
- Neuropathology involves amyloid plaques and neurofibrillary tangles.

Vascular dementia (Option C):

- This form of dementia results from cerebrovascular disease, often associated with a stepwise decline in cognitive function due to multiple infarcts or small vessel disease.
- While some modifiable risk factors like hypertension can be controlled, the established dementia is not typically reversible .

Creutzfeldt-Jakob disease (Option D):

- A rapidly progressive dementia caused by prion disease.
- It leads to severe neurological deficits, including myoclonus and ataxia, with death typically occurring within a year.
- It is not reversible.

Solution for Question 13:

Correct Answer: C) Rapid onset and fluctuation of symptoms

Explanation:

- Rapid onset and fluctuation of symptoms is NOT typical of dementia but is more characteristic of delirium, which is an acute, fluctuating disturbance in attention and cognition.

Dementia/Major Neurocognitive Disorder is characterized by a progressive decline in cognitive abilities . (Option A) which involves the impairment of at least one (usually two or more) cognitive domains such as:

- Memory (Option B)
- Attention
- Executive function
- Language
- Perceptual-motor abilities
- Social cognition

Etiology:

Dementia can be caused by several underlying conditions, including:

Alzheimer's Disease Vascular Dementia Dementia with Lewy Bodies	Frontotemporal Dementia Traumatic Brain Injury (TBI) HIV-associated Dementia	Prion Diseases Parkinson's Disease Huntington's Disease
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Epidemiology: With the aging population, dementia has become increasingly common, especially in older adults.

- Approximately 5% of individuals over the age of 65 suffer from moderate to severe dementia
- The prevalence rises sharply in the elderly, affecting 20-40% of individuals over the age of 85
- Alzheimer's disease is the most common cause, alone or in combination with other etiologies, accounts for about 70% to 80% of dementias.
- Dementia with Lewy Bodies (DLB) is characterized by the same neuropathologic changes as Alzheimer disease, with neurofibrillary tangles and plaques, plus Lewy bodies in cortical and brain stem regions, and accounts for about 15% to 35% of dementias.
- Vascular disease, both small vessel and large vessel disease affecting cortical or subcortical regions accounts for about 5% to 20% of dementias.
- It associated with risk factors like hypertension and cerebrovascular disease
- Other causes of dementia, such as those related to alcohol, head trauma, and movement disorders like Huntington's and Parkinson's diseases, each represent around 1-5% of cases

Symptoms:

Dementia presents with a broad range of cognitive and behavioral symptoms, which may vary based on the underlying cause but generally include:

Memory Impairment	Early signs include difficulty recalling recent events, conversations, or tasks (e.g., forgetting appointments, misplacing items). Over time, even long-term memories may be lost , and patients may forget personal details like their birthplace or the names of close relatives.
Impaired Attention and Executive Functioning	Patients struggle with tasks requiring planning, decision-making, and problem-solving. This impairment affects their ability to perform activities of daily living , such as managing finances, cooking, or following a schedule.

Memory Impairment	Early signs include difficulty recalling recent events, conversations, or tasks (e.g., forgetting appointments, misplacing items). Over time, even long-term memories may be lost, and patients may forget personal details like their birthplace or the names of close relatives.
Disorientation	Patients may become disoriented to time, place, or person. For example, they may forget where they are or be unable to find their way back to their room after using the bathroom.
Language Deficits (Aphasia)	Difficulty in finding the right words, naming objects, or following a conversation is common, especially in cortical dementias like Alzheimer's disease. As the condition progresses, speech may become nonsensical or fragmented.
Personality and Behavioral Changes (Option D)	Some patients become socially withdrawn, while others exhibit agitation, irritability, or aggression. Disinhibition and inappropriate behavior can occur. Patients with psychotic symptoms may develop paranoid delusions or hallucinations. (Most common is the delusion of persecution).
Mood and Emotional Changes	Depression and anxiety are common. Patients may also experience emotional lability, swinging between laughing and crying without obvious triggers.
Motor Symptoms	Motor symptoms such as tremors, rigidity, and bradykinesia can accompany cognitive decline in certain types of dementia, like Parkinson's disease or Lewy body dementia. Neurologic findings such as seizures or primitive reflexes (e.g., grasp reflex) may appear in some patients.

Solution for Question 14:

Correct Answer: A) Dementia

Explanation:

- Dementia is characterized by progressive cognitive decline, with memory impairment (especially recent memory) and sometimes paranoia or delusions.
- Dementia is characterized by:
- In this case, the patient has had progressive memory loss over a year and has become suspicious, consistent with the cognitive and behavioral changes seen in dementia■.

Feature	Delirium	Dementia	Schizophrenia
Onset	Acute (hours to days)	Gradual (months to years)	Gradual (late adolescence/adulthood)
Course	Fluctuates (worse at night)	Progressive, stable day-to-day	Chronic, can have relapses
Attention	Impaired	Initially normal, impaired later	Usually intact
Orientation	Disoriented	Mostly intact until late stages	Intact
Hallucinations	Visual, tactile (common)	Rare	Auditory (common)
Memory	Short-term impaired	Long-term impaired	Intact (except in psychosis)
Consciousness	Fluctuating	Normal until late stages	Normal

Schizophrenia (Option B) is less likely because the onset of schizophrenia is typically earlier in life (late adolescence to adulthood). Although paranoia is a common feature of schizophrenia, the absence of auditory hallucinations and the late onset of symptoms makes schizophrenia an unlikely diagnosis■.

Delirium (Option C) is an acute condition, associated with fluctuating levels of consciousness and disorientation, which are not present in this case■.

Major Depressive Disorder (Option D) with psychotic features could cause paranoia, but the absence of other classic depressive symptoms such as a low mood and loss of interest makes this diagnosis less likely■.

Solution for Question 15:

Correct Answer: C) Rule out metabolic and infectious causes

Explanation:

This patient presents with typical clinical features of delirium , including acute onset, fluctuating course, disorientation, hallucinations, and altered attention.

- Delirium is a common condition in hospitalized elderly patients and is often secondary to underlying medical issues such as infections, metabolic imbalances, or medication side effects.
- The primary step in managing delirium is to identify and treat the underlying cause , which in this case could include pneumonia or other metabolic disturbances.

Assessment :

Delirium is assessed primarily through clinical observation and mental status examinations .

Specific tools for assessment:

- The Confusion Assessment Method (CAM) is commonly used to diagnose delirium based on four features: acute onset, fluctuating course, inattention, and disorganized thinking or altered consciousness.
- The Mini-Mental State Examination (MMSE) can help document cognitive impairments and establish a baseline to monitor changes over time■.

Physical and Neurological Examination :

Common findings include:

- Fever (suggestive of infection), tachycardia (metabolic disturbances or hyperthyroidism), or bradycardia (intracranial pressure, hypothyroidism).
- Neurological signs : Tremor, asterixis, incoordination, nystagmus, and reflex abnormalities (e.g., Babinski sign) may point toward neurological causes.

Laboratory and Imaging Workup : Laboratory tests help to rule out metabolic causes:

- Blood chemistries (electrolytes, renal and hepatic function, glucose levels)
- Complete blood count (to detect infection or anemia)
- Thyroid function tests
- Toxicology screens (for substance-induced delirium)
- If indicated, imaging studies such as CT or MRI are performed to rule out structural brain lesions. EEG may be done to differentiate delirium from other psychiatric conditions, as it often shows diffuse slowing in delirium ■.

Treatment :

The treatment of delirium has two major components: addressing the underlying cause and providing symptomatic relief .

- Identifying and reversing the underlying cause is the most important step. Causes can range from infections (such as pneumonia or urinary tract infections) to metabolic imbalances, dehydration, or drug toxicity.
- Patients should be in a safe and calm environment , free from both sensory deprivation and overstimulation. Measures include: Repeated orientation to time, place, and person Having familiar objects or people around (e.g., family, familiar pictures) Adequate lighting, presence of clocks and calendars
- Repeated orientation to time, place, and person
- Having familiar objects or people around (e.g., family, familiar pictures)
- Adequate lighting, presence of clocks and calendars
- Repeated orientation to time, place, and person
- Having familiar objects or people around (e.g., family, familiar pictures)
- Adequate lighting, presence of clocks and calendars
- Antipsychotics - Haloperidol (commonly used drug), Olanzapine - for managing psychosis and agitation.
- Sodium valproate or dexmedetomidine - for agitation when antipsychotics fail or are contraindicated.
- Benzodiazepines - DOC in Alcohol Withdrawal (Delirium tremens)
- Insomnia - Benzodiazepines worsen confusion in non-alcohol-related delirium so non-pharmacologic strategies like sleep hygiene practices are preferred.

Administer antipsychotic medication immediately (Option A) : They are not the first-line treatment and should only be used after the underlying cause of delirium has been identified.

Discontinue all medications (Option B) : Abruptly stopping all medications can cause harm.

- A more measured approach is required where medications with a potential role in causing delirium (e.g., anticholinergics, opioids) are reviewed and possibly adjusted or discontinued after careful consideration.

Start long-term benzodiazepine therapy (Option C) : Benzodiazepines are contraindicated in most forms of delirium except for those caused by alcohol or benzodiazepine withdrawal.

- They can worsen confusion and increase the risk of falls in elderly patients.

Solution for Question 16:

Correct Answer: C) Delirium

Explanation:

Delirium is a life-threatening but potentially reversible disorder that involves an acute decline in attention, awareness, and cognition, often triggered by medical conditions, medications, or substance withdrawal.

Neurochemical and Neuroanatomical Basis of Delirium:

- Decreased Ach
- The major neuroanatomical area involved is the reticular formation which regulates attention and arousal
- The major pathway implicated is the dorsal tegmental pathway, which projects from the mesencephalic reticular formation to the tectum and thalamus.

Clinical Features and Clinical Diagnosis of Delirium:

Delirium is characterized by an acute onset of altered consciousness and cognitive impairment, with a hallmark feature being impaired attention. It often presents with abnormalities of mood, perception, and behavior.

Common clinical features include:

- Fluctuating course : Symptoms tend to wax and wane, and the patient may have periods of lucidity followed by severe cognitive disorganization.
- Impaired attention : Patients have difficulty focusing, sustaining, or shifting attention.
- Cognitive dysfunctions: These include disorientation (especially to time and place), memory impairment, and disturbances in language and perception.
- Motor disturbances: Hyperactive or Hypoactive
- Mood and behavioral changes: Abnormalities in mood (such as agitation or lethargy) and perception (hallucinations or delusions) are frequent.
- Sleep disturbances: Decreased sleep, reversal of sleep-wake cycle.
- Neurological signs: Tremors, asterixis, nystagmus, and incoordination may be observed.
- Sundowning - symptoms worsen at night.
- Course: Acute : hours to days Persistent : weeks to months
- Acute : hours to days
- Persistent : weeks to months
- Acute : hours to days

- Persistent : weeks to months

Diagnosis is primarily clinical , with a focus on history and mental status examination.

- The Mini-Mental State Examination (MMSE) can be utilized to assess cognitive impairment and establish a baseline to monitor the patient's clinical course.
- The Confusion Assessment Method (CAM) is used for diagnosing delirium that involves the presence of
 - Acute onset and fluctuating course,
 - Inattention, and
 - Disorganized thinking
 - Altered level of consciousness.

A positive diagnosis is made if features 1 and 2 are present, along with either feature 3 or feature 4.

- Additionally, an EEG may show diffuse slowing, which helps distinguish delirium from psychiatric disorders like depression or psychosis.

Dementia (Option A): While dementia can present with cognitive deficits, it is a chronic, progressive condition with a gradual decline in memory, thinking, and problem-solving. Unlike delirium, dementia does not have an acute onset or fluctuating course.

Feature	Delirium	Dementia
Onset	Acute (hours to days)	Insidious, progressive (months to years)
Duration	Short-term (hours to weeks)	Long-term (months to years)
Course	Fluctuating, worse at night (sundowning)	Steady, progressive deterioration
Consciousness	Impaired, fluctuating levels of alertness	Clear until late stages
Attention	Markedly impaired	Normal until later stages
Memory Impairment	Short-term memory is severely impaired	Short-term memory loss in the early stages, long-term memory affected later
Hallucinations/ Delusions	Common, often visual or tactile	Uncommon in early stages, may occur in advanced stages
Sleep-Wake Cycle	Disrupted (e.g., insomnia, daytime drowsiness, reversal of sleep cycle)	Sleep disturbances occur late in the disease

Feature	Delirium	Dementia
EEG Findings	Abnormal, showing diffuse slowing	Normal in early stages; slowing in late stages

Schizophrenia (Option B): Schizophrenia is a chronic psychiatric disorder characterized by hallucinations, delusions, disorganized thinking, and negative symptoms such as flat affect.

- It does not present with an acute onset or fluctuating course of cognitive impairment, and attention is generally preserved outside of psychotic episodes.

Major depressive disorder with psychotic features (Option D): This condition can include psychosis, such as hallucinations or delusions, but the primary feature is a depressed mood lasting for at least two weeks.

- The cognitive disturbances in depression are more static and do not exhibit the fluctuating, acute course seen in delirium.

Solution for Question 17:

Correct Answer: B) Fluctuating levels of consciousness

Explanation:

The patient's symptoms of impaired consciousness, confusion, inattention, disorganized thinking, and visual hallucinations—with symptoms worsening at night—are highly suggestive of delirium.

Fluctuating levels of consciousness are a hallmark of delirium. Patients often exhibit periods of confusion and altered awareness that can change throughout the day. It develops quickly and has a fluctuating course with lucid intervals (waxing and waning)

- Delirium is an acute decline in both the level of consciousness and cognition with particular impairment in attention.
- A life-threatening yet potentially reversible disorder of the central nervous system (CNS), delirium often involves perceptual disturbances, abnormal psychomotor activity, and sleep cycle impairment.
- Abnormalities of mood, perception, and behavior are common psychiatric symptoms.
- Tremor, asterixis, nystagmus, incoordination, and urinary incontinence are common neurologic symptoms.

- Classically, delirium has a sudden onset (hours or days), a brief and fluctuating course, and rapid improvement when we can identify the cause and eliminate it, but each of these characteristic features can vary in individual patients.

Feature	Delirium	Dementia
Onset	Acute (hours to days)	Insidious, progressive (months to years)
Duration	Short-term (hours to weeks)	Long-term (months to years)
Course	Fluctuating, worse at night (sundowning)	Steady, progressive deterioration
Consciousness	Impaired, fluctuating levels of alertness	Clear until late stages
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Memory Impairment	Short-term memory is severely impaired	Short-term memory loss in the early stages, long-term memory affected later
Hallucinations/Delusions	Common, often visual or tactile	Uncommon in early stages, may occur in advanced stages
Sleep-Wake Cycle	Disrupted (e.g., insomnia, daytime drowsiness, reversal of sleep cycle)	Sleep disturbances occur late in the disease
EEG Findings	Abnormal, showing diffuse slowing	Normal in early stages; slowing in late stages

Predisposing factors:

- Elderly age
- History of substance abuse or alcohol withdrawal (Delirium Tremens)
- Association with a chronic medical illness
- History of surgical illness
- Post-operative period
- Sensory deprivation
- Poly-pharmacy

Etiologies:

Drugs	Alcohol Sedatives Antidepressants
Endocrinal	Hypothyroidism/hyperthyroidism Hypopituitarism/hyperpituitarism
Liver	Hepatic encephalopathy
Infections	Septicemia Pneumonia
Renal	Uremic Encephalopathy
Intracranial	Epilepsy Tumors Injury
Others	Heat Sleep deprivation
Metabolic	Hypoxia Hypoglycemia Electrolyte imbalance

Type of Delirium	Causes
Hypoactive Delirium	Electrolyte imbalances, hypoglycemia, hepatic encephalopathy, uremia Urinary tract infections, pneumonia, sepsis Stroke, traumatic brain injury, dementia Sedatives (benzodiazepines, opioids), anticholinergics, psychotropic medications Heart failure, respiratory failure, dehydration
Hyperactive Delirium	Alcohol withdrawal, withdrawal from benzodiazepines or sedatives Stimulants (cocaine, amphetamines), hallucinogens (LSD, PCP) Hyperthyroidism, fever or infection (sepsis) Traumatic brain injury, encephalitis Anticholinergics, stimulants
Mixed Delirium	Acute severe infections (pneumonia, sepsis), post-surgical states Mixed electrolyte imbalances, liver or renal failure Combination of sedatives and stimulants. Dementia with acute exacerbation, delirium superimposed on chronic cognitive decline

Gradual onset of cognitive decline (Option A): This feature is more typical of dementia rather than delirium, which usually presents with a more acute change in mental status.

Stable and coherent thought processes (Option C): This is characteristic of normal cognition or dementia, not delirium, where thought processes are often disorganized and incoherent.

Fixed and focused attention (Option D): In delirium, attention is typically impaired, making this option incorrect. Patients may have difficulty sustaining attention or may become easily distracted.

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